SMITHIAN FOLD THEORY V3

From Fold to Mathematics

An Exact, Parameter-Free and Machine-Closed Derivation of Mathematical Foundations from Smithian Fold Theory

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> ***Publication control.*** *This is a new versioned successor. It does not rewrite the preceding paper, its DOI record, historical receipts or evidence. No push, upload, DOI action, release or publication is authorised until Maria Smith explicitly confirms the exact final files and hashes.*
>
> ***Lean verification integration.*** *The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.*

Abstract

This successor presents the complete current `mathematics` paper surface: 341 live model-admitted claims, 101,888 generated candidates, 341 unique survivors and 1,364 passed controls. It integrates 18 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

The field reconstruction covers exact arithmetic and number structure; algebraic extensions; combinatorics; graphs, networks and matroids; linear, multilinear and tensor structure; algebraic systems; order, lattices and domains; geometry; topology; calculus correspondence; analysis correspondence; equation structures; measure and integration; deterministic-support probability and statistics; optimisation and operations research; dynamical systems; logic and foundations; category, type and compositional structures; numerical mathematics; symbolic and constructive mathematics; cross-disciplinary interfaces; the complete validation vector; and one-owner handoffs. The previously admitted Smithian Fold Scientific Calculator remains the accessible translation surface and does not become an alternate admission route.

The proof domain admits no conventional axiom as an SFT premise, no fitted or free parameter, no semantic numerical zero, negative proof magnitude, irrational or imaginary proof scalar, floating proof quantity, completed infinity, ungenerated continuum or ontic randomness. Displayed `0` denotes structural absence; opposition is typed orientation; non-rational conventional objects enter only through exact constructions or certified enclosures. The Validation Grand Lock partitions all 305 pre-validation receipts exactly once across 22 families and preserves all 305 favourable, adverse,

absent, unresolved and boundary records. The final six handoffs bring the branch to 323/323 without transferring empirical ownership to Mathematics or claiming permanent closure.

Version 1.5 preserves the complete version 1.4 paper and its foundational, prior-corpus and calculator derivations, then executes the roadmap that version 1.4 registered. No application selected these laws. No engine or protected-verifier source was modified. Every later extension must enter as a new registered question and pass the same public protocol.

Keywords: Smithian Fold Theory; complete-field mathematics; exact arithmetic; algebra; combinatorics; graph theory; geometry; topology; calculus; analysis; probability; optimisation; dynamics; logic; category theory; numerical mathematics; symbolic mathematics; scientific calculator; computational proof; open science.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 341 current claim(s) in its declared branch ownership boundary; 18 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** `SFT-MATH-OAI26-SPHERE-PACKING-001` - SFT reconstruction of sharp Cohn-Elkies sphere-packing asymptotics; `SFT-MATH-OAI26-BINARY-CODE-MRRW-002` - SFT reconstruction of the strict binary-code MRRW improvement; `SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-003` - SFT reconstruction of the strict spherical-code hierarchy; `SFT-MATH-OAI26-NONSOFIC-GROUP-004` - SFT reconstruction of a finitely presented non-sofic group; `SFT-MATH-OAI26-CONNES-RIGIDITY-005` - SFT reconstruction of the Connes-rigidity counterexample family; `SFT-MATH-OAI26-EHRHART-VOLUME-006` - SFT reconstruction of the sharp Ehrhart volume inequality; `SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-007` - SFT reconstruction of Erdos problem 183; `SFT-MATH-OAI26-COMPACTNESS-008` - SFT reconstruction of the quantitative compactness counterexample; `SFT-MATH-OAI26-TWO-DEGENERATE-009` - SFT reconstruction of the two-degenerate extremal counterexample; `SFT-MATH-OAI26-SPHERE-PACKING-VALIDITY-001` - SFT disproof of source-artifact validity: PackingBounds.sharpFullCohnElkiesManuscriptConclusions; `SFT-MATH-OAI26-BINARY-CODE-MRRW-VALIDITY-002` - SFT disproof of source-artifact validity: MetricCodes.Johnson.binaryRate_lt_mrrw; `SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-VALIDITY-003` - SFT disproof of source-artifact validity: MetricCodes.Spherical.HigherHierarchy.strict_hierarchy; `SFT-MATH-OAI26-NONSOFIC-GROUP-VALIDITY-004` - SFT disproof of source-artifact validity: SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group; `SFT-MATH-OAI26-CONNES-RIGIDITY-VALIDITY-005` - SFT disproof of source-artifact validity: ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors; `SFT-MATH-OAI26-EHRHART-VOLUME-VALIDITY-006` - SFT disproof of source-artifact validity: Ehrhart.Volume.ehrhart_volume_inequality_for_sets; `SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-VALIDITY-007` - SFT disproof of source-artifact validity: ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit; `SFT-MATH-OAI26-COMPACTNESS-VALIDITY-008` - SFT disproof of source-artifact validity: CompactnessConjecture.quantitativeCompactnessCounterexample; `SFT-MATH-OAI26-TWO-DEGENERATE-VALIDITY-009` - SFT disproof of source-artifact validity: TwoDegenerateGraphs.twoDegenerateExtremalCounterexample.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Results first: what this branch changes

Headline result	Exact executed result	Scientific consequence
Complete-field Mathematics	323/323 frozen obligations are receipt-backed across 24 families, with 97,280 enumerated candidates, 323 unique survivors and 1,292 passed controls.	The former roadmap is now an executed, independently replayable mathematical corpus rather than a statement of future intent.
Arithmetic without hidden continuum primitives	Exact positive finite generation, normalised fractions, ordered gaps, interval enclosures and structural absence replace semantic zero, negative proof magnitude and floating proof values.	Ordinary notation remains available at the interface, but every proof value retains an exact Fold construction or certified enclosure.
Full structural span	Combinatorics, graphs, linear and algebraic structures, order, geometry, topology, calculus, analysis, equations, measure, probability, optimisation, dynamics, logic, category, numerical and symbolic mathematics are separately owned and dependency ordered.	No familiar discipline name imports its conventional axioms; each law has its own generated grammar, eliminations, survivor and certificate.
Numerical and symbolic computation	Exact rounding custody, interval propagation, stability, conditioning, convergence, root isolation, solvers, quadrature, recurrences, canonical syntax, rewrite provenance, transforms and constructive certificates are admitted.	Numerical and symbolic tools can be audited without hiding floating, signed, imaginary or oracle-selected proof values.
Validation Grand Lock	All 305 pre-validation claims are partitioned exactly once across 22 families, with 305 adverse and boundary records retained.	Completeness is a reproducible reconciliation claim, not a label applied after favourable results.
Dated, extension-open completion	Six final handoffs distinguish mathematical structure from downstream meaning, measurement, conventional correspondence and engineering use.	Mathematics is 100.0% complete to this frozen census while remaining open to correction, falsification and lawful new discoveries.
Public proof surface	Every claim exposes its registration, candidate census, decisions, controls, certificate, external record, receipt and executable entry point.	Independent researchers can reproduce or challenge the work without credential, paywall or institutional permission selecting the outcome.

Current status, evidence language and reader map

Status field	Current position
Branch and completion boundary	Mathematics: 323/323 frozen obligations; 323 live claims. This is dated, versioned completion and remains open to lawful extension.
Formal status	Every live claim represented here has a current model-admitted receipt. Formal admission does not by itself imply empirical confirmation.
Empirical status	Claim-specific. Blind, non-blind, development-observed, holdout, adverse, unresolved, unavailable and formal-only distinctions remain exactly as recorded in the claim ledger and evidence packages.
Chronology	Claim-specific registration, seal, custody and observation order governs. Later evidence does not retroactively become an earlier prediction.
Publication status	Version 1.5.0 is a final publication candidate awaiting Maria Smith's approval. It is not yet deposited.
Existing record lineage	Current record DOI `10.5281/zenodo.21627708`; concept DOI `10.5281/zenodo.21516145`. Publication is permitted only through the existing record's new-version route.
What is not claimed	Permanent closure of science, universal empirical confirmation, transfer of another branch's ownership or permission to replace an adverse or unresolved record.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the specified formal, computational or empirical test; its kind must be named.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications. None silently substitutes for another.
Foundational closure	Completion of the registered foundation only. It does not imply field-wide closure.
Field-wide closure	Completion of the frozen or registered dated field census named in this paper.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may be added by a later version without rewriting existing receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relationships; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The cited repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. Those files remain authoritative where prose abbreviates their display.

Editorial change control

This version may improve expression, order, typography and navigation. It may not change scientific meaning, scope, chronology, evidence class, ownership, claim status or machine identity. Any apparent conflict between authoritative records must be traced and either resolved by explicit supersession or recorded for Maria Smith; prose alone cannot manufacture agreement.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Mathematics, this standard means that familiarity, elegance and textbook authority cannot admit a law. A proof must expose its carrier, generated alternatives, exact witness, generality certificate and falsification boundary. The calculator applies the same rule to everyday use: a friendly decimal display may communicate an exact result, but it cannot replace or weaken the underlying certificate.

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Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through https://discord.gg/ucwGryVxGr, or inspect the public project at https://github.com/MettaMazza.

1. Central scientific claim

The exact claim of this version is:

> *Within the frozen SFT V3 Mathematics census dated 29 July 2026, all 323 obligations in all 24 registered families are individually admitted through the untouched engine, independently reconstructed, controlled, observed after target-free registration and exactly reconciled. The current closed count is 323, the open count is structural absence, and completion remains open to lawful versioned extension.*

The frozen census identity is `sha256:10e2399fbecd99ae9106436660edbc8c0f84fad6b9dde75139099d202d58c36c`. The final reconciliation identity is `sha256:526c6d78a3ff77c21c6d4dd71c266499f5b234a7763cb223e5d009fa9f101c64`. The branch contains 97,280 candidate decisions, 323 unique survivors and 1,292 passing controls. Every registration has an empty axiom list and an empty free-parameter list. Each dependency chain reaches `SFT-ROOT-THERE-IS-NO-NOTHING` through actual admitted receipts rather than through narrative citation.

This claim is bounded precisely. It does not assert a completed infinite universe, convert structural absence into a numerical object, import irrational or imaginary proof scalars, or assign measured physical meaning to a mathematical structure. Completion means every question in the frozen dated census has passed the full protocol. A new question, counterexample or stronger certificate remains admissible only as a visible versioned extension.

2. Standalone dependency foundation

This paper is standalone in exposition but not dependency-free in the scientific census. The prior Foundation branch supplies sixteen exact receipts, including operational occurrence, structural One, complete positive finite count, exact held/whole part coordinates, the minimal Fold, cross-partition equivalence, Fold assembly, finite form grammar, canonical identity, exact cast/Fold/Take, half-One, Fold dynamics, extended primitive-map uniqueness, root-bound replay, admission enforcement and the one-way measurement boundary. Those are cited as admitted dependencies, not copied assumptions. The published Foundation paper remains a separate work and is not modified by this paper.

The sole root theorem remains *there is no nothing*. Every dependency path terminates at its admitted receipt. Each Mathematics registration contains empty axiom and free-parameter tuples. Prior SFT versions, conventional theorems and correspondence names are excluded from derivational source manifests.

3. Exact mathematical constitution

The admitted domain contains structural One, complete positive finite generation traces, exact positive held/whole parts, canonical labels, fibres, branches, words, pair cells, relations, forms and proof traces. The domain does not contain semantic numerical zero, a negative quantity, an irrational or imaginary proof value, floating proof arithmetic, a completed infinite set or an ungenerated continuum. An empty selection, identity path or no-transition duration is the structural empty-One form. Opposed direction is a distinct held orientation with its positive remainder trace.

Python host integers, Boolean checks, empty containers and process status codes are implementation mechanics. They do not become SFT proof objects. The scientific sources explicitly return structural forms or exact positive part relations at the boundary where a conventional implementation might otherwise introduce zero, negative or floating values. The admission engine independently rejects registered axioms, free parameters, missing dependencies, incomplete censuses, multiple survivors, failed controls and source drift.

4. What forced, closed and independently validated mean

A declaration is not forced because it is elegant or familiar. Every claim specifies a product grammar of structural questions. Each coordinate supplies at least one explicitly rejected alternative and exactly one all-preserving alternative. The complete Cartesian product is generated in canonical order. Candidate identities, exact forms and trace hashes are recorded. Every candidate receives one survival decision, elimination reason and proof hash. Exactly one candidate must survive.

Closure then requires more than the finite run. Minimality shows that replacing any preserving coordinate loses a registered requirement and that no extra rule remains. Named-shape uniqueness shows that the full product contains only one all-preserving coordinate. A claim-specific structural One base and successor certificate extends the classification to every generated finite depth. This is depth-independent closure over finite generation, not a completed infinite object.

Four controls are mandatory. A false premise must be rejected; a changed official source identity must be detected; a missing, duplicated or additional survivor must fail; and an excluded object or answer-producing model must halt at the boundary. After the derivation is sealed, a separate Python process whose file hash is not part of the scientific implementation regenerates the literal candidate product, decision vector, unique survivor, closure flags and controls. Only the shared admission engine can add the receipt to the census.

5. Foundational and calculator execution preserved from version 1.4

> ***Version 1.5 scope note.*** *The twenty-seven rows in this preserved section are the original foundational and calculator evidence surface, not the current branch denominator. The controlling complete-field census is 323/323 and is documented claim by claim in Section 40.*

Order	Claim	Candidate structures	Dimensions	Closure
1	`SFT-MATH-EXACT-ARITHMETIC-001`	256	8	depth-independent
2	`SFT-MATH-DISCRETE-001`	256	8	depth-independent

Order	Claim	Candidate structures	Dimensions	Closure
3	SFT-MATH-COMBINATORICS-001	256	8	depth-independent
4	SFT-MATH-GRAPH-NETWORK-001	512	9	depth-independent
5	SFT-MATH-ALGEBRA-001	512	9	depth-independent
6	SFT-MATH-ORDER-LATTICE-001	512	9	depth-independent
7	SFT-MATH-GEOMETRY-TOPOLOGY-001	1,024	10	depth-independent
8	SFT-MATH-PROBABILITY-STATISTICS-001	512	9	depth-independent
9	SFT-MATH-OPTIMIZATION-001	512	9	depth-independent
10	SFT-MATH-DYNAMICAL-SYSTEMS-001	1,024	10	depth-independent
11	SFT-MATH-LOGIC-PROOF-001	1,024	10	depth-independent
12	SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	1,024	10	depth-independent
13	SFT-MATH-EXACT-RELATIONS-002	256	8	depth-independent
14	SFT-MATH-ORBIT-NUMBER-THEORY-002	256	8	depth-independent
15	SFT-MATH-LIMIT-CONTINUUM-002	256	8	depth-independent
16	SFT-MATH-ALGEBRAIC-BALANCE-002	256	8	depth-independent
17	SFT-MATH-BOUNDED-N-BODY-002	256	8	depth-independent
18	SFT-MATH-FLOORED-FLUID-REGULARITY-002	256	8	depth-independent
19	SFT-MATH-PRIME-PAIR-CENSUS-002	256	8	depth-independent
20	SFT-MATH-RIEMANN-MIRROR-002	256	8	depth-independent
21	SFT-MATH-COLLATZ-FINITE-CENSUS-002	256	8	depth-independent
22	SFT-MATH-SELF-SIMILAR-CONVERGENCE-002	256	8	depth-independent
23	SFT-MATH-SCIENTIFIC-CALCULATOR-003	1,024	10	depth-independent; superseded adverse evidence
24	SFT-MATH-SCIENTIFIC-CALCULATOR-004	1,024	10	depth-independent; current
25	SFT-MATH-SCIENTIFIC-CALCULATOR-005	8,192	13	depth-independent; current
26	SFT-MATH-SCIENTIFIC-CALCULATOR-006	1,024	10	depth-independent; current
27	SFT-MATH-SCIENTIFIC-CALCULATOR-007	256	8	depth-independent; current

The version 1.3 Mathematics evidence surface preserves **21,504** generated structures across twenty-seven registered records. The twenty-two foundational claims contribute 9,984; the calculator lineage contributes 11,520. Claim 003 is retained as the exact historical record superseded after the endpoint-parity defect was found; it is not counted as a current law. Claims 004 through 007 and the original twenty-two claims form the twenty-six current non-superseded results.

The finite product count reports representation classes executed by the registered claim grammars. It is not a count of every mathematical object producible by the laws. Generality is carried by the induction certificate stated for each claim below.

6. Derivation 1: Exact arithmetic and generated number structure

Claim identity: `SFT-MATH-EXACT-ARITHMETIC-001`

6.1 Question and necessity

Arithmetic is required before later branches can count arrangements, compare structures or assign exact support weights, but importing a conventional number field would violate the clean-room boundary.

The exact theorem statement is:

> *Every admitted arithmetic relation on generated positive finite traces and exact parts has one parameter-free structural kernel: disjoint junction for addition, complete pair-cell refinement for product, finite repeated composition for powers, common-refinement pairing for exact quotient and comparison, and held orientation for unmatched remainder; no semantic zero, negative, irrational, floating or completed-infinite value enters the law.*

Admitted dependencies: `SFT-FOUNDATION-COUNT-001`, `SFT-FOUNDATION-PART-001`, `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`.

6.2 Generated grammar and exact boundary

Count supplies complete positive traces; Part supplies held/whole coordinates; Part Equivalence supplies common refinement and bijection. Exhausting the eight representation choices leaves one kernel that preserves every source identity without introducing a scale or forbidden value.

Generation rule: Generate the complete product of trace coverage, junction, pair-cell product, exact quotient, comparison, remainder orientation, finite generality and extra-rule status.

Exact grammar boundary: All arithmetic kernels constructible from admitted positive finite traces, exact held/whole parts, common refinement, canonical form identity and no imported scalar field.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:e5594c9df6b591e6d7bac2b5afb86bcb32a41d450fb06727426740b982a57500`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-EXACT-ARITHMETIC-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`coverage`	`partial-trace`	A partial trace omits an admitted generated occurrence.	`complete-trace` - The complete trace retains every generated occurrence once.
`junction`	`overlap-merge`	Overlap repeats a shared occurrence and loses disjoint provenance.	`disjoint-junction` - Disjoint junction preserves both complete positive traces.
`product`	`repeated-unlabelled`	Unlabelled repetition erases the two source coordinates.	`pair-cell-refinement` - Each generated left label is paired once with each generated right label.
`quotient`	`rounded-measure`	A rounded or decimal value is measured rather than exact derivational evidence.	`common-refinement-pairing` - A complete common refinement and bijection witnesses the exact part relation.
`comparison`	`borrowed-scalar-order`	A borrowed scalar order imports the answer-producing number model.	`complete-trace-pairing` - Pairing identifies equality and a held unmatched trace identifies orientation.
`difference`	`signed-magnitude`	A signed magnitude installs forbidden negative quantity.	`held-oriented-remainder` - A distinct held label records which trace retains the unmatched positive remainder.
`generality`	`bounded-answer-table`	A finite answer table has no depth-independent arithmetic certificate.	`base-successor-induction` - The One base and generated successor preserve each operation at every finite depth.
`addition`	`has-extra-rule`	An extra scale, constant or field is not supplied by the dependencies.	`no-extra-rule` - The kernel uses only admitted traces, parts, refinement, pairing and held labels.

6.3 Unique survivor, minimality and laws

> *The exact arithmetic kernel is complete positive-trace coverage with disjoint junction, pair-cell product, common-refinement quotient, complete pairing comparison, held-oriented remainder, base/successor generality and no extra rule.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- disjoint junction is associative on mutually disjoint generated traces.
- pair-cell product distributes over disjoint junction by generated cell provenance.
- pair-cell product composition is associative after canonical path flattening.
- exact quotient exists only with a complete refinement-pairing witness.
- comparison returns same-form or a held positive remainder orientation.

6.4 Operational witnesses and unfavourable checks

- `junction-associativity` - (a joined b) joined c and a joined (b joined c) retain the same canonical trace. Result: `PASS`.
- `pair-cell-cardinality` - Two generated cells paired with three generated cells produce the complete six labelled pair cells. Result: `PASS`.
- `product-associativity` - Canonical flattened pair paths are independent of binary grouping. Result: `PASS`.
- `exact-pairing` - Equal complete traces admit one-to-one and onto pairing; unequal traces do not. Result: `PASS`.
- `held-remainder` - A larger trace reports a positive held remainder instead of a negative value. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial trace omits an admitted generated occurrence. Result: `PASS`; receipt `sha256:33e3d5d241540fd337807b34e03db365ce48e9e3d0ea4d102d153a58df8199c6`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not a value; negative magnitude is replaced by held orientation; irrational, imaginary and floating proof values are excluded; completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:19138ef235a0e43f75285e80856edccff9d19b3aced53986c83f5de9fa2f4fca`.

6.5 Depth-independent closure

Base: The structural One supplies the first nonempty trace and exact self-whole.

Successor: Appending one fresh generated occurrence preserves complete trace identity; junction appends it once, pair-cell refinement appends one labelled cell per opposite trace element, and pairing either extends with one mate or records the unmatched held remainder.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

6.6 Meaning and scientific consequence

This result changes the role of arithmetic. A number is not admitted as an unexplained occupant of a pre-existing field. Exact positive magnitude is the identity of a complete generated trace. Addition is the preservation of two disjoint traces in their junction; product is the complete provenance-retaining pair-cell refinement. An exact quotient is not a division oracle but the existence of a complete pairing over a common refinement. Comparison never requires a negative result: equality is complete pairing and difference is a held orientation together with the unmatched positive trace.

6.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: positive integers, rational numbers, addition, multiplication, division, order, signed difference. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

6.8 Exact exclusions and limitations

- semantic numerical zero is not a value.
- negative magnitude is replaced by held orientation.
- irrational, imaginary and floating proof values are excluded.
- completed infinity and an ungenerated continuum are excluded.

This law closes exact generated finite arithmetic and positive rational part relations. It does not admit a completed infinite set, continuum, irrational field or negative proof magnitude. Conventional notation may describe the result only in correspondence.

6.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:db8d9a5838f2f84cd04cd3425e0faa5232ec8afb6461f1a7fe62ab9f6263246f`.
- Derivation seal: `sha256:1254abf40a91886525ad8a846896b75a1023fd785907f233bcf817db2c744dc4`.
- Independent implementation: `sha256:9d0af694427b46f2b09ed218d83d0d87f30b16601b03c3694ac71738254bf164`.
- Independent certificate: `sha256:73b24906c1005bd7f42f36000f396bb5ec88f633a4a69b6159d5cc349bb3be2a`.
- External validation: `sha256:9a05361b10ccce837c8a32def2b24dd0ec714cb2f45d397a9e6b72fedec306a7`.
- Engine receipt: `sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-EXACT-ARITHMETIC-001-28252bae62373d86.json`.

7. Derivation 2: Generated discrete objects, relations and induction

Claim identity: `SFT-MATH-DISCRETE-001`

7.1 Question and necessity

Combinatorics, graphs, algebra, logic and computation require exact collections and relations, but an imported set-theoretic universe would add axioms and completed infinities outside the SFT domain.

The exact theorem statement is:

> *Every admitted discrete object is a complete canonical generated finite collection or held selection; membership is retained occurrence, relations are held selections of complete pair-cell support, maps are total single-valued relations, sequences retain generated order, and general laws close by structural One/base-successor induction.*

Admitted dependencies: SFT-FOUNDATION-COUNT-001, SFT-FOUNDATION-FORM-GRAMMAR-001, SFT-FOUNDATION-FORM-ENFORCEMENT-001, SFT-MATH-EXACT-ARITHMETIC-001.

7.2 Generated grammar and exact boundary

Canonical form enforcement supplies identity; count supplies complete finite traces; exact arithmetic supplies lawful trace composition. Product enumeration of the eight representational questions forces the collection-selection-relation-map-induction kernel.

Generation rule: Generate the complete product of collection coverage, canonical identity, membership, relation support, map law, sequence order, induction closure and extra-constructor status.

Exact grammar boundary: All discrete structures generated from finite canonical One/Fold forms, exact positive traces, held selections, pair-cell relations and successor construction.

The complete product contains 256 candidates. Its completeness-certificate identity is sha256:0103778e8716a64ec9ed8d8b243af9af9bc7fed90360339b4be989a03d263302. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in claims/SFT-MATH-DISCRETE-001/candidate_census.json.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
collection	partial-or-duplicated	Omission or duplication changes the generated collection trace.	complete-unique-trace - Every generated member occurs once in the canonical collection trace.
identity	presentation-alias	Presentation names can merge distinct forms or split one form.	canonical-form-identity - Canonical construction traces supply exact member identity.
membership	unrecorded-assertion	An unrecorded assertion has no source-bound occurrence witness.	retained-occurrence - Membership retains the canonical member occurrence within the source trace.
relation	free-pair-list	A free pair list has no complete source-product boundary.	held-pair-cell-selection - A relation holds a generated selection from complete source pair-cell support.
map	partial-or-multivalued	A missing or repeated image does not define one image for every source member.	total-single-valued - Every source member has exactly one retained target mate.
sequence	unordered-carrier	An unordered carrier erases the generated predecessor relation.	held-generation-order - The canonical trace retains each predecessor-successor position.
induction	sampled-depths	Sampled depths do not establish the successor step.	One-base-successor - The One base and freshness-preserving successor cover every generated finite trace.
addition	extra-constructor	An extra constructor is not supplied by the Foundation grammar.	no-extra-constructor - All objects decompose into admitted forms, selections, pair cells and successors.

7.3 Unique survivor, minimality and laws

> *The discrete kernel is complete unique canonical collections with retained membership, pair-cell-selected relations, total single-valued maps, held sequence order, One/base-successor induction and no extra constructor.*

Shared claim-record clause MATH-S001 applies. The full, unchanged clause is preserved in the shared-clause appendix.

- canonical membership is decidable for every generated finite collection.
- held selection never introduces an element outside its source collection.
- relation complement is a held complementary selection of the same complete pair support.
- map composition preserves total single-valuedness when interfaces agree.
- structural induction covers every generated finite collection trace.

7.4 Operational witnesses and unfavourable checks

- unique-collection - A generated collection retains each canonical member once. Result: PASS.
- held-membership - A held selection contains only members of its registered source. Result: PASS.
- empty-One-selection - A held-empty selection returns the empty One form, not numerical zero. Result: PASS.
- relation-boundary - Every retained relation pair belongs to the complete generated pair-cell support. Result: PASS.

- `total-map` - The example relation gives every source form exactly one image. Result: `PASS`.
- `fresh-successor` - A successor appends exactly one fresh canonical form. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or duplication changes the generated collection trace. Result: `PASS`; receipt `sha256:bfcb72c0f0a5cfdb140547bb18d237268d4ae9231dbf57f6be92112190ed979a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no completed infinite set; no ungenerated universal collection; no semantic numerical zero for the empty selection; no presentation alias may replace canonical form identity Result: `PASS`; receipt `sha256:fdcae55e3e8ce516575f15e1d490505002d0dffbdc3dc372c54ed70cd69b236e`.

7.5 Depth-independent closure

Base: The structural One is the first canonical generated member and the empty One marks a held-empty selection without numerical zero.

Successor: Given a complete unique canonical collection, append one fresh canonical form once; membership, relation pair support, map-image obligation and predecessor order extend by their local generated clauses.

Shared claim-record clause `MATH-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

7.6 Meaning and scientific consequence

The theorem reconstructs the working core normally supplied by an ambient set universe. A collection has a finite generated boundary, canonical member identities and no unrestricted comprehension. Membership is a retained occurrence; a relation is a held selection from complete pair support; and a map is a relation that passes totality and single-valuedness. This makes every later discrete object auditable at its source.

7.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: finite set, subset, membership, relation, function, sequence, structural induction. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

7.8 Exact exclusions and limitations

- no completed infinite set.
- no ungenerated universal collection.
- no semantic numerical zero for the empty selection.
- no presentation alias may replace canonical form identity.

The theorem concerns generated finite objects. It does not install an ungenerated universe, completed infinite set or unrestricted comprehension principle.

7.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:41a82043fd37d0c054d490f9c426ad37c23d6bb8ea2e4f0e9a5af6fb8fe30285`.

- Derivation seal: `sha256:1d3b73c941e235c5ffe473ccafa9cd59b800ddd33d43e4feea7a6821ed8f45c9`.
- Independent implementation: `sha256:c808241fc28e1fa7bfbbea0e67fc594eab9b957575534f2f0b88af57def8dbae`.
- Independent certificate: `sha256:2f9da4efdc67b6f08074be8d3ddc47595444a9300fd00e507fb04c228686f44b`.
- External validation: `sha256:5dad446dc09d74caf9325f27b314d7f886275164df426cd561a687d9283cd109`.
- Engine receipt: `sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-DISCRETE-001-632de46c995329ed.json`.

8. Derivation 3: Exact finite combinatorial generation

Claim identity: `SFT-MATH-COMBINATORICS-001`

8.1 Question and necessity

Counting arrangements by quoting familiar formulas would import their answers. The generated family itself must be primary so every count, symmetry reduction and recurrence remains inspectable.

The exact theorem statement is:

> *Every admitted finite combinatorial family is generated by an explicit held-choice recurrence over a complete canonical carrier, preserves construction order and provenance, removes duplicates only by canonical form identity, and counts the resulting complete trace; selections include the structural empty One form rather than numerical zero.*

Admitted dependencies: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`.

8.2 Generated grammar and exact boundary

Discrete canonical collections provide the carrier and held selection; Fold words provide exact choice traces; arithmetic counts the complete result. Exhausting the eight assembly choices forces explicit held-choice recurrence and provenance-preserving class accounting.

Generation rule: Generate the complete product of carrier coverage, choice rule, provenance, duplication policy, symmetry identity, class accounting, recurrence closure and extra-formula status.

Exact grammar boundary: All arrangement, selection and finite classification families produced from a registered canonical finite carrier by held choice, remainder and successor recurrence.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b5a4fe7ce476fb3f349f0ac30a1ccfbd35c2637505be5ef9c4022d90e65a403b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-COMBINATORICS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`partial-carrier`	A partial carrier silently omits possible generated choices.	`complete-canonical-carrier` - Every distinct generated source member is present once.
`choice`	`sampled-choices`	Sampling does not exhaust the registered choice positions.	`held-choice-recurrence` - Each available member is held once at the next position and recursion continues on the exact remainder.
`provenance`	`erased-origin`	Erasing origin prevents reconstruction and duplicate diagnosis.	`complete-choice-trace` - Every result retains the ordered held-choice construction trace.
`duplicates`	`presentation-deduplication`	Presentation equality can merge structurally distinct results.	`canonical-identity-deduplication` - Only identical canonical construction traces are identified.
`symmetry`	`assumed-symmetry`	An assumed quotient may remove distinguishable held labels.	`generated-bijection-witness` - A declared label-preserving bijection is required before forms share a symmetry class.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`accounting`	`signed-inclusion-subtraction`	Signed subtraction imports negative quantities.	`disjoint-provenance-classes` - Each generated form is assigned once to a held disjoint class before counts are joined.
`generality`	`finite-size-table`	A table of small sizes does not force the next recurrence.	`carrier-successor-recurrence` - Adding one fresh carrier member partitions new forms by whether and where it is held.
`addition`	`imported-formula`	A stored factorial, binomial or inclusion formula selects the conventional answer.	`no-imported-formula` - Counts are identities of the complete generated output traces.

8.3 Unique survivor, minimality and laws

> *The combinatorial kernel is complete canonical carrier generation by held-choice recurrence with complete choice provenance, canonical deduplication, witnessed symmetry, disjoint class accounting, successor generality and no imported formula.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- every arrangement holds each carrier member exactly once.
- every selection is produced once by the fresh-member absent/held recurrence.
- canonical trace equality is the only duplicate-elimination rule.
- disjoint class ledgers conserve the complete generated family.
- enumerated count is the identity of the complete output trace.

8.4 Operational witnesses and unfavourable checks

- `arrangement-completeness` - Three distinct held labels generate six unique ordered arrangements. Result: `PASS`.
- `arrangement-coverage` - Every arrangement holds every carrier label exactly once. Result: `PASS`.
- `selection-recurrence` - Three held labels generate the structural empty One and every absent/held combination once. Result: `PASS`.
- `disjoint-accounting` - Parity-named witness classes partition the carrier without overlap or omission. Result: `PASS`.
- `fresh-extension` - Fresh-member recurrence creates a disjoint absent/held copy of every prior selection. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial carrier silently omits possible generated choices. Result: `PASS`; receipt `sha256:c1dbfffb55e34b1990cfc2d3bd521fd163d147f13c5598b09021959fcee3cfe3`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no random sampling as completeness evidence; no factorial or binomial answer table; no negative inclusion-exclusion proof value; no completed infinite combinatorial family Result: `PASS`; receipt `sha256:5e256307e37a27f3cb51ece4a149427a8131c602ec3eb31051eb5b10fdb3dced`.

8.5 Depth-independent closure

Base: A one-member carrier has one complete arrangement and two structural selection forms: empty One and held member.

Successor: For a fresh carrier member, arrangements partition by its held position and recurse on the prior remainder; selections partition into prior forms and the same forms with the fresh member held. These disjoint generated classes exhaust the successor carrier without an imported formula.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

8.6 Meaning and scientific consequence

Combinatorial values arise after, not before, generation. The arrangement and selection families are constructed by exhaustive held choices with each remainder retained. Their counts are identities of the resulting complete traces. Symmetry reduction requires an explicit label-preserving bijection, and overlap is handled by disjoint provenance classes instead of negative inclusion-exclusion proof quantities.

8.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: permutation, combination, recurrence, symmetry class, pigeonhole counting, inclusion-exclusion. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

8.8 Exact exclusions and limitations

- no random sampling as completeness evidence.
- no factorial or binomial answer table.
- no negative inclusion-exclusion proof value.
- no completed infinite combinatorial family.

The law closes the generative foundation of finite combinatorics. Named advanced enumerations require their own derived generators; none may be admitted by quoting a conventional closed-form answer.

8.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:7d7554f879a30b739c05c4f9aeba7c5812c4ceddea2a1d032a71ecb8fa10ba1a`.
- Derivation seal: `sha256:3177572ea372062f65a7969fdf4e498c86c9af558e37c048165e1d950cf1d942`.
- Independent implementation: `sha256:00ac1dfb12aa98fba2ec8b0ab7eae0b940cee7ca8c55b75fa07678864e9b4086`.
- Independent certificate: `sha256:ea22b95c87c9f336726c9df87d1e88fb38b1ad27519789c334a5052f5f864ae6`.
- External validation: `sha256:dbfc2711bbabde9be3bf4a4cc3c5de3cd62440c9fbeea20b2fceed6dcfb1275b`.
- Engine receipt: `sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-COMBINATORICS-001-52e1db94bb786527.json`.

9. Derivation 4: Exact finite graph and network structure

Claim identity: `SFT-MATH-GRAPH-NETWORK-001`

9.1 Question and necessity

Graphs are the minimal mathematics of explicit relations and paths. Their laws must follow from generated carriers and pair support rather than imported matrices, weights or probabilistic ensembles.

The exact theorem statement is:

> *Every admitted finite graph is a complete canonical node collection plus a held selection of generated ordered node-pair relations; paths retain every adjacent transition, reachability is a complete finite path witness, cycles retain return, trees are connected cycle-free forms, cuts are exact crossing-edge selections, and internal network conservation is complete ingress/egress pairing.*

Admitted dependencies: `SFT-FOUNDATION-FORM-GRAMMAR-001`, `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-COMBINATORICS-001`.

9.2 Generated grammar and exact boundary

Discrete mathematics supplies canonical carriers and relations; combinatorics supplies complete path-family generation. Exhausting nine structural choices leaves the relation/path/return/pairing graph kernel.

Generation rule: Generate the complete product of node coverage, edge provenance, orientation, path trace, connectivity, cycle return, cut support, internal balance and extra-network-rule status.

Exact grammar boundary: All finite graphs and networks generated from canonical finite form carriers, held pair-cell relations, complete path traces, return relations and exact positive flow-token pairings.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:447b14b12f08e4bcf5c21bd3b34a7e06cbccd3ad90d3669153c9d05306221bfd`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-GRAPH-NETWORK-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`nodes`	`partial-or-aliased-nodes`	Missing or aliased nodes change the graph carrier.	`complete-canonical-nodes` - Every generated node form occurs once with canonical identity.
`edges`	`free-endpoint-list`	A free list can contain ungenerated endpoints or duplicate relations.	`held-pair-cell-relation` - Edges are a held duplicate-free selection of complete node pair support.
`orientation`	`signed-edge-weight`	A signed weight imports negative magnitude and does not identify endpoints.	`held-endpoint-order` - The ordered endpoint labels retain orientation; complementary order represents reversal.
`path`	`endpoint-only`	Endpoints alone omit the intervening adjacency requirements.	`complete-adjacent-trace` - Every node and edge transition in the path is retained in order.
`connectivity`	`assumed-connected`	An assertion without a path omits the connecting structure.	`generated-path-witness` - Reachability requires a complete generated adjacent path.
`cycle`	`repeated-label-only`	A repeated label without a transition trace does not prove return.	`explicit-return-path` - A cycle is a nontrivial complete path whose terminal relation returns to a held prior node.
`cut`	`borrowed-scalar-threshold`	A scalar threshold imports an external coordinate.	`held-crossing-selection` - A held node selection forces exactly the edges with one held and one complementary endpoint.
`balance`	`unaccounted-net-value`	A net signed value hides unmatched tokens.	`ingress-egress-pairing` - Every incoming positive token has one outgoing mate and conversely.
`addition`	`extra-network-rule`	An extra weight, metric or random law is not supplied by the graph dependencies.	`no-extra-network-rule` - All graph properties are witnessed by carriers, relations, paths, returns, selections and pairings.

9.3 Unique survivor, minimality and laws

> *The graph/network kernel is complete canonical nodes with held ordered pair-cell edges, complete adjacent paths, witnessed reachability and return, held crossing cuts, ingress/egress pairing and no extra rule.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- path concatenation is lawful exactly when the interface node agrees.
- reachability is reflexive by the retained identity path and transitive by path composition.
- a tree is a connected generated graph with no nontrivial return path.
- a cut contains every and only relation crossing a held/complementary node partition.
- network balance preserves token identity rather than cancelling signed magnitudes.

9.4 Operational witnesses and unfavourable checks

- `valid-path` - The chain retains both required adjacent edges. Result: `PASS`.
- `missing-edge-control` - An endpoint sequence with a missing adjacency is not a path. Result: `PASS`.
- `reachability` - Generated path search reaches c from a and does not reach a from c in the directed chain. Result: `PASS`.

- `cycle-return` - The explicit c-to-a return distinguishes the cycle from the chain. Result: `PASS`.
- `cut-selection` - Holding the first chain node selects exactly its crossing edge. Result: `PASS`.
- `paired-balance` - Equal unique ingress and egress token traces witness internal balance. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Missing or aliased nodes change the graph carrier. Result: `PASS`; receipt `sha256:2c881c5d8c3f3ab8e894f11a4fce432fbbfa4e49226ddb271ea8023ed8e3228b`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ungenerated vertex or edge; no negative edge or flow quantity; no stochastic graph law; no infinite graph as a completed object Result: `PASS`; receipt `sha256:3f199f993e865f79dd3ff17ba48418bbfca2470b4a79c8a8697148409b3ba7be`.

9.5 Depth-independent closure

Base: One canonical node supplies the identity path and contains no nontrivial edge cycle.

Successor: Appending one fresh node generates exactly its ordered pair cells with prior nodes; any held edge selection extends paths locally, and connectivity, cycle, cut and balance witnesses update only through those new cells.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

9.6 Meaning and scientific consequence

Graph theory becomes the exact mathematics of generated relations. Vertices are canonical forms, edges are held ordered pair cells, and paths preserve every adjacency. Reachability, cycles and trees are not labels but path predicates. A cut is forced by a held/complementary node selection. Network conservation retains the identities of paired ingress and egress tokens instead of cancelling signed scalar totals.

9.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: directed graph, path, reachability, cycle, tree, cut, network flow. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

9.8 Exact exclusions and limitations

- no ungenerated vertex or edge.
- no negative edge or flow quantity.
- no stochastic graph law.
- no infinite graph as a completed object.

This law closes exact generated finite graph structure. Weighted geometry, random graphs, infinite graph limits and algorithmic graph complexity require later derived laws.

9.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:e9e22d5262250c7cb551e278121b29431ef59d58cb53b28c9188f2d912979d8c`.

- Derivation seal: `sha256:f1f74c52072189f08134293f1f55aaff63aeb73fc045d79ac09a9372554c7ad9`.
- Independent implementation: `sha256:82cc03abc10d06d9d288a85c6171bc60db065b82a074e9fc39745f5b1e59ebe4`.
- Independent certificate: `sha256:531d0c9a5ec7a97b8e97c4a1664d67af65d7fab17127bcf042e76a86386a9210`.
- External validation: `sha256:9f9e2483be55ff773cf570f6fc3aee074f68f44fde118492f152023767fef5c8`.
- Engine receipt: `sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-GRAPH-NETWORK-001-b4072f93147c8cd9.json`.

10. Derivation 5: Generated finite algebraic structures

Claim identity: `SFT-MATH-ALGEBRA-001`

10.1 Question and necessity

Algebra is forced when generated operations are composed and compared. The exhaustive table boundary prevents familiar algebraic classifications from supplying unproved laws.

The exact theorem statement is:

> *An admitted finite algebraic structure is a complete canonical carrier with a complete, single-valued, closed pair-cell operation relation. Identity, associative composition, reversible return mates, commutation, substructure and structure-preserving maps are admitted exactly when their exhaustive carrier witnesses pass; none is installed as an ungenerated axiom.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`.

10.2 Generated grammar and exact boundary

Discrete pair support supplies all operation inputs; exact canonical equality tests results; form enforcement supplies carrier identity. The nine-way census leaves the witness-governed operation kernel.

Generation rule: Generate the complete product of carrier, operation coverage, closure, identity, association, return, property-admission, structure-map and extra-law status.

Exact grammar boundary: All finite operation structures generated from canonical carriers, complete pair-cell relations, canonical path composition, held reversal and exhaustive property witnesses.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:ccae1cb3b71ff2b6cffd4acead871b01722969c069b797ad61e9cb2050f09701`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ALGEBRA-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`partial-or-aliased-carrier`	Omission or aliasing changes which operation inputs exist.	`complete-canonical-carrier` - Every generated carrier form occurs once with canonical identity.
`operation`	`partial-operation-sample`	A partial sample does not assign every generated input pair.	`complete-single-valued-table` - Every ordered carrier pair has exactly one retained result.
`closure`	`external-result`	An external result silently enlarges the registered carrier.	`carrier-closed-result` - Every result is a canonical member of the same carrier.
`identity`	`named-identity`	A name alone does not prove left and right preservation.	`exhaustive-identity-witness` - One carrier form preserves every member on both sides.
`association`	`assumed-parenthesis-law`	An assumed equation imports an algebraic axiom.	`complete-triple-witness` - Every carrier triple has equal canonical flattened composites.
`return`	`signed-inverse-value`	A signed magnitude imports negative quantity.	`unique-held-return-mate` - A retained complementary carrier mate returns on both sides to identity.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`properties`	`property-by-name`	A familiar structure name cannot supply commutation or return.	`property-by-exhaustive-witness` - Each property is registered only after every required carrier tuple passes.
`maps`	`arbitrary-carrier-map`	An arbitrary map can erase the operation relation.	`operation-preserving-map` - The complete image relation commutes with every source operation cell.
`addition`	`extra-algebraic-axiom`	An extra axiom is not forced by the registered operation traces.	`no-extra-algebraic-axiom` - All admitted properties have explicit generated witnesses.

10.3 Unique survivor, minimality and laws

The algebraic kernel is a complete canonical carrier and complete closed single-valued operation table, with identity, association, returns, special properties and maps admitted only by exhaustive witnesses.

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- closure is exact membership of every operation result in the registered carrier.
- identity is a two-sided exhaustive preservation witness.
- association is equality of both complete composite traces for every carrier triple.
- return is a unique held mate relation and never a negative proof value.
- homomorphism is complete operation preservation, not resemblance of presentations.

10.4 Operational witnesses and unfavourable checks

- `closed-table` - Every complete input pair in the witness table has one carrier result. Result: `PASS`.
- `identity` - The structural One form preserves both witness carrier forms. Result: `PASS`.
- `association` - Both parenthesizations agree for every witness triple. Result: `PASS`.
- `return-mates` - Every witness form has one held two-sided return mate. Result: `PASS`.
- `commutation-is-conditional` - Commutation passes for the symmetric witness and fails for a lawful noncommuting table. Result: `PASS`.
- `structure-map` - The canonical self-map preserves every operation cell. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or aliasing changes which operation inputs exist. Result: `PASS`; receipt `sha256:b0cb6ed9032c814991aae189ba084f58a0344948ecdeba10de30448360699e08`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported group, ring or field axioms; no negative or irrational carrier requirement; no property inferred from a conventional name; no completed infinite carrier Result: `PASS`; receipt `sha256:e7630fd960384d7d35b15f2e1c900d720e648b0dd23ec4cea7ac6ce9dadc209c`.

10.5 Depth-independent closure

Base: The One carrier has one pair cell, its result remains One, and identity and association coincide.

Successor: Adding one fresh carrier form generates exactly its left, right and self pair cells. A proposed extension is admitted only after results remain in the extended carrier and every newly formed identity, triple, return and map obligation is checked.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

10.6 Meaning and scientific consequence

The algebra theorem deliberately separates an operation kernel from named algebraic species. A complete closed operation table is forced first. Identity, association, return, commutation and homomorphism status are then admitted only if their complete carrier obligations pass. Thus the words group, monoid or commutative structure cannot import properties; they are correspondence classifications of an already sealed witness.

10.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: magma, monoid, group, commutativity, subalgebra, homomorphism. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

10.8 Exact exclusions and limitations

- no imported group, ring or field axioms.
- no negative or irrational carrier requirement.
- no property inferred from a conventional name.
- no completed infinite carrier.

This closes the finite algebraic kernel and its property tests. Particular richer algebraic species require their own generated carriers and independently witnessed additional operations.

10.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:c45fbdd44b40d4d5d007c619fbd6dd03554a43e980396d1d3158b5f9791b34a3`.
- Derivation seal: `sha256:ae92defb9a52942af89734332b791cb1921ae655499b56e808a6250f539d6396`.
- Independent implementation: `sha256:280231864879a04285ec75e17ecf961cb20c43bcf39c228fabf05b3ccd58e033`.
- Independent certificate: `sha256:b41bcba0ab04cfad3eb95cd55ae4286b9ea1c52748ae444b33e2330ec7b52bf4`.
- External validation: `sha256:5443f0a807dfeff488bfd29b7059fbbd086799f212e974d12867670b9f644654`.
- Engine receipt: `sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ALGEBRA-001-1f08bfaf9f602a38.json`.

11. Derivation 6: Exact finite order and conditional lattice structure

Claim identity: `SFT-MATH-ORDER-LATTICE-001`

11.1 Question and necessity

Order is needed for comparison, optimisation, proof strength and topology. Retaining incomparability prevents a conventional scalar line from being imported where the Fold structure does not force one.

The exact theorem statement is:

> *An admitted finite order is a held carrier relation whose reflexive, antisymmetric and transitive cells are exhaustively witnessed. Totality is admitted only when every generated pair is comparable. Lower and upper bounds are complete relation selections; meet and join exist only as unique greatest-lower and least-upper witnesses, and monotone maps preserve every retained comparison.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-ALGEBRA-001`.

11.2 Generated grammar and exact boundary

Discrete relations supply the comparison cells; algebra supplies lawful composition; exact identity supplies antisymmetry. Exhausting nine choices forces witnessed partial order and conditional lattice operations.

Generation rule: Generate the complete product of carrier identity, relation provenance, partial-order laws, comparability, bound generation, meet uniqueness, join uniqueness, monotonicity and extra-order status.

Exact grammar boundary: All finite order structures generated from canonical carriers, held pair-cell relations, exhaustive relation closure, complete bound selections and unique extremal witnesses.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:06bc1ab2205eecb2fd980877791e2617c87d47e067be6162d904c903370cfeec`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ORDER-LATTICE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`carrier`	`presentation-carrier`	Presentation aliases do not fix exact ordered identity.	`canonical-carrier` - Every ordered form has canonical generated identity.
`relation`	`borrowed-scalar-comparison`	A scalar comparison imports an answer-producing order.	`held-pair-relation` - Each comparison is a retained generated carrier pair.
`order`	`named-order`	A name does not establish reflexivity, antisymmetry and transitivity.	`exhaustive-partial-order-witness` - Every required relation cell and composite is checked.
`comparability`	`assumed-totality`	Partial orders can contain incomparable forms.	`pairwise-totality-witness` - Totality is recorded only when every generated pair is comparable.
`bounds`	`selected-bound`	Selecting a familiar candidate can omit another held condition.	`complete-bound-selection` - Every carrier form is tested against every held target.
`meet`	`assumed-meet`	A lower bound need not be greatest or unique.	`unique-greatest-lower-witness` - Exactly one lower bound is above every other lower bound.
`join`	`assumed-join`	An upper bound need not be least or unique.	`unique-least-upper-witness` - Exactly one upper bound is below every other upper bound.
`maps`	`arbitrary-map`	An arbitrary map can reverse a retained comparison.	`comparison-preserving-map` - Every source relation cell maps to a target relation cell.
`addition`	`extra-order-axiom`	An extra totality or completeness axiom is not structurally forced.	`no-extra-order-axiom` - Only exhaustively witnessed order properties are admitted.

11.3 Unique survivor, minimality and laws

> *The order/lattice kernel is a canonical carrier with an exhaustively witnessed partial-order relation, conditional totality, complete bound selections, uniquely witnessed meet/join and monotone maps.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- partial order requires reflexive, antisymmetric and transitive relation closure.
- incomparability is retained and never forced into a borrowed total scale.
- meet and join are partial operations unless unique extremal witnesses exist.
- a finite lattice is exactly an admitted partial order with meet and join for every generated pair.
- monotonicity is preservation of every registered relation cell.

11.4 Operational witnesses and unfavourable checks

- `partial-order` - The diamond relation passes all three partial-order obligations. Result: `PASS`.
- `incomparability-retained` - The diamond retains left and right as incomparable. Result: `PASS`.
- `conditional-totality` - The explicit chain is total while the diamond is not. Result: `PASS`.
- `meet` - The two incomparable diamond arms have structural One as their unique meet. Result: `PASS`.

- `join` - The two incomparable diamond arms have whole as their unique join. Result: `PASS`.
- `monotone-identity` - The canonical identity map preserves every diamond comparison. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Presentation aliases do not fix exact ordered identity. Result: `PASS`; receipt `sha256:8b99f164faa8fd4a22432325f9d4c0afd5c8882ed01e4d35781cbb3544f51254`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported real-number line; no assumed total order; no lattice completeness without generated meets and joins; no completed infinite chain Result: `PASS`; receipt `sha256:484dd6f1f271f8054f6f3e5e74696abdbe274c36cb08b223628218b599f5da79`.

11.5 Depth-independent closure

Base: The One carrier has its identity relation and One is both unique lower and upper bound of itself.

Successor: Adding one fresh canonical form generates all comparisons to prior forms. Order closure, comparability, bounds and extremal uniqueness are rechecked only for pairs and composites touching the fresh form.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

11.6 Meaning and scientific consequence

Order is relational, not borrowed from a numerical line. Reflexivity, antisymmetry and transitivity are exhausted over the carrier. Incomparability is positive retained information and is not forced into a total ranking. Bounds are complete selections. A meet or join exists only when the respective greatest-lower or least-upper witness is unique, so lattice structure remains conditional on generated evidence.

11.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: partial order, total order, poset, meet, join, lattice, monotone map. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

11.8 Exact exclusions and limitations

- no imported real-number line.
- no assumed total order.
- no lattice completeness without generated meets and joins.
- no completed infinite chain.

This theorem closes finite generated orders. It does not assert totality, lattice status or completeness for a carrier unless the corresponding finite witness is present.

11.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:fb4627050dca9b64d628dfc773d8de1d30656f40689d1cb8a0ae8dd3105940bd`.
- Derivation seal: `sha256:d4421e8aaf34c239bda0fcacfe2a6aa770fac55ec3afe50c0ba5085d83946527`.

- Independent implementation: `sha256:e0b750be248ae4a9da52c952d8988d01fa236ccdd6a994063a12f297d2088a8f`.
- Independent certificate: `sha256:f1b8cbb06e153828d4410413baf3b6af6f075df5a5f3f297a1f6dee4d1c51bce`.
- External validation: `sha256:f2b2f4bbab7729985711dbdff1cbc25cd7d16f0423a9752f8abdba8caebdd136`.
- Engine receipt: `sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ORDER-LATTICE-001-d0e20f40782f51cc.json`.

12. Derivation 7: Exact finite geometry and topology for computation

Claim identity: `SFT-MATH-GEOMETRY-TOPOLOGY-001`

12.1 Question and necessity

Algorithms need incidence, locality, paths, neighborhoods and continuity, but do not require importing a real-coordinate continuum. The finite structural form is the exact computationally required kernel.

The exact theorem statement is:

> *Computational geometry is forced as canonical finite cells plus an acyclic held boundary-incidence relation; dimension is retained boundary depth, adjacency is shared incidence, and distance is a shortest generated path with self-distance represented by empty One rather than numerical zero. Computational topology is a complete generated open-family closed under generated joins and pairwise intersections; continuity is exact inverse-image preservation.*

Admitted dependencies: `SFT-FOUNDATION-PART-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`.

12.2 Generated grammar and exact boundary

Graphs supply paths and incidence; orders supply joins and intersections; exact parts supply held/whole selection. The ten-dimensional census forces the finite incidence-and-open-family kernel.

Generation rule: Generate the complete product of cell carrier, incidence, dimension, adjacency, path distance, open-family closure, connectedness, continuity, deformation and extra-geometric status.

Exact grammar boundary: All finite incidence complexes, path geometries and finite open families generated from canonical forms, held boundary relations, exact path traces, selections, joins and intersections.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:be96c202d0f654cb38088fbd1c4afc210f949ce35b52525cdc7d3e45ad189ed0`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-GEOMETRY-TOPOLOGY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`cells`	`coordinate-samples`	Samples from a borrowed coordinate field omit structural provenance.	`canonical-generated-cells` - Every cell is an exact canonical generated form.
`incidence`	`assumed-containment`	Assumed containment has no exact face-to-cell witness.	`held-acyclic-boundary` - Every face relation is retained and boundary descent forbids return.
`dimension`	`borrowed-coordinate-count`	A coordinate count imports an ambient space.	`boundary-chain-depth` - Dimension is the longest retained face-descent trace.
`adjacency`	`metric-threshold`	A threshold imports a free scale.	`shared-incidence-witness` - Cells are adjacent exactly when they share a generated face.
`distance`	`floating-metric`	A floating metric imports precision and possibly irrational values.	`shortest-generated-path` - Distance is the complete least-length path trace; identity is empty One.
`opens`	`named-neighborhoods`	Named neighborhoods need not satisfy closure.	`generated-open-family-closure` - Empty One, whole, generated joins and pairwise intersections remain in the family.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`connectedness`	`visual-continuity`	A presentation cannot establish structural connection.	`no-disjoint-open-separation` - No two nonempty disjoint admitted opens jointly exhaust the carrier.
`continuity`	`epsilon-scale`	An imported scale is neither forced nor needed in the finite grammar.	`inverse-open-preservation` - Every target open has an admitted source inverse image.
`deformation`	`untracked-shape-change`	An untracked change can erase incidence.	`reversible-incidence-trace` - Each elementary change retains a lawful reverse and preserves registered incidence invariants.
`addition`	`continuum-or-extra-metric`	A continuum or metric field is outside the generated computational need.	`no-ambient-continuum` - The law uses only finite generated cells, paths and open families.

12.3 Unique survivor, minimality and laws

> *The geometry/topology kernel is finite canonical cells with held acyclic incidence, boundary-depth dimension, shared-face adjacency, exact shortest paths, closed finite open families, inverse-image continuity and reversible incidence-preserving deformation.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- cell dimension is determined by complete acyclic boundary descent.
- path distance retains a shortest generated transition trace and uses empty One at identity.
- finite topological closure is exhaustively decidable from the registered open family.
- continuity is exact inverse-image preservation of every admitted target open.
- geometric equivalence requires a reversible invariant-preserving transformation trace.

12.4 Operational witnesses and unfavourable checks

- `incidence` - Every witness face and cell belongs to the finite carrier. Result: `PASS`.
- `dimension-depth` - The witness triangle has one more boundary layer than its edge. Result: `PASS`.
- `adjacency` - The two witness edges are adjacent exactly through their shared q face. Result: `PASS`.
- `path-distance` - Shortest path retains a-to-b-to-c and identity returns empty One. Result: `PASS`.
- `finite-topology` - The complete two-point selection family is closed. Result: `PASS`.
- `continuity` - The identity map preserves every witness open by inverse image. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Samples from a borrowed coordinate field omit structural provenance. Result: `PASS`; receipt `sha256:602d7acdc1da5b0de53c819a78f381ccac9858cd0bba3688edd642448a2ff375`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ungenerated continuum; no irrational or floating proof coordinate; no free metric threshold; no completed infinite topological family Result: `PASS`; receipt `sha256:2133cc61a907c35bdc264f3d462617d553589263450769cb5527f812c4268469`.

12.5 Depth-independent closure

Base: One point is a canonical cell; its open family consists of empty One and the whole point carrier.

Successor: Adding one fresh cell generates its possible boundary incidences, shared-face adjacencies and path edges; adding one fresh point generates exactly the open selections needed for closure checks. Every new boundary, path, intersection, join and inverse image is then exhaustively tested.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

12.6 Meaning and scientific consequence

The theorem derives exactly the geometry and topology required by finite computation without an ambient continuum. Geometry is incidence, boundary depth, shared-face adjacency and shortest retained path. Topology is a generated family of opens closed under the operations available at the finite boundary. Continuity is inverse-open preservation, while deformation requires a reversible incidence-preserving trace. Smooth or real-coordinate descriptions may later correspond to approximations but do not select this kernel.

12.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: incidence complex, dimension, adjacency, metric, topology, continuity, homotopy. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

12.8 Exact exclusions and limitations

- no ungenerated continuum.
- no irrational or floating proof coordinate.
- no free metric threshold.
- no completed infinite topological family.

This closes geometry and topology where finite computation requires them. Smooth, differential, measure and continuum correspondences are not premises and require separate generated approximation laws if used.

12.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:36d95c3b1ee446c17b5b4e4f609e10ce1f679f164c720eebd8e482ab68873eaa`.
- Derivation seal: `sha256:7afff182127d9b93f9705212b997801eafd5a3629fd2a8a56753ce674832d0ad`.
- Independent implementation: `sha256:27f1aa8512b811cfef6ccf01fb51ebe844c79f99414e59d4783bc293980bb132`.
- Independent certificate: `sha256:8b8595f6ef14fa36929ff546752015da701e4a49af5213d54f0f6a1d4559ce68`.
- External validation: `sha256:35658ed3991ba98df705a53a75bacc5c7e813bde5f0505e5d20b549ed8cf3dd5`.
- Engine receipt: `sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-GEOMETRY-TOPOLOGY-001-a04f7de0ede6e234.json`.

13. Derivation 8: Exact finite probability and statistics from Fold observation

Claim identity: `SFT-MATH-PROBABILITY-STATISTICS-001`

13.1 Question and necessity

Prediction and experiment require uncertainty and statistics, but a superdeterministic Fold model cannot import stochastic dynamics. Exact observation classes supply the required epistemic structure.

The exact theorem statement is:

> *Probability is an exact held-support/whole-support part relation over a complete deterministic generated microstate support; uncertainty records distinctions closed by an observation class and is not a causal random premise. Conditional weight is exact common-refinement restriction, independence is complete pair-cell factorization, and statistics are provenance-retaining finite trace summaries. The empty event is empty One, never numerical zero.*

Admitted dependencies: `SFT-FOUNDATION-PART-EQUIVALENCE-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`.

13.2 Generated grammar and exact boundary

Parts supply exact held/whole quantities; combinatorics supplies complete support; the measurement boundary isolates data. Nine structural choices force the observation-support account without a prior.

Generation rule: Generate the complete product of support coverage, event formation, weight, uncertainty status, conditioning, independence, statistics, empirical isolation and extra-random-law status.

Exact grammar boundary: All exact finite probability spaces and statistical traces generated from complete microstate support, held selections, observation classes, common refinement, exact parts and sealed empirical measurements.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:0702345917ea32913e5c17e13d86f16d7a885f34e09f8941c7e90be0963f286b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-PROBABILITY-STATISTICS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`support`	`sampled-or-partial-support`	A partial support silently changes every derived weight.	`complete-generated-support` - Every registered deterministic microstate occurs once.
`event`	`named-outcome`	A name without retained membership does not identify support.	`held-support-selection` - An event is a duplicate-free held selection of exact support.
`weight`	`floating-propensity`	A floating propensity imports precision and a stochastic cause.	`exact-held-whole-part` - Weight is the exact held count over the complete support count.
`uncertainty`	`ontic-random-cause`	A causal random source is neither generated nor required.	`closed-observation-distinction` - Uncertainty records which deterministic microstate distinctions observation does not retain.
`conditioning`	`renormalized-guess`	A guessed renormalization can omit common support.	`exact-common-refinement` - Restrict to the complete held condition and count its exact event intersection.
`independence`	`uncorrelated-assertion`	An assertion does not prove factorized joint support.	`pair-cell-factorization` - Complete joint support is the exact product and event weights multiply by bijection.
`statistics`	`opaque-estimator`	An opaque scalar erases source rows and tie structure.	`provenance-retaining-summary` - Every summary is recomputable from the complete finite trace and retains ties.
`empirical`	`target-informed-fit`	Using target rows during derivation allows them to select the law.	`post-seal-blind-check` - Natural measurements enter only after formal sealing under registered isolation.
`addition`	`extra-random-parameter`	A seed distribution or prior is a free parameter.	`no-extra-random-parameter` - All weights follow from support and observation without stochastic dynamics.

13.3 Unique survivor, minimality and laws

> *The probability/statistics kernel is complete deterministic support, held events, exact held/whole weights, observation-class uncertainty, common-refinement conditioning, pair-cell independence, transparent finite summaries and post-seal empirical checking without random parameters.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- event weight is an exact positive part relation; empty event is empty One.

- conditional weight is exact restriction to a nonempty held condition.
- independence is a bijective complete joint-support factorization.
- uncertainty belongs to an observation partition and does not alter deterministic microstate evolution.
- statistical summaries retain exact provenance and all tied alternatives.

13.4 Operational witnesses and unfavourable checks

- `exact-weight` - Two held states in four complete states have the exact one-over-two part. Result: `PASS`.
- `empty-event` - The empty event is structural empty One rather than numerical zero. Result: `PASS`.
- `conditional` - Conditioning on the two a-prefix states retains one selected state as one-over-two. Result: `PASS`.
- `independence` - The complete two-by-two pair support factorizes exactly. Result: `PASS`.
- `ties-retained` - An exact modal summary preserves both equally supported alternatives. Result: `PASS`.
- `observation-classes` - A prefix observation retains two exact two-state classes. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A partial support silently changes every derived weight. Result: `PASS`; receipt `sha256:646eb932b825e9c344756ec999063ca07df35072a81f697e0b591039c293685a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ontic stochastic premise; no floating probability in proof evidence; no Bayesian or distributional prior parameter; no target data may select a formal law Result: `PASS`; receipt `sha256:6f88285aa8b2bb6873871ac8acd897d8ea1b6e120bfad06b20ca3c3f9ad50133`.

13.5 Depth-independent closure

Base: A One-state support has whole certainty; its held-empty event is represented by empty One.

Successor: Adding one fresh microstate extends whole support once and either extends or leaves each held event. Exact parts, observation classes, intersections and summaries update from that retained membership without a prior.

Shared claim-record clause `MATH-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

13.6 Meaning and scientific consequence

Probability is compatible with superdeterminism because it is not installed as a random cause. The complete microstate support is deterministic; an observation closes some distinctions and retains others. Exact event weight is the held-support/whole-support part relation. Conditioning is exact restriction to a held common refinement, and independence is a bijective product-support fact. Statistics remain recomputable from all source rows and preserve ties. Natural data may test a sealed consequence but cannot choose it.

13.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: probability space, conditional probability, independence, random variable, statistic, sampling. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

13.8 Exact exclusions and limitations

- no ontic stochastic premise.
- no floating probability in proof evidence.
- no Bayesian or distributional prior parameter.
- no target data may select a formal law.

This closes exact finite probability and descriptive statistics. Inferential claims about natural data remain empirical claims and require preregistered blind validation through the empirical engine.

13.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:45f28c927c6994a7817ce2fd870441d41ee99e030be73063970674b092e6bab4`.
- Derivation seal: `sha256:9ad2c0aa860b03d0fe11c390e7e1fada396f48565fd89d055f06fec28b32ed67`.
- Independent implementation: `sha256:cdb15cdfaa90a02a1222dcbb0e8000e6ff93fddf67172c0ee26db74d236a17a9`.
- Independent certificate: `sha256:d80527e03481be10b16d4765742d717ebcefaa9f1830e2c1c37d6e40869a62e9`.
- External validation: `sha256:39850e79359dbd3ae8e41fe032474bcf197212f6ac0214cdfd90605919561761`.
- Engine receipt: `sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-PROBABILITY-STATISTICS-001-2a86d644881cdacb.json`.

14. Derivation 9: Exact finite optimisation and retained optima

Claim identity: `SFT-MATH-OPTIMIZATION-001`

14.1 Question and necessity

Search and decision require lawful selection among alternatives. Exact dominance and tie retention prevent a familiar score, tolerance or traversal order from selecting the result.

The exact theorem statement is:

> *An admitted optimization problem is a complete canonical candidate carrier, a source-bound exact feasibility relation and a witnessed preference relation. Solutions are the complete undominated held selection; uniqueness is claimed only for a singleton, while ties and incomparable Pareto alternatives remain explicit. Constraints, decomposition and approximation are exact trace relations without floating objectives, negative penalties or free tuning parameters.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ORDER-LATTICE-001`.

14.2 Generated grammar and exact boundary

Orders supply witnessed preference and incomparability; graphs supply search relations and decomposable interfaces; exact arithmetic supplies part bounds. Nine choices force the complete undominated kernel.

Generation rule: Generate the complete product of candidate coverage, feasibility, preference, optimum selection, tie retention, constraint provenance, decomposition, approximation and extra-objective status.

Exact grammar boundary: All finite optimisation instances generated from canonical candidates, exact predicates, held preference relations, complete dominance checks, interface-preserving decomposition and exact positive part bounds.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:a15ea2f582138e72734b4421a5d69ae92089f444f70597f7539532587a30ddbe`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-OPTIMIZATION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`candidates`	`sampled-candidates`	Sampling cannot prove that a better generated candidate was not omitted.	`complete-canonical-candidates` - Every registered candidate form occurs once.
`feasibility`	`implicit-constraint`	An implicit constraint has no candidate-level check trace.	`source-bound-verdicts` - Every candidate receives one exact verdict from each registered constraint.
`preference`	`floating-score`	A floating score imports scale and precision choices.	`witnessed-order-relation` - Every preference is an exact retained candidate pair.
`solution`	`first-found`	First-found depends on traversal presentation.	`complete-undominated-selection` - Every feasible candidate is compared and all undominated forms are retained.
`ties`	`silent-tie-break`	A silent tie-break adds a free preference.	`retained-equivalence-class` - All jointly optimal or incomparable surviving forms remain explicit.
`constraints`	`penalty-parameter`	A tunable penalty is a free parameter and can select the answer.	`exact-membership-predicate` - Each constraint is a source-bound exact allowed-form relation.
`decomposition`	`assumed-separability`	Shared constraints can invalidate separate optima.	`exact-interface-factorization` - Subproblems compose only when all shared interface relations are retained.
`approximation`	`decimal-error`	A decimal error imports rounding and a scale.	`exact-positive-part-bound` - The candidate/optimum relation is a generated exact part certificate.
`addition`	`extra-objective-or-tie-rule`	An added score or tie rule is not forced by the instance.	`no-extra-objective` - Only registered feasibility and preference relations select solutions.

14.3 Unique survivor, minimality and laws

> *The optimization kernel is complete canonical candidates, source-bound feasibility, exact preference, complete undominated selection, retained ties, exact constraints and interfaces, and positive-part bounds without an added objective.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- optimisation may eliminate a candidate only with a retained feasible dominator witness.
- all undominated candidates survive and uniqueness is a separately checked singleton property.
- Pareto comparison retains incomparability across exact criteria.
- decomposition is lawful only with complete shared-interface factorization.
- approximation is an exact generated part relation rather than a rounded scalar.

14.4 Operational witnesses and unfavourable checks

- `feasible-selection` - The complete verdict relation retains exactly a and b. Result: `PASS`.
- `unique-optimum` - A witnessed a-over-b relation leaves a as the singleton optimum. Result: `PASS`.
- `ties-retained` - With no forced preference both feasible forms survive explicitly. Result: `PASS`.
- `constraint-membership` - The witness candidate is checked against every exact allowed set. Result: `PASS`.
- `pareto-incomparability` - Opposed exact criteria retain both incomparable forms. Result: `PASS`.
- `exact-bound` - Three generated cells against a two-cell optimum produce exact two-over-three. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot prove that a better generated candidate was not omitted. Result: `PASS`; receipt `sha256:4fca7bb33759009f032e930b863d7c4fce12d4ce19f11686801c02efdbba245f`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.

- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no floating objective or tolerance; no negative penalty term; no tunable regularizer or tie-break parameter; no unenumerated candidate domain Result: `PASS`; receipt `sha256:40faa5eb68ab3bcadce5ee1937d3b89e679a8f1fa2ba12fca91be6318f92420c`.

14.5 Depth-independent closure

Base: A One-candidate feasible carrier has that candidate as its unique complete undominated selection.

Successor: Adding one fresh candidate generates its feasibility verdicts and every preference pair with prior feasible forms. It is either eliminated by a retained dominator, eliminates witnessed dominated forms, or joins the retained optimal class; no prior optimum is silently discarded.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.6 Meaning and scientific consequence

Optimisation is forced as elimination by exact witness rather than maximization of an imported floating score. Every candidate and feasibility verdict is present. A candidate is removed only when a retained feasible dominator exists. The complete undominated selection is the solution; singleton status alone licenses uniqueness. Multiple optimal or Pareto-incomparable forms remain visible rather than being broken by an undeclared convention.

14.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: feasible set, objective, optimum, Pareto frontier, constraint, approximation ratio. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

14.8 Exact exclusions and limitations

- no floating objective or tolerance.
- no negative penalty term.
- no tunable regularizer or tie-break parameter.
- no unenumerated candidate domain.

This closes exact finite optimisation structure. Complexity of finding the complete frontier and laws for particular objective families belong to the algorithms and complexity branches.

14.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:435743b4bc243523b8c405514232bdc88a02fd48764040fa26c914085ae0f813`.
- Derivation seal: `sha256:11d1d87d66d732bca3efc296de283e8e5e11ed9d291dd083c10f86d6801277e9`.
- Independent implementation: `sha256:19c3f41f797dbf136a28392518b1ef3ff431214a314099a6080c8c5ea04edd95`.
- Independent certificate: `sha256:9cb0790543dac4fb7c8ff8bce76f783ccb1c96e84176f8cf59c203060920bfa4`.
- External validation: `sha256:7f6345c8109261bf2f63108a2613767ff27fae5f9d411bace11a33cc9046ddb3`.
- Engine receipt: `sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849`.

- Engine receipt path:
  `receipts/engine/model_admitted/SFT-MATH-OPTIMIZATION-001-3385414ceea0fd42.json`.

15. Derivation 10: Exact generated finite dynamical systems

Claim identity: `SFT-MATH-DYNAMICAL-SYSTEMS-001`

15.1 Question and necessity

Computation and natural modelling both require change through state. Exact transitions expose the laws of time, recurrence and information loss without importing continuous equations or stochastic causes.

The exact theorem statement is:

> *An admitted dynamical system is a complete canonical state carrier with a held transition relation. A trajectory retains every state and transition; time is the positive transition trace and the identity trajectory is empty One. Fixed forms, returns, recurrence, basins, reversibility and stability are admitted only through exhaustive generated witnesses. Merged predecessors remain recoverable only when their exact records are retained.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-OPTIMIZATION-001`.

15.2 Generated grammar and exact boundary

Graph paths supply evolution; optimisation supplies complete invariant and basin selection; canonical forms supply state identity. Ten choices force the finite transition-and-record dynamics kernel.

Generation rule: Generate the complete product of state coverage, transition provenance, trajectory trace, time, fixed forms, return, basin/recurrence, reversibility, stability and extra-dynamic status.

Exact grammar boundary: All finite dynamics generated from canonical state carriers, held transition relations, complete paths, return traces, reachability basins, retained predecessor records and registered finite perturbation classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:4a398993d220a557338f9d3f2e0592838df6bccbb7f1d20274106c1a406ad684`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-DYNAMICAL-SYSTEMS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`states`	`partial-or-aliased-state`	Omitted or aliased states change possible evolution.	`complete-canonical-states` - Every generated state occurs once with canonical identity.
`transitions`	`opaque-update-rule`	An opaque update hides which relation cells exist.	`held-state-relation` - Every allowed state transition is a retained generated pair.
`trajectory`	`endpoint-only`	Endpoints erase intervening states and cannot prove reachability.	`complete-transition-trace` - Every adjacent transition is retained in order.
`time`	`continuous-clock-parameter`	A free real clock imports scale and continuum.	`positive-transition-count` - Elapsed time is the exact trace of generated transitions; identity is empty One.
`fixed`	`visual-sameness`	Presentation similarity does not prove dynamical identity.	`retained-identity-transition` - A fixed form has an explicit self relation.
`return`	`periodic-assertion`	A period name without a trace omits return structure.	`explicit-first-return-trace` - A retained nonempty path returns to a prior canonical state.
`basin`	`sampled-neighborhood`	Sampling cannot establish reachability for the registered perturbation class.	`complete-reachability-selection` - Every registered state is exhaustively tested for reachability to the invariant form.
`reversibility`	`infer-unrecorded-predecessor`	A merged image does not identify which predecessor occurred.	`retained-predecessor-record` - The exact source-to-image distinction is held alongside the step.
`stability`	`epsilon-parameter`	A chosen epsilon imports a free scale.	`registered-perturbation-closure` - Every explicitly generated perturbation form satisfies the invariant reachability condition.
`addition`	`extra-force-or-random-law`	An added force, differential equation or random update is not supplied by the state relation.	`no-extra-dynamic-law` - All properties follow from exact finite transition structure.

15.3 Unique survivor, minimality and laws

The dynamical kernel is complete canonical states, held transitions, complete trajectories, transition-count time, witnessed fixed/return/basin structure, retained-record reversibility and registered perturbation stability.

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- trajectory composition is lawful exactly at an equal canonical interface state.
- time is a positive transition trace and identity duration is empty One.
- fixed, cyclic and recurrent behaviour require explicit transition witnesses.
- many-to-one evolution loses predecessor distinction unless a source record is retained.
- stability is quantified only over a complete registered finite perturbation class.

15.4 Operational witnesses and unfavourable checks

- `well-formed` - Every witness transition remains in the complete state carrier. Result: `PASS`.
- `trajectory` - The a-to-b-to-terminal trace retains both allowed transitions. Result: `PASS`.
- `time` - Two transitions are retained exactly and an identity trajectory returns empty One. Result: `PASS`.
- `fixed-form` - The terminal witness has an explicit identity transition. Result: `PASS`.
- `first-return` - A repeated canonical state yields its exact first return trace. Result: `PASS`.
- `predecessor-retention` - Both merged predecessors remain distinct and an exact retained ledger reverses them. Result: `PASS`.
- `registered-stability` - Every registered witness perturbation reaches terminal. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omitted or aliased states change possible evolution. Result: `PASS`; receipt `sha256:644913b363fe507365e0c82e18a6847f33d91874acd59dea32a28f49e1ed33b4`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no continuous real time premise; no differential equation imported as a law; no stochastic transition cause; no unrecorded predecessor reconstruction Result: `PASS`; receipt `sha256:5c5f8dd163823e87bd822f6397adc6f8723ee60d3aa2defdfb5cc33a3ff76649`.

15.5 Depth-independent closure

Base: One state supplies its identity trajectory; no transition time or predecessor ambiguity is introduced.

Successor: Adding one transition appends one exact state pair to every extended trajectory. Fixed, return, basin, predecessor and stability obligations change only for paths that use the new relation and are rechecked completely.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

15.6 Meaning and scientific consequence

Dynamics is state-transition structure before any differential equation. Time is the exact positive trace of transitions, with the identity trajectory represented structurally by empty One. Fixed forms and cycles require explicit returns. Basins and finite stability classes are exhaustive reachability selections. If distinct predecessors merge, reversal requires the precise predecessor record; the image alone cannot recreate a distinction it does not contain.

15.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: state space, trajectory, fixed point, cycle, attractor, reversibility, stability. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

15.8 Exact exclusions and limitations

- no continuous real time premise.
- no differential equation imported as a law.
- no stochastic transition cause.
- no unrecorded predecessor reconstruction.

This closes exact finite dynamical systems. Continuum differential models may later appear only as measured or finite approximation correspondences, never as premises of this theorem.

15.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:366c10101f60725e0e8bc8f29c974d4a7707b564125c8b7608d655e2092ad5a2`.
- Derivation seal: `sha256:972a41752e953bd0b4d12f5c3c7d8e68c2aa2107a5aacaf9e330a2278b1a3a51`.
- Independent implementation: `sha256:6c685056fb0e01ef2133db2ab947796fbade3bd1e246c56479bbac46055751c0`.
- Independent certificate: `sha256:c8e806f9f4e6b74927f0bfe79e2c5b4351113323176e59ecde786d4121290172`.
- External validation: `sha256:f93adcb26bb1c7baeec979059b40325d370b5f2f3d5f50cd6885a5159b10362f`.
- Engine receipt: `sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-DYNAMICAL-SYSTEMS-001-3e0e44b65fbe660f.json`.

16. Derivation 11: Exact finite logic, inference and proof boundaries

Claim identity: `SFT-MATH-LOGIC-PROOF-001`

16.1 Question and necessity

Every branch depends on distinguishable valid and invalid derivations. The proof law makes admission mechanical while preventing conventional logical axioms or institutional authority from entering silently.

The exact theorem statement is:

> *An admitted proposition is a canonical distinguished form with a held orientation; denial is the retained complementary orientation, not a negative number. Conjunction retains joint proofs, disjunction retains a chosen generated branch and its alternatives, implication is a registered premise-to-conclusion transition, and a proof is a complete dependency trace of accepted rules. Consistency, soundness and completeness are machine-checkable only relative to the exact registered finite grammar and rule set.*

Admitted dependencies: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`, `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-ORDER-LATTICE-001`.

16.2 Generated grammar and exact boundary

Canonical forms supply proposition identity; held fibres supply opposed orientations; discrete relations supply rules; order supplies proof-strength relations. Ten choices force the replayable finite proof kernel.

Generation rule: Generate the complete product of proposition identity, denial, conjunction, disjunction, implication, inference provenance, consistency, soundness, grammar completeness and extra-logical status.

Exact grammar boundary: All propositions and finite proofs generated from canonical forms, two held orientations, joint and alternative constructions, registered inference transitions and complete dependency traces.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:c3171696c5a96a3fa85bb809773cca0fc438e1d7196b6657aa3f94ce06e1e622`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-LOGIC-PROOF-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`proposition`	`presentation-sentence`	A sentence presentation can alias or hide the judged form.	`canonical-distinguished-form` - The exact generated form is the proposition identity.
`denial`	`negative-truth-value`	A negative magnitude is outside the admitted domain.	`complementary-held-orientation` - Denial retains the same form with the other exact fibre orientation.
`conjunction`	`single-side-proof`	One side cannot establish the joint claim.	`joint-complete-proof` - Both constituent proof traces are retained together.
`disjunction`	`unidentified-choice`	An unidentified choice erases which alternative is supported.	`held-alternative-with-support` - The held branch and complete generated alternative family are retained.
`implication`	`truth-table-import`	A borrowed table installs conventional logical values.	`proof-carrying-transition` - A registered rule maps exact premises to an exact conclusion.
`inference`	`conclusion-only`	A bare conclusion erases its premises and rules.	`complete-dependency-trace` - Every step retains available premises, rule identity and conclusion.
`consistency`	`assumed-consistent`	An assertion does not search for opposed admitted orientations.	`no-opposed-admission` - No form and its complementary orientation are jointly admitted.
`soundness`	`authority-or-name`	Authority cannot replace validation against registered rules.	`rule-and-witness-preservation` - Every accepted step matches a registered source-bound rule and preserves its witnesses.
`completeness`	`unbounded-global-claim`	A finite proof kernel cannot establish an ungenerated universal domain.	`registered-grammar-exhaustion` - Every proposition in the exact generated grammar has its declared decision evidence.
`addition`	`extra-logical-axiom`	An extra axiom is not supplied by the canonical form and rule traces.	`no-extra-logical-axiom` - All admitted inference is registered and witnessed.

16.3 Unique survivor, minimality and laws

> *The logic/proof kernel is canonical distinguished propositions, complementary held denial, joint and held alternative proof forms, proof-carrying implication, complete inference traces, explicit consistency and soundness, and completeness only over the exhausted registered grammar.*

Shared claim-record clause `MATH-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

- denial changes held orientation while preserving proposition identity.
- conjunction and disjunction retain their complete source proof structure.
- an implication is admitted through a registered proof-carrying transition.
- a valid proof is replayable step by step from its premises and rule registry.
- completeness is always indexed by an explicit generated grammar boundary.

16.4 Operational witnesses and unfavourable checks

- `held-denial` - Denial preserves P while changing only its held orientation. Result: `PASS`.
- `joint-proof` - Conjunction retains both exact constituent proofs. Result: `PASS`.
- `held-disjunction` - Disjunction retains the supported branch and both alternatives. Result: `PASS`.
- `proof-replay` - The registered P-to-Q step replays from its available premise. Result: `PASS`.

- `tampered-rule-control` - Changing the rule identity invalidates the same conclusion trace. Result: `PASS`.
- `consistency` - P and Q are consistent, while P and its held denial are not. Result: `PASS`.
- `bounded-completeness` - Both registered proposition forms have explicit held decisions. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A sentence presentation can alias or hide the judged form. Result: `PASS`; receipt `sha256:6d035b68865a940e61e80880013f484689afa3fbfab68a65b3604a51f203a98c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported truth-value arithmetic; no negative truth magnitude; no authority-based proof admission; no global completeness beyond a generated grammar Result: `PASS`; receipt `sha256:5dff4ae767ffc5d54e6bf8debb3ec71db9285d631376f5ba4b148be47b1cda93`.

16.5 Depth-independent closure

Base: One proposition form and its two orientations provide the first decidable grammar cell.

Successor: Adding one proposition or rule generates its complementary orientation, joint and alternative combinations, and all new premise tuples. Consistency and proof replay are rechecked for precisely those new forms and steps.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

16.6 Meaning and scientific consequence

Logic is reconstructed as distinguished canonical forms and proof-carrying transitions. Denial changes a held orientation; it is not negative truth magnitude. Conjunction retains both supporting traces and disjunction retains the supported alternative and the full alternative family. A proof is valid only by stepwise replay against registered rules. Soundness and completeness are therefore exact but explicitly relative to the generated grammar; no unbounded global completeness claim is smuggled into the result.

16.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: proposition, negation, conjunction, disjunction, implication, proof, soundness, completeness. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

16.8 Exact exclusions and limitations

- no imported truth-value arithmetic.
- no negative truth magnitude.
- no authority-based proof admission.
- no global completeness beyond a generated grammar.

This closes finite generated proof theory. Computability, self-reference, undecidability and incompleteness boundaries require the later formal-computation branch and are not claimed here.

16.9 Evidence identities

- Registered status: `independently_replicated`.

- Source manifest: `sha256:5b549044ca00a139f499155286a06ed09af0735799cf371650de8fd8d336a66d`.
- Derivation seal: `sha256:849294dd48e23a2aefd379804094c01132234adad33b937a3164b3c73f8230bf`.
- Independent implementation: `sha256:636f912fdacfb994061d4018dcbfa82b20c5f99693af0f3f5e9be0ee84b13a42`.
- Independent certificate: `sha256:dfeb338769fcf4b551277891fd7c5576f15ce79e533addbb15e6e4eed87a288a`.
- External validation: `sha256:d3cc342c7a7bf4b50450ae303b15dd250328f761236f869150456ba1550c3cb9`.
- Engine receipt: `sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-LOGIC-PROOF-001-3d366d57b26d002c.json`.

17. Derivation 12: Forced category, type and compositional structure

Claim identity: `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`

17.1 Question and necessity

Every later science must compose local derivations without losing interfaces or provenance. The forced path kernel supplies this language while keeping category and type concepts downstream of Fold structure.

The exact theorem statement is:

> *Canonical Fold forms act as objects and proof-carrying transitions as arrows. Identity is the empty-One return, composition is forced exactly by equal interfaces, and association follows from canonical complete path flattening. Types are source-bound predicates over generated carriers; products are complete joint pair support and sums retain held alternative labels. Functors preserve interfaces, identity and composition, while naturality is equality of the two complete composite traces.*

Admitted dependencies: `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ALGEBRA-001`, `SFT-MATH-LOGIC-PROOF-001`.

17.2 Generated grammar and exact boundary

Graphs supply paths, algebra supplies operation association, and logic supplies proof-carrying transitions. Ten choices force exact interfaces, identities, flat composition, types and preservation structures.

Generation rule: Generate the complete product of objects, arrows, identity, composition, association, types, product/sum structure, functoriality, naturality and extra-composition status.

Exact grammar boundary: All finite compositional structures generated from canonical objects, proof-carrying arrows, exact interface matching, empty-One returns, flattened paths, source-bound type predicates and held joint/alternative forms.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:61559f26b9f1269604ea49148c48493a1565fa04b140255215af667d0a20aa20`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-CATEGORY-TYPE-COMPOSITION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`objects`	`presentation-objects`	Presentation names can alias distinct structures.	`canonical-Fold-objects` - Every object is an exact canonical generated form.
`arrows`	`endpoint-only-arrow`	Endpoints alone erase the transformation proof.	`proof-carrying-transition` - An arrow retains source, target and complete transition trace.
`identity`	`assumed-neutral-map`	A named neutral map has no return witness.	`empty-One-return` - The zero-step structural return preserves the same object without numerical zero.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`composition`	`presentation-concatenation`	Textual concatenation can join unequal interfaces.	`exact-interface-match` - The first target and second source must be the same canonical object.
`association`	`borrowed-category-axiom`	Importing an axiom would reverse the derivation order.	`canonical-path-flattening` - Both groupings retain the same ordered elementary transition trace.
`types`	`informal-class-name`	A class name need not decide carrier membership.	`source-bound-form-predicate` - Every generated carrier form receives one exact membership verdict.
`constructors`	`borrowed-set-constructor`	A borrowed constructor imports its ambient universe.	`joint-pairs-and-held-alternatives` - Products are complete pair support; sums retain left/right held provenance.
`functor`	`arbitrary-renaming`	Renaming can break interfaces and composites.	`identity-and-composition-preservation` - Object and arrow maps retain sources, targets, returns and composites.
`naturality`	`diagram-by-appearance`	A drawn square cannot prove both routes equal.	`equal-complete-paths` - Both interface-compatible composite routes have the same canonical trace.
`addition`	`extra-composition-axiom`	An extra universal property is not supplied by the registered traces.	`no-extra-composition-axiom` - Only generated interfaces, paths and preservation witnesses are admitted.

17.3 Unique survivor, minimality and laws

> *The compositional kernel is canonical Fold objects, proof-carrying arrows, empty-One identity, exact-interface composition, path-flattening association, source-bound types, joint products, held sums, preserving functors and equal-path naturality.*

Shared claim-record clause `MATH-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

- identity preserves source and target and contributes no nonidentity transition.
- composition exists exactly at a canonical interface match.
- associativity is equality of flattened complete transition traces.
- products preserve both coordinates and sums preserve the held source alternative.
- functoriality and naturality are finite replayable preservation witnesses.

17.4 Operational witnesses and unfavourable checks

- `interface-composition` - f then g composes from A to C with both traces retained. Result: `PASS`.
- `identity` - Empty-One return preserves the interface and adds no elementary transition. Result: `PASS`.
- `association` - Both groupings of f, g and h flatten to the same complete path. Result: `PASS`.
- `type-predicate` - The witness predicate retains exactly its admitted carrier forms. Result: `PASS`.
- `product-sum` - Joint products and held sums retain both source coordinates. Result: `PASS`.
- `functor-interface` - The identity transport preserves the witness arrow interface. Result: `PASS`.
- `naturality-path` - Two independently assembled routes with the same elementary trace commute. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Presentation names can alias distinct structures. Result: `PASS`; receipt `sha256:0b1412200b24f35693db81fac968726a12b5f6d4a14d7866a33dfdfa597ffe8a`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported category axioms; no ungenerated universe of all objects; no impredicative type constructor; no universal property without an exhaustive finite witness Result: `PASS`; receipt

`sha256:b3ab5d7231f463ee56b4759ebf5e5a9ec0c2a2f15ff312709eed57a7ba33281d`.

17.5 Depth-independent closure

Base: One canonical object supplies its empty-One identity arrow and the first singleton type.

Successor: Adding one object or arrow generates its identities, every exact interface-compatible composite, product and sum cells, type verdicts and preservation squares. Closure follows by appending its trace to prior flat paths.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

17.6 Meaning and scientific consequence

Composition is the final mathematical unifier of the branch. Canonical forms are objects and complete proof-carrying transitions are arrows. Equal interfaces force lawful composition; empty-One return supplies identity; and associativity is equality of flattened elementary paths. Types are exact predicates over a registered carrier. Products retain joint coordinates, sums retain held origin, and functorial or natural structure is admitted by replayable path-preservation witnesses.

17.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: category, object, morphism, identity, composition, type, product, sum, functor, natural transformation. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

17.8 Exact exclusions and limitations

- no imported category axioms.
- no ungenerated universe of all objects.
- no impredicative type constructor.
- no universal property without an exhaustive finite witness.

This closes the finite compositional kernel. Any stronger universal construction must be separately generated and exhaustively witness its claimed mapping property inside an explicit grammar.

17.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:8962e0d82e8291e69ae4b1477518316402aea5b3e663f08b418e88bec287d135`.
- Derivation seal: `sha256:756d7d6dce776fc1fb4b3906e2203e4e2da4e1ded6cf733eb041334eb7cb5c02`.
- Independent implementation: `sha256:ced0eda6a320565b205b93618d604af7b494f3ad001044d71c87df0f4c281f90`.
- Independent certificate: `sha256:1d380a6b0d0ce9d32cfdff52fae121315257e3e561da9379ab5509c835680dad`.
- External validation: `sha256:fc7b4ce671b53f4d62da07705c4fc5d92b0195d5218a30f2a40a773a2fb7c981`.
- Engine receipt: `sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-CATEGORY-TYPE-COMPOSITION-001-47873a83a90e29fa.json`.

18. Derivation 13: Exact ratio, separation, reciprocal, threshold and cycle composition

Claim identity: `SFT-MATH-EXACT-RELATIONS-002`

18.1 Question and necessity

V1/V2 separately registered ratio, separation, threshold, beat and joint-cycle laws that the broad arithmetic claim did not enumerate atomically.

The exact theorem statement is:

> *Exact generated relations force positive ratios, shorter-part separation, reciprocal return to the One, the unique m-fold holding complement, local Fold separation transport, telescoping relative views, and least-common-return composition for every supplied finite family of exact cycles.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`.

18.2 Generated grammar and exact boundary

Common refinement forces exact ratios; held orientation preserves a positive remainder; Fold supplies its own transport count; exhaustive returns force the least common return.

Generation rule: Exhaust the provenance, relation, orientation, threshold, transport, composition and generality choices.

Exact grammar boundary: Exact positive generated parts, held orientation and supplied positive finite cycle families.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:8ea91d0dcba9fca561d79bb2a7cc1dcc43af1caa1861db2123c98f81a5366bc0`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-EXACT-RELATIONS-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`provenance`	`borrowed-field`	A borrowed field imports the result.	`generated-parts` - Both terms retain generated part provenance.
`relation`	`rounded-scalar`	Rounding destroys exact identity.	`exact-ratio` - Common refinement forces the positive ratio.
`orientation`	`signed-gap`	Signed magnitude is outside the domain.	`held-shorter-gap` - A held side records orientation and the shorter positive part records separation.
`threshold`	`selected-angle`	A selected angle adds a parameter.	`unique-complement` - The m-fold complement is the unique (m less One)-of-m part.
`transport`	`fitted-rate`	A fitted rate is not derived.	`fold-factor` - One Fold transports an uncast local gap by the generated fibre count.
`composition`	`sampled-return`	Sampling cannot prove joint return.	`least-common-return` - The least common multiple is the first shared return.
`generality`	`fixed-table`	A fixed table has no successor certificate.	`finite-family-induction` - Adding one cycle composes its period by one further least-common return.
`addition`	`extra-rule`	An extra rule violates zero-parameter closure.	`no-extra-rule` - Only admitted part and cycle operations occur.

18.3 Unique survivor, minimality and laws

> *The unique exact relation kernel is generated-part ratio, held shorter separation, unique complement threshold, Fold-factor transport and least-common-return composition with finite-family induction.*

Shared claim-record clause `MATH-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

- a positive ratio composed with its reciprocal returns the One.
- relative views telescope by exact ratio composition.
- the first common return of finite periods is their iterated least common multiple.
- count times the corresponding equal part reconstructs the One.

18.4 Operational witnesses and unfavourable checks

- `reciprocal` - The ratio three-of-two composed with two-of-three returns the One. Result: `PASS`.
- `count-measure` - Eight equal one-of-eight parts reconstruct the One. Result: `PASS`.
- `telescoping` - The exact views five-of-three and three-of-two compose to five-of-two. Result: `PASS`.

- `threshold` - The binary holding complement is the unique one-of-two part. Result: `PASS`.
- `joint-return` - Periods two, three and four first return together at twelve. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A borrowed field imports the result. Result: `PASS`; receipt `sha256:f4a735a031ef37a2a648b155e7a8b2f3c18457f33a93f23ab569081113e28c09`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

18.5 Depth-independent closure

Base: One exact cycle returns after its own generated period.

Successor: Composing one more positive finite cycle replaces the current joint period by its least common return with the new period.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

18.6 Meaning and scientific consequence

This reconstruction joins the relation laws that V1/V2 stated separately. Ratio is a common-refinement identity; separation is a positive shorter part with held orientation; reciprocal composition returns the One; the m-fold threshold is its unique complement; and finite cycles return jointly at the least common period. The result is stronger than a table because adding one cycle extends the same return certificate.

18.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: ratio, distance, reciprocal, threshold, least common multiple, beat period. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

18.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The cycle theorem applies to supplied positive finite exact periods and never constructs a completed infinite family.

18.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:57cdb2201f91bc9edd24a9a610305d92426d3d47ee69b2f4c70283343738c3d2`.
- Derivation seal: `sha256:516d208341bde6e161b32070c8051f089f2d0f11ccc3748c75680c6c4e4d1dc6`.

- Independent implementation: `sha256:9b3d9ce6df6053f3f2a7b6935480a45bd0a5daaca31c76fb3beaeadba9f79530`.
- Independent certificate: `sha256:2ac8d699989a021d04fe3f921d3858795423d28863d83fcfb06542408a254a36`.
- External validation: `sha256:3d32599f5a7999ee642da792bdd7e0fbde43faf5e6a8a73468ed802b380396b1`.
- Engine receipt: `sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-EXACT-RELATIONS-002-8141546a120b73aa.json`.

19. Derivation 14: Fold orbit number theory and exact prime-spectrum structure

Claim identity: `SFT-MATH-ORBIT-NUMBER-THEORY-002`

19.1 Question and necessity

The orbit/order identity, prime-period division, cyclotomic tiling and finite Artin census were explicit prior Mathematics results.

The exact theorem statement is:

> *For every reduced part over an odd positive finite denominator, Fold orbit length is exactly the multiplicative order of the binary generator; reduced residues partition into equal cyclotomic orbits, prime periods divide one-less-than-prime, and a power-of-two factor supplies exactly its counted transient depth.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`.

19.2 Generated grammar and exact boundary

Fold return of p/q requires a first k with p times 2^k congruent to p; reduction cancels p and leaves exactly the definition of the binary multiplicative order.

Generation rule: Exhaust denominator domain, reduction, transition, period, partition, transient, generality and extra-rule choices.

Exact grammar boundary: All supplied positive finite reduced rational parts, with odd cores and counted binary factors.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:e0aad69a9214fa26f8480e30923fd5a5399c6cbcc15ac6ec0a5ecad661f97921`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ORBIT-NUMBER-THEORY-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`domain`	`arbitrary-real`	An arbitrary real imports infinite information.	`finite-rational` - Generated parts have finite whole numerator and denominator traces.
`reduction`	`untracked-factor`	An untracked common factor duplicates state identity.	`lowest-terms` - The reduced denominator fixes the orbit class.
`transition`	`sampled-map`	A sampled map is not the Fold.	`double-and-cast` - Fold doubles the residue and casts complete denominators.
`period`	`observed-cycle`	An observed cycle has no arithmetic identity.	`multiplicative-order` - Return is exactly the first binary power congruent to the One residue.
`partition`	`overlap-cover`	Overlapping cycles duplicate residues.	`cyclotomic-tiling` - Disjoint equal-length orbits tile all reduced residues.
`transient`	`eternal-even-core`	A binary factor cannot remain under repeated Fold.	`counted-binary-depth` - Each Fold removes one binary denominator factor before the odd cycle.
`generality`	`bounded-prime-list`	A bounded list is not the theorem.	`definition-identity` - The Fold return predicate and multiplicative-order predicate are the same remainder identity for every supplied denominator.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
addition	extra-number-law	An imported number-theory theorem would select the answer.	no-extra-rule - Only Fold, whole remainder and exact counting occur.

19.3 Unique survivor, minimality and laws

Fold return and binary multiplicative order are one exact remainder identity; odd reduced residues tile into cyclotomic orbits and each binary denominator factor contributes one transient Fold.

Shared claim-record clause MATH-S001 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- orbit length equals the multiplicative order of the binary generator modulo the reduced odd denominator.
- the reduced-residue count is the product of orbit length and orbit count.
- for a prime denominator the period divides one less than that prime.
- binary valuation is the exact transient depth to the odd recurrent core.

19.4 Operational witnesses and unfavourable checks

- orders - The independently counted orders for denominators three, five, seven, nine and eleven are two, four, three, six and ten. Result: PASS.
- all-reduced-ranks - Every reduced rank through odd denominator ninety-nine has orbit length equal to its denominator order. Result: PASS.
- prime-division - Each tested prime period divides one less than the prime. Result: PASS.
- orbit-tiling - Reduced residues for denominator seventy-three tile into eight disjoint orbits of length nine. Result: PASS.

The engine-level adverse controls are:

- false_premise - expected: reject a generated form missing the first required structural condition Observed: An arbitrary real imports infinite information. Result: PASS; receipt sha256:fea1540fb44cc68fbfc743b66ed93b6afee3da4b0032929ab3b8cc8ce3b8c976.
- tampered_source - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: PASS; receipt sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237.

19.5 Depth-independent closure

Base: The first reduced odd denominator above the One executes one complete finite orbit.

Successor: For any next supplied odd denominator the finite residue permutation independently partitions into cycles; adjoining one binary factor prepends exactly one transient Fold.

Shared claim-record clause MATH-S002 **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

19.6 Meaning and scientific consequence

Number theory appears here as Fold dynamics, not as imported doctrine. Returning p/q after k Folds and requiring the binary generator to have order k modulo the reduced odd denominator are literally the same remainder identity. That identity forces the cyclotomic partition, prime-period divisor law and exact binary transient depth. The Artin and asymptotic prime questions remain visible boundaries rather than inherited claims.

19.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: multiplicative order, Fermat period, cyclotomic coset, primitive root, prime spectrum. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

19.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The theorem identifies the exact orbit skeleton. It does not claim an asymptotic prime-counting law or prove Artin's conjecture.

19.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:beec43280b54374b36d3da29ec4ca906cd28cee4decfcb7cc49d6435fbc499f4`.
- Derivation seal: `sha256:8c00984dae257d972ebc6ed4e059a3c958f43e2c29b39f9775f78883dd376086`.
- Independent implementation: `sha256:66273410b8c810940888e97f3fd03d86752ca8ca1f450ae3cf1fcfa9b5af5142`.
- Independent certificate: `sha256:b47923bcf2b3c784ecd0689286d35a111a91faa6f86ec1c1c8971414772bcc6d`.
- External validation: `sha256:1ff8387a5a11bb9bd5bf273921e948e63554192e6cd284b33ef50734a2254174`.
- Engine receipt: `sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ORBIT-NUMBER-THEORY-002-8725bce36fe77923.json`.

20. Derivation 15: Potential infinity, exact refinement and the continuum boundary

Claim identity: `SFT-MATH-LIMIT-CONTINUUM-002`

20.1 Question and necessity

V1/V2 registered continuum-limit, potential-infinite, continuum-hypothesis and finite-inventory statements that require one exact object boundary.

The exact theorem statement is:

> *Every generated rung is a finite exact positive part with a counted successor; refinement can continue without a greatest finite depth, exact rational sequences can carry monotone bounds and convergence certificates, and no completed infinity, unbounded denominator or continuum is admitted as a proof object.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-ORDER-LATTICE-001`.

20.2 Generated grammar and exact boundary

Fold assembly supplies a base and successor but never an already-completed unbounded collection. Exact convergence therefore lives in replayable rational bounds, not an imported limit value.

Generation rule: Exhaust object, successor, refinement, convergence, continuum, cardinality, generality and extra-rule choices.

Exact grammar boundary: Generated positive finite depth traces and exact rational bounds at every supplied finite stage.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b146824353456f9812e707aca84e39c4f1c7aaf50beda6e2d64bab7a2f94090b`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-LIMIT-CONTINUUM-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`object`	`completed-infinity`	A completed unbounded totality is not generated.	`finite-stage` - Every admitted stage has a complete finite trace.
`successor`	`largest-stage`	No generated rule supplies a largest finite stage.	`always-next` - Appending one Fold supplies the next finite stage.
`refinement`	`vanishing-zero`	Numerical zero is outside the domain.	`positive-rung` - Every refinement remains a positive exact part.
`convergence`	`limit-as-value`	An ungenerated limit cannot enter as a value.	`nested-rational-bounds` - Exact nested rational bounds certify approach and error.
`continuum`	`ambient-real-line`	An ambient continuum imports forbidden objects.	`unbounded-process-boundary` - The continuum is correspondence for an unbounded refinement process.
`cardinality`	`completed-comparison`	Cardinality of a nonobject has no SFT witness.	`finite-bound-count` - Every reached bound has an exact generated count.
`generality`	`fixed-depth`	One fixed depth does not close the successor law.	`base-successor` - The positive-rung property is preserved by every next finite refinement.
`addition`	`extra-limit-rule`	A limit oracle adds an axiom.	`no-extra-rule` - Only exact parts, bounds and successor traces occur.

20.3 Unique survivor, minimality and laws

> *Potential infinity is the never-terminal generated successor process; every reached object is finite and positive, convergence is an exact nested-bound certificate, and completed infinity or continuum is outside the admitted object language.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- every dyadic rung reaches the One in exactly its counted depth.
- every finite depth has a lawful next depth.
- exact finite-difference sequences may converge by shrinking positive rational error bounds.
- no completed continuum cardinality claim is admitted without a generated object.

20.4 Operational witnesses and unfavourable checks

- `finite-rungs` - Every generated dyadic rung through depth fourteen is positive and returns after its counted depth. Result: `PASS`.
- `quadratic-curvature` - The centred second difference of x squared is exactly two at every tested rational spacing. Result: `PASS`.
- `cubic-convergence` - The forward second-difference curvature of x cubed at the One has exact error six times the spacing, halving at each refinement. Result: `PASS`.
- `finite-closure` - The first five dyadic rungs plus the boundary rung reconstruct the One. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A completed unbounded totality is not generated. Result: `PASS`; receipt `sha256:579bf8f8ad064ae312c49bd3194500e06ac9a4dff8dc68b6498b2dcc8c1d9233`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.

- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

20.5 Depth-independent closure

Base: The One is the complete finite base and its first refinement is a positive exact part.

Successor: Appending one generated Fold depth replaces a positive one-of-b^k rung by the positive one-of-b^(k+1) rung and records one additional return step.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

20.6 Meaning and scientific consequence

The theorem distinguishes an indefinitely extendable procedure from an already completed infinite object. Every reached rung is finite, exact and positive, and every rung has a lawful successor. Convergence is carried by nested rational bounds and shrinking exact errors. The continuum can describe that open-ended refinement at correspondence, but cannot enter the derivation as an ungenerated container or answer-producing cardinality.

20.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: potential infinity, constructive limit, continuum, continuum hypothesis, finite difference convergence. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

20.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This is a theorem about the SFT object language and exact finite certificates; it does not deny the utility of continuum correspondence outside derivation.

20.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:3e1480da83b7864de42150a1dfe2616b954a893ff1f782267ae8de12e497682b`.
- Derivation seal: `sha256:b4d6ada858b868c1e51dbc259fafd1672b962526efc836f55e2d8e4958729e94`.
- Independent implementation: `sha256:cb2b5e3aa8dfc0485cffc29b26c677a7abad0c1e2a8cec704353005ff760820f`.
- Independent certificate: `sha256:cd1fbdbe8cc78f9e9a0ef75dc32657fab1b1173c5a44cb9c1e871dfd05321f3c`.
- External validation: `sha256:5873d50654bda8941184c01fe4fabee288a50208f9b899a81f529dfbf2c05ef6`.
- Engine receipt: `sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-LIMIT-CONTINUUM-002-a32c6d6ee891f97a.json`.

21. Derivation 16: Positive polynomial-balance certificates for algebraic magnitudes

Claim identity: `SFT-MATH-ALGEBRAIC-BALANCE-002`

21.1 Question and necessity

The V1 algebraic-magnitude engine and V2 Vieta cross-check require an explicit representation consistent with the prohibition on irrational proof values.

The exact theorem statement is:

> *A magnitude not itself admitted as an irrational value may be identified by two positive-coefficient polynomial sides, an exact rational bracket in which their order swaps, and a generated bisection trace that narrows that bracket without ever importing the root as a proof value.*

Admitted dependencies: `SFT-MATH-ALGEBRA-001`, `SFT-MATH-ORDER-LATTICE-001`.

21.2 Generated grammar and exact boundary

Exact arithmetic evaluates two positive polynomial sides; exact order supplies a bracket; repeated common refinement supplies a replayable enclosure without adding the missing scalar field.

Generation rule: Exhaust representation, coefficients, enclosure, refinement, identity, reconstruction, generality and extra-rule choices.

Exact grammar boundary: Positive polynomial-side evaluations and exact rational brackets at supplied finite refinement depth.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:143a3ac73bd3e1ce5da20a137c0b51e4aad53f7d86e2a29717a115408881616d`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-ALGEBRAIC-BALANCE-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`representation`	`irrational-scalar`	The scalar is forbidden as a proof value.	`balance-certificate` - The object is the equality condition between two positive sides.
`coefficients`	`signed-polynomial`	Negative coefficients import signed magnitude.	`positive-separated-sides` - Terms are moved to two positive sides with held orientation.
`enclosure`	`decimal-guess`	A decimal guess is not exact.	`rational-order-swap` - Exact endpoint comparisons bracket the balance.
`refinement`	`floating-iteration`	Floating iteration loses exact provenance.	`exact-bisection` - Each midpoint is an exact rational part.
`identity`	`approximate-equality`	Tolerance cannot define the algebraic object.	`polynomial-balance` - The defining identity is exact equality of the two sides.
`reconstruction`	`unverified-root`	An asserted root has no certificate.	`bracket-trace` - The full nested bracket trace reconstructs every decision.
`generality`	`named-radicals`	A radical list omits higher-degree forms.	`finite-polynomial-induction` - Each supplied finite polynomial pair is evaluated by the same exact trace law.
`addition`	`extra-field`	An algebraic field is not supplied by Foundation.	`no-extra-rule` - Only arithmetic, order and exact parts occur.

21.3 Unique survivor, minimality and laws

> *The unique admitted algebraic-magnitude representation is a positive polynomial-balance identity with exact rational order-swap bracket and replayable bisection trace; the irrational root is never a proof value.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- separating signed terms into two positive sides preserves the defining balance.
- an endpoint order swap plus exact bisection yields nested rational enclosures.
- Vieta relations provide an implementation-distinct coefficient cross-check for a bracketed root family.

21.4 Operational witnesses and unfavourable checks

- `square-balance` - The positive sides x squared and two swap order across seven-of-five and three-of-two. Result: `PASS`.
- `exact-midpoints` - Twenty bisections of that bracket retain exact rational endpoints and an order swap. Result: `PASS`.
- `vieta-rational` - The polynomial with roots one-of-two, one-of-three and one-of-six has exact sum One and exact pair sum eleven-of-thirty-six. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: The scalar is forbidden as a proof value. Result: `PASS`; receipt `sha256:98b59157c376c2de4e3224bf91fa0fa031bc85c840cfea93fe45c1599a773101`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

21.5 Depth-independent closure

Base: One exact rational bracket with opposite side order is a complete first enclosure.

Successor: The exact midpoint retains the order swap in exactly one half-bracket unless it is itself the exact balance, so the certificate narrows at every supplied finite step.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

21.6 Meaning and scientific consequence

The no-irrational rule does not erase algebraic geometry or spectral magnitude. It changes the proof object. Instead of storing an irrational scalar, SFT stores the exact equality of two positive polynomial sides and a rational bracket whose order reverses. Exact bisection narrows the enclosure, while Vieta identities cross-check whole root families. The object is its replayable balance certificate, never a rounded decimal.

21.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: algebraic number, root isolation, bisection, Vieta identities, Eudoxan magnitude. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

21.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The certificate identifies an algebraic balance to any supplied finite rational enclosure depth; it does not admit the incommensurable magnitude as a scalar proof object.

21.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:e12152b418d417621346edd4eabd39be17b1dabe6741c85682aa51290b3c8736`.
- Derivation seal: `sha256:5e3dbc95200564d41c0418f508c6fb35eaadfd1a0077f49a9faae4aa1eaeaf14`.
- Independent implementation: `sha256:e9a7d54155ece0dcb7d43f8027e3f531eb4b2f2f67d5433cd29e4a82b4b4a32a`.
- Independent certificate: `sha256:d309c7ff8f819486d4e24671c0e8a7587328436e35e6b0f0343f9d37b6618202`.
- External validation: `sha256:e2d92d98d73d97f48673bbc2181cdf7a54b3db60afd4e0ef58bd2d4022d5efff`.
- Engine receipt: `sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-ALGEBRAIC-BALANCE-002-ba975fdfe0c8946c.json`.

22. Derivation 17: Exact recurrence of every finite componentwise Fold configuration

Claim identity: `SFT-MATH-BOUNDED-N-BODY-002`

22.1 Question and necessity

The V1/V2 three-body and general n-body results were explicitly restricted to Fold-built bounded-denominator configurations.

The exact theorem statement is:

> *Every supplied positive finite tuple of reduced rational Fold states has a finite exact joint support; after its transient binary depths, the componentwise Fold configuration recurs with period equal to the least common multiple of the odd-core component periods.*

Admitted dependencies: `SFT-MATH-ORBIT-NUMBER-THEORY-002`, `SFT-MATH-EXACT-RELATIONS-002`.

22.2 Generated grammar and exact boundary

Each component has a finite transient and odd-core cycle. Their Cartesian product is finite and deterministic; exact return occurs at the iterated least common period.

Generation rule: Exhaust state domain, component rule, denominator custody, support, recurrence, period, generality and extra-dynamics choices.

Exact grammar boundary: Finite tuples whose dynamics is exactly componentwise Fold on generated rational parts.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:ba9434365f40862645ffa648cf1c71c99e659011a45ccdb7f7745a13425a4d6d`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-BOUNDED-N-BODY-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`state_domain`	`continuum-coordinates`	Continuum coordinates are outside the object language.	`finite-rational-tuple` - Each component has a finite exact trace.
`component_rule`	`imported-force-law`	An imported force law changes the system.	`componentwise-fold` - Each component advances only by Fold.
`denominator`	`unbounded-drift`	Fold cannot introduce a new denominator factor.	`conserved-odd-core` - The reduced odd denominator core is invariant.
`support`	`uncounted-space`	An uncounted space cannot establish recurrence.	`finite-product-support` - The joint support is the finite product of component supports.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`recurrence`	`chaos-label`	A label does not defeat a finite deterministic return certificate.	`exact-return` - The joint tuple is replayed exactly.
`period`	`selected-period`	A selected period is not forced.	`least-common-period` - The first joint return is the least common component return.
`generality`	`three-body-table`	One tuple size is not the theorem.	`tuple-successor` - Appending one component extends the period by one least-common return.
`addition`	`external-dynamics`	External gravitational dynamics is a different claim.	`no-extra-rule` - The theorem contains only componentwise Fold.

22.3 Unique survivor, minimality and laws

All finite componentwise Fold tuples are eventually recurrent; their recurrent joint period is the iterated least common multiple of component odd-core periods.

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- Fold preserves each reduced odd denominator core.
- a finite product of finite deterministic component orbits is finite.
- the first joint recurrent return is the least common component period.

22.4 Operational witnesses and unfavourable checks

- `three-cycle` - The tuple one-, two- and four-of-seven returns after three componentwise Folds. Result: `PASS`.
- `four-cycle` - The four ranks over denominator five rotate and return after four Folds. Result: `PASS`.
- `joint-period` - Components over denominators three, five and seven return jointly after twelve Folds. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Continuum coordinates are outside the object language. Result: `PASS`; receipt `sha256:44a51a0546406e551657eda4a8017def7dcf9eb869401c2fb7045d7e03f36db6`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

22.5 Depth-independent closure

Base: One rational Fold component is transient-to-periodic by its denominator decomposition.

Successor: Appending one component preserves finite product support and replaces the recurrent period by its least common return with the new component period.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

22.6 Meaning and scientific consequence

The n-body theorem is exact for the model V1/V2 actually names: a finite tuple of rational states advanced componentwise by Fold. Each component has a finite transient and odd-core cycle; their product is finite and its first joint return is the least common period. This does not swap in continuum Newtonian dynamics after the fact. It demonstrates precisely how the SFT native dynamics makes every finite tuple replayable and recurrent.

22.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: finite dynamical system, three-body choreography, n-body recurrence, joint period. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

22.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This theorem is not a claim about arbitrary continuum Newtonian or relativistic n-body differential equations; it closes the native SFT componentwise Fold model stated by the prior record.

22.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:cdbbb4b8d5888934ecf54a47c059ec62c8dcc975af4bd9c5d0e50509cdfb2aec`.
- Derivation seal: `sha256:847c6bb6921ac26739ce553890f204b1a5b39c3cd085a008554db882434fc93d`.
- Independent implementation: `sha256:a631e8eb34441bc501703dcc24f92b1c5949aa0d93b9b58769a2463326d81a8c`.
- Independent certificate: `sha256:1d3d888f14db2f03b13c38a139422f63da47c5f5753dfd7483c543aec36280d3`.
- External validation: `sha256:456731eecc5f2e6ff6ffaebf17f952836f77a5c93554fc3f4d8d140a5a90b481`.
- Engine receipt: `sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-BOUNDED-N-BODY-002-e355bc5aa649df8f.json`.

23. Derivation 18: Depth-independent discrete-gradient bound for floored Fold fluids

Claim identity: `SFT-MATH-FLOORED-FLUID-REGULARITY-002`

23.1 Question and necessity

The prior Navier-Stokes and CFD statements require the exact native-lattice regularity theorem and an explicit boundary against the distinct continuum Millennium grammar.

The exact theorem statement is:

> *At every supplied positive finite Fold depth k, an exact lattice with minimum spacing one-of-b^k and velocity separation at most the One has discrete gradient and vorticity magnitude bounded by b^k; at depth five the exact bound is thirty-two, so no finite-depth Fold-lattice blow-up is expressible.*

Admitted dependencies: `SFT-MATH-GEOMETRY-TOPOLOGY-001`, `SFT-MATH-LIMIT-CONTINUUM-002`.

23.2 Generated grammar and exact boundary

Generated geometry supplies a positive minimum edge; the One ceiling supplies the maximum positive velocity gap; exact quotient forces their reciprocal-floor bound.

Generation rule: Exhaust lattice, spacing, velocity, derivative, bound, conservation, generality and continuum-scope choices.

Exact grammar boundary: Exact finite Fold lattices with positive minimum spacing and bounded positive velocity differences.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:b0c0b232760703024bf387b6944b8719cca03fcc99d223697cc4e1d8c8c51639`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-FLOORED-FLUID-REGULARITY-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`lattice`	`continuum-points`	An ungenerated continuum has no smallest spacing.	`generated-grid` - Every site and edge is generated.
`spacing`	`vanishing-length`	Numerical zero is outside the domain.	`positive-floor` - Minimum spacing is one exact rung.
`velocity`	`unbounded-speed`	Unbounded speed violates the One ceiling.	`one-bounded-gap` - Velocity separation is at most the One.
`derivative`	`limit-derivative`	A continuum limit changes the claim.	`edge-quotient` - The discrete gradient is exact gap over exact edge spacing.
`bound`	`inserted-cap`	An inserted cap would be an engineering parameter.	`derived-reciprocal-floor` - The maximum quotient is forced by One over the minimum rung.
`conservation`	`floating-rescale`	Floating rescaling cannot prove exact transport.	`edge-ledger` - Every transferred part retains source and destination identity.
`generality`	`depth-five-only`	One depth is only an example.	`depth-successor` - At k plus One the reciprocal-floor bound multiplies by the Fold count.
`scope`	`continuum-millennium`	The classical continuum problem is not the same grammar.	`fold-lattice-regularity` - The theorem is exactly the native finite-depth Fold lattice.

23.3 Unique survivor, minimality and laws

> *The native Fold-lattice gradient bound is b^k at depth k, attained by a One velocity gap across one minimum edge; the depth-five value is thirty-two and every finite-depth field remains bounded.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- positive minimum spacing prevents a vanishing denominator.
- One-bounded velocity separation divided by one-of-b^k is at most b^k.
- the successor depth multiplies the bound by the generated Fold count.

23.4 Operational witnesses and unfavourable checks

- `depth-five` - The exact reciprocal of the depth-five binary floor is thirty-two. Result: `PASS`.
- `depth-table` - Every depth one through fourteen has reciprocal-floor bound two-to-depth. Result: `PASS`.
- `mass-ledger` - An exact transport split preserves a sample whole mass ledger. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An ungenerated continuum has no smallest spacing. Result: `PASS`; receipt `sha256:8afa8db26073931de8036d7069b515191f55870645ca256d0c09fba76b1fee7d`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

23.5 Depth-independent closure

Base: At the first Fold depth, the positive floor is one-of-b and the edge-gradient bound is b.

Successor: Refining one Fold divides the floor by b while preserving the One velocity ceiling, so the exact finite bound is multiplied by b.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

23.6 Meaning and scientific consequence

Regularity is forced from two structural bounds: velocity separation cannot exceed the One and lattice spacing cannot fall below its declared positive rung. Their exact quotient is b^k at depth k, giving 32 at depth five. The number is therefore not an artificial CFD stabilizer. The paper keeps the categorical distinction between this depth-independent Fold-lattice theorem and the separate classical continuum Millennium problem.

23.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: discrete fluid, vorticity bound, Navier-Stokes, finite-volume lattice, regularity. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

23.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This closes SFT floored-lattice regularity. It does not claim a proof of the classical continuum Navier-Stokes existence-and-smoothness problem.

23.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:80e3b092ab9de1ffc5d8cdff13dc5f12f42cd26c5f0cd360366228db1b277135`.
- Derivation seal: `sha256:18460e9edef12ead62e6409f1a92e8cc5a6e8dec68cde2e1c44e36aa76d20ff2`.
- Independent implementation: `sha256:e51e73101b8bd2730f2aa03837b17acfbd5adc8c861034f703dba1b45359d6ec`.
- Independent certificate: `sha256:49330684436f47d6548aa71275bc43bef3c45ca37528b6d574ddf24fb4d430d2`.
- External validation: `sha256:4b299d88a28e233a46659148ceb5fe0662ef1714e84a8e23302ea7be4765abc2`.
- Engine receipt: `sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-FLOORED-FLUID-REGULARITY-002-6fb3f40f5631367f.json`.

24. Derivation 19: Exact bounded Goldbach and twin-prime census

Claim identity: `SFT-MATH-PRIME-PAIR-CENSUS-002`

24.1 Question and necessity

V2 Step 278 registered exact bounded counts; an unrestricted Goldbach or twin-prime claim was not machine-established and is explicitly excluded.

The exact theorem statement is:

> *Exhaustive exact whole-number enumeration proves that every even whole from four through ten thousand has at least one prime complementary pair, that this range contains exactly four-thousand-nine-hundred-ninety-nine tested evens with no failure, and that exactly two-hundred-five twin-prime pairs have upper member at most ten thousand.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`.

24.2 Generated grammar and exact boundary

Generated whole arithmetic supplies the interval, complements and divisor certificates; exhaustive enumeration supplies exact finite coverage without an imported prime table.

Generation rule: Exhaust range, primality, pair generation, complement, coverage, twin adjacency, boundary and extra-theorem choices.

Exact grammar boundary: Positive finite whole-number census with even inputs four through ten thousand inclusive.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:c101ca3690ea2f183271329f9aa7671c7441160017c056df669fa3d70748debc`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-PRIME-PAIR-CENSUS-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`range`	`sampled-evens`	Sampling cannot close the declared range.	`all-declared-evens` - Every even input in the frozen range is executed.
`primality`	`borrowed-table`	A borrowed table hides verification.	`trial-certificate` - Each prime status is independently decided by complete divisor search to the square bound.
`pairs`	`one-guessed-pair`	A guess omits alternatives and failures.	`all-complements` - Every positive complementary split is generated.
`complement`	`signed-sum`	Signed arithmetic is unnecessary.	`positive-junction` - Both prime parts join exactly to the even whole.
`coverage`	`spot-check`	Spot checks do not prove the bounded census.	`exhaustive-census` - All 4,999 declared evens receive a verdict.
`twins`	`approximate-density`	A density estimate is not an exact count.	`exact-gap-two` - Every prime pair at whole gap two is counted once.
`boundary`	`unrestricted-conjecture`	The finite run cannot prove an unrestricted conjecture.	`explicit-ten-thousand` - The exact boundary is sealed into the claim.
`addition`	`external-answer`	An external published count cannot enter generation.	`no-extra-rule` - Only exact arithmetic and complete enumeration occur.

24.3 Unique survivor, minimality and laws

> *The exhaustive 4 through 10,000 census contains 4,999 even inputs, zero Goldbach failures and exactly 205 twin-prime pairs with upper member at most 10,000.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- every declared even input is partitioned by all positive complementary pairs.
- prime status is decided by a complete finite divisor certificate.
- the finite boundary is part of the theorem and cannot be erased.

24.4 Operational witnesses and unfavourable checks

- `goldbach-range` - Every even whole from four through ten thousand has a generated prime complement pair. Result: `PASS`.
- `even-count` - The declared range contains exactly four thousand nine hundred ninety-nine even inputs. Result: `PASS`.
- `spot-counts` - Twelve has one unordered prime pair and one hundred has six. Result: `PASS`.
- `twin-count` - The complete twin-prime count through ten thousand is two hundred five. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot close the declared range. Result: `PASS`; receipt `sha256:f0d8ec988eaf6a70e795c33155c0ec30332c2558947a51a3fcc305cdcdadb26c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

24.5 Depth-independent closure

Base: The first declared even whole, four, is the junction of two and two.

Successor: The census successor advances to the next even whole, exhausts every complementary split and records either a prime witness or an explicit failure.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

24.6 Meaning and scientific consequence

This is a genuine exhaustive empirical proof over a frozen finite mathematical population. Every one of 4,999 even wholes is generated, every complement is checked and every prime status is independently recomputed. The result has zero failures and the twin census is exactly 205. Its strength comes from total declared coverage, not from pretending that a boundary of 10,000 proves an unrestricted conjecture.

24.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Goldbach census, twin primes, primality certificate, finite exhaustive proof. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

24.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This is an exact finite theorem through 10,000, not an unrestricted proof of Goldbach's conjecture or the twin-prime conjecture.

24.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:a5a78fe935ed3c85b51d8123cbae2d6fc773a1d384e74f1ea0379ac7effb151c`.
- Derivation seal: `sha256:670c35bc2174d10dd88d35ee6892f424ce2a6c039ea35b2d1c22652511398f80`.
- Independent implementation: `sha256:50dd6ed96c8f0d57441b24b3d3a605e43c57146cb5c82112a757a01058a7ba03`.

- Independent certificate: `sha256:ef0331a6ee13bc25a810cc2bebef35fb073d1d91c695a60e6963b02fe9b1d572`.
- External validation: `sha256:a3643b08d7428ba160534292675aba56bd828fa8b80dd84f938a492633b42fc1`.
- Engine receipt: `sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-PRIME-PAIR-CENSUS-002-52fcc090c336360d.json`.

25. Derivation 20: Unique half-One reflection axis and Riemann correspondence boundary

Claim identity: `SFT-MATH-RIEMANN-MIRROR-002`

25.1 Question and necessity

V1 XII-2 records the prime skeleton and half-One mirroring while expressly leaving complex zero location outside the framework; V2 Step 132 requires the fixed-axis result and its adverse logical control.

The exact theorem statement is:

> *The exact complement involution on positive parts has one and only one self-partner, the half-One; every other admitted part forms a distinct complementary pair. This forces the unique SFT reflection axis corresponding to the classical zeta functional equation's one-half symmetry, while complex zero location remains outside the no-imaginary proof language.*

Admitted dependencies: `SFT-MATH-EXACT-RELATIONS-002`, `SFT-FOUNDATION-HALF-ONE-001`.

25.2 Generated grammar and exact boundary

Exact complement is an involution. Solving self-complementarity by positive junction forces two equal parts to be the One, whose unique first Fold split is half-One.

Generation rule: Exhaust domain, involution, fixed point, off-axis pairing, prime tie, complex boundary, generality and extra-analytic choices.

Exact grammar boundary: Exact positive rational parts under complement within the One; no complex analytic continuation object.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:89db3b94da1ed133bcb1ef1b66c505b79d2f35dd5623b8b7d4c6cac95f3616a2`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-RIEMANN-MIRROR-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`domain`	`complex-plane`	Imaginary values are not admitted proof objects.	`positive-parts` - The involution acts on exact parts of the One.
`involution`	`imported-functional-equation`	An external equation cannot derive the Fold law.	`one-complement` - The map is exact complement within the One.
`fixed_point`	`selected-half`	Selecting one-half would not prove uniqueness.	`unique-self-complement` - Exact self-complement forces two equal parts to reconstruct the One.
`off_axis`	`unpaired-values`	Complement is total on the declared part domain.	`distinct-pairs` - Every nonfixed part has one distinct complementary partner.
`prime_tie`	`zeta-import`	The analytic zeta object is not an SFT input.	`orbit-prime-skeleton` - Prime correspondence comes from independently derived Fold orders.
`zero_location`	`claim-classical-rh`	Symmetry alone cannot locate every classical complex zero.	`explicit-language-boundary` - Only the symmetry axis is claimed inside SFT.
`generality`	`bounded-denominator-only`	A fixed denominator table does not prove the involution law.	`exact-part-identity` - The equality p=One-p has the unique exact solution two-p=One for every admitted part.
`addition`	`extra-analysis`	Complex analysis is outside the registered dependencies.	`no-extra-rule` - Only complement, equality and prior orbit number theory occur.

25.3 Unique survivor, minimality and laws

> *Half-One is the unique fixed axis of exact complement; every other positive part is paired. This closes the SFT Riemann-mirroring claim and explicitly does not assert classical complex zero location.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- complement composed twice is identity.
- self-complementarity uniquely forces the half-One.
- reflection symmetry does not by itself imply that all points of an invariant set lie on the fixed axis.

25.4 Operational witnesses and unfavourable checks

- `fixed-half` - Half-One is equal to its exact complement. Result: `PASS`.
- `off-axis-pairs` - Quarter/three-quarter and one-third/two-thirds are distinct complement pairs. Result: `PASS`.
- `symmetry-control` - A symmetric two-point set can lie off the fixed axis, so symmetry alone cannot prove zero location. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Imaginary values are not admitted proof objects. Result: `PASS`; receipt `sha256:47986d182ec9dcb7e128ba21b6556b7f1a6cd9e96b2a99e5d6aef6ae62254001`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

25.5 Depth-independent closure

Base: The Foundation half-One certificate supplies the self-complementary witness.

Successor: Every newly generated non-half part is paired with its exact complement and neither becomes a second fixed point.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.6 Meaning and scientific consequence

The complement involution has exactly one fixed point: the half-One. That is the forced SFT mirror corresponding to the one-half axis of the classical functional equation. An unfavourable control constructs an off-axis symmetric pair, proving that symmetry alone cannot locate all members of a symmetric set. This preserves the strongest detailed V1 statement: prime-orbit structure and the mirror close, while classical complex zero location does not.

25.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Riemann functional symmetry, critical line, reflection involution, Riemann hypothesis boundary. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

25.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.

- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

This theorem does not prove the classical Riemann hypothesis; it proves the SFT arithmetic prime skeleton and unique reflection-axis correspondence stated in the stronger detailed V1 record.

25.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:3b19a68ec191cac8e3039f672c938c16213165cc8ca33c752c1be7b1a15136c0`.
- Derivation seal: `sha256:100e6f21f2a33c3c374d80f8d96672c9351cf1678caed4084e3535e6c0b3d89a`.
- Independent implementation: `sha256:743035dbc946735329c39331a72b4bff57f4ff7fbd4f850703367923ebca7f6a`.
- Independent certificate: `sha256:e66aa5200f0ef07b92629cd5e0b9b18c210ab9b371a1522e100ed681e7200870`.
- External validation: `sha256:dee17f0c705cb0506767a9ed5f9bc61294d4d624f8bbc1ddcbdf8cb82cbd285c`.
- Engine receipt: `sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6e6ca001a24622e55364dfc4`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-RIEMANN-MIRROR-002-27847629dbd2fb1d.json`.

26. Derivation 21: Exact bounded Collatz census and contraction-control correction

Claim identity: `SFT-MATH-COLLATZ-FINITE-CENSUS-002`

26.1 Question and necessity

V2 Step 277 contains an exact bounded census and a separate contraction rationale. V3 must reproduce the former and mechanically reject the latter where it is false.

The exact theorem statement is:

> *Every positive whole start from the One through one-hundred-thousand reaches the 1-4-2 cycle under the declared Collatz transition; start twenty-seven takes exactly one-hundred-eleven steps. The exact census also rejects the prior constant-three-quarter pointwise contraction shortcut: a lawful odd macrostep may rise, so only the bounded execution result is admitted.*

Admitted dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`.

26.2 Generated grammar and exact boundary

Exact whole arithmetic supplies the transition and full finite traces. Exhaustive execution proves the declared bound; the adverse start three disproves a constant pointwise contraction premise.

Generation rule: Exhaust range, transition, parity, trace, cycle, contraction, boundary and extra-theorem choices.

Exact grammar boundary: Positive whole starts one through one hundred thousand under the exact conventional Collatz transition.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:018b733207ce8a309044f81d99745b01d33beae74730ced7f35aa611a3899a38`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-COLLATZ-FINITE-CENSUS-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`range`	`sampled-starts`	Sampling cannot close the declared range.	`all-declared-starts` - Every start in the frozen bound is executed.
`transition`	`fold-analogy`	An analogy is not the conventional Collatz map.	`exact-collatz` - Even starts halve and odd starts map to three-times plus One.
`parity`	`assumed-even`	Parity must be checked from exact whole arithmetic.	`computed-parity` - Every transition computes its parity exactly.
`trace`	`terminal-only`	A terminal result without its path cannot be replayed.	`complete-step-trace` - Every successor is counted until the cycle.
`cycle`	`zero-sink`	Zero is not in the declared positive domain.	`one-four-two` - The terminal recurrent class is the exact 1-4-2 cycle.
`contraction`	`constant-three-quarters`	The shortcut is false: the odd macrostep from three reaches five after removing all binary factors.	`executed-variable-descent` - No pointwise contraction premise enters; the exact trace decides the bounded result.
`boundary`	`unrestricted-collatz`	A finite census cannot establish the unrestricted conjecture.	`explicit-one-hundred-thousand` - The sealed maximum is part of the claim.
`addition`	`external-sequence`	A published trajectory cannot enter the execution.	`no-extra-rule` - Only exact whole operations drive the census.

26.3 Unique survivor, minimality and laws

The complete starts One through 100,000 census has zero failures, start 27 takes 111 steps, the terminal cycle is 1-4-2, and the constant-3/4 pointwise contraction premise is rejected.

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- every odd three-times-plus-One result is even.
- every declared start has a finite replay trace to the terminal cycle.
- the claim boundary remains finite and does not become the unrestricted Collatz conjecture.
- a false contraction heuristic cannot serve as proof even when the bounded census passes.

26.4 Operational witnesses and unfavourable checks

- `odd-even` - Every positive odd start through ten thousand one maps to an even three-times-plus-One result. Result: `PASS`.
- `bounded-census` - Every start through one hundred thousand reaches the One. Result: `PASS`.
- `twenty-seven` - Start twenty-seven takes exactly one hundred eleven steps. Result: `PASS`.
- `contraction-reject` - The accelerated odd step from three reaches five and therefore is not a pointwise contraction. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Sampling cannot close the declared range. Result: `PASS`; receipt `sha256:f0d8ec988eaf6a70e795c33155c0ec30332c2558947a51a3fcc305cdcdadb26c`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

26.5 Depth-independent closure

Base: The One lies on the declared 1-4-2 recurrent cycle.

Successor: For each next positive whole start, execute the exact transition until an already certified cycle member appears and retain the full prefix.

Shared claim-record clause `MATH-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

26.6 Meaning and scientific consequence

All starts through 100,000 execute and reach the 1-4-2 cycle; start 27 takes 111 steps. Crucially, V3 also tests the explanation attached to the old count. Start three maps to five after the first required halving, so a constant three-quarter pointwise contraction is false. The engine retains the exact census and records the failed shortcut, showing how clean-room reconstruction can strengthen a result without rewriting its evidential history.

26.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Collatz map, hailstone sequence, finite census, cycle detection. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

26.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The result is exact for starts through 100,000. The unrestricted Collatz conjecture remains outside this finite certificate.

26.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:b23dfdfbf1bd53b640539bdc1dbdd900c5786659755b5cb5622bbe362cc8081f`.
- Derivation seal: `sha256:e57a27f1b95bade1301c14d5bfc05434c7c63751e8d32cf64aad2e29f8958be8`.
- Independent implementation: `sha256:9d533f192814a4720b16e34cfe3c49acbe02de80dce91c0d60b1be9c01c02421`.
- Independent certificate: `sha256:83e49654996c1fc161e8b8d3abfe7e78342759ed823d227809aa1f80e9034eda`.
- External validation: `sha256:e52f71a74ebea2b23714d471b78148eed57a6163e4b3ca9f522c20bb0e82a101`.
- Engine receipt: `sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cbd40c4f9a045`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-COLLATZ-FINITE-CENSUS-002-3bb3a9fe47901503.json`.

27. Derivation 22: Exact self-similarity, chaotic rate and convergent-series laws

Claim identity: `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`

27.1 Question and necessity

V1/V2 separately registered Lyapunov/entropy antilogs, covering, scale depth, unit power, exact convergence rate and bounded accumulated separation.

The exact theorem statement is:

> *Fold forces binary local separation expansion, one closed distinction per step, the unique unit exponent under rank doubling, exact m^d support growth, and rational convergence certificates whose tails shrink by a generated factor without importing logarithms or irrational limit values.*

Admitted dependencies: `SFT-MATH-EXACT-RELATIONS-002`, `SFT-MATH-LIMIT-CONTINUUM-002`.

27.2 Generated grammar and exact boundary

Fold supplies the only scale successor and predecessor merge; exact support recurrence and rational tail comparison force the registered rates without analytic functions.

Generation rule: Exhaust scale action, separation, information, power exponent, support, series, generality and extra-function choices.

Exact grammar boundary: Exact generated finite scales, supports and rational partial sums.

The complete product contains 256 candidates. Its completeness-certificate identity is `sha256:bb9aec7646934eff006d7d0d640b9a605b82eb46f40bc3b18374c5e37238e7e6`. The following table gives every grammar axis, its explicit failure alternative and its forced coordinate. Their Cartesian product is the complete candidate census; the full row-level list is preserved in `claims/SFT-MATH-SELF-SIMILAR-CONVERGENCE-002/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
`scale`	`continuous-rescale`	A continuous scale group is not generated.	`fold-doubling` - The scale successor is the Fold count.
`separation`	`fitted-lyapunov`	A fitted rate is not a theorem.	`exact-binary-expansion` - Before cast, local separation is multiplied by the Fold count.
`information`	`logarithmic-value`	A logarithm is outside exact proof arithmetic.	`one-closed-label` - One Fold merge closes exactly one binary predecessor distinction.
`power`	`selected-exponent`	A selected exponent is a parameter.	`unit-self-similar` - Rank doubling and count halving uniquely select the One exponent in the enumerated exponent grammar.
`support`	`binary-only`	The successor law is not tied to one label count.	`m-to-depth` - An m-label successor multiplies support by m at every depth.
`series`	`irrational-limit`	The limit cannot enter as a proof value.	`rational-tail-bound` - Exact partial sums and a shrinking rational tail prove finiteness.
`generality`	`finite-table`	A depth table lacks a recurrence.	`base-successor` - Every law carries a structural base and exact successor.
`addition`	`extra-analytic-function`	Logarithm or exponential would add a model.	`no-extra-rule` - Only exact ratios, counts and recurrences occur.

27.3 Unique survivor, minimality and laws

> *The exact Fold rate is binary separation expansion and one closed label per step; unit rank power is the unique Fold-self-similar exponent, m-labelled support is m^d, and decreasing rational terms close convergence without importing their limit as a value.*

Shared claim-record clause `MATH-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

- local uncast separation multiplies by the Fold count.
- one predecessor label is closed per Fold merge.
- rank doubling paired with count halving selects the unit power among the generated nonnegative whole exponent forms.
- m-labelled successor support obeys the base/successor count m^d.
- a geometric rational tail bound proves finite accumulated separation.

27.4 Operational witnesses and unfavourable checks

- `chaotic-rate` - The separation two-of-thirty-five advances to four-of-thirty-five before cast. Result: `PASS`.
- `support` - Ternary support has three, nine and twenty-seven states at depths one, two and three. Result: `PASS`.
- `unit-power` - Only exponent One among whole exponents One through four gives exact halving when rank doubles. Result: `PASS`.
- `gap-formula` - The exact gap formula is positive and strictly decreases through depth fourteen. Result: `PASS`.
- `finite-bracket` - The first eleven exact gap terms sum below eleven-of-sixty. Result: `PASS`.

The engine-level adverse controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A continuous scale group is not generated. Result: `PASS`; receipt `sha256:e3868d8a6f435a5025d3289a24b18b2bef31ca89fa78f2d5992aadece72362ad`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded Result: `PASS`; receipt `sha256:d3e9e102d09c15d45a65651b202a39c4339861878490c72915593ef97661f237`.

27.5 Depth-independent closure

Base: At depth One an m-labelled successor has m states and one Fold closes one binary distinction.

Successor: Appending one label multiplies support by m; applying one Fold multiplies an uncast local gap by b and closes one predecessor label; one further rational term preserves the tail bound.

Shared claim-record clause `MATH-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

27.6 Meaning and scientific consequence

One Fold doubles an uncast local gap and closes one binary predecessor label. The same successor forces m^d support, selects the unit rank exponent from the declared whole-exponent grammar, and gives exact rational convergence rates. Finite accumulated separation is proved by partial sums and a rational tail bracket; no logarithm, exponential or irrational limit value has to be smuggled into the proof language.

27.7 Correspondence after sealing

Only after the SFT survivor and receipt are fixed may the following established terms describe the result: Lyapunov antilog, KS entropy, power law, covering number, convergent series. These terms do not occur as dependencies and do not supply an answer table. Agreement identifies a translation boundary; disagreement would remain visible rather than being repaired by changing the sealed SFT law.

27.8 Exact exclusions and limitations

- semantic numerical zero is not an admitted value.
- negative magnitude is represented only by held orientation.
- irrational, imaginary and floating proof values are excluded.
- a completed infinity and an ungenerated continuum are excluded.

The unit exponent is unique within the explicitly enumerated positive whole exponent grammar; natural-system correspondence is tested only after the formal seal in owning empirical branches.

27.9 Evidence identities

- Registered status: `independently_replicated`.
- Source manifest: `sha256:a10ca3b6a55f0437152c829d3f7362cb209922b9d65a00cea34005c6929aa238`.
- Derivation seal: `sha256:3b5b67ae9a24eac5e74f0b60feb92e8f66224134eb397f911876b3e37eec91d1`.
- Independent implementation: `sha256:06101b52023de1f7ae1ec661673cf97b543f66fa9d46b2071557fc41fe6b828a`.
- Independent certificate: `sha256:ace930a00105674c923aa54768e10d198ce879d8b3cf8828c92b2eb5bd27363d`.
- External validation: `sha256:fad1024f01bba59f703064314f4b493e5386d5811e1d3e2d5ed01e848184bf1c`.
- Engine receipt: `sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fb606d5b1`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SELF-SIMILAR-CONVERGENCE-002-0abf8d557108f37f.json`.

28. Preserved version 1.2 extension: the Smithian Fold Scientific Calculator

The calculator is not a conventional floating-point calculator carrying SFT labels. It is an executable translation of the admitted Mathematics value law. Familiar keys and notation form a human boundary; every accepted expression is parsed into exact SFT runtime types, every operation is routed through one exact evaluator, and every result that would require a prohibited scalar halts before it can enter the answer, memory or history. The browser surface contains no arithmetic implementation.

Five claim records are included because transparent correction is part of the scientific result. Claim 003 passed its declared grammar but a later operational audit found reversed odd/even endpoints in one alternating arctangent enclosure. Its artifacts and receipt remain preserved as superseded adverse evidence. Claim 004 re-derived the core with corrected endpoint parity and an exact tangent-composition check. Claim 005 extended that core to the declared scientific-calculator language and stateful application. Claim 006 completed the calculator-first app, law explorer, resource controls and active-file coverage. Direct manual testing then exposed a blank deprecated desktop widget on macOS and usability failures on phones. Claim 007 held the claim-006 evaluator immutable and forced the accessible standards-rendered local web adapter.

28.1 Exact traced SFT-native scientific calculator

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-003`

Publication status: `superseded_adverse_evidence`. The original machine receipt is preserved, but this claim is not the active calculator law. Its later-discovered parity defect is the reason claim 004 was derived independently rather than mutating claim 003.

WHY. A calculator makes the Mathematics laws directly inspectable, but it may not smuggle the conventional signed real or complex number model back into the proof domain through its user interface.

DERIVATION. Exhausting the ten representation and execution choices leaves one compositional machine: exact entered parts, empty-One, held orientations, admitted arithmetic recurrences, rational balance enclosures, typed orthogonal fibres, mandatory halts and complete traces. Each supported scientific function is then a finite composition of those structures, not a new axiom or fitted parameter.

CHECK. Execute all 1,024 generated semantics, require the sole all-preserving survivor, run exact arithmetic, root, combinatorial, transcendental, orthogonal, parser, trace and halt witnesses, run four adverse controls, and independently regenerate both the product and operational examples.

Exact statement:

> *A scientific-calculator interface is admitted in SFT only when every entered decimal is translated character-by-character to an exact rational part, displayed zero is structural empty-One, subtraction uses held orientation, all algebraic and transcendental non-rational results remain exact rational enclosures with replayable bounds, orthogonal values remain*

> *typed Fold fibres, every operation is traced, and every undefined or uncertified operation halts.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`.

Grammar boundary: All calculator semantics formed from the ten declared binary representation and execution choices, with exact finite rational recurrences and a declared finite resource bound.

Generation rule: Generate the complete product of input-value, empty, orientation, exact-operation, non-rational, transcendental, orthogonal, domain, trace and extra-rule choices.

The complete Cartesian product contains 1,024 candidate forms and has completeness identity `sha256:e6ff9fd1094ac119184dc42e13e19c44ecdbed05f9aa48a7d074d81408645e3c`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`input`	`host-float` - A host float loses the exact entered part before proof begins.	`character-exact-part` - Characters generate the exact counted numerator, denominator and exponent.
`empty`	`numeric-zero` - Numerical zero is not an admitted SFT value.	`structural-empty-One` - The empty-One form records complete cancellation or empty magnitude.
`orientation`	`negative-scalar` - A negative scalar imports a forbidden signed field.	`held-fibre-orientation` - One of two generated held labels records orientation with a positive magnitude.
`operations`	`host-arithmetic-oracle` - A host arithmetic oracle erases the generated Fold construction.	`generated-exact-kernels` - Junction, pairing, refinement and recurrence execute the operation exactly.
`nonrational`	`irrational-scalar` - An irrational scalar is outside the proof domain.	`rational-balance-enclosure` - An exact rational bracket and polynomial balance certify the result without importing a scalar.
`transcendental`	`black-box-library-value` - A black-box value supplies no Fold derivation or remainder certificate.	`finite-rational-recurrence-bound` - A finite exact recurrence and positive tail bound enclose the function value.
`orthogonal`	`imaginary-scalar` - An imaginary scalar is not an admitted proof value.	`typed-orthogonal-fibre-pair` - Two exact Fold fibres retain real and orthogonal coordinates compositionally.
`domain`	`nan-infinity-continuation` - NaN or infinity silently continues beyond the admitted domain.	`explicit-lawful-halt` - The machine halts before returning an uncertified value.
`trace`	`answer-only` - An answer without provenance cannot be replayed or independently checked.	`complete-proof-resource-trace` - Input translation, operations, result and counted resources are retained.
`addition`	`has-extra-rule` - A fitted constant or answer-selecting rule is not supplied by the dependencies.	`no-extra-rule` - The calculator composes only already-admitted exact structures and proofs.

Unique survivor and exact result:

> *The unique admitted calculator semantics uses exact character-generated rational parts, structural empty-One, held orientation, generated exact kernels, certified rational enclosures, finite rational recurrences with explicit remainder bounds, typed orthogonal fibres, mandatory domain halts, complete proof/resource traces and no extra rule.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- decimal and scientific notation are exact rational boundary encodings, never floating proof values.
- complete cancellation returns structural empty-One and unmatched remainder retains held orientation.
- every exact operation composes admitted arithmetic, combinatorial and order structures.
- every non-rational result is an exact rational enclosure carrying its generating recurrence and bound.
- an unclosed enclosure, exhausted resource bound or invalid domain halts without a result.
- orthogonal composition is a typed two-fibre product and introduces no imaginary scalar.
- decimal output is a downstream display projection and never proof evidence.

Executed witnesses:

- `exact-decimal` - 0.1 plus 0.2 is the exact rational part 3/10, not a host floating result. Result: `PASS`.
- `empty-cancellation` - One substituted from One yields structural empty-One. Result: `PASS`.
- `held-remainder` - One substituted by three yields counter-held positive magnitude two. Result: `PASS`.
- `root-balance` - The exact sqrt(2) bracket has lower square below two and upper square at or above two. Result: `PASS`.
- `combinatorial-recurrence` - Factorial and selection recurrence reproduce exact finite arrangements. Result: `PASS`.
- `transcendental-enclosure` - exp, ln, sin and circle operations return ordered rational enclosures with certificates. Result: `PASS`.
- `orthogonal-composition` - The square of the unit orthogonal fibre is counter-held real One with empty orthogonal fibre. Result: `PASS`.
- `domain-halt` - Division by structural empty-One halts without returning NaN or infinity. Result: `PASS`.
- `complete-trace` - A parsed calculation retains boundary translation, operation and terminal records. Result: `PASS`.
- `scientific-notation` - Scientific notation is translated exactly without a floating intermediate. Result: `PASS`.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A host float loses the exact entered part before proof begins. Result: `PASS`; receipt `sha256:f1cae5eb43509a204e1458eacacacc8b1011649fb3e24e2004e089a9a431f8d5`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only Result: `PASS`; receipt `sha256:bdef05251b3846ffaedc057704bd0da3f30f3d8ddb10ea21de6236ba416c59f5`.

Depth-independent base: One entered digit is translated to an exact counted part; the structural empty-One and structural One supply cancellation and identity bases.

Depth-independent successor: Appending a generated digit refines the exact denominator by one counted tenfold composition; appending an operator composes two already-certified values, and appending one recurrence term retains the previous exact sum plus one exact part while replacing the remainder by a proven smaller positive bound.

Exact exclusions:

- semantic numerical zero, NaN and infinity are not admitted values.
- negative, irrational and imaginary scalars are replaced by typed Fold structures.
- binary floating-point and opaque host-library answers are excluded from proof.
- uncertified roots, series, singularities and exhausted resource bounds halt.
- decimal projections are correspondence displays only.

Limitation: Closure is for the declared calculator grammar and finite resource bounds. A displayed decimal is not proof evidence. Enclosures report their exact bounds; requests that cannot close inside the declared bound halt rather than returning a guess. Additional named functions remain lawful extensions only when they reduce to admitted exact recurrences and pass the same engine route.

Evidence identities:

- Source manifest: `sha256:5e8d18b5f8595c1ba6292a3bb736436d28fcc4813cd47aa6bc32744c300d8ac0`.
- Derivation seal: `sha256:8c6a64e9113e32d23e6bf5c08a1fad24692788433b322c7d4b55f7828f81ee18`.
- Independent implementation: `sha256:0a52eca4b5694848230b3671ed2b0c5ff4b2d170f394686aee38e98a60f8f8b5`.
- Independent certificate: `sha256:f26d7f99675fa8d4bd035214d0817ca65309d23fe4d4bb53b71b907fa9a46b22`.
- External validation: `sha256:0fa836f9219c8cae0ba1b24cb0fea4e2fbd0c530ed93fd996f4551c44a5e78e6`.
- Engine receipt: `sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-003-f95ac235ac96429b.json`.

28.2 Exact traced SFT-native scientific calculator with corrected enclosure parity

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-004`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. A calculator makes the Mathematics laws directly inspectable, but it may not smuggle the conventional signed real or complex number model back into the proof domain through its user interface.

DERIVATION. Exhausting the ten representation and execution choices leaves one compositional machine: exact entered parts, empty-One, held orientations, admitted arithmetic recurrences, rational balance enclosures, typed orthogonal fibres, mandatory halts and complete traces. Each supported scientific function is then a finite composition of those structures, not a new axiom or fitted parameter.

CHECK. Execute all 1,024 generated semantics, require the sole all-preserving survivor, run the exact arithmetic, root, combinatorial, transcendental, orthogonal, parser, trace and halt witnesses, verify alternating-series endpoint parity and the exact tangent-composition identity, run four adverse controls, and independently regenerate both the full product and operational certificates.

Exact statement:

> *A scientific-calculator interface is admitted in SFT only when every entered decimal is translated character-by-character to an exact rational part, displayed zero is structural empty-One, subtraction uses held orientation, all algebraic and transcendental non-rational results remain exact rational enclosures with replayable bounds, orthogonal values remain typed Fold fibres, every operation is traced, and every undefined or uncertified operation halts.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`.

Grammar boundary: All calculator semantics formed from the ten declared binary representation and execution choices, with exact finite rational recurrences and a declared finite resource bound.

Generation rule: Generate the complete product of input-value, empty, orientation, exact-operation, non-rational, transcendental, orthogonal, domain, trace and extra-rule choices.

The complete Cartesian product contains 1,024 candidate forms and has completeness identity `sha256:e6ff9fd1094ac119184dc42e13e19c44ecdbed05f9aa48a7d074d81408645e3c`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`input`	`host-float` - A host float loses the exact entered part before proof begins.	`character-exact-part` - Characters generate the exact counted numerator, denominator and exponent.
`empty`	`numeric-zero` - Numerical zero is not an admitted SFT value.	`structural-empty-One` - The empty-One form records complete cancellation or empty magnitude.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`orientation`	`negative-scalar` - A negative scalar imports a forbidden signed field.	`held-fibre-orientation` - One of two generated held labels records orientation with a positive magnitude.
`operations`	`host-arithmetic-oracle` - A host arithmetic oracle erases the generated Fold construction.	`generated-exact-kernels` - Junction, pairing, refinement and recurrence execute the operation exactly.
`nonrational`	`irrational-scalar` - An irrational scalar is outside the proof domain.	`rational-balance-enclosure` - An exact rational bracket and polynomial balance certify the result without importing a scalar.
`transcendental`	`black-box-library-value` - A black-box value supplies no Fold derivation or remainder certificate.	`finite-rational-recurrence-bound` - A finite exact recurrence and positive tail bound enclose the function value.
`orthogonal`	`imaginary-scalar` - An imaginary scalar is not an admitted proof value.	`typed-orthogonal-fibre-pair` - Two exact Fold fibres retain real and orthogonal coordinates compositionally.
`domain`	`nan-infinity-continuation` - NaN or infinity silently continues beyond the admitted domain.	`explicit-lawful-halt` - The machine halts before returning an uncertified value.
`trace`	`answer-only` - An answer without provenance cannot be replayed or independently checked.	`complete-proof-resource-trace` - Input translation, operations, result and counted resources are retained.
`addition`	`has-extra-rule` - A fitted constant or answer-selecting rule is not supplied by the dependencies.	`no-extra-rule` - The calculator composes only already-admitted exact structures and proofs.

Unique survivor and exact result:

> *The unique admitted calculator semantics uses exact character-generated rational parts, structural empty-One, held orientation, generated exact kernels, certified rational enclosures, finite rational recurrences with explicit remainder bounds, typed orthogonal fibres, mandatory domain halts, complete proof/resource traces and no extra rule.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- decimal and scientific notation are exact rational boundary encodings, never floating proof values.
- complete cancellation returns structural empty-One and unmatched remainder retains held orientation.
- every exact operation composes admitted arithmetic, combinatorial and order structures.
- every non-rational result is an exact rational enclosure carrying its generating recurrence and bound.
- an unclosed enclosure, exhausted resource bound or invalid domain halts without a result.
- orthogonal composition is a typed two-fibre product and introduces no imaginary scalar.
- decimal output is a downstream display projection and never proof evidence.

Executed witnesses:

- `exact-decimal` - 0.1 plus 0.2 is the exact rational part 3/10, not a host floating result. Result: `PASS`.
- `empty-cancellation` - One substituted from One yields structural empty-One. Result: `PASS`.
- `held-remainder` - One substituted by three yields counter-held positive magnitude two. Result: `PASS`.
- `root-balance` - The exact sqrt(2) bracket has lower square below two and upper square at or above two. Result: `PASS`.
- `combinatorial-recurrence` - Factorial and selection recurrence reproduce exact finite arrangements. Result: `PASS`.
- `orthogonal-composition` - The square of the unit orthogonal fibre is counter-held real One with empty orthogonal fibre. Result: `PASS`.
- `domain-halt` - Division by structural empty-One halts without returning NaN or infinity. Result: `PASS`.
- `complete-trace` - A parsed calculation retains boundary translation, operation and terminal records. Result: `PASS`.
- `scientific-notation` - Scientific notation is translated exactly without a floating intermediate. Result: `PASS`.
- `alternating-parity-certificate` - Odd arctangent partial counts supply upper bounds; even counts supply lower bounds, with the next exact part closing the opposite endpoint. Result: `PASS`.
- `exact-angle-composition` - Two exact tangent doublings give 120/119 and subtraction of atan(1/239) gives tangent One, certifying the quarter-turn identity without decimal target selection. Result: `PASS`.

- `corrected-circle-enclosure` - The circle constant has ordered exact rational endpoints carrying the corrected alternating parity and exact angle-composition trace. Result: `PASS`.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A host float loses the exact entered part before proof begins. Result: `PASS`; receipt `sha256:f1cae5eb43509a204e1458eacacacc8b1011649fb3e24e2004e089a9a431f8d5`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only Result: `PASS`; receipt `sha256:bdef05251b3846ffaedc057704bd0da3f30f3d8ddb10ea21de6236ba416c59f5`.

Depth-independent base: One entered digit is translated to an exact counted part; the structural empty-One and structural One supply cancellation and identity bases.

Depth-independent successor: Appending a generated digit refines the exact denominator by one counted tenfold composition; appending an operator composes two already-certified values, and appending one recurrence term retains the previous exact sum plus one exact part while replacing the remainder by a proven smaller positive bound.

Exact exclusions:

- semantic numerical zero, NaN and infinity are not admitted values.
- negative, irrational and imaginary scalars are replaced by typed Fold structures.
- binary floating-point and opaque host-library answers are excluded from proof.
- uncertified roots, series, singularities and exhausted resource bounds halt.
- decimal projections are correspondence displays only.

Limitation: Closure is for the declared calculator grammar and finite resource bounds. A displayed decimal is not proof evidence. Enclosures report their exact bounds; requests that cannot close inside the declared bound halt rather than returning a guess. Additional named functions remain lawful extensions only when they reduce to admitted exact recurrences and pass the same engine route. The earlier 003 receipt is preserved as adverse evidence because its arctangent endpoint parity was reversed; this 004 law supersedes it and does not depend on it.

Evidence identities:

- Source manifest: `sha256:8969c3c7e5c60d8f5978f45f734e109ba530f00f838bf32d9d9beb938adbf104`.
- Derivation seal: `sha256:4fea624cffb8780f8e96d0a85eccb81c8fb609c1e712983a339a6f3f9b57d0d9`.
- Independent implementation: `sha256:5d2c4b11b1cb3ba9a7ba092af4d6d622ac73caf25b682f3bfb32ad7a000e6983`.
- Independent certificate: `sha256:500be40f0488c5003ebf92841385cec6f50ef8405b48e0d02288c7d7726f56c9`.
- External validation: `sha256:da76fedc1a4a70de18e6a51ab3f10065f18e5f04e09cf810ee913a4d11149c1f`.
- Engine receipt: `sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b`.

- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-004-2d243dbd08260c48.json`.

28.3 Complete exact SFT scientific calculator and cross-platform learning app

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-005`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. A 1-to-1 scientific calculator must be usable like the familiar desktop calculator, not merely expose a narrow proof API. Accessibility cannot weaken the exact SFT value law or hide its certificates.

DERIVATION. Claim 004 supplies the admitted exact core. Exhausting the three extension choices forces complete declared function coverage, explicit session state and a cross-platform evidence-bearing app. Every new function is then reduced to exact arithmetic, rational recurrence, certified enclosure, typed fibre composition or a mandatory domain halt; UI controls map one-for-one to those operations.

CHECK. Execute all 8,192 combined semantics, require one survivor, replay every core witness, exercise ordinary equals, general powers, all function families, three angle modes, state and memory, terminal execution and the complete GUI control map; run adverse controls and independently regenerate the product and operational examples.

Exact statement:

> *The admitted exact calculator core extends uniquely to the declared mainstream scientific-calculator surface: terminal equals execution; editable expressions; exact and certified general powers, roots, logarithmic, exponential, trigonometric, inverse-trigonometric, hyperbolic, combinatorial and statistical operations; radian, degree and gradian modes; retained answer, memory and history; and one cross-platform desktop application exposing decimal projection, exact evidence, proof trace and SFT learning guidance. No interface action may bypass the same exact fail-closed evaluator.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`, `SFT-MATH-SCIENTIFIC-CALCULATOR-004`.

Grammar boundary: The complete thirteen-coordinate product formed by the ten exact core semantics plus the declared mainstream scientific function manifest, session interaction law and standard-library desktop evidence surface.

Generation rule: Generate the complete product of input-value, empty, orientation, exact-operation, non-rational, transcendental, orthogonal, domain, trace and extra-rule choices. Extend that complete product by declared scientific-function coverage, stateful interaction, and cross-platform application/evidence-surface choices.

The complete Cartesian product contains 8,192 candidate forms and has completeness identity `sha256:081b1dcd0e2431b4d583b8dbfa65cccf17502ee1d570b06609baf708f2878903`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`input`	`host-float` - A host float loses the exact entered part before proof begins.	`character-exact-part` - Characters generate the exact counted numerator, denominator and exponent.
`empty`	`numeric-zero` - Numerical zero is not an admitted SFT value.	`structural-empty-One` - The empty-One form records complete cancellation or empty magnitude.
`orientation`	`negative-scalar` - A negative scalar imports a forbidden signed field.	`held-fibre-orientation` - One of two generated held labels records orientation with a positive magnitude.
`operations`	`host-arithmetic-oracle` - A host arithmetic oracle erases the generated Fold construction.	`generated-exact-kernels` - Junction, pairing, refinement and recurrence execute the operation exactly.
`nonrational`	`irrational-scalar` - An irrational scalar is outside the proof domain.	`rational-balance-enclosure` - An exact rational bracket and polynomial balance certify the result without importing a scalar.
`transcendental`	`black-box-library-value` - A black-box value supplies no Fold derivation or remainder certificate.	`finite-rational-recurrence-bound` - A finite exact recurrence and positive tail bound enclose the function value.
`orthogonal`	`imaginary-scalar` - An imaginary scalar is not an admitted proof value.	`typed-orthogonal-fibre-pair` - Two exact Fold fibres retain real and orthogonal coordinates compositionally.
`domain`	`nan-infinity-continuation` - NaN or infinity silently continues beyond the admitted domain.	`explicit-lawful-halt` - The machine halts before returning an uncertified value.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`trace`	`answer-only` - An answer without provenance cannot be replayed or independently checked.	`complete-proof-resource-trace` - Input translation, operations, result and counted resources are retained.
`addition`	`has-extra-rule` - A fitted constant or answer-selecting rule is not supplied by the dependencies.	`no-extra-rule` - The calculator composes only already-admitted exact structures and proofs.
`scientific_surface`	`partial-basic-function-subset` - A basic subset is not the requested scientific calculator.	`complete-declared-scientific-surface` - Arithmetic, rational/non-rational powers, roots, logs, exponentials, circular/inverse/hyperbolic functions, combinatorics, exact statistics and typed orthogonal operations are all routed through certified kernels.
`session`	`stateless-expression-only` - A stateless expression evaluator lacks terminal equals, answer, memory and history behaviour.	`equals-answer-memory-history` - The terminal equals key executes; Ans, memory and ordered history retain explicit typed state without altering laws.
`application`	`terminal-only-opaque-answer` - A terminal-only answer surface is not a clean general-user calculator app and hides evidence.	`cross-platform-app-trace-and-guide` - One standard-library desktop app provides editable display, keypad, angle selector, memory, history, exact detail, proof trace and embedded SFT guidance.

Unique survivor and exact result:

> *The unique expanded calculator is the exact core plus the complete declared scientific-function surface, terminal equals/Ans/memory/history state, RAD/DEG/GRAD translation, and one cross-platform desktop app whose buttons and keyboard invoke only the same fail-closed evaluator. Every exact result retains its Fold type; every non-rational result remains a certified rational enclosure; decimal output is display only.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- decimal and scientific notation are exact rational boundary encodings, never floating proof values.
- complete cancellation returns structural empty-One and unmatched remainder retains held orientation.
- every exact operation composes admitted arithmetic, combinatorial and order structures.
- every non-rational result is an exact rational enclosure carrying its generating recurrence and bound.
- an unclosed enclosure, exhausted resource bound or invalid domain halts without a result.
- orthogonal composition is a typed two-fibre product and introduces no imaginary scalar.
- decimal output is a downstream display projection and never proof evidence.
- a rational exponent is counted root followed by counted composition; a certified exponent uses exp of the certified product with ln.
- degree and gradian inputs translate through the independently certified circle enclosure before circular evaluation.
- inverse circular and hyperbolic operations are exact recurrence compositions with explicit domain halts.
- memory, answer and history retain values but never mutate operation law or proof certificates.
- terminal and desktop interfaces are two views of one evaluator and cannot create an untraced answer.

Executed witnesses:

- `exact-decimal` - 0.1 plus 0.2 is the exact rational part 3/10, not a host floating result. Result: `PASS`.
- `empty-cancellation` - One substituted from One yields structural empty-One. Result: `PASS`.
- `held-remainder` - One substituted by three yields counter-held positive magnitude two. Result: `PASS`.
- `root-balance` - The exact sqrt(2) bracket has lower square below two and upper square at or above two. Result: `PASS`.
- `combinatorial-recurrence` - Factorial and selection recurrence reproduce exact finite arrangements. Result: `PASS`.
- `orthogonal-composition` - The square of the unit orthogonal fibre is counter-held real One with empty orthogonal fibre. Result: `PASS`.
- `domain-halt` - Division by structural empty-One halts without returning NaN or infinity. Result: `PASS`.

- `complete-trace` - A parsed calculation retains boundary translation, operation and terminal records. Result: `PASS`.
- `scientific-notation` - Scientific notation is translated exactly without a floating intermediate. Result: `PASS`.
- `alternating-parity-certificate` - Odd arctangent partial counts supply upper bounds; even counts supply lower bounds, with the next exact part closing the opposite endpoint. Result: `PASS`.
- `exact-angle-composition` - Two exact tangent doublings give 120/119 and subtraction of atan(1/239) gives tangent One, certifying the quarter-turn identity without decimal target selection. Result: `PASS`.
- `corrected-circle-enclosure` - The circle constant has ordered exact rational endpoints carrying the corrected alternating parity and exact angle-composition trace. Result: `PASS`.
- `ordinary-equals-calculation` - A user may enter 1+1= and receive exact forward-held two. Result: `PASS`.
- `general-power-surface` - Rational and certified non-rational exponents both return exact rational enclosures. Result: `PASS`.
- `angle-mode-correspondence` - DEG sin(30) encloses exact half-One and DEG atan(One) encloses exact forty-five. Result: `PASS`.
- `hyperbolic-base` - cosh(empty-One) returns an enclosure containing structural One. Result: `PASS`.
- `exact-statistics` - Mean and population variance retain exact rational parts. Result: `PASS`.
- `memory-answer-history` - Stored answer, recalled memory and ordered history execute as typed session state. Result: `PASS`.
- `discoverable-desktop-surface` - The app exposes the ordinary keypad, scientific functions, equals, memory, history evidence and SFT guide. Result: `PASS`.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: A host float loses the exact entered part before proof begins. Result: `PASS`; receipt `sha256:f1cae5eb43509a204e1458eacacacc8b1011649fb3e24e2004e089a9a431f8d5`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only; no GUI button, memory action, angle mode or display conversion may bypass the exact evaluator; no host random generator, floating transcendental library, silent complex promotion or hidden calculator state; no claim to functions outside the explicit feature manifest without a new versioned extension Result: `PASS`; receipt `sha256:627845b4d6543d567c0cd5487a9f699ffe05b60347b66670929301ef27028896`.

Depth-independent base: One entered digit is translated to an exact counted part; the structural empty-One and structural One supply cancellation and identity bases.

Depth-independent successor: Appending a generated digit refines the exact denominator by one counted tenfold composition; appending an operator composes two already-certified values, and appending one recurrence term retains the previous exact sum plus one exact part while replacing the remainder by a proven smaller positive bound. Each additional named function is admitted only when its finite expression tree terminates in already certified kernels; each session action appends a typed state record; each GUI action maps to exactly one parser token or state transition.

Exact exclusions:

- semantic numerical zero, NaN and infinity are not admitted values.

- negative, irrational and imaginary scalars are replaced by typed Fold structures.
- binary floating-point and opaque host-library answers are excluded from proof.
- uncertified roots, series, singularities and exhausted resource bounds halt.
- decimal projections are correspondence displays only.
- no GUI button, memory action, angle mode or display conversion may bypass the exact evaluator.
- no host random generator, floating transcendental library, silent complex promotion or hidden calculator state.
- no claim to functions outside the explicit feature manifest without a new versioned extension.

Limitation: Closure covers the explicit feature manifest documented by the app and README, not every proprietary button ever shipped on every calculator. Additional functions remain open lawful extensions. Physical-unit conversion tables and random-number generation are excluded because they require separately registered empirical units or a deterministic Fold-randomness law rather than an opaque host oracle.

Evidence identities:

- Source manifest: `sha256:682bc8e380e0b70795ab3c5868cd2cf8ca71c65b26cab9624d9491d54205c990`.
- Derivation seal: `sha256:2fd07cdc7dc1f857c209c3814c709f68ab21def91f3dee750f097614add757d9`.
- Independent implementation: `sha256:a5ed2ee9a44574a28530382f7b74c0b0f39446a683fd36979342248f574c5828`.
- Independent certificate: `sha256:1e9b190d79b01ebaa1a13c76556fd3d0ad80e9bac8e7cf65570bd0e1e07b1db2`.
- External validation: `sha256:24d777eb642d44e737201e495de6c64aa30ad0c484913f11be86cf405b9561a9`.
- Engine receipt: `sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-005-75b0b5df0397c0ae.json`.

28.4 Fully realised SFT scientific calculator and complete Mathematics law explorer

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-006`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. Exact Mathematics is most useful when any person can calculate with it without surrendering the derivation chain. Familiarity, complete current-knowledge reach and machine-inspectable evidence must therefore coexist.

DERIVATION. The twenty-four dependencies fix every current Mathematics law and the corrected exact calculator. Exhausting the ten completion distinctions eliminates an API-only tool, prohibited host values, partial functions, omitted branches, opaque output, overwhelming presentation, platform lock-in, unreduced large-angle exhaustion, silent resource failure and incomplete tests. The only surviving form is the familiar exact app plus its progressively disclosed law explorer.

CHECK. Enumerate all 1,024 completion forms; require one all-admitted survivor; execute ordinary, prohibited-value, certified nonrational, large-angle, result-chain, current-census, proof-output, visible-control, advanced-discovery and exact coverage witnesses; pass four adverse controls; independently regenerate the product, replay all twenty-four Mathematics families and rerun the declared active file coverage gate.

Exact statement:

> *The admitted exact calculator extends uniquely to a familiar, progressively disclosed and cross-platform application that evaluates the complete declared scientific expression language, exposes every currently registered predecessor Mathematics family, emits engine-comparable but explicitly non-admission proof output, halts at every exact value/domain/resource boundary, and is completely statement-and-branch tested.*

Dependencies: `SFT-MATH-EXACT-ARITHMETIC-001`, `SFT-MATH-DISCRETE-001`, `SFT-MATH-COMBINATORICS-001`, `SFT-MATH-GRAPH-NETWORK-001`, `SFT-MATH-ALGEBRA-001`,

`SFT-MATH-ORDER-LATTICE-001`, `SFT-MATH-GEOMETRY-TOPOLOGY-001`, `SFT-MATH-PROBABILITY-STATISTICS-001`, `SFT-MATH-OPTIMIZATION-001`, `SFT-MATH-DYNAMICAL-SYSTEMS-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`, `SFT-MATH-EXACT-RELATIONS-002`, `SFT-MATH-ORBIT-NUMBER-THEORY-002`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-MATH-ALGEBRAIC-BALANCE-002`, `SFT-MATH-BOUNDED-N-BODY-002`, `SFT-MATH-FLOORED-FLUID-REGULARITY-002`, `SFT-MATH-PRIME-PAIR-CENSUS-002`, `SFT-MATH-RIEMANN-MIRROR-002`, `SFT-MATH-COLLATZ-FINITE-CENSUS-002`, `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002`, `SFT-MATH-SCIENTIFIC-CALCULATOR-004`, `SFT-MATH-SCIENTIFIC-CALCULATOR-005`.

Grammar boundary: The complete ten-coordinate binary product of application completions over the immutable claim-005 calculator and the exact set of twenty-four registered predecessor Mathematics claims.

Generation rule: Hold all twenty-four admitted predecessor Mathematics claims fixed, then exhaust the ten independent completion choices for interaction, value law, scientific surface, branch census, proof output, accessibility, portability, periodic scale, fail-closed resources and complete testing.

The complete Cartesian product contains 1,024 candidate forms and has completeness identity `sha256:1bcc8b271c6aaea78b627a79be19833dafce9c2f98d4df0ce66e0b2eb8938b21`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`interaction`	`expression-api-only` - An API alone is not a familiar calculator.	`familiar-keypad-and-result-chaining` - Digits, keypad, terminal equals, Ans chaining, memory, clear-entry, all-clear and backspace are all explicit.
`value_law`	`projected-host-scalars` - Host zero, negative, irrational, imaginary or floating proof values violate the admitted Mathematics boundary.	`exact-SFT-runtime-types` - Every answer is empty-One, a held positive rational, a certified rational enclosure or typed Fold fibres.
`scientific_surface`	`partial-function-subset` - A partial subset cannot realise the declared scientific calculator.	`complete-declared-expression-language` - Every documented arithmetic, power, root, circular, hyperbolic, logarithmic, combinatorial, statistical and typed-fibre function is routed through an exact kernel.
`mathematics_census`	`calculator-only-scalar-view` - A scalar-only view omits registered structured and theorem-level Mathematics families.	`all-current-predecessor-families` - All twenty-four predecessor claims have a one-to-one structured/scalar translation and local enumeration replay.
`proof_output`	`opaque-friendly-number` - A projected number without exact evidence hides the calculation law.	`typed-certificate-trace-and-resources` - Exact type, rational parts, certificate, operation trace, resources, dependencies and official receipt reference remain inspectable.
`accessibility`	`all-evidence-visible-at-once` - Dumping proof detail onto first-time users makes ordinary calculator use inaccessible.	`familiar-first-progressive-disclosure` - The standard calculator is the default; proof, functions, learning and law exploration open on demand.
`portability`	`platform-specific-or-heavy-runtime` - A platform-specific or container-dependent app blocks independent access.	`standard-library-mac-Windows-Linux` - One Python-standard-library application plus installed command and three operating-system launchers provides the same evaluator.
`periodic_scale`	`unreduced-direct-series-only` - A direct series alone exhausts resources on ordinary large circular arguments.	`certified-whole-turn-reduction` - Large angles subtract an exact whole count of certified turns before rational recurrence.
`resource_boundary`	`unbounded-or-silent-failure` - Unbounded construction or NaN/infinity hides an unclosed computation.	`counted-early-check-and-mandatory-halt` - Digits, exponents and counted operations are checked early and every invalid or unclosed expression halts without a value.
`testing`	`passing-examples-with-gaps` - Passing examples do not execute every implementation path.	`complete-statement-and-branch-coverage` - Every statement and branch in the declared active inherited and completion implementation is executed, alongside CLI, GUI, launcher, control and law-replay tests.

Unique survivor and exact result:

> *The unique completion is the familiar cross-platform app whose ordinary input invokes only exact SFT runtime types; whose advanced view exposes the complete current Mathematics census and engine-comparable proof; whose large circular inputs are certified by whole-turn reduction; and whose declared active implementation records no missing statement or branch.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- human boundary notation is translated before evaluation and never becomes a prohibited proof scalar.
- every button, keyboard action, memory transition and terminal expression uses one exact evaluator.
- a local Mathematics replay can reproduce a registered census but cannot issue or impersonate an engine admission receipt.
- progressive disclosure changes presentation only and cannot change the exact retained answer.
- large circular input is equivalent after subtraction of any generated whole number of complete turns.
- the current expression census is one-to-one with the registered predecessor catalogue and halts if that catalogue changes without a translation.
- complete implementation coverage is necessary but remains distinct from engine forcing and independent validation.

Executed witnesses:

- `ordinary-use` - 1+1= returns exact forward-held two. Result: `PASS`.
- `prohibited-value-translation` - Correspondence zero and negative are empty-One and counter-held positive magnitude. Result: `PASS`.
- `nonrational-certificate` - sqrt(2) remains a replayable rational enclosure. Result: `PASS`.
- `large-angle-closure` - A million-radian sine closes after certified whole-turn reduction. Result: `PASS`.
- `familiar-result-chain` - 2 followed by +3 returns readable five through retained Ans. Result: `PASS`.
- `complete-predecessor-census` - The exact dependency set contains twenty-four unique current Mathematics predecessors. Result: `PASS`.
- `engine-comparable-not-admission` - Proof output contains exact checks and explicitly denies issuing an engine admission. Result: `PASS`.
- `familiar-visible-controls` - Every ordinary calculator control is present once. Result: `PASS`.
- `advanced-function-discovery` - Aggregate and typed-fibre operations remain discoverable without crowding the keypad. Result: `PASS`.
- `complete-coverage` - The declared active implementation has no missing statement or branch. Result: `PASS`.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An API alone is not a familiar calculator. Result: `PASS`; receipt `sha256:ac04d355305d1827450c11aba9b90a9b2ae9bf5aeb23b9fcccd3efc6acf7c1a2`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no numerical zero, negative magnitude, irrational scalar, imaginary scalar or floating proof value; no fitted scientific parameter, host transcendental answer, random oracle, NaN or infinity; no GUI or terminal path may bypass the exact evaluator; no calculator replay is labelled an admission receipt; no claim to future Mathematics families not present in the exact predecessor census; no proprietary calculator-specific surface is claimed beyond the declared expression language Result: `PASS`; receipt `sha256:eefb6e9e942e0a51c61bd23f963e704839a3ef7cb8507308620dae96ab4f04fc`.

Depth-independent base: Claim 005 supplies one exact scalar calculation and one desktop evidence surface; the twenty-four fixed predecessors supply the complete current Mathematics catalogue.

Depth-independent successor: Adding one expression, interaction, platform view or Mathematics family is lawful only when it maps to an admitted kernel, retains exact typed evidence, adds a generated census coordinate or registered translation, and introduces no uncovered execution branch.

Exact exclusions:

- no numerical zero, negative magnitude, irrational scalar, imaginary scalar or floating proof value.
- no fitted scientific parameter, host transcendental answer, random oracle, NaN or infinity.
- no GUI or terminal path may bypass the exact evaluator.
- no calculator replay is labelled an admission receipt.
- no claim to future Mathematics families not present in the exact predecessor census.
- no proprietary calculator-specific surface is claimed beyond the declared expression language.

Limitation: Closure covers the exact current predecessor catalogue, declared scientific expression language and application behaviour. It does not claim that a calculation is a new theorem, that every future Mathematics discovery is already translated, or that every proprietary calculator button is required. Lawful future additions remain open versioned extensions.

Evidence identities:

- Source manifest: `sha256:033235dcc6970ac7ca58141f8497d4e43edf6cb15ce2d7ba31d2d84e7b9e6aca`.
- Derivation seal: `sha256:e14a12e172dab6232491cb9591dc4b82928760b721e3a99aaabba68a2857b054`.
- Independent implementation: `sha256:89732fb0c989727f66e8f370ac133fd2472ba0faf7e4af904f8f01e915eab6be`.
- Independent certificate: `sha256:1a5a1a0a29eae41f6822d1f1184279361ad4058e9b1afdb83baa9089c6a3a62e`.
- External validation: `sha256:d2eaedf0621c01b32b0e669c6428e62b622057e53b21bb5fb1773ab1d5ace28f`.
- Engine receipt: `sha256:addeb561438c9913313824ba43eb8fceccb53c574f5a9853b3c9c1fac4e4e4b7`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-006-addeb561438c9913.json`.

28.5 Accessible SFT-native scientific calculator application

Claim identity: `SFT-MATH-SCIENTIFIC-CALCULATOR-007`

Publication status: `independently_replicated`; current versioned calculator lineage.

WHY. A proof-producing calculator is accessible only when the ordinary surface remains familiar and the complicated distinctions stay available underneath. A painted, responsive and state-isolated surface is therefore part of the application boundary.

DERIVATION. Claim 006 fixes every mathematical operation. Exhausting the eight presentation distinctions eliminates the blank deprecated widget path, arbitrary keypad reflow, conventional prohibited projections, duplicate arithmetic, shared visitor state, computer-only reach, heavy dependencies and geometry-only testing. The remaining adapter changes no law.

CHECK. Enumerate all 256 adapters; require one all-admitted survivor; execute ordinary, cancellation, negative-halt, exact-certificate, fresh-session, complete-control, standard-layout, responsive-contract, single-kernel and complete-coverage witnesses; then independently start the application as a subprocess and replay its HTTP boundary.

Exact statement:

> *The admitted exact calculator extends to one dependency-free standards-rendered application whose familiar calculator notation never selects the mathematics: displayed zero retains empty-One semantics; negative-result attempts halt transactionally; certified intervals and orthogonal fibres remain typed exact objects; every page has an independent session; the conventional standard pad remains structurally coherent; scientific functions are progressively disclosed; and the same application is reachable on macOS, Windows, Linux and same-network phones.*

Dependencies: `SFT-MATH-SCIENTIFIC-CALCULATOR-006`.

Grammar boundary: The complete eight-coordinate binary product of presentation adapters over the immutable claim-006 controller; no coordinate may alter arithmetic, value construction, expression parsing, proof law or engine protocol.

Generation rule: Hold the immutable claim-006 evaluator fixed and exhaust eight independent presentation completions: rendered surface, keypad organisation, value boundary, kernel route, session state, network reach, dependency boundary and validation.

The complete Cartesian product contains 256 candidate forms and has completeness identity `sha256:06ce9a083d2cd80333cfb1ab854033ad171d986829472dd33b7ef0006536ec1e`. Every coordinate is shown below; the candidate census and elimination receipt preserve every product member and decision.

Axis	Rejected coordinate and reason	Forced coordinate and reason
`rendered_surface`	`deprecated-widget-geometry-only` - Mapped geometry did not ensure painted controls on the target macOS runtime.	`standards-rendered-browser-surface` - The installed browser paints semantic HTML controls through a local standard-library service.
`keypad_organisation`	`flattened-function-grid` - Arbitrary reflow destroys the familiar four-column calculator grammar.	`standard-pad-plus-scientific-panel` - Memory, standard four-column keys and scientific functions retain separate coherent organisations.
`value_boundary`	`conventional-prohibited-projection` - Displaying a negative or irrational-style decimal silently reimports a prohibited scalar.	`familiar-notation-with-SFT-halt-and-types` - Zero is familiar empty-One notation, negative-result attempts halt, and non-scalar results remain exact typed certificates.
`kernel_route`	`client-side-duplicate-arithmetic` - A second arithmetic implementation could diverge from the admitted evaluator.	`server-side-immutable-claim-006-controller` - Every visible action reaches the same admitted controller and the browser performs no arithmetic.
`session_state`	`shared-global-visitor-state` - One visitor or test can contaminate another visitor's landing value.	`fresh-independent-page-session` - Every page load receives a fresh retained controller identity while its own history and memory persist.
`network_reach`	`desktop-loopback-only` - A computer-only default blocks ordinary same-network phone use.	`same-network-default-with-private-option` - The default binds to the local network and displays its phone address; private mode remains explicit.
`dependency_boundary`	`heavy-GUI-or-container-runtime` - A heavy platform runtime violates the accessibility requirement.	`Python-standard-library-and-installed-browser` - HTTP service, exact controller and page assets require no third-party runtime or container.
`validation`	`hidden-widget-dimensions-only` - Widget dimensions did not detect the observed blank window.	`API-render-responsive-adverse-and-complete-coverage` - The corrected route checks HTTP behaviour, semantic rendering, mobile structure, prohibited results, session isolation and complete active-file coverage.

Unique survivor and exact result:

> *The unique completion is a familiar calculator-first local web application with a coherent standard pad, optional scientific and proof panels, fresh per-page state, same-network default access and one immutable exact SFT evaluation route.*

Replacing any forced coordinate by its enumerated alternative loses a registered requirement; adding an undeclared rule violates the no-extra-rule boundary. The unique product survivor therefore closes the declared grammar by minimality and named-shape uniqueness. Its operational laws are:

- presentation notation may describe an admitted value but cannot create a new value type.
- displayed 0 denotes structural empty One and does not admit a numerical-zero proof object.
- a result containing any counter-held scalar or bound halts before answer, memory or history mutation.
- a certified rational interval is a typed certificate and is never displayed as one irrational decimal scalar.
- each browser page owns one fresh controller state and cannot inherit another page's tested answer.
- the standard four-column keypad grammar is invariant under phone-width reflow.
- touch-keypad actions never focus the editable field or summon the phone keyboard.
- the browser performs interaction and presentation only; all evaluation remains in claim 006.

Executed witnesses:

- `ordinary-result` - 1+1 displays familiar exact 2. Result: `PASS`.

- `empty-one-display` - 1-1 displays familiar 0 while the evaluator retains empty One. Result: `PASS`.
- `negative-result-halt` - 1-4 halts without retaining history. Result: `PASS`.
- `certificate-not-irrational` - sqrt(2) displays exact rational bounds without an irrational-style decimal. Result: `PASS`.
- `fresh-session` - A calculation in one page leaves the next page at 0. Result: `PASS`.
- `complete-key-partition` - All sixty-four controls occur exactly once across memory, standard and scientific organisations. Result: `PASS`.
- `standard-keypad` - The visible standard calculator is six complete four-key rows. Result: `PASS`.
- `responsive-contract` - The page declares phone disclosure, four-column standard layout and horizontal-overflow prevention. Result: `PASS`.
- `touch-focus-contract` - Phone keypad taps do not automatically open the keyboard. Result: `PASS`.
- `one-kernel-route` - The browser sends actions to the local API and defines no JavaScript arithmetic model. Result: `PASS`.
- `complete-active-coverage` - Every active adapter statement and branch is executed. Result: `PASS`.

Adverse controls:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Mapped geometry did not ensure painted controls on the target macOS runtime. Result: `PASS`; receipt `sha256:7695430e13d324c741370ca77b432f700184cdda88fded52e646b1bb7554aa87`.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: `PASS`; receipt `sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: `PASS`; receipt `sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no edit to the frozen engine or immutable claim-006 source; no browser-side arithmetic or alternate evaluator; no displayed negative, irrational scalar, imaginary scalar, NaN or infinity; no counter-held result retained in answer, memory or history; no shared landing-state contamination between page sessions; no automatic expression focus on a touch-width calculator; no heavy GUI framework, container or network service beyond the user's local network Result: `PASS`; receipt `sha256:b3a4601336eba4f37db8cf7851d8743b5a75a8be9e1793ed1f984b360c1cb4e7`.

Depth-independent base: Claim 006 supplies the exact evaluator, controller, proof trace and complete scientific expression language; one fresh page begins with its familiar displayed 0 and empty history.

Depth-independent successor: Adding one interaction, viewport or visitor is lawful only when it preserves the standard keypad organisation, routes to the same exact controller, retains a distinct session and rejects every prohibited result transactionally.

Exact exclusions:

- no edit to the frozen engine or immutable claim-006 source.
- no browser-side arithmetic or alternate evaluator.
- no displayed negative, irrational scalar, imaginary scalar, NaN or infinity.
- no counter-held result retained in answer, memory or history.
- no shared landing-state contamination between page sessions.
- no automatic expression focus on a touch-width calculator.
- no heavy GUI framework, container or network service beyond the user's local network.

Limitation: Closure covers the current calculator presentation and declared browser-standard route. It does not claim every future browser or device has been physically tested. Manual review remains a separate publication gate, and lawful extensions remain open.

Evidence identities:

- Source manifest: `sha256:ee871d9a0d471b7525254af0fed9dff8f54a5f9254e7e2cc642c5150284b1822`.
- Derivation seal: `sha256:fdb9d7eb5203c03e24df26187742b0650bfef090233f9f4fcbd46331a29b2fa6`.
- Independent implementation: `sha256:f554a6058ae2fc15ea6ff60ddbcfd95409601eac16cec182d3d737f9268b2f5e`.
- Independent certificate: `sha256:9576ee0eee57fe8d2ab6e7245012100ddfbbb25d4eefe3b1b3cd761952fa80fb`.
- External validation: `sha256:a1402278d191f89ea10fde58a863df1f12a30c8ec97f891892efdd0e587f487d`.
- Engine receipt: `sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3e0fe5dbf0882aac9867`.
- Engine receipt path: `receipts/engine/model_admitted/SFT-MATH-SCIENTIFIC-CALCULATOR-007-feeb1879d97191ed.json`.

28.6 Exact value translation and fail-closed semantics

Familiar surface	Exact calculator meaning	Boundary behaviour
`0`	structural empty One	displayable and usable as the empty/identity form; never stored as a conventional numerical-zero proof scalar
positive integer or decimal	exact positive whole or rational Fold part, parsed character by character	no binary floating conversion is admitted
subtraction with a nonnegative result	exact trace pairing and retained forward remainder	cancellation displays `0` with empty-One semantics
subtraction whose result crosses below empty One	prohibited negative/counter-held numeric result	transactional `HALT`; prior answer, memory and history are restored
exact rational root or function result	exact Fold scalar	exact numerator/denominator retained
non-rational correspondence such as `sqrt(2)`	certified rational lower and upper bounds plus replay trace	no irrational decimal is promoted to a scalar
orthogonal correspondence	typed real and orthogonal Fold fibres	no imaginary scalar or silent complex promotion
undefined domain, singularity, exhausted resource or unclosed certificate	no admitted result	mandatory `HALT`; no NaN or infinity

This distinction is why `1-1` may visibly return `0` while `1-4` must halt. The first is a structural empty result. The second would require a prohibited scalar to be admitted as the calculator answer. Claim 007 snapshots the complete controller state before every action and restores that snapshot if answer, memory or any history row contains a counter-held scalar. The rejected expression and explanation remain visible, but the prohibited value does not.

28.7 Declared calculation language

The expression grammar accepts exact integers, decimal and scientific notation, parentheses, addition, subtraction, multiplication, division, powers, postfix factorial and percent, and optional terminal equals. It supports result chaining, `Ans`, memory clear/recall/store/add/subtract, ordered history, clear-entry, all-clear, backspace and explicit RAD, DEG and GRAD angle modes. Every button and typed expression reaches the same evaluator.

The declared scientific functions are:

```
abs recip sqrt cbrt root pow
sin cos tan asin acos atan
sinh cosh tanh asinh acosh atanh
exp ln log log10 log2
ncr npr complex conj
sum prod mean variance stddev
floor ceil gcd lcm mod hypot
```

Constants and retained state are `pi`, `tau`, `e`, `phi`, `empty`, `ans` and `mem`. Circle, exponential, logarithmic, inverse and hyperbolic operations are not delegated to opaque host transcendental functions. They produce exact rational enclosure objects with finite recurrence evidence. Large circular inputs undergo exact whole-turn reduction before enclosure, preventing magnitude from silently exhausting the recurrence. Operation limits and requested display precision are counted interface/resource boundaries; they do not tune a mathematical law.

28.8 One evaluator, three operating systems and same-network phones

The current public application uses only Python's standard library and an installed standards-compliant browser. The local server binds to the machine's local network by default, advertises the phone URL, and offers an explicit private mode. Each page load receives a fresh controller identity; one visitor cannot inherit another visitor's answer, memory or history. A request token and page-session identifier bind local actions, requests are size bounded, responses disable storage, and the content-security policy prevents external script or content dependencies.

The visible standard pad is always six rows of four keys. Memory controls are a separate five-key row and thirty-five scientific controls remain a separate seven-by-five organisation, giving sixty-four controls exactly once. On phone widths the standard pad appears first and scientific functions are disclosed by one button. Keypad taps do not focus the expression field or summon the software keyboard; the keyboard opens only when the user deliberately taps the editable expression. The layout was rendered and exercised at a 390-by-844 phone viewport without horizontal overflow.

Launch routes:

```
python3 -m sft.mathematics.calculator_browser
calculator_launchers/Launch Smithian Fold Calculator.command   # macOS
calculator_launchers/Launch Smithian Fold Calculator.bat       # Windows
calculator_launchers/launch-smithian-fold-calculator.sh         # Linux
```

28.9 Verification, coverage and direct-use evidence

Claim 006 records 1,869 executable statements and 696 branches across its declared active calculator implementation, all covered with no missing statement, branch or partial branch. Claim 007 records 293 executable statements and 86 branches across the active browser adapter, again with no missing statement, branch or partial branch. These are statement-and-branch coverage facts over the declared files, not a claim that every physical browser or device in existence has been tested.

The executed checks cover exact values; every operation and parser branch; invalid domains and resource exhaustion; expression-machine, session, memory and history integration; pure controller state; command-line execution; operating-system launchers; server API behaviour; request bounds and authorisation; fresh-session isolation; full control partition; semantic page structure; phone reflow and focus behaviour; exact interval presentation; negative-result rollback; and implementation-distinct HTTP replay. Direct use supplied an additional unfavourable route: the first desktop widget visibly opened blank even though hidden geometry tests passed. That failure is preserved in the 007 false-premise control and is why the standards-rendered route, rather than the deprecated widget path, survives.

Representative end-to-end outcomes:

Input	Visible outcome	Exact significance
`1+1`	`2`	exact disjoint-junction result
`1-1`	`0`	structural empty One
`1-4`	`HALT`	prohibited negative result rejected transactionally, history unchanged
`0.1+0.2`	`3/10` or its familiar exact display	decimal characters translated to exact rational parts
`sqrt(2)`	certified rational interval	no `1.414...` scalar and no approximation glyph in proof output
`complex(2,3)`	real and orthogonal Fold fibres	typed composition, not an imaginary proof scalar

28.10 Proof output, engine boundary and lawful transfer

A calculator result carries its exact expression, typed result, rational numerator/denominator or enclosure bounds, certificate, complete operation trace, tokens read, operations executed, calculator claim dependency and official calculator receipt identity. That output is computational evidence that an admitted calculation law evaluated the given expression. It is not, by itself, a receipt admitting a new theorem or scientific claim.

A researcher may transfer the exact expression, result object and trace into a proposed claim package, but official admission still requires a registered statement and dependencies, complete candidate generation, uniqueness, minimality, adverse controls, independent validation, execution through the untouched `SFTAdmissionEngine` and the engine-issued receipt. The calculator cannot mint that receipt and does not call or modify the engine. Version 1.3 does not claim a one-click source-bound evidence-export/replay bridge; that would be a separate lawful extension. This boundary prevents a useful calculation tool from becoming a shadow admission route.

The frozen engine tree remains `ad30f4866c18b2adbade95a0b2de40d5caa61308`. The calculator work did not alter that tree. Claims 006 and 007 are immutable admitted dependencies at their recorded source identities; future calculator work must use a new claim and cannot rewrite either receipt.

29. Complete V1/V2 Mathematics reconciliation

The clean-room rule does not permit the new branch boundary to make an older result disappear. The audit reads all 356 V1 theorem-manifest rows and all 407 V2 numbered results as observational reconstruction requirements, decomposes composite rows into atomic scientific obligations and assigns each atomic obligation to exactly one categorical owner. Prior executables and answer artifacts never enter the V3 derivation runtime.

The Mathematics ledger identifies 71 owned atomic obligations across 29 V1 rows and 38 V2 steps. Every atom maps to one or more of the twenty-two foundational receipts and every mapping is closed. The remaining 327 V1 rows and 369 V2 steps are recorded in an exact exclusion identity as reviewed non-Mathematics entries, not silently ignored. The authoritative file is `census/mathematics_prior_obligations.json`.

Reconciliation result	Exact outcome
V1 rows reviewed	356
V2 numbered results reviewed	407
Mathematics atomic obligations	71
Closed Mathematics atomic obligations	71
Open Mathematics atomic obligations	0
Explicit corrections or invalidations retained	4

Four explicit reconciliation outcomes deserve front-page visibility because they demonstrate that V3 is neither a rubber stamp nor a historical rewrite. First, old signed and zero host storage is retained as implementation history but replaced at the admitted boundary by held orientation and structural empty One. Second, V1's detailed Riemann statement governs: the prime skeleton and unique reflection axis close, while an unrestricted complex-zero theorem does not arise from symmetry alone. Third, the prior constant-three-quarter Collatz shortcut fails the start-three control even though the complete 100,000-start census succeeds. Fourth, the depth-five fluid value 32 is derived as One divided by the positive floor, not installed as numerical damping. Physical interpretations of formal rational thresholds remain assigned to their empirical owning branches rather than feeding back into Mathematics.

30. Cross-derivation synthesis

The twenty-two foundational theorems form one dependency chain rather than a list of renamed fields. Exact arithmetic gives lawful trace junction, pair-cell product, refinement and comparison. Discrete mathematics turns those operations into canonical carriers, selections, relations, maps and induction. Combinatorics then generates complete families of choices; graph theory holds a relation from complete pair support and promotes it to path, cycle, cut and network structure. Algebra asks which complete operation relations close, associate, return and map. Order retains only witnessed comparison and derives conditional extremal operations.

Geometry and topology use graph paths, incidence and order closure to derive the finite structures computation actually requires. Probability uses complete combinatorial support and exact parts to quantify observation classes without altering deterministic state. Optimisation uses order to preserve feasible undominated alternatives. Dynamics uses graph transitions to define trajectory, return, basin and record-dependent reversal. Logic then turns distinguished forms and relations into proof-carrying inference. Category, type and composition close the chain by showing how exact objects and transformations compose without losing interface provenance.

No result licenses backward importation. Because category-like composition is derived last, it cannot be an axiom used to select arithmetic. Because probability is downstream of complete deterministic support, it cannot install an ontic random cause in the earlier state law. Because geometry is finite incidence before continuum correspondence, an irrational coordinate cannot retroactively become foundational proof evidence.

The ten lineage reconstructions then make the archived named claims explicit. Relation composition supplies the beat and joint-return laws; the same remainder identity unifies Fold orbit and multiplicative order; exact finite successor traces separate potential infinity from a completed continuum; polynomial balance retains algebraic magnitude without irrational proof values; finite product support closes native n-body recurrence; reciprocal-floor geometry closes native fluid regularity; complete bounded censuses establish their declared populations; and adverse controls prevent Riemann symmetry or Collatz heuristics from being inflated beyond what the execution establishes.

31. Exact arithmetic without zero, negatives or irrational proof values

The prohibition on semantic numerical zero does not make empty, identity or no-transition cases inexpressible. It forces them to retain their structural origin. Empty selection is empty One; identical traces return same-form; an identity arrow is the empty-One return; identity duration is the one-state trajectory with no transition. These forms are typed structural records, not a scalar that can be silently substituted across domains.

Negative quantities are likewise unnecessary for exact comparison. If two traces pair completely they are the same form. Otherwise the unmatched positive occurrences are retained and a held label records which trace owns them. Network balance pairs positive token identities. Logical denial changes held orientation. Algebraic return uses a complementary carrier mate. No proof relies on subtracting a larger magnitude from a smaller one.

Irrational, imaginary and floating values are excluded from formal admission because the generated Foundation supplies exact positive parts and finite refinement, not an ungenerated completed field. This does not deny that conventional sciences record such descriptions. It establishes that they may enter only as correspondence, finite approximation or measured values after the law is sealed. A later approximation theorem must state its generated sequence, exact error relation, convergence boundary and measurement custody.

32. Probability, statistics and superdeterminism

The probability theorem is often the point at which a deterministic model is tempted to import an incompatible prior. SFT does not do so. A microstate is one exact generated form. The registered support contains every such form once. Observation is a total classification relation from microstates to retained observation labels. States with the same image form an observation class. Uncertainty is precisely the collection of distinctions present in support but closed by that observation; the underlying transition law remains deterministic.

An event is a held support selection. For a nonempty event, probability quantity is the exact positive count of held states relative to the exact positive count of the complete support. The empty event remains empty One and never passes through numerical-zero arithmetic. Conditioning restricts both event and whole to a nonempty held condition before forming the exact part. Independence is not lack of a fitted correlation; it is the structural fact that joint support is the complete pair-cell product and the event pairing factorizes.

Statistical summaries are allowed because they are transparent functions of complete finite records. Ties are retained. A natural-data claim would require a separate empirical registration, frozen source and target custody. The target must remain closed until after the derivation seal; all rows, including failures, must be preserved. No natural dataset was needed or used to admit the formal probability theorem in this paper.

33. Formal proof and empirical constitution

The twenty-two foundational claims and five calculator records are formal structural theorems, exhaustive finite mathematical censuses or executable translations. Their empirical content in this paper is computational execution and direct interface testing: the candidate products are actually generated, decisions actually run, controls are deliberately perturbed, separate validator processes execute and receipts are independently replayed. This execution evidence demonstrates that the declared finite grammars and certificates have the reported machine behaviour. It is not observational evidence about an external natural system, and the paper does not mislabel it as such.

When a later branch asserts a relation to natural data, the Foundation measurement boundary applies. Formal law selection and target evaluation are different phases with different custody. A blind opaque predictor cannot replace a derivation chain because a score alone does not identify premises, alternatives, eliminations, source identity, failure conditions or whether the target influenced the law. An empirical SFT claim must expose all of those objects and halt if custody or capability isolation fails.

34. Machine implementation and complete reproduction

The definitive cross-platform logical command is `sft verify-all`, invoked with the host's standard Python launcher:

```
python3 -m sft verify-all   # macOS and most Linux systems
py -m sft verify-all        # standard Windows Python launcher
```

No Docker image, virtual machine, network service or third-party Python package is required for the scientific verification path. The command validates repository structure, runs all unit, unfavourable-control and end-to-end tests under line coverage, requires 100 percent executable-line coverage of the 15-module admission engine, loads the ordered execution manifest, rebuilds each source manifest, reruns every candidate and control, launches every independent validator and compares each new receipt byte-for-structure with the admitted stored receipt.

The proportionate local report for this successor paper is:

```
MATHEMATICS PRIOR-OBLIGATION LEDGER: 71/71 CLOSED
MATHEMATICS REGISTERED CLAIM RECORDS: 27/27
MATHEMATICS CURRENT NON-SUPERSEDED CLAIMS: 26
MATHEMATICS GENERATED CANDIDATES PRESERVED: 21,504
CALCULATOR CLAIMS 004 THROUGH 007: CURRENT
CALCULATOR CLAIM 003: PRESERVED SUPERSEDED ADVERSE EVIDENCE
BRANCH PUBLICATION GATE: run once after final render
```

The coverage statement concerns the engine implementation; mathematical closure comes from the claim-specific candidate grammar, decision vector, unique survivor, minimality, named-shape uniqueness and induction certificate. Neither substitutes for the other.

35. Independent criticism and invalidation routes

A reviewer need not accept the vocabulary or conclusions to reproduce the result. The strongest direct criticism is to identify a candidate inside a declared boundary that the registered product omits. That attacks completeness. A second route is to show that a rejected coordinate actually preserves every requirement, defeating minimality, or that another product member survives, defeating uniqueness. A third is to invalidate the base or successor certificate. A fourth is to provide an input inside the declared domain that violates an operational law.

The repository also supports mechanical attacks: alter a source after registration, change a candidate identity, remove a decision, add a survivor, fail a control, tamper with a certificate, reorder the execution manifest or change a receipt. The engine halts at the exact violated gate. Hashes do not establish truth; they establish which exact sources and artifacts were checked so an altered object cannot silently inherit a prior status.

36. Consequences for the remaining knowledge tree

The current-evidence-complete branch supplies the exact resources downstream branches may cite: symbols can be canonical distinguished forms; encodings can be complete maps; information quantity can be an exact relation between support distinctions and observation; channels can be relations; entropy and uncertainty must respect deterministic support and exact parts; coding can use combinatorial word families, graph paths and algebraic operations. None of those downstream laws is admitted by this paper merely because its mathematical vocabulary now exists.

Formal computation may build states from canonical forms, transitions from relations, machines from dynamical systems, resource orders from exact positive traces, semantics from proof-carrying composition and quantum support from complete Fold word families. Physics and later natural sciences may use finite geometry, dynamics, exact probability and optimisation, but must separately derive their laws and pass blind empirical validation where observation is required. Applications remain frontier translations until the relevant science branches close.

37. Scope, limitations and lawful extension

Current-evidence completion is exact at the versioned generated-finite Mathematics boundary. It does not assert a completed universe of all sets, a real or complex continuum, an infinite-dimensional space or every named theorem of conventional mathematics. It derives the general structural kernels needed to generate and test finite exact instances. Named special structures must still expose their carrier and property evidence. Continuum expressions can enter only through separately registered finite approximation or measurement claims.

No official natural empirical claim is made in this branch. The calculator is an executable mathematical translation and validation surface, not a natural-data experiment. Information Science and the computation branches have their own later papers and receipts; after this Mathematics 1.5 publication the active clean-room programme returns to Physics. None of those downstream results can retroactively select a Mathematics law.

38. Open-science status, rights and participation

Mathematical validity must not depend on whether an institution funded the question, whether a journal selected it, whether a reviewer recognizes the author's credential, or whether a prevailing programme finds the conclusion comfortable. A proof either exposes a lawful chain at its declared boundary or it does not. Institutional review can supply valuable criticism, but it cannot substitute for generated completeness, unique survival, exact witnesses and a reproducible certificate. The same rule binds Maria Smith, Ernos Labs, universities, publishers and anonymous critics.

This positioning is not rhetoric detached from evidence. Sponsor-associated outcome differences and sponsor influence on research agendas are documented; grant-review studies report limited agreement, panel sensitivity and unequal outcomes; a Cochrane review found insufficient evidence that grant peer review improves funded-research quality; and null or negative results remain underrepresented in visible literatures. UNESCO's Recommendation on Open Science identifies paywalls and high publication charges as sources of inequality and calls for methods, software, source code and outputs to be available for scrutiny. These findings do not accuse every institution or reviewer of misconduct. They establish that prestige, funding and publication are selection systems with incentives and failure modes, not neutral certificates of mathematical truth.

The exclusion cost is intellectual as well as economic. Scarce grants, credential filters, prestige markets, subscription access and author charges shape who can devote time to a problem, which problems appear respectable, what enters the searchable record and who can inspect it. Maria Smith developed SFT outside conventional academic education and funding. That fact is not evidence that any theorem is correct. It is an indictment of a gate that would have treated the speaker's status as a proxy for whether the work deserved to be heard. The unknown number of minds and questions lost to that proxy is precisely why admission here attaches to public evidence rather than biography.

Opaque computational prediction reproduces the same authority problem in machine form. A hidden model can be a useful instrument and may establish bounded predictive reliability in a genuinely blind trial. A score alone does not reveal the premises, candidate space, exclusions, leakage path, failure boundary or derivation of a law. NIST treats validity, reliability, transparency, accountability, explainability and interpretability as related features of trustworthy AI. For formal mathematics, an oracle answer is therefore never a proof; for empirical science, blindness and transparency must coexist. The target stays closed until the law seals, and the full source-to-result route remains inspectable afterward.

The scientific platform is openly inspectable, reusable and redistributable under its published licences. Maria Smith retains copyright and authorship; that preserves attribution and does not close the knowledge. Paper and documentation text is CC BY 4.0 and code is Apache-2.0, permitting copying, forking, testing, modification, redistribution and criticism under their terms. The Ernos Labs designation is a separate conformance mark requiring the published scientific constitution, transparent evidence, fail-closed engine rules and community conduct policy.

Credentials cannot rescue a failed gate, and lack of credentials cannot prevent a reproducible criticism from being evaluated. Scientific authority is narrower than authorship: it is a currently accepted receipt at an explicit boundary and is revoked when a broken dependency, omitted candidate, second survivor, failed control or contradictory complete census is reproduced. Independent replications, omissions, counterexamples and submissions are invited through Maria.Smith.Sftoe@gmail.com and https://discord.gg/ucwGryVxGr.

39. Conclusion

Mathematics is complete to its frozen 29 July 2026 census: **323/323 obligations, 24/24 families, 97,280 complete candidate decisions, 323 unique survivors, 1,292 passed adverse controls, 323 implementation-distinct reconstructions and no open registered obligation**. The final reconciliation identity is `sha256:526c6d78a3ff77c21c6d4dd71c266499f5b234a7763cb223e5d009fa9f101c64`. The current lightweight repository status records 2,751 admitted V3 claims across the full platform, while the canonical engine seal and verification-authority seal remain unchanged.

The scientific result is both mathematical and methodological. Arithmetic, algebra, discrete structure, geometry, topology, analysis, probability, optimisation, dynamics, logic, compositional mathematics, numerical analysis and symbolic construction are expressed through one exact generated constitution. Each familiar correspondence is downstream of a sealed SFT derivation. Each empty case is structural absence, each opposed direction is typed, each non-rational conventional object is represented by an exact construction or enclosure, and every claimed generality has a finite-successor or explicit boundary certificate.

The Validation Grand Lock does not erase adverse evidence. It preserves all 305 pre-validation boundary records and partitions every receipt exactly once. The final handoff family then prevents Mathematics from absorbing the empirical meanings owned by Physics, Chemistry, Biology, Social Science or Engineering. This is dated completion, not permanent closure: future discoveries remain welcome, but they do not enter by editorial preference, reputation or fit. They enter by registration, enumeration, uniqueness, falsification, independent reconstruction, observation and the unchanged engine.

Appendix A. Foundational and calculator receipt identities preserved from version 1.4

Claim	Engine receipt
SFT-MATH-EXACT-ARITHMETIC-001	sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418
SFT-MATH-DISCRETE-001	sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb
SFT-MATH-COMBINATORICS-001	sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968
SFT-MATH-GRAPH-NETWORK-001	sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a
SFT-MATH-ALGEBRA-001	sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd
SFT-MATH-ORDER-LATTICE-001	sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6
SFT-MATH-GEOMETRY-TOPOLOGY-001	sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e
SFT-MATH-PROBABILITY-STATISTICS-001	sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b
SFT-MATH-OPTIMIZATION-001	sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849
SFT-MATH-DYNAMICAL-SYSTEMS-001	sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe
SFT-MATH-LOGIC-PROOF-001	sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813
SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8
SFT-MATH-EXACT-RELATIONS-002	sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238
SFT-MATH-ORBIT-NUMBER-THEORY-002	sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff
SFT-MATH-LIMIT-CONTINUUM-002	sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098
SFT-MATH-ALGEBRAIC-BALANCE-002	sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d
SFT-MATH-BOUNDED-N-BODY-002	sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4
SFT-MATH-FLOORED-FLUID-REGULARITY-002	sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0
SFT-MATH-PRIME-PAIR-CENSUS-002	sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6
SFT-MATH-RIEMANN-MIRROR-002	sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6e6ca001a24622e55364dfc4
SFT-MATH-COLLATZ-FINITE-CENSUS-002	sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cbd40c4f9a045
SFT-MATH-SELF-SIMILAR-CONVERGENCE-002	sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fb606d5b1
SFT-MATH-SCIENTIFIC-CALCULATOR-003	sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c
SFT-MATH-SCIENTIFIC-CALCULATOR-004	sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b
SFT-MATH-SCIENTIFIC-CALCULATOR-005	sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb
SFT-MATH-SCIENTIFIC-CALCULATOR-006	sha256:addeb561438c9913313824ba43eb8fceccb53c574f5a9853b3c9c1fac4e4e4b7
SFT-MATH-SCIENTIFIC-CALCULATOR-007	sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3e0fe5dbf0882aac9867

Appendix B. Complete evidence route

For every claim identifier `<CLAIM>`, the complete review route is:

```
claims/<CLAIM>/registration.json
claims/<CLAIM>/WHY_DERIVATION_CHECK.md
claims/<CLAIM>/candidate_census.json
claims/<CLAIM>/elimination_receipt.json
claims/<CLAIM>/controls.json
```

```
claims/<CLAIM>/certificate.json
claims/<CLAIM>/execution.py
generated/<BRANCH>/<VERSIONED-INDEPENDENT-VALIDATOR>.py
receipts/engine/model_admitted/<CLAIM>-<receipt-prefix>.json
```

Executable law sources are under `sft/mathematics/`. `sft/mathematics/catalog.py` fixes the foundational and calculator-kernel order; the versioned claim registrations and execution manifest retain the 007 presentation extension. `census/claims.json` is the model-admitted projection; `census/execution_manifest.json` is the complete replay order; `publications/inventories/mathematics.json` freezes scope. The separate paper evidence map binds every derivation section to exact source, claim-package and receipt hashes.

Appendix C. Reproducibility interpretation

A successful rerun proves that the checked sources deterministically reproduce the registered censuses, decisions, closures, controls, independent certificates and receipts on the reviewing host. It does not compel agreement with the grammar boundary. Scientific review therefore has two complementary tasks: reproduce the artifact and scrutinize whether the declared grammar exhausts the stated boundary. The repository makes both tasks explicit.

Family results, complete claim inventory and claim-level audit records

The dependency-ordered ledger below is the complete human-readable Mathematics claim inventory for this version. Each claim section exposes its claim identity, family, formal and empirical status, exact statement, dependency route, carrier and boundary, candidate grammar and count, unique survivor, elimination logic, falsification condition, controls, source and custody records, chronology, scientific meaning and receipt identities. Where a generic clause is referenced, its full unchanged wording is retained in the shared claim-record appendix. The machine packages remain authoritative for complete candidates, decisions and executable traces.

40. Complete-field Mathematics execution - version 1.5

Version 1.4 froze the complete-field roadmap. Version 1.5 executes it. This section is controlling for the current census and preserves the earlier foundational derivations rather than rewriting their historical receipts.

40.1 Complete family census

Order	Family	Claims	Candidates	Controls	Status
1	BASE	27	21,504	108	complete, exact-replayed, extension-open
2	ARITH	18	4,608	72	complete, exact-replayed, extension-open
3	ALEXT	10	2,560	40	complete, exact-replayed, extension-open
4	COMB	12	3,072	48	complete, exact-replayed, extension-open
5	GRAPH	14	3,584	56	complete, exact-replayed, extension-open
6	LINEAR	14	3,584	56	complete, exact-replayed, extension-open
7	ALG	16	4,096	64	complete, exact-replayed, extension-open
8	ORDER	12	3,072	48	complete, exact-replayed, extension-open
9	GEOM	16	4,096	64	complete, exact-replayed, extension-open
10	TOPO	14	3,584	56	complete, exact-replayed, extension-open
11	CALC	12	3,072	48	complete, exact-replayed, extension-open
12	ANAL	16	4,096	64	complete, exact-replayed, extension-open
13	EQN	12	3,072	48	complete, exact-replayed, extension-open
14	MEAS	10	2,560	40	complete, exact-replayed, extension-open
15	PROB	16	4,096	64	complete, exact-replayed, extension-open
16	OPT	16	4,096	64	complete, exact-replayed, extension-open
17	DYN	12	3,072	48	complete, exact-replayed, extension-open
18	LOGIC	16	4,096	64	complete, exact-replayed, extension-open
19	CAT	12	3,072	48	complete, exact-replayed, extension-open
20	NUM	12	3,072	48	complete, exact-replayed, extension-open
21	SYMB	10	2,560	40	complete, exact-replayed, extension-open
22	XINT	8	2,048	32	complete, exact-replayed, extension-open
23	VALID	12	3,072	48	complete, exact-replayed, extension-open
24	HAND	6	1,536	24	complete, exact-replayed, extension-open
Total	**24 families**	**323**	**97,280**	**1,292**	**323/323**

40.2 Admission constitution applied to every claim

Every entry below records the owned question, exact statement, unique survivor, exhaustive candidate and control counts, post-registry observation, source custody, dependency edge and immutable receipt. The machine-readable packages remain controlling; the paper makes their scientific content visible rather than substituting hashes for findings.

40.3 Family `BASE` - 27/27 complete

This family contributes 21,504 completely decided candidates, 27 unique survivors and 108 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-BASE-001` - Exact arithmetic and generated number structure

- **Claim:** `SFT-MATH-EXACT-ARITHMETIC-001` - Exact arithmetic and generated number structure.
- **Forced law:** Every admitted arithmetic relation on generated positive finite traces and exact parts has one parameter-free structural kernel: disjoint junction for addition, complete pair-cell refinement for product, finite repeated composition for powers, common-refinement pairing for exact quotient and comparison, and held orientation for unmatched remainder; no semantic zero, negative, irrational, floating or completed-infinite value enters the law.
- **Unique survivor:** The exact arithmetic kernel is complete positive-trace coverage with disjoint junction, pair-cell product, common-refinement quotient, complete pairing comparison, held-oriented remainder, base/successor generality and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-COUNT-001 -> SFT-FOUNDATION-PART-001 -> SFT-FOUNDATION-PART-EQUIVALENCE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not a value; negative magnitude is replaced by held orientation; irrational, imaginary and floating proof values are excluded; completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:1254abf40a91886525ad8a846896b75a1023fd785907f233bcf817db2c744dc4`; independent `sha256:73b24906c1005bd7f42f36000f396bb5ec88f633a4a69b6159d5cc349bb3be2a`; empirical `sha256:9a05361b10ccce837c8a32def2b24dd0ec714cb2f45d397a9e6b72fedec306a7`; engine receipt `sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418`.

`SFT-MATH-OBL-BASE-002` - Generated discrete objects, relations and induction

- **Claim:** `SFT-MATH-DISCRETE-001` - Generated discrete objects, relations and induction.
- **Forced law:** Every admitted discrete object is a complete canonical generated finite collection or held selection; membership is retained occurrence, relations are held selections of complete pair-cell support, maps are total single-valued relations, sequences retain generated order, and general laws close by structural One/base-successor induction.
- **Unique survivor:** The discrete kernel is complete unique canonical collections with retained membership, pair-cell-selected relations, total single-valued maps, held sequence order, One/base-successor induction and no extra constructor.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-COUNT-001 -> SFT-FOUNDATION-FORM-GRAMMAR-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-EXACT-ARITHMETIC-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no completed infinite set; no ungenerated universal collection; no semantic numerical zero for the empty selection; no presentation alias may replace canonical form identity.
- **Certificates:** derivation `sha256:1d3b73c941e235c5ffe473ccafa9cd59b800ddd33d43e4feea7a6821ed8f45c9`; independent `sha256:2f9da4efdc67b6f08074be8d3ddc47595444a9300fd00e507fb04c228686f44b`; empirical

sha256:5dad446dc09d74caf9325f27b314d7f886275164df426cd561a687d9283cd109; engine receipt sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb.

SFT-MATH-OBL-BASE-003 - Exact finite combinatorial generation

- **Claim:** SFT-MATH-COMBINATORICS-001 - Exact finite combinatorial generation.
- **Forced law:** Every admitted finite combinatorial family is generated by an explicit held-choice recurrence over a complete canonical carrier, preserves construction order and provenance, removes duplicates only by canonical form identity, and counts the resulting complete trace; selections include the structural empty One form rather than numerical zero.
- **Unique survivor:** The combinatorial kernel is complete canonical carrier generation by held-choice recurrence with complete choice provenance, canonical deduplication, witnessed symmetry, disjoint class accounting, successor generality and no imported formula.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status independently_replicated.
- **Root lineage:** SFT-FOUNDATION-FOLD-ASSEMBLY-001 -> SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DISCRETE-001; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no random sampling as completeness evidence; no factorial or binomial answer table; no negative inclusion-exclusion proof value; no completed infinite combinatorial family.
- **Certificates:** derivation sha256:3177572ea372062f65a7969fdf4e498c86c9af558e37c048165e1d950cf1d942; independent sha256:ea22b95c87c9f336726c9df87d1e88fb38b1ad27519789c334a5052f5f864ae6; empirical sha256:dbfc2711bbabde9be3bf4a4cc3c5de3cd62440c9fbeea20b2fceed6dcfb1275b; engine receipt sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968.

SFT-MATH-OBL-BASE-004 - Exact finite graph and network structure

- **Claim:** SFT-MATH-GRAPH-NETWORK-001 - Exact finite graph and network structure.
- **Forced law:** Every admitted finite graph is a complete canonical node collection plus a held selection of generated ordered node-pair relations; paths retain every adjacent transition, reachability is a complete finite path witness, cycles retain return, trees are connected cycle-free forms, cuts are exact crossing-edge selections, and internal network conservation is complete ingress/egress pairing.
- **Unique survivor:** The graph/network kernel is complete canonical nodes with held ordered pair-cell edges, complete adjacent paths, witnessed reachability and return, held crossing cuts, ingress/egress pairing and no extra rule.
- **Enumeration and falsification:** 512 candidates, one survivor, 4 controls; closure depth_independent and external status independently_replicated.
- **Root lineage:** SFT-FOUNDATION-FORM-GRAMMAR-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-COMBINATORICS-001; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no ungenerated vertex or edge; no negative edge or flow quantity; no stochastic graph law; no infinite graph as a completed object.
- **Certificates:** derivation sha256:f1f74c52072189f08134293f1f55aaff63aeb73fc045d79ac09a9372554c7ad9; independent sha256:531d0c9a5ec7a97b8e97c4a1664d67af65d7fab17127bcf042e76a86386a9210; empirical sha256:9f9e2483be55ff773cf570f6fc3aee074f68f44fde118492f152023767fef5c8; engine receipt sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a.

`SFT-MATH-OBL-BASE-005` - Generated finite algebraic structures

- **Claim:** `SFT-MATH-ALGEBRA-001` - Generated finite algebraic structures.
- **Forced law:** An admitted finite algebraic structure is a complete canonical carrier with a complete, single-valued, closed pair-cell operation relation. Identity, associative composition, reversible return mates, commutation, substructure and structure-preserving maps are admitted exactly when their exhaustive carrier witnesses pass; none is installed as an ungenerated axiom.
- **Unique survivor:** The algebraic kernel is a complete canonical carrier and complete closed single-valued operation table, with identity, association, returns, special properties and maps admitted only by exhaustive witnesses.
- **Enumeration and falsification:** 512 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DISCRETE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no imported group, ring or field axioms; no negative or irrational carrier requirement; no property inferred from a conventional name; no completed infinite carrier.
- **Certificates:** derivation `sha256:ae92defb9a52942af89734332b791cb1921ae655499b56e808a6250f539d6396`; independent `sha256:b41bcba0ab04cfad3eb95cd55ae4286b9ea1c52748ae444b33e2330ec7b52bf4`; empirical `sha256:5443f0a807dfeff488bfd29b7059fbbd086799f212e974d12867670b9f644654`; engine receipt `sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd`.

`SFT-MATH-OBL-BASE-006` - Exact finite order and conditional lattice structure

- **Claim:** `SFT-MATH-ORDER-LATTICE-001` - Exact finite order and conditional lattice structure.
- **Forced law:** An admitted finite order is a held carrier relation whose reflexive, antisymmetric and transitive cells are exhaustively witnessed. Totality is admitted only when every generated pair is comparable. Lower and upper bounds are complete relation selections; meet and join exist only as unique greatest-lower and least-upper witnesses, and monotone maps preserve every retained comparison.
- **Unique survivor:** The order/lattice kernel is a canonical carrier with an exhaustively witnessed partial-order relation, conditional totality, complete bound selections, uniquely witnessed meet/join and monotone maps.
- **Enumeration and falsification:** 512 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-ALGEBRA-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no imported real-number line; no assumed total order; no lattice completeness without generated meets and joins; no completed infinite chain.
- **Certificates:** derivation `sha256:d4421e8aaf34c239bda0fcacfe2a6aa770fac55ec3afe50c0ba5085d83946527`; independent `sha256:f1b8cbb06e153828d4410413baf3b6af6f075df5a5f3f297a1f6dee4d1c51bce`; empirical `sha256:f2b2f4bbab7729985711dbdff1cbc25cd7d16f0423a9752f8abdba8caebdd136`; engine receipt `sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6`.

`SFT-MATH-OBL-BASE-007` - Exact finite geometry and topology for computation

- **Claim:** `SFT-MATH-GEOMETRY-TOPOLOGY-001` - Exact finite geometry and topology for computation.
- **Forced law:** Computational geometry is forced as canonical finite cells plus an acyclic held boundary-incidence relation; dimension is retained boundary depth, adjacency is shared incidence, and distance is a shortest generated path with

self-distance represented by empty One rather than numerical zero. Computational topology is a complete generated open-family closed under generated joins and pairwise intersections; continuity is exact inverse-image preservation.

- **Unique survivor:** The geometry/topology kernel is finite canonical cells with held acyclic incidence, boundary-depth dimension, shared-face adjacency, exact shortest paths, closed finite open families, inverse-image continuity and reversible incidence-preserving deformation.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-PART-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-ORDER-LATTICE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no ungenerated continuum; no irrational or floating proof coordinate; no free metric threshold; no completed infinite topological family.
- **Certificates:** derivation `sha256:7afff182127d9b93f9705212b997801eafd5a3629fd2a8a56753ce674832d0ad`; independent `sha256:8b8595f6ef14fa36929ff546752015da701e4a49af5213d54f0f6a1d4559ce68`; empirical `sha256:35658ed3991ba98df705a53a75bacc5c7e813bde5f0505e5d20b549ed8cf3dd5`; engine receipt `sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e`.

`SFT-MATH-OBL-BASE-008` - Exact finite probability and statistics from Fold observation

- **Claim:** `SFT-MATH-PROBABILITY-STATISTICS-001` - Exact finite probability and statistics from Fold observation.
- **Forced law:** Probability is an exact held-support/whole-support part relation over a complete deterministic generated microstate support; uncertainty records distinctions closed by an observation class and is not a causal random premise. Conditional weight is exact common-refinement restriction, independence is complete pair-cell factorization, and statistics are provenance-retaining finite trace summaries. The empty event is empty One, never numerical zero.
- **Unique survivor:** The probability/statistics kernel is complete deterministic support, held events, exact held/whole weights, observation-class uncertainty, common-refinement conditioning, pair-cell independence, transparent finite summaries and post-seal empirical checking without random parameters.
- **Enumeration and falsification:** 512 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-PART-EQUIVALENCE-001 -> SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001 -> SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-COMBINATORICS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no ontic stochastic premise; no floating probability in proof evidence; no Bayesian or distributional prior parameter; no target data may select a formal law.
- **Certificates:** derivation `sha256:9ad2c0aa860b03d0fe11c390e7e1fada396f48565fd89d055f06fec28b32ed67`; independent `sha256:d80527e03481be10b16d4765742d717ebcefaa9f1830e2c1c37d6e40869a62e9`; empirical `sha256:39850e79359dbd3ae8e41fe032474bcf197212f6ac0214cdfd90605919561761`; engine receipt `sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b`.

`SFT-MATH-OBL-BASE-009` - Exact finite optimization and retained optima

- **Claim:** `SFT-MATH-OPTIMIZATION-001` - Exact finite optimization and retained optima.
- **Forced law:** An admitted optimization problem is a complete canonical candidate carrier, a source-bound exact feasibility relation and a witnessed preference relation. Solutions are the complete undominated held selection; uniqueness is claimed

only for a singleton, while ties and incomparable Pareto alternatives remain explicit. Constraints, decomposition and approximation are exact trace relations without floating objectives, negative penalties or free tuning parameters.

- **Unique survivor:** The optimization kernel is complete canonical candidates, source-bound feasibility, exact preference, complete undominated selection, retained ties, exact constraints and interfaces, and positive-part bounds without an added objective.
- **Enumeration and falsification:** 512 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-ORDER-LATTICE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no floating objective or tolerance; no negative penalty term; no tunable regularizer or tie-break parameter; no unenumerated candidate domain.
- **Certificates:** derivation `sha256:11d1d87d66d732bca3efc296de283e8e5e11ed9d291dd083c10f86d6801277e9`; independent `sha256:9cb0790543dac4fb7c8ff8bce76f783ccb1c96e84176f8cf59c203060920bfa4`; empirical `sha256:7f6345c8109261bf2f63108a2613767ff27fae5f9d411bace11a33cc9046ddb3`; engine receipt `sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849`.

`SFT-MATH-OBL-BASE-010` - Exact generated finite dynamical systems

- **Claim:** `SFT-MATH-DYNAMICAL-SYSTEMS-001` - Exact generated finite dynamical systems.
- **Forced law:** An admitted dynamical system is a complete canonical state carrier with a held transition relation. A trajectory retains every state and transition; time is the positive transition trace and the identity trajectory is empty One. Fixed forms, returns, recurrence, basins, reversibility and stability are admitted only through exhaustive generated witnesses. Merged predecessors remain recoverable only when their exact records are retained.
- **Unique survivor:** The dynamical kernel is complete canonical states, held transitions, complete trajectories, transition-count time, witnessed fixed/return/basin structure, retained-record reversibility and registered perturbation stability.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-OPTIMIZATION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no continuous real time premise; no differential equation imported as a law; no stochastic transition cause; no unrecorded predecessor reconstruction.
- **Certificates:** derivation `sha256:972a41752e953bd0b4d12f5c3c7d8e68c2aa2107a5aacaf9e330a2278b1a3a51`; independent `sha256:c8e806f9f4e6b74927f0bfe79e2c5b4351113323176e59ecde786d4121290172`; empirical `sha256:f93adcb26bb1c7baeec979059b40325d370b5f2f3d5f50cd6885a5159b10362f`; engine receipt `sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe`.

`SFT-MATH-OBL-BASE-011` - Exact finite logic, inference and proof boundaries

- **Claim:** `SFT-MATH-LOGIC-PROOF-001` - Exact finite logic, inference and proof boundaries.
- **Forced law:** An admitted proposition is a canonical distinguished form with a held orientation; denial is the retained complementary orientation, not a negative number. Conjunction retains joint proofs, disjunction retains a chosen generated branch and its alternatives, implication is a registered premise-to-conclusion transition, and a proof is a complete

dependency trace of accepted rules. Consistency, soundness and completeness are machine-checkable only relative to the exact registered finite grammar and rule set.

- **Unique survivor:** The logic/proof kernel is canonical distinguished propositions, complementary held denial, joint and held alternative proof forms, proof-carrying implication, complete inference traces, explicit consistency and soundness, and completeness only over the exhausted registered grammar.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-ORDER-LATTICE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no imported truth-value arithmetic; no negative truth magnitude; no authority-based proof admission; no global completeness beyond a generated grammar.
- **Certificates:** derivation `sha256:849294dd48e23a2aefd379804094c01132234adad33b937a3164b3c73f8230bf`; independent `sha256:dfeb338769fcf4b551277891fd7c5576f15ce79e533addbb15e6e4eed87a288a`; empirical `sha256:d3cc342c7a7bf4b50450ae303b15dd250328f761236f869150456ba1550c3cb9`; engine receipt `sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813`.

`SFT-MATH-OBL-BASE-012` - Forced category, type and compositional structure

- **Claim:** `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001` - Forced category, type and compositional structure.
- **Forced law:** Canonical Fold forms act as objects and proof-carrying transitions as arrows. Identity is the empty-One return, composition is forced exactly by equal interfaces, and association follows from canonical complete path flattening. Types are source-bound predicates over generated carriers; products are complete joint pair support and sums retain held alternative labels. Functors preserve interfaces, identity and composition, while naturality is equality of the two complete composite traces.
- **Unique survivor:** The compositional kernel is canonical Fold objects, proof-carrying arrows, empty-One identity, exact-interface composition, path-flattening association, source-bound types, joint products, held sums, preserving functors and equal-path naturality.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-ALGEBRA-001 -> SFT-MATH-LOGIC-PROOF-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no imported category axioms; no ungenerated universe of all objects; no impredicative type constructor; no universal property without an exhaustive finite witness.
- **Certificates:** derivation `sha256:756d7d6dce776fc1fb4b3906e2203e4e2da4e1ded6cf733eb041334eb7cb5c02`; independent `sha256:1d380a6b0d0ce9d32cfdff52fae121315257e3e561da9379ab5509c835680dad`; empirical `sha256:fc7b4ce671b53f4d62da07705c4fc5d92b0195d5218a30f2a40a773a2fb7c981`; engine receipt `sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8`.

`SFT-MATH-OBL-BASE-013` - Exact ratio, separation, reciprocal, threshold and cycle composition

- **Claim:** `SFT-MATH-EXACT-RELATIONS-002` - Exact ratio, separation, reciprocal, threshold and cycle composition.

- **Forced law:** Exact generated relations force positive ratios, shorter-part separation, reciprocal return to the One, the unique m-fold holding complement, local Fold separation transport, telescoping relative views, and least-common-return composition for every supplied finite family of exact cycles.
- **Unique survivor:** The unique exact relation kernel is generated-part ratio, held shorter separation, unique complement threshold, Fold-factor transport and least-common-return composition with finite-family induction.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DYNAMICAL-SYSTEMS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:516d208341bde6e161b32070c8051f089f2d0f11ccc3748c75680c6c4e4d1dc6`; independent `sha256:2ac8d699989a021d04fe3f921d3858795423d28863d83fcfb06542408a254a36`; empirical `None`; engine receipt `sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238`.

`SFT-MATH-OBL-BASE-014` - Fold orbit number theory and exact prime-spectrum structure

- **Claim:** `SFT-MATH-ORBIT-NUMBER-THEORY-002` - Fold orbit number theory and exact prime-spectrum structure.
- **Forced law:** For every reduced part over an odd positive finite denominator, Fold orbit length is exactly the multiplicative order of the binary generator; reduced residues partition into equal cyclotomic orbits, prime periods divide one-less-than-prime, and a power-of-two factor supplies exactly its counted transient depth.
- **Unique survivor:** Fold return and binary multiplicative order are one exact remainder identity; odd reduced residues tile into cyclotomic orbits and each binary denominator factor contributes one transient Fold.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DYNAMICAL-SYSTEMS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:8c00984dae257d972ebc6ed4e059a3c958f43e2c29b39f9775f78883dd376086`; independent `sha256:b47923bcf2b3c784ecd0689286d35a111a91faa6f86ec1c1c8971414772bcc6d`; empirical `None`; engine receipt `sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff`.

`SFT-MATH-OBL-BASE-015` - Potential infinity, exact refinement and the continuum boundary

- **Claim:** `SFT-MATH-LIMIT-CONTINUUM-002` - Potential infinity, exact refinement and the continuum boundary.
- **Forced law:** Every generated rung is a finite exact positive part with a counted successor; refinement can continue without a greatest finite depth, exact rational sequences can carry monotone bounds and convergence certificates, and no completed infinity, unbounded denominator or continuum is admitted as a proof object.
- **Unique survivor:** Potential infinity is the never-terminal generated successor process; every reached object is finite and positive, convergence is an exact nested-bound certificate, and completed infinity or continuum is outside the admitted

object language.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-ORDER-LATTICE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:b4d6ada858b868c1e51dbc259fafd1672b962526efc836f55e2d8e4958729e94`; independent `sha256:cd1fbdbe8cc78f9e9a0ef75dc32657fab1b1173c5a44cb9c1e871dfd05321f3c`; empirical `None`; engine receipt `sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098`.

`SFT-MATH-OBL-BASE-016` - Positive polynomial-balance certificates for algebraic magnitudes

- **Claim:** `SFT-MATH-ALGEBRAIC-BALANCE-002` - Positive polynomial-balance certificates for algebraic magnitudes.
- **Forced law:** A magnitude not itself admitted as an irrational value may be identified by two positive-coefficient polynomial sides, an exact rational bracket in which their order swaps, and a generated bisection trace that narrows that bracket without ever importing the root as a proof value.
- **Unique survivor:** The unique admitted algebraic-magnitude representation is a positive polynomial-balance identity with exact rational order-swap bracket and replayable bisection trace; the irrational root is never a proof value.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-ALGEBRA-001 -> SFT-MATH-ORDER-LATTICE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:5e3dbc95200564d41c0418f508c6fb35eaadfd1a0077f49a9faae4aa1eaeaf14`; independent `sha256:d309c7ff8f819486d4e24671c0e8a7587328436e35e6b0f0343f9d37b6618202`; empirical `None`; engine receipt `sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d`.

`SFT-MATH-OBL-BASE-017` - Exact recurrence of every finite componentwise Fold configuration

- **Claim:** `SFT-MATH-BOUNDED-N-BODY-002` - Exact recurrence of every finite componentwise Fold configuration.
- **Forced law:** Every supplied positive finite tuple of reduced rational Fold states has a finite exact joint support; after its transient binary depths, the componentwise Fold configuration recurs with period equal to the least common multiple of the odd-core component periods.
- **Unique survivor:** All finite componentwise Fold tuples are eventually recurrent; their recurrent joint period is the iterated least common multiple of component odd-core periods.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.

- **Root lineage:** `SFT-MATH-ORBIT-NUMBER-THEORY-002 -> SFT-MATH-EXACT-RELATIONS-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:847c6bb6921ac26739ce553890f204b1a5b39c3cd085a008554db882434fc93d`; independent `sha256:1d3d888f14db2f03b13c38a139422f63da47c5f5753dfd7483c543aec36280d3`; empirical `None`; engine receipt `sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4`.

`SFT-MATH-OBL-BASE-018` - Depth-independent discrete-gradient bound for floored Fold fluids

- **Claim:** `SFT-MATH-FLOORED-FLUID-REGULARITY-002` - Depth-independent discrete-gradient bound for floored Fold fluids.
- **Forced law:** At every supplied positive finite Fold depth k, an exact lattice with minimum spacing one-of-b^k and velocity separation at most the One has discrete gradient and vorticity magnitude bounded by b^k; at depth five the exact bound is thirty-two, so no finite-depth Fold-lattice blow-up is expressible.
- **Unique survivor:** The native Fold-lattice gradient bound is b^k at depth k, attained by a One velocity gap across one minimum edge; the depth-five value is thirty-two and every finite-depth field remains bounded.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOMETRY-TOPOLOGY-001 -> SFT-MATH-LIMIT-CONTINUUM-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:18460e9edef12ead62e6409f1a92e8cc5a6e8dec68cde2e1c44e36aa76d20ff2`; independent `sha256:49330684436f47d6548aa71275bc43bef3c45ca37528b6d574ddf24fb4d430d2`; empirical `None`; engine receipt `sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0`.

`SFT-MATH-OBL-BASE-019` - Exact bounded Goldbach and twin-prime census

- **Claim:** `SFT-MATH-PRIME-PAIR-CENSUS-002` - Exact bounded Goldbach and twin-prime census.
- **Forced law:** Exhaustive exact whole-number enumeration proves that every even whole from four through ten thousand has at least one prime complementary pair, that this range contains exactly four-thousand-nine-hundred-ninety-nine tested evens with no failure, and that exactly two-hundred-five twin-prime pairs have upper member at most ten thousand.
- **Unique survivor:** The exhaustive 4..10,000 census contains 4,999 even inputs, zero Goldbach failures and exactly 205 twin-prime pairs with upper member at most 10,000.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-COMBINATORICS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.

- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:670c35bc2174d10dd88d35ee6892f424ce2a6c039ea35b2d1c22652511398f80`; independent `sha256:ef0331a6ee13bc25a810cc2bebef35fb073d1d91c695a60e6963b02fe9b1d572`; empirical `None`; engine receipt `sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6`.

`SFT-MATH-OBL-BASE-020` - Unique half-One reflection axis and Riemann correspondence boundary

- **Claim:** `SFT-MATH-RIEMANN-MIRROR-002` - Unique half-One reflection axis and Riemann correspondence boundary.
- **Forced law:** The exact complement involution on positive parts has one and only one self-partner, the half-One; every other admitted part forms a distinct complementary pair. This forces the unique SFT reflection axis corresponding to the classical zeta functional equation's one-half symmetry, while complex zero location remains outside the no-imaginary proof language.
- **Unique survivor:** Half-One is the unique fixed axis of exact complement; every other positive part is paired. This closes the SFT Riemann-mirroring claim and explicitly does not assert classical complex zero location.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-RELATIONS-002 -> SFT-FOUNDATION-HALF-ONE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:100e6f21f2a33c3c374d80f8d96672c9351cf1678caed4084e3535e6c0b3d89a`; independent `sha256:e66aa5200f0ef07b92629cd5e0b9b18c210ab9b371a1522e100ed681e7200870`; empirical `None`; engine receipt `sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6e6ca001a24622e55364dfc4`.

`SFT-MATH-OBL-BASE-021` - Exact bounded Collatz census and contraction-control correction

- **Claim:** `SFT-MATH-COLLATZ-FINITE-CENSUS-002` - Exact bounded Collatz census and contraction-control correction.
- **Forced law:** Every positive whole start from the One through one-hundred-thousand reaches the 1-4-2 cycle under the declared Collatz transition; start twenty-seven takes exactly one-hundred-eleven steps. The exact census also rejects the prior constant-three-quarter pointwise contraction shortcut: a lawful odd macrostep may rise, so only the bounded execution result is admitted.
- **Unique survivor:** The complete starts 1..100,000 census has zero failures, start 27 takes 111 steps, the terminal cycle is 1-4-2, and the constant-3/4 pointwise contraction premise is rejected.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DYNAMICAL-SYSTEMS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values

are excluded; a completed infinity and an ungenerated continuum are excluded.

- **Certificates:** derivation `sha256:e57a27f1b95bade1301c14d5bfc05434c7c63751e8d32cf64aad2e29f8958be8`; independent `sha256:83e49654996c1fc161e8b8d3abfe7e78342759ed823d227809aa1f80e9034eda`; empirical `None`; engine receipt `sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cbd40c4f9a045`.

`SFT-MATH-OBL-BASE-022` - Exact self-similarity, chaotic rate and convergent-series laws

- **Claim:** `SFT-MATH-SELF-SIMILAR-CONVERGENCE-002` - Exact self-similarity, chaotic rate and convergent-series laws.
- **Forced law:** Fold forces binary local separation expansion, one closed distinction per step, the unique unit exponent under rank doubling, exact m^d support growth, and rational convergence certificates whose tails shrink by a generated factor without importing logarithms or irrational limit values.
- **Unique survivor:** The exact Fold rate is binary separation expansion and one closed label per step; unit rank power is the unique Fold-self-similar exponent, m-labelled support is m^d, and decreasing rational terms close convergence without importing their limit as a value.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-RELATIONS-002 -> SFT-MATH-LIMIT-CONTINUUM-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero is not an admitted value; negative magnitude is represented only by held orientation; irrational, imaginary and floating proof values are excluded; a completed infinity and an ungenerated continuum are excluded.
- **Certificates:** derivation `sha256:3b5b67ae9a24eac5e74f0b60feb92e8f66224134eb397f911876b3e37eec91d1`; independent `sha256:ace930a00105674c923aa54768e10d198ce879d8b3cf8828c92b2eb5bd27363d`; empirical `None`; engine receipt `sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fb606d5b1`.

`SFT-MATH-OBL-BASE-023` - Exact traced SFT-native scientific calculator

- **Claim:** `SFT-MATH-SCIENTIFIC-CALCULATOR-003` - Exact traced SFT-native scientific calculator.
- **Forced law:** A scientific-calculator interface is admitted in SFT only when every entered decimal is translated character-by-character to an exact rational part, displayed zero is structural empty-One, subtraction uses held orientation, all algebraic and transcendental non-rational results remain exact rational enclosures with replayable bounds, orthogonal values remain typed Fold fibres, every operation is traced, and every undefined or uncertified operation halts.
- **Unique survivor:** The unique admitted calculator semantics uses exact character-generated rational parts, structural empty-One, held orientation, generated exact kernels, certified rational enclosures, finite rational recurrences with explicit remainder bounds, typed orthogonal fibres, mandatory domain halts, complete proof/resource traces and no extra rule.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-COMBINATORICS-001 -> SFT-MATH-ALGEBRAIC-BALANCE-002 -> SFT-MATH-LIMIT-CONTINUUM-002 -> SFT-MATH-DYNAMICAL-SYSTEMS-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.

- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only.
- **Certificates:** derivation `sha256:8c6a64e9113e32d23e6bf5c08a1fad24692788433b322c7d4b55f7828f81ee18`; independent `sha256:f26d7f99675fa8d4bd035214d0817ca65309d23fe4d4bb53b71b907fa9a46b22`; empirical `None`; engine receipt `sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c`.

`SFT-MATH-OBL-BASE-024` - Exact traced SFT-native scientific calculator with corrected enclosure parity

- **Claim:** `SFT-MATH-SCIENTIFIC-CALCULATOR-004` - Exact traced SFT-native scientific calculator with corrected enclosure parity.
- **Forced law:** A scientific-calculator interface is admitted in SFT only when every entered decimal is translated character-by-character to an exact rational part, displayed zero is structural empty-One, subtraction uses held orientation, all algebraic and transcendental non-rational results remain exact rational enclosures with replayable bounds, orthogonal values remain typed Fold fibres, every operation is traced, and every undefined or uncertified operation halts.
- **Unique survivor:** The unique admitted calculator semantics uses exact character-generated rational parts, structural empty-One, held orientation, generated exact kernels, certified rational enclosures, finite rational recurrences with explicit remainder bounds, typed orthogonal fibres, mandatory domain halts, complete proof/resource traces and no extra rule.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-COMBINATORICS-001 -> SFT-MATH-ALGEBRAIC-BALANCE-002 -> SFT-MATH-LIMIT-CONTINUUM-002 -> SFT-MATH-DYNAMICAL-SYSTEMS-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only.
- **Certificates:** derivation `sha256:4fea624cffb8780f8e96d0a85eccb81c8fb609c1e712983a339a6f3f9b57d0d9`; independent `sha256:500be40f0488c5003ebf92841385cec6f50ef8405b48e0d02288c7d7726f56c9`; empirical `None`; engine receipt `sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b`.

`SFT-MATH-OBL-BASE-025` - Complete exact SFT scientific calculator and cross-platform learning app

- **Claim:** `SFT-MATH-SCIENTIFIC-CALCULATOR-005` - Complete exact SFT scientific calculator and cross-platform learning app.
- **Forced law:** The admitted exact calculator core extends uniquely to the declared mainstream scientific-calculator surface: terminal equals execution; editable expressions; exact and certified general powers, roots, logarithmic, exponential, trigonometric, inverse-trigonometric, hyperbolic, combinatorial and statistical operations; radian, degree and gradian modes; retained answer, memory and history; and one cross-platform desktop application exposing decimal projection, exact evidence, proof trace and SFT learning guidance. No interface action may bypass the same exact fail-closed evaluator.
- **Unique survivor:** The unique expanded calculator is the exact core plus the complete declared scientific-function surface, terminal equals/Ans/memory/history state, RAD/DEG/GRAD translation, and one cross-platform desktop app whose buttons and keyboard invoke only the same fail-closed evaluator. Every exact result retains its Fold type; every non-rational

result remains a certified rational enclosure; decimal output is display only.

- **Enumeration and falsification:** 8,192 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-COMBINATORICS-001 -> SFT-MATH-ALGEBRAIC-BALANCE-002 -> SFT-MATH-LIMIT-CONTINUUM-002 -> SFT-MATH-DYNAMICAL-SYSTEMS-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-MATH-SCIENTIFIC-CALCULATOR-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: semantic numerical zero, NaN and infinity are not admitted values; negative, irrational and imaginary scalars are replaced by typed Fold structures; binary floating-point and opaque host-library answers are excluded from proof; uncertified roots, series, singularities and exhausted resource bounds halt; decimal projections are correspondence displays only; no GUI button, memory action, angle mode or display conversion may bypass the exact evaluator; no host random generator, floating transcendental library, silent complex promotion or hidden calculator state; no claim to functions outside the explicit feature manifest without a new versioned extension.
- **Certificates:** derivation `sha256:2fd07cdc7dc1f857c209c3814c709f68ab21def91f3dee750f097614add757d9`; independent `sha256:1e9b190d79b01ebaa1a13c76556fd3d0ad80e9bac8e7cf65570bd0e1e07b1db2`; empirical `None`; engine receipt `sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb`.

`SFT-MATH-OBL-BASE-026` - Fully realised SFT scientific calculator and complete Mathematics law explorer

- **Claim:** `SFT-MATH-SCIENTIFIC-CALCULATOR-006` - Fully realised SFT scientific calculator and complete Mathematics law explorer.
- **Forced law:** The admitted exact calculator extends uniquely to a familiar, progressively disclosed and cross-platform application that evaluates the complete declared scientific expression language, exposes every currently registered predecessor Mathematics family, emits engine-comparable but explicitly non-admission proof output, halts at every exact value/domain/resource boundary, and is completely statement-and-branch tested.
- **Unique survivor:** The unique completion is the familiar cross-platform app whose ordinary input invokes only exact SFT runtime types; whose advanced view exposes the complete current Mathematics census and engine-comparable proof; whose large circular inputs are certified by whole-turn reduction; and whose declared active implementation records no missing statement or branch.
- **Enumeration and falsification:** 1,024 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-COMBINATORICS-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-ALGEBRA-001 -> SFT-MATH-ORDER-LATTICE-001 -> SFT-MATH-GEOMETRY-TOPOLOGY-001 -> SFT-MATH-PROBABILITY-STATISTICS-001 -> SFT-MATH-OPTIMIZATION-001 -> SFT-MATH-DYNAMICAL-SYSTEMS-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-MATH-EXACT-RELATIONS-002 -> SFT-MATH-ORBIT-NUMBER-THEORY-002 -> SFT-MATH-LIMIT-CONTINUUM-002 -> SFT-MATH-ALGEBRAIC-BALANCE-002 -> SFT-MATH-BOUNDED-N-BODY-002 -> SFT-MATH-FLOORED-FLUID-REGULARITY-002 -> SFT-MATH-PRIME-PAIR-CENSUS-002 -> SFT-MATH-RIEMANN-MIRROR-002 -> SFT-MATH-COLLATZ-FINITE-CENSUS-002 -> SFT-MATH-SELF-SIMILAR-CONVERGENCE-002 -> SFT-MATH-SCIENTIFIC-CALCULATOR-004 -> SFT-MATH-SCIENTIFIC-CALCULATOR-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.

- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no numerical zero, negative magnitude, irrational scalar, imaginary scalar or floating proof value; no fitted scientific parameter, host transcendental answer, random oracle, NaN or infinity; no GUI or terminal path may bypass the exact evaluator; no calculator replay is labelled an admission receipt; no claim to future Mathematics families not present in the exact predecessor census; no proprietary calculator-specific surface is claimed beyond the declared expression language.
- **Certificates:** derivation `sha256:e14a12e172dab6232491cb9591dc4b82928760b721e3a99aaabba68a2857b054`; independent `sha256:1a5a1a0a29eae41f6822d1f1184279361ad4058e9b1afdb83baa9089c6a3a62e`; empirical `None`; engine receipt `sha256:addeb561438c9913313824ba43eb8fceccb53c574f5a9853b3c9c1fac4e4e4b7`.

`SFT-MATH-OBL-BASE-027` - Accessible SFT-native scientific calculator application

- **Claim:** `SFT-MATH-SCIENTIFIC-CALCULATOR-007` - Accessible SFT-native scientific calculator application.
- **Forced law:** The admitted exact calculator extends to one dependency-free standards-rendered application whose familiar calculator notation never selects the mathematics: displayed zero retains empty-One semantics; negative-result attempts halt transactionally; certified intervals and orthogonal fibres remain typed exact objects; every page has an independent session; the conventional standard pad remains structurally coherent; scientific functions are progressively disclosed; and the same application is reachable on macOS, Windows, Linux and same-network phones.
- **Unique survivor:** The unique completion is a familiar calculator-first local web application with a coherent standard pad, optional scientific and proof panels, fresh per-page state, same-network default access and one immutable exact SFT evaluation route.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `independently_replicated`.
- **Root lineage:** `SFT-MATH-SCIENTIFIC-CALCULATOR-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** the preserved foundational package carries its versioned external-validation identity; no later target was imported into its historical source manifest.
- **Sources and boundaries:** registered internal exact observation corpus. Exclusions: no edit to the frozen engine or immutable claim-006 source; no browser-side arithmetic or alternate evaluator; no displayed negative, irrational scalar, imaginary scalar, NaN or infinity; no counter-held result retained in answer, memory or history; no shared landing-state contamination between page sessions; no automatic expression focus on a touch-width calculator; no heavy GUI framework, container or network service beyond the user's local network.
- **Certificates:** derivation `sha256:fdb9d7eb5203c03e24df26187742b0650bfef090233f9f4fcbd46331a29b2fa6`; independent `sha256:9576ee0eee57fe8d2ab6e7245012100ddfbbb25d4eefe3b1b3cd761952fa80fb`; empirical `None`; engine receipt `sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3e0fe5dbf0882aac9867`.

40.4 Family `ARITH` - 18/18 complete

This family contributes 4,608 completely decided candidates, 18 unique survivors and 72 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-ARITH-001` - Generated whole succession and exact induction

- **Claim:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001` - Generated whole succession and exact induction.
- **Forced law:** Every generated whole is a finite counted trace beginning at the One; adjoining one complete new unit is the unique successor and preserves exact induction.
- **Unique survivor:** ARITH-001 uniquely retains generated-one-successor-trace with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-001: successor; exact observation: [1,2,3,4,5,6,7,8]; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e5137a26ec62099fffe783b950bd43a0440b99cd63c208abd2912f18b857b4d4`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:d3291e5807001f8a01f8b19f4a09dff9ae1e18ed782d85dd690c55d2a0e007a1`; engine receipt `sha256:d3229eb70862bcc3a5d03f42c9abfc650ab90781d45fc2d5090abf8322d2ca08`.

`SFT-MATH-OBL-ARITH-002` - Addition and disjoint-junction arithmetic

- **Claim:** `SFT-MATH-ARITH-JUNCTION-ADDITION-002` - Addition and disjoint-junction arithmetic.
- **Forced law:** Addition is the complete junction of two held-disjoint generated traces, with every source unit retained exactly once.
- **Unique survivor:** ARITH-002 uniquely retains complete-disjoint-junction with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-GENERATED-SUCCESSION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-002: junction; exact observation: {"left":3,"right":5,"whole":8}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4ffd74e6676430a4c3c039d2a0c0b706ad4b810be8f486212f54bc33f85a11cc`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:b1027c7eaea0f8db1fe062f008a04b1e98ddb61fdf311717d93bdc2fa34997e9`; engine receipt `sha256:ecbb32a37fb10022315429bb39d88f49437fc856396d0b8db42fca49b2032493`.

`SFT-MATH-OBL-ARITH-003` - Multiplication and complete pair-cell arithmetic

- **Claim:** `SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003` - Multiplication and complete pair-cell arithmetic.
- **Forced law:** Multiplication is the complete product of every unit of one generated trace with every unit of the other, producing one pair cell per ordered incidence.
- **Unique survivor:** ARITH-003 uniquely retains complete-pair-cell-product with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-003: pair_cells; exact observation: {"cells":12,"columns":4,"rows":3}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f02cbeef4cb6ef18af4d1f5a8dad8709ffa696231ab05092bf60c7fc1a2c29b5`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:556adfc964a4cec9c3db61903a2bedf75c78b71812540c77fb263d8e14fd9965`; engine receipt `sha256:e25cde93469d6d813be366bc228e55cb13e704a965e60711ab396c6f7a36f476`.

`SFT-MATH-OBL-ARITH-004` - Divisibility, common divisors and common multiples

- **Claim:** `SFT-MATH-ARITH-DIVISIBILITY-GCD-LCM-004` - Divisibility, common divisors and common multiples.
- **Forced law:** Divisibility is exact complete grouping; greatest common grouping and least common return are forced by exhaustive positive divisor and multiple ledgers.
- **Unique survivor:** ARITH-004 uniquely retains complete-divisor-common-return-ledger with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-004: common_structure; exact observation: {"gcd_18_24":6,"lcm_18_24":72}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:242eaa8be1c6ad19a2e95419505bd3536e46862466de72da225e749b84cabd85`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:8752fe2d56e45cd94c015b46d99d503411bfe33f5c020c2d4f580a3cac3afb3e`; engine receipt `sha256:e02b5bc7444746ceaf5a115853d012583abae63804f6531b39727fc504424e7e`.

`SFT-MATH-OBL-ARITH-005` - Exact quotient and oriented remainder

- **Claim:** `SFT-MATH-ARITH-QUOTIENT-REMAINDER-005` - Exact quotient and oriented remainder.
- **Forced law:** Exact quotient is the maximal count of complete divisor groups, with any unmatched positive part retained as a held remainder and structural absence displayed by host 0 only.

- **Unique survivor:** ARITH-005 uniquely retains maximal-complete-groups-held-remainder with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-DIVISIBILITY-GCD-LCM-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-005: quotient_remainder; exact observation: {"part":5,"quotient":3,"remainder":2,"whole":17}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ce3cadef25ffadd3d9bb765504c1effb735fd36fb3e304bdc242195ed65c862c`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:2b13431564a6cff71b9d362331053a4d6927e12f37db2c0644927f8bd2301820`; engine receipt `sha256:4b082b2a5fd4bed326e2ebaa180ff23aa3d7bcb0475947732bedf9d65f821af6`.

`SFT-MATH-OBL-ARITH-006` - Prime and irreducible whole structure

- **Claim:** `SFT-MATH-ARITH-PRIME-IRREDUCIBLE-006` - Prime and irreducible whole structure.
- **Forced law:** A prime whole is exactly a generated whole above the One whose complete positive divisor ledger retains only the One and itself.
- **Unique survivor:** ARITH-006 uniquely retains only-one-and-self-complete-divisors with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-QUOTIENT-REMAINDER-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-006: primes_through_30; exact observation: [2,3,5,7,11,13,17,19,23,29]; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bfa9c3c6f6e17be8158753387137dc211af47e8419b959fcdc5d5841381988b1`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:cd82e1fdd851c2716987331f81e42ab87b2a6056561d30168ec8bfd94b761141`; engine receipt `sha256:d74f03c498a284dcb6a576786ce45a09c87dd921ac78db3ce71bcd54247bc018`.

`SFT-MATH-OBL-ARITH-007` - Unique finite factorization certificate

- **Claim:** `SFT-MATH-ARITH-UNIQUE-FACTORIZATION-007` - Unique finite factorization certificate.

- **Forced law:** Repeated least complete divisor extraction forces a finite prime-factor trace whose product reconstructs the original whole; any distinct trace is eliminated by the first least-divisor disagreement.
- **Unique survivor:** ARITH-007 uniquely retains least-divisor-prime-factor-trace with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-PRIME-IRREDUCIBLE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-007: factorizations_2_20; exact observation: {"10":[2,5],"11":[11],"12":[2,2,3],"13":[13],"14":[2,7],"15":[3,5],"16":[2,2,2,2],"17":[17],"18":[2,3,3],"19":[19],"2":[2],"20":[2,2,5],"3":[3],"4":[2,2],"5":[5],"6":[2,3],"7":[7],"8":[2,2,2],"9":[3,3]}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:67799fe8f79b955973a8d2eb3c26b026504328a73c03bbf72febf1a4982114b8`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:294512563c572bf4834ef96585bb0ec57a833c6d3278008adf496442d624afa6`; engine receipt `sha256:12882efe837fb967097e5f5540d48077cd12b48bb0c37662fbdd6e0052c797ac`.

`SFT-MATH-OBL-ARITH-008` - Canonical exact fractions and common refinement

- **Claim:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008` - Canonical exact fractions and common refinement.
- **Forced law:** An exact fraction is a held part of a generated whole in lowest terms; equality and arithmetic are forced by common refinement and canonical reduction.
- **Unique survivor:** ARITH-008 uniquely retains reduced-parts-common-refinement with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-UNIQUE-FACTORIZATION-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-008: fractions; exact observation: {"reduced_6_8":[3,4],"sum_1_3_1_4":[7,12]}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a9f1a0d04583152c75b6ec6276b0d5f6d9247117c76952d447c9429af9408c01`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:6527e06bb4a4287c6dfefc1a163c324e8e9bf0c6c75c3cef958ff2559842b31e`; engine receipt

sha256:45b93a4d1bb7a137599e21c74e7e9522054f6b741433365d957dd06fc703a3a1.

SFT-MATH-OBL-ARITH-009 - Finite continued-fraction correspondence

- **Claim:** SFT-MATH-ARITH-CONTINUED-FRACTION-009 - Finite continued-fraction correspondence.
- **Forced law:** Every supplied exact fraction has a finite quotient-remainder expansion; its coefficient trace reconstructs the same part without importing an irrational limit.
- **Unique survivor:** ARITH-009 uniquely retains finite-quotient-remainder-expansion with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-009: continued_fraction_355_113; exact observation: [3,7,16]; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2fba324171a5a5fa66783899a7e8f4aa2c7c8f975c4fb78cf2d0a7cf4f009cbb`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:194553afefe2203bc3a75c8b7615df4223999d6d54b16489ec07377efbd7ab98`; engine receipt `sha256:b18a07076f76362a892ce1ed995e8b67e1af5c41ecdde90b16ea34a35dfa886d`.

SFT-MATH-OBL-ARITH-010 - Residue classes and congruence

- **Claim:** SFT-MATH-ARITH-CONGRUENCE-010 - Residue classes and congruence.
- **Forced law:** Two wholes are congruent at a generated modulus exactly when complete grouping leaves the same held remainder class.
- **Unique survivor:** ARITH-010 uniquely retains same-held-remainder-class with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-CONTINUED-FRACTION-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-010: residue_mod_5; exact observation: {"17":2,"18":3,"2":2}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:f743ef14d4215122f5f65c2c32d2439647a222d44ba6a74ce336c5b4351144f2`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:7c0e40454c392c3b91ffac9cc4a59805a224b0151b779e6fdbf72260567110fb`; engine receipt `sha256:dfc025633b4bc068fab1d51dd8596eaf531d428a8b8d83d5c49b015b1311b62a`.

`SFT-MATH-OBL-ARITH-011` - Compatible congruence composition

- **Claim:** `SFT-MATH-ARITH-COMPATIBLE-CONGRUENCE-011` - Compatible congruence composition.
- **Forced law:** A compatible finite family of congruence records forces one residue class over the least common return; incompatibility remains an explicit halt.
- **Unique survivor:** ARITH-011 uniquely retains least-common-period-compatible-residue with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-CONGRUENCE-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-011: compatible_congruence; exact observation: [23]; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3c65c79f55fbecb36aba96a0a6dcec3020784315b34fb95ce45669a00785d233`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:115e6a0305cc20f054c011aed21ff8302c1e15c38e107f1f5481956ec180ac88`; engine receipt `sha256:f15dd91147438a821fcedb1cd0abcab032151e1393fea1766f02d97fe5d4d64f`.

`SFT-MATH-OBL-ARITH-012` - Prime-power valuation and divisibility depth

- **Claim:** `SFT-MATH-ARITH-VALUATION-012` - Prime-power valuation and divisibility depth.
- **Forced law:** Prime-power valuation is the exact number of successive complete prime groupings before divisibility ceases, with no logarithm or continuum premise.
- **Unique survivor:** ARITH-012 uniquely retains counted-prime-factor-depth with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-COMPATIBLE-CONGRUENCE-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-012: valuations; exact observation: {"v2_40":3,"v3_81":4}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity,

ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:81a14c2fca1602fdae4a8b7ae95d639812887b471d16ec4b28a633bb4bbe21a2`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:842d1d86b7fead857e5744795de55a71f7237a77e6285c2e5ce26359f07925d0`; engine receipt `sha256:3f73120fde18553beedeec0de54e9ee4834b9eb9ebe19976ddb38b251d1bffac`.

`SFT-MATH-OBL-ARITH-013` - Diophantine relation enumeration

- **Claim:** `SFT-MATH-ARITH-DIOPHANTINE-ENUMERATION-013` - Diophantine relation enumeration.
- **Forced law:** A Diophantine relation is closed only by a complete declared positive-whole candidate census, exact substitution and retention of every satisfying tuple.
- **Unique survivor:** ARITH-013 uniquely retains complete-positive-whole-solution-census with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-VALUATION-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-013: diophantine; exact observation: {"positive_x_plus_y_8":7,"three_four_five":true}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:21f1cd588434c90fbc4b854759d3fcda183147d8d61b823eb259b94a15b0c38a`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:a4db244f6a7fd7cdc721b332be3b800ff720a5dabd2756a0479cf639c85f7bdf`; engine receipt `sha256:2a078b638cd00e98a59fb4ccd6a47e9b691c4a713710bdbb4112062441c41085`.

`SFT-MATH-OBL-ARITH-014` - Recurrence laws and exact sequences

- **Claim:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014` - Recurrence laws and exact sequences.
- **Forced law:** An exact sequence is forced by a complete initial record plus one generated recurrence rule; each next term retains the entire prior dependency trace.
- **Unique survivor:** ARITH-014 uniquely retains initial-record-rule-successor-sequence with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-DIOPHANTINE-ENUMERATION-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-014: recurrence; exact observation: [1,1,2,3,5,8,13,21,34,55]; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:63cbd919dd71af239017beb4af757a841c56b69ba63a7e6c7b71969eddf83f77`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:730e7e284286b7ca41cc300415321a660c034d7b90dbbce8c01047f6993fcef0`; engine receipt `sha256:a8a9c91bf09055eafe7c8946783af33d8cbf408241c7a0cc65b022fc8a85393d`.

`SFT-MATH-OBL-ARITH-015` - Finite generating-function correspondence

- **Claim:** `SFT-MATH-ARITH-GENERATING-FUNCTION-015` - Finite generating-function correspondence.
- **Forced law:** A generating expression is an exact finite support ledger whose product coefficients count complete compositional incidences; no infinite formal sum is admitted as an object.
- **Unique survivor:** ARITH-015 uniquely retains coefficient-support-composition with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-015: geometric_pair_coefficients; exact observation: [1,2,3,4,5,6,7,8]; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:b2e94f4ac6398ab601e78a9d206a58a9a7e495f9dc55400618f0f2cbee87a7e1`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:b244016840493249a89133375749ba8318ac1fdc5164e48bf8e3252136fe2c8f`; engine receipt `sha256:18143d49ddbc35dfaa29e6333b0529d9c93c269483c06206f79a2bd465d2bbfb`.

`SFT-MATH-OBL-ARITH-016` - Whole partitions and ordered compositions

- **Claim:** `SFT-MATH-ARITH-PARTITION-COMPOSITION-016` - Whole partitions and ordered compositions.
- **Forced law:** Partitions and compositions are complete positive-part decompositions of one generated whole, distinguished respectively by erased or retained order.
- **Unique survivor:** ARITH-016 uniquely retains complete-partition-composition-census with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-GENERATING-FUNCTION-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** ARITH-016: decompositions_8; exact observation: {"compositions":128,"partitions":22}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:48a46a0a2357fbff856317659c458537081533476278 32c2b04d96127b3521cf`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:31a2e8fa29dba26f037c7e336d34f178ed3a6c1bbe3458c6f0b7cb16ac2e2f83`; engine receipt `sha256:322408091620c6dfed8b960578f5744a202f7f05536bc0b348f9e62cccb1b9fb`.

`SFT-MATH-OBL-ARITH-017` - Arithmetic functions and divisor ledgers

- **Claim:** `SFT-MATH-ARITH-ARITHMETIC-FUNCTIONS-017` - Arithmetic functions and divisor ledgers.

- **Forced law:** Divisor count, divisor accumulation and coprime count are exact functions of the complete positive divisor and residue ledgers.

- **Unique survivor:** ARITH-017 uniquely retains complete-divisor-derived-functions with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-PARTITION-COMPOSITION-016`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** ARITH-017: arithmetic_functions_12; exact observation: {"phi":4,"sigma":28,"tau":6}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:51e10a837a5c3e859bc1a4f82f9b8d40fd00005900b96bd4579c01491ef20de7`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:0e24b98f71a5878df7b28bf9523fae42127e7fa3aa8af145ed8705bffbe15798`; engine receipt `sha256:a0126b838d3f636788373c0a95adc8ff48e8a9d9b1d7c0797aae6403fda8473e`.

`SFT-MATH-OBL-ARITH-018` - Finite prime-distribution and growth enclosures

- **Claim:** `SFT-MATH-ARITH-PRIME-DISTRIBUTION-ENCLOSURE-018` - Finite prime-distribution and growth enclosures.

- **Forced law:** Prime distribution is admitted as exact finite censuses and proved successor enclosures at each generated bound; no unsupported completed asymptotic object enters.

- **Unique survivor:** ARITH-018 uniquely retains finite-prime-census-successor-enclosures with canonical identity, complete enumeration, root-bound forcing, post-registry exact observation, finite-successor closure and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-EXACT-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-ARITH-ARITHMETIC-FUNCTIONS-017`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ARITH-018: prime_enclosures; exact observation: {"between_n_2n_through_50":true,"prime_count_100":25}; complete family observation custody: 18 records; all rows preserved; structural absence display and artifact counters remain outside admitted proof values
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ARITHMETIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no semantic numerical zero, negative magnitude, irrational, imaginary or floating proof scalar; structural absence is empty One; host 0 may only display absence or count artifacts; no completed infinity, ungenerated continuum or infinite formal series is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ee28441ff7a60f8dfb90d921b866cc07ec6475a946bfc5e688adff9f993964f6`; independent `sha256:3ce99c3aec4dd29e95f3520ab1b94246996525755a977bd1dbdf03c1a2a4ca0f`; empirical `sha256:f533f200dab5bde6d5c8f1e83c9dca7e4abdd9a67de85cfe52e3610f8c9e3985`; engine receipt `sha256:2d9cb180af9554925c3ff156e2b05d1545b9dba2206a24b60c0e17023e0e9fe2`.

40.5 Family `ALEXT` - 10/10 complete

This family contributes 2,560 completely decided candidates, 10 unique survivors and 40 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-ALEXT-001` - Polynomial identity and exact root isolation

- **Claim:** `SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001` - Polynomial identity and exact root isolation.
- **Forced law:** A polynomial root is admitted through an exact defining balance and nested rational brackets whose endpoint order swaps, never as an imported irrational scalar.
- **Unique survivor:** ALEXT-001 uniquely retains positive-polynomial-side-order-swap, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-001: square_root_two_bracket; exact observation: {"lower":[7,5],"upper":[3,2]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3bd3c47022683ec450823b0288cc689829085d70394a7503edb2e3858f531162`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:166fd480bad9c76fe70038e0f62717850cc5ae98eb39c2007b8bda15ca5fb691`; engine receipt `sha256:37fd63da2da1d8bff586bea60bebad91ef43d0115437e9efd7e360dbf05b94f0`.

`SFT-MATH-OBL-ALEXT-002` - Algebraic-magnitude balance certificates

- **Claim:** `SFT-MATH-ALEXT-ALGEBRAIC-BALANCE-002` - Algebraic-magnitude balance certificates.
- **Forced law:** An algebraic magnitude is the exact equality condition between two positive polynomial sides plus a replayable rational enclosure certificate.

- **Unique survivor:** ALEXT-002 uniquely retains exact-algebraic-balance-enclosure, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-002: cube_root_two_bracket; exact observation: {"lower":[5,4],"upper":[4,3]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:abab55524337c48c1be35aac53cfe92ea32362af118d7ada40faf510f0f0917b`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:7453b8deea57bf6af44addcaa61eb8578d5550176ce56a42ce8eacd12601c5fa`; engine receipt `sha256:41100b9b02e60bc1321f4ffa7590f3f2c5ef6223f3983e323a4177d53bde4a4e`.

`SFT-MATH-OBL-ALEXT-003` - Exact finite extension towers

- **Claim:** `SFT-MATH-ALEXT-EXTENSION-TOWER-003` - Exact finite extension towers.
- **Forced law:** A finite extension is a generated basis-label carrier with exact reduction identities and complete coefficient custody at every composition.
- **Unique survivor:** ALEXT-003 uniquely retains finite-basis-reduction-tower, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-ALGEBRAIC-BALANCE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-003: beta_square_two_product; exact observation: {"left":[1,1],"result":[3,2],"right":[1,1]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3a8a62cbcb132f9b8460b3c89d6e6b517480e6c76707de7bbd150a6d250ae03a`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:c7487379e9500b2adacaca4dc332e7e914201856bb58d3b5f10c62d3db9d0cb8`; engine receipt `sha256:b4db0d57fd93d9489c8d94fe9bae1167b9ffa29c775db04f909dc9c707cdb922`.

`SFT-MATH-OBL-ALEXT-004` - Finite-field correspondence

- **Claim:** `SFT-MATH-ALEXT-FINITE-FIELD-004` - Finite-field correspondence.
- **Forced law:** A finite-field correspondence requires complete addition and multiplication tables, one identity, exact closure and one inverse for every nonabsence label.

- **Unique survivor:** ALEXT-004 uniquely retains finite-label-field-table, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-EXTENSION-TOWER-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-004: field_mod_five; exact observation: {"labels":5,"nonabsence_inverses":[1,3,2,4]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f2d21fa1e9e0b6f98ca6c955dfcc96a4897471f4b30bb562aeeff0a85783138b`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:e537681880946b9983cc42e684f3bbf56e23cf52887b36dd2efcd1e8feca4ada`; engine receipt `sha256:3e417f42c092e58b78ceda9477a4d4c3303b8a1cb0896c6d3623f387edc24824`.

`SFT-MATH-OBL-ALEXT-005` - Finite Galois-orbit correspondence

- **Claim:** `SFT-MATH-ALEXT-GALOIS-ORBIT-005` - Finite Galois-orbit correspondence.
- **Forced law:** A finite Galois correspondence is the complete orbit of exact root labels under structure-preserving finite permutations, not an imported theorem name.
- **Unique survivor:** ALEXT-005 uniquely retains root-label-automorphism-orbit, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-FINITE-FIELD-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-005: finite_root_orbit; exact observation: {"labels":[1,2,4],"square_action":[1,4,2]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:865352dc86625e7642c37fbec07c1321954afd4b352342b6aefbc4770cfa4fc2`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:aabc680b24f95f944df0997efc23ef0482fe779c8b4d26796a60b5f3236b9d4c`; engine receipt `sha256:2dff8bb2232a8572a0c20e34c05fc873f83f2e9cb74d008cc32b7b8ff2d3d26b`.

`SFT-MATH-OBL-ALEXT-006` - Cyclotomic and root-of-unity correspondence

- **Claim:** `SFT-MATH-ALEXT-CYCLOTOMIC-CORRESPONDENCE-006` - Cyclotomic and root-of-unity correspondence.
- **Forced law:** Roots of unity are represented by exact periodic phase labels whose generated action returns after the counted period; no imaginary scalar is admitted.

- **Unique survivor:** ALEXT-006 uniquely retains periodic-phase-label-cycle, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-GALOIS-ORBIT-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-006: five_phase_cycle; exact observation: [1,2,3,4,0]; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:336b4c8409c8ace2569f6a6d2f6a26896b4fddf2744a3f8354cc6c00e9955e8a`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:2457a1efde05513902286a69eaf3e9b2015897a230ea11646dd3bd382e27e60e`; engine receipt `sha256:81f843b80e1f07e44b1f212eeefd448f604aef89255cfe08dd26ff5fd8c5c253`.

`SFT-MATH-OBL-ALEXT-007` - Held-pair complex-number correspondence

- **Claim:** `SFT-MATH-ALEXT-HELD-PAIR-COMPLEX-007` - Held-pair complex-number correspondence.
- **Forced law:** Complex correspondence is an ordered pair of exact magnitudes with held orthogonal orientation and exact pair composition; structural absence replaces numerical zero.
- **Unique survivor:** ALEXT-007 uniquely retains held-orthogonal-pair-arithmetic, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-CYCLOTOMIC-CORRESPONDENCE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-007: held_pair_product; exact observation: {"imaginary_orientation_magnitude":2,"left":[1,1],"real":"structural-absence","right":[1,1]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:8679b9f81cc22074cbd29db0fe6377cc5d2120a93f966f17724c4ed564b3ed7e`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:3c3c4577ead2d585bd95ce8ecb0570fc8a9ad2d3f2a83d4417ad9a0b047015eb`; engine receipt `sha256:ccaaa6636f523d2e6bc6cd13996d49a0c748dbc4378568188f6e307de4bfc48d`.

`SFT-MATH-OBL-ALEXT-008` - Exact real-algebraic ordering

- **Claim:** `SFT-MATH-ALEXT-REAL-ALGEBRAIC-ORDER-008` - Exact real-algebraic ordering.

- **Forced law:** Real-algebraic order is forced by disjoint exact rational enclosures and polynomial-side comparisons, without evaluating an irrational proof scalar.
- **Unique survivor:** ALEXT-008 uniquely retains disjoint-rational-enclosure-order, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-HELD-PAIR-COMPLEX-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-008: algebraic_order; exact observation: {"cube_root_two_upper":[4,3],"disjoint":true,"square_root_two_lower":[7,5]}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:01957d2b92163d4c8796addce43e30ff4156cd23f9624accb5db5a521c6532fe`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:85e0a543862759662f1b4d73622cde3be4f617b525fb886e483bf31ef3f02311`; engine receipt `sha256:1492db9e71b8fbd66da325cd1c27997d144df3e1674818eda01f3b9e11547d2b`.

`SFT-MATH-OBL-ALEXT-009` - Prime-adic valuation correspondence

- **Claim:** `SFT-MATH-ALEXT-PRIME-ADIC-VALUATION-009` - Prime-adic valuation correspondence.
- **Forced law:** Prime-adic closeness is the counted prime-power divisibility depth of a held difference, not a negative, infinite or continuum-valued distance.
- **Unique survivor:** ALEXT-009 uniquely retains prime-power-closeness-depth, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-REAL-ALGEBRAIC-ORDER-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-009: ternary_valuation; exact observation: {"depth":4,"held_difference":81}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a317a46521b2221cd956eda2c542824e6074e9f77be587f9eae5999cdbf83a7a`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:b8feb2741aba195ced03be9c30380e55eeba5e1862ccb9b99e8a71e3733b4b69`; engine receipt `sha256:c38b6cbec7e61e8fbfa7be7ca0a1ab4eb3af572576521a37c9cf03acde3bb413`.

`SFT-MATH-OBL-ALEXT-010` - Transcendental and nonrepresentability boundary

- **Claim:** `SFT-MATH-ALEXT-TRANSCENDENTAL-BOUNDARY-010` - Transcendental and nonrepresentability boundary.
- **Forced law:** When no registered finite rational or algebraic certificate exists, the value remains an explicit unrepresented boundary; finite search never licenses an unsupported transcendence claim.
- **Unique survivor:** ALEXT-010 uniquely retains explicit-unrepresented-scalar-boundary, exact identity, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-ARITH-VALUATION-012 -> SFT-MATH-ALEXT-PRIME-ADIC-VALUATION-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALEXT-010: nonrepresentability_boundary; exact observation: {"exact_match_present":false,"rational_square_search_denominator_bound":20,"transcendence_claimed":false}; complete family observation custody: 10 records; all rows preserved; irrational and imaginary scalars remain outside admitted proof values; exact structures and enclosures are retained
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRAIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; no irrational or imaginary object enters as a proof scalar; negative magnitude is held orientation and structural absence is typed empty One; no completed infinite extension, continuum or unbounded field is admitted; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:43099af77c8b4467b77a71bda5c0d7ff2b85ca61013528e8e178708d920ddd2e`; independent `sha256:e0b858bd749c311a8878f32a66dec94a8545ff67ec81619032b95dce4e7ffee0`; empirical `sha256:349dc2fb17d85705868c581df4f90e545c4a2cb52e9956da64f9c4e96d255bf8`; engine receipt `sha256:674c5e710359cd7da9382006ad39c9539591fc9f0ba8d7bd9728ca3eb804a010`.

40.6 Family `COMB` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-COMB-001` - Product, sum and bijection counting laws

- **Claim:** `SFT-MATH-COMB-COUNTING-LAWS-001` - Product, sum and bijection counting laws.
- **Forced law:** Finite counting is forced by complete disjoint junction, complete pair incidence and exact reversible pairing; no object is omitted or counted twice.
- **Unique survivor:** COMB-001 uniquely retains disjoint-sum-product-bijection-count, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-001: sum_product; exact observation: {"product":12,"sum":7}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space;

no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:d8804c8b8d02d716adfd440b178c87c38e745288cb15638ff2da7950eea8cd23`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:c7cfd31bcaedf3f490350f678282b8d3f233cc5772f97cc35f6e336e50dfbd9e`; engine receipt `sha256:afca8ab23f33cbad39e5aa6e51d5df24bfddd946cfa5568f46e6c781b5503d4f`.

`SFT-MATH-OBL-COMB-002` - Permutation and combination enumeration

- **Claim:** `SFT-MATH-COMB-PERMUTATION-COMBINATION-002` - Permutation and combination enumeration.
- **Forced law:** Permutations retain every order distinction while combinations quotient only order; both require complete generated selection censuses.
- **Unique survivor:** COMB-002 uniquely retains ordered-and-unordered-selection-census, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-COUNTING-LAWS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-002: permutation_combination; exact observation: {"combinations_5_2":10,"permutations_5":120}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a948c4fc828429807eaf8b84f56b598ff887bfb613caf648b9f8bbe43c5333c5`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:bb7cd6c1af3fc9f1be1eb4ca4c3a9a23cdaee1f5cb1000d27961e5921d8ed4c0`; engine receipt `sha256:9b95112577a4816525ad072bf79acf3c8a573c98847872d16c79f5161d11625a`.

`SFT-MATH-OBL-COMB-003` - Inclusion-exclusion with complete overlap custody

- **Claim:** `SFT-MATH-COMB-INCLUSION-EXCLUSION-003` - Inclusion-exclusion with complete overlap custody.
- **Forced law:** Union count is the complete support ledger with each overlap retained and corrected exactly once at every finite depth.
- **Unique survivor:** COMB-003 uniquely retains overlap-corrected-support-ledger, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-PERMUTATION-COMBINATION-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-003: inclusion_exclusion; exact observation: {"left":6,"overlap":2,"right":4,"union":8}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f84451c1b4f9347d6ba1ecd8ddd95b7d9b0561f1f0280b00f04e59ef21c4f790`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:3332737c4e62c5a6b89b190bb6df658719c916214c1bb5705f77ec8658b3acac`; engine receipt `sha256:5d43c44b342b930315faa5159e3ab7ec026fc538d9b3f77bdd56ff7826ce098e`.

`SFT-MATH-OBL-COMB-004` - Pigeonhole and occupancy forcing

- **Claim:** `SFT-MATH-COMB-PIGEONHOLE-OCCUPANCY-004` - Pigeonhole and occupancy forcing.
- **Forced law:** When more generated carriers occupy fewer boxes, complete enumeration forces a box whose occupancy reaches the exact quotient ceiling correspondence.
- **Unique survivor:** COMB-004 uniquely retains complete-occupancy-lower-bound, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-INCLUSION-EXCLUSION-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-004: occupancy; exact observation: {"boxes":3,"forced_minimum_maximum":3,"objects":7}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f15b3d7b997a61e4188e1032cbe3212746d2a1de728829539e57e7333f6a8439`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:60ce883be389759d13178b1d0c8b05ccfded798646c22a18fd781331fab97ca0`; engine receipt `sha256:c40a635564aae5ef85402c7b69e06ec78ed0024112d2084b58fcc0a2ccf7e004`.

`SFT-MATH-OBL-COMB-005` - Recurrence and generating-function counting

- **Claim:** `SFT-MATH-COMB-RECURRENCE-GENERATING-005` - Recurrence and generating-function counting.
- **Forced law:** A combinatorial recurrence is a disjoint compositional decomposition whose successor counts exactly reconstruct the next generated family.
- **Unique survivor:** COMB-005 uniquely retains compositional-count-recurrence, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-PIGEONHOLE-OCCUPANCY-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`;

axioms `0`; free parameters `0`.

- **Post-registry observations:** COMB-005: binary_recurrence; exact observation: [2,4,8,16,32,64,128,256]; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5c2ad967cb2ae85ece7ed7e905379b2beb4194b571de18ae1f4d0bdbbc3d063a`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:ced245e087bb65cc89b2a4cb8799a615635ced18d08a7d2db0d2716550e11383`; engine receipt `sha256:ba8380cefcf6fd4b157b57674588bacdf9c5565bc3e848f3638882df1bcc3a1f`.

`SFT-MATH-OBL-COMB-006` - Integer partitions and Young-type incidence

- **Claim:** `SFT-MATH-COMB-PARTITION-INCIDENCE-006` - Integer partitions and Young-type incidence.
- **Forced law:** An integer partition is a complete unordered positive-part decomposition with every cell and incidence retained.
- **Unique survivor:** COMB-006 uniquely retains unordered-positive-part-incidence, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-RECURRENCE-GENERATING-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-006: partitions_five; exact observation: {"cell_count":5,"count":7}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c3acbdfd5e41e2171fb5ab37408defd78183f6d33cdf87a92eab6257ef6c69b4`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:3f2b44350fea9286d4d257f2c67dde35922cde4739eecb96f35035ab78595d7a`; engine receipt `sha256:2d097fe60b69c55a86accb8d417c083c3bdbe07041e0eb058081887689943664`.

`SFT-MATH-OBL-COMB-007` - Extremal finite-set systems

- **Claim:** `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007` - Extremal finite-set systems.
- **Forced law:** An extremal set result requires exhaustive feasible-family generation, exact constraint testing and retention of every maximizing witness.
- **Unique survivor:** COMB-007 uniquely retains complete-feasible-family-extremum, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-PARTITION-INCIDENCE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-007: antichain_four; exact observation: {"maximum":6,"middle_layer":6}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a192869192fb3a1ed54b9cfda1fc325f2785a5348ab3866e95aed13b2fa1285e`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:5df089c3c43e0518edc953fdcfccff2e2892f3ec5101f69e854e6ab0662f020c`; engine receipt `sha256:e01c8059812a3e920d2f722ef9f47e2e3375ae255da534d20c5efcab589e17c2`.

`SFT-MATH-OBL-COMB-008` - Probabilistic-method correspondence without ontic randomness

- **Claim:** `SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008` - Probabilistic-method correspondence without ontic randomness.
- **Forced law:** An average over a complete deterministic support forces existence of a member at least as large as the exact average; no ontic randomness is required.
- **Unique survivor:** COMB-008 uniquely retains complete-support-average-existence, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-008: deterministic_support_average; exact observation: {"average":[3,2],"existence_at_least":2,"support":8,"total_held_labels":12}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:803050b5cc73b64b0a53fb149db319a917420b1b649787be5f1dc086b2978cbe`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:5e7f26ab0153b28366e287c07b76cc49e153f144cc771c1c07c2068856d162c7`; engine receipt `sha256:d650fb44b26a7778b1431e877854dbcbaed78f8612c57d47d6e80118fc5f37d6`.

`SFT-MATH-OBL-COMB-009` - Design, block and incidence structures

- **Claim:** `SFT-MATH-COMB-DESIGN-INCIDENCE-009` - Design, block and incidence structures.

- **Forced law:** A finite design is a complete block-incidence carrier whose point and subset multiplicities satisfy one exact balanced ledger.
- **Unique survivor:** COMB-009 uniquely retains balanced-complete-incidence-design, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-009: fano_incidence; exact observation: {"block_size":3,"blocks":7,"pair_degree":1,"point_degree":3,"points":7}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bfe1086309fada9224fa6cbe2cb9702f7de1c502e55dff3ffd6e4fd7e44a3eb1`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:6dcb8817650fa6725e231f09288b06e82d8ffcd7bf179d9863513894e92f799c`; engine receipt `sha256:d1e03a25a63c3099f5d2a8f414ad7815c8471db3b30581e48a08d58a39dcf982`.

`SFT-MATH-OBL-COMB-010` - Coding and packing combinatorics

- **Claim:** `SFT-MATH-COMB-CODING-PACKING-010` - Coding and packing combinatorics.
- **Forced law:** A finite code bound is the exact maximum of the complete word-subset census under the registered distinguishability distance.
- **Unique survivor:** COMB-010 uniquely retains complete-distance-packing-census, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-DESIGN-INCIDENCE-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** COMB-010: binary_code; exact observation: {"maximum_size":2,"minimum_distance":3,"word_length":3}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:acb913996b198ea4a667443568563d841a0306a98207d9ab79c6cc6b8bfd1fce`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical

sha256:cf362ef2d305105021a6385196c0fd5c29b2d93db1a4579602936822971ad755; engine receipt sha256:b5ede9884b69fb563c6774eacd7bd3df38cd803bb164cb775e000e7d1cb8c2ac.

SFT-MATH-OBL-COMB-011 - Ramsey-type finite forcing boundaries

- **Claim:** SFT-MATH-COMB-RAMSEY-FORCING-011 - Ramsey-type finite forcing boundaries.
- **Forced law:** A Ramsey boundary is closed only when every colouring in the declared finite census contains the registered monochromatic substructure and the preceding boundary has a counterexample where claimed.
- **Unique survivor:** COMB-011 uniquely retains complete-colouring-forced-substructure, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-CODING-PACKING-010; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** COMB-011: ramsey_six; exact observation: {"edge_colourings":32768,"monochromatic_triangle_forced":true}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:1c27db22f7f053a18874e51e7ee493d2ad5a88177ab27988b999ce9c1af23d82; independent sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6; empirical sha256:1ebffece285129cd85bbd56109aa25a2fb87ccd2d2d71d9c67e7eeec5180ec34; engine receipt sha256:2a8c17b72cdefc363e819e2019edbffaecf13d12ea569b64b0bee4dfa8f01cf2.

SFT-MATH-OBL-COMB-012 - Species and compositional enumeration

- **Claim:** SFT-MATH-COMB-SPECIES-COMPOSITION-012 - Species and compositional enumeration.
- **Forced law:** A combinatorial species is the complete set of structures transported by label bijections, with composition retaining every component and symmetry quotient.
- **Unique survivor:** COMB-012 uniquely retains structure-species-composition-census, declared symmetry, complete enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ARITH-GENERATED-SUCCESSION-001 -> SFT-MATH-ARITH-JUNCTION-ADDITION-002 -> SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003 -> SFT-MATH-COMB-RAMSEY-FORCING-011; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** COMB-012: set_partition_species_four; exact observation: {"structures":15}; complete family observation custody: 12 records; all rows preserved; complete deterministic enumeration supplies the observation; no ontic-random premise or fitted count enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMBINATORICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target count, imported theorem answer, fitted parameter or probabilistic oracle selects the law; host 0 only displays absence or counts artifacts; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite family or continuum sample space; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:210239200288cd66ef188e2d64ebdd85ed3322b3adf981f87a7997a496187c77`; independent `sha256:948188852db8153c34954a43aa68c6c3424f72a8d52bee5d2699fe75ac3bf1c6`; empirical `sha256:d58cf8390986a10d7e5455ce57b64dda0bd19f873009d87aa51ae81c890f6210`; engine receipt `sha256:c7ab86cf76ff13e5c47488f640d8d34afa30fb2008fc16e4a273562b3f4e2c98`.

40.7 Family `GRAPH` - 14/14 complete

This family contributes 3,584 completely decided candidates, 14 unique survivors and 56 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-GRAPH-001` - Graph identity, adjacency and isomorphism

- **Claim:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001` - Graph identity, adjacency and isomorphism.
- **Forced law:** A graph is the exact carrier-incidence Fold structure; isomorphism is a reversible carrier relabelling that preserves every adjacency and non-adjacency distinction.
- **Unique survivor:** GRAPH-001 uniquely retains exact-adjacency-bijection-isomorphism, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-001: graph_identity_isomorphism; exact observation: {"edges":4,"isomorphic_relabellings":8,"vertices":4}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:72ff6381941027ad0ef2a51d7acb40351acc6d404cafd311fa2553fb47cfee26`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:9840f74199987617039af73e2310ee58ed75b581b5b74327097f8ba4661f1c97`; engine receipt `sha256:489635c31e29a277498395071437690e60c00fd3287af337754ed86cd2d1c9a8`.

`SFT-MATH-OBL-GRAPH-002` - Paths, walks, reachability and cycles

- **Claim:** `SFT-MATH-GRAPH-PATH-REACHABILITY-CYCLE-002` - Paths, walks, reachability and cycles.
- **Forced law:** Paths are composable adjacent distinctions, reachability is their finite closure, and a cycle is a nonempty closed path with retained intermediate carriers.
- **Unique survivor:** GRAPH-002 uniquely retains generated-incidence-path-closure, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-002: path_reachability_cycle; exact observation: {"least_cycle_length":4,"reachable_order":[1,2,3,4]}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4da989b3f4dc9fae1e5e31a0e88f7358ca5527adfcb58fb38e9c52d51ea8d3ec`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:56cf52b7395fb9d6d3bffc0b45a7d1b7ba94cf819639f9da631f77d3a533ec3e`; engine receipt `sha256:164d8d5708f6238d6c7864f2b2261dea4a4097d12c335f68d56765cea778cc01`.

`SFT-MATH-OBL-GRAPH-003` - Connectivity, cuts and flow support

- **Claim:** `SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003` - Connectivity, cuts and flow support.
- **Forced law:** Connection, separation and flow are forced by the same complete incidence ledger: every transported unit crosses every separating cut exactly once.
- **Unique survivor:** GRAPH-003 uniquely retains cut-flow-dual-custody, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-PATH-REACHABILITY-CYCLE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-003: connectivity_cut_flow; exact observation: {"maximum_integral_flow":3,"minimum_cut_capacity":3}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:13e49cf9371e8afacd373f9ed1091429ddb3503ccc9b44e8e53155fd56085279`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:8d81978bc666c0e0b2c7896042f546193be806344bf60ae9dfe76a564ce3f688`; engine receipt `sha256:383274c9576c7f228b386a5eb969fb995d051ca71d963c8068014712304be26c`.

`SFT-MATH-OBL-GRAPH-004` - Trees, forests and spanning structure

- **Claim:** `SFT-MATH-GRAPH-TREE-FOREST-SPANNING-004` - Trees, forests and spanning structure.
- **Forced law:** A tree is the unique conjunction of connected carrier support and cycle absence; forests and spanning structures retain the same incidence boundary componentwise.
- **Unique survivor:** GRAPH-004 uniquely retains acyclic-connected-spanning-incidence, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-004: tree_forest_spanning; exact observation: {"complete_graph_vertices":4,"edges_per_tree":3,"spanning_trees":16}; complete family observation custody: 14 records;

all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:287777e7449516de3013535504d80a29638f42d81b29c2ec0d797149d183d7dc`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:78e17ec40a5ccca26b63d799109230f4f55377ab3ccdb9498e18580e62cd17c2`; engine receipt `sha256:f81889343d13708d3cc085df21f33b4636619dea455cbf58cd4df9abada48e57`.

`SFT-MATH-OBL-GRAPH-005` - Planarity and embedding correspondence

- **Claim:** `SFT-MATH-GRAPH-PLANARITY-EMBEDDING-005` - Planarity and embedding correspondence.

- **Forced law:** A finite embedding retains cyclic incidence and face custody; planarity is admitted only with an exact embedding witness or an exact obstruction at the declared boundary.

- **Unique survivor:** GRAPH-005 uniquely retains face-edge-vertex-embedding-custody, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-TREE-FOREST-SPANNING-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** GRAPH-005: planarity_embedding; exact observation: {"euler_total":2,"k4_edges":6,"k4_faces":4,"k4_vertices":4,"k5_edge_bound_exceeded":true}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:3e0e4217fe0ba244b29bb431baac87bd1983b7a230fd4ee50c74d782cb1857b6`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:5305ac98dbf2c285a0535e923e8d678bc4d84932191c3d520cbf9b36adce571a`; engine receipt `sha256:9b310cbbfe57e1eca969de9a2e73c7136ce86428d741a10cff9e7e1170fa24c4`.

`SFT-MATH-OBL-GRAPH-006` - Colouring and constraint partitions

- **Claim:** `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` - Colouring and constraint partitions.

- **Forced law:** A colouring is an exact label partition whose adjacent carriers remain distinguishable; the chromatic boundary is the least complete label support that survives every constraint.

- **Unique survivor:** GRAPH-006 uniquely retains least-lawful-distinction-partition, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-PLANARITY-EMBEDDING-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** GRAPH-006: colouring_constraints; exact observation: {"chromatic_number":3,"cycle_vertices":5}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:98ebbcf6456abfb3a6dcce68736b71ab67c1e7a24c928b11446f90399f0df394`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:f0dee66e8a412527916a6b16c998f36daa9bef0f88d38e439a70ddebe711a236`; engine receipt `sha256:361fb44c8f989543dacde61ce4b3d03f6f6096ab9fba773664b339627f275671`.

`SFT-MATH-OBL-GRAPH-007` - Matching, covering and packing

- **Claim:** `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` - Matching, covering and packing.
- **Forced law:** Matching packs pairwise disjoint incidences while covering touches every incidence; exact extrema require complete candidate enumeration and retained witnesses.
- **Unique survivor:** GRAPH-007 uniquely retains disjoint-incidence-packing-cover-duality, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-007: matching_covering_packing; exact observation: {"bipartite_part_size":3,"maximum_matching":3,"minimum_vertex_cover":3}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2f9271892d41ef7cb7cb4d2d9415f2e4a6f4ffbd643929cbb7f986c031a64b07`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:bef029b06bdf103257d0a6a116a72653b6f45646dcbe1fd06923dc0792ccea26`; engine receipt `sha256:283c65166202c674101581fbcf1b718bab3f5ad45ddcf38b4375f34480fa11f9`.

`SFT-MATH-OBL-GRAPH-008` - Directed graphs and causal reachability

- **Claim:** `SFT-MATH-GRAPH-DIRECTED-CAUSAL-REACHABILITY-008` - Directed graphs and causal reachability.
- **Forced law:** Directed incidence retains source and target labels, so causal reachability composes only orientation-compatible transitions.
- **Unique survivor:** GRAPH-008 uniquely retains orientation-retained-causal-closure, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-008: directed_causal_reachability; exact observation: {"causal_pairs":9,"topological_order":[1,2,3,4,5],"vertices":5}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ddc1e1b21eeb795b860d7ee38912000586b692f3a1ca11ee7282dd76b950c864`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:cf2d7d512d3066a55bb5e20ddc7d0e874409f5e34600cf99988159d19628e519`; engine receipt `sha256:a3ea7e43de3c0a16186153b9855e4b9f90acc8b096a2767bc84d0002b4f5e66c`.

`SFT-MATH-OBL-GRAPH-009` - Weighted network correspondence with exact parts

- **Claim:** `SFT-MATH-GRAPH-WEIGHTED-NETWORK-EXACT-PARTS-009` - Weighted network correspondence with exact parts.
- **Forced law:** A weighted network assigns lawful exact parts to incidences and compares paths only by exact accumulation without floating approximation.
- **Unique survivor:** GRAPH-009 uniquely retains exact-part-path-accumulation, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-DIRECTED-CAUSAL-REACHABILITY-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-009: weighted_exact_parts; exact observation: {"alternative_weight":{"denominator":3,"numerator":4},"path_weight":{"denominator":6,"numerator":7}}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:d312cdeb42c0f9eed26d0ffd024d5cc72c47018e89f9a9588b53afa5f1734803`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:d206e2841f91345e04992ba075426429420fcd8812b8c7ae832119f5c51765f5`; engine receipt `sha256:95711a8c91c539d061e323813786738ecf6f65f5b4d800ce7a7619e1a64bc993`.

`SFT-MATH-OBL-GRAPH-010` - Hypergraphs and higher incidence

- **Claim:** `SFT-MATH-GRAPH-HYPERGRAPH-HIGHER-INCIDENCE-010` - Hypergraphs and higher incidence.
- **Forced law:** A hyperedge is one held incidence over a generated carrier subset; rank and degree follow from complete higher-incidence custody.

- **Unique survivor:** GRAPH-010 uniquely retains multi-carrier-incidence-custody, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-WEIGHTED-NETWORK-EXACT-PARTS-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-010: hypergraph_higher_incidence; exact observation: {"hyperedges":4,"rank":3,"vertex_degree":2,"vertices":6}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:21a970469487805e0e6841839950d3ddd1f018507c531c24945b8e024a5afef1`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:2ad2463f9eca076c919e998f2fcf3b71cc2e3e8bd564330754456f922b3c10e8`; engine receipt `sha256:9ce82e3125329f6738cca2ab5b89ef6bbd92f237af6cf1619160351d5d220b69`.

`SFT-MATH-OBL-GRAPH-011` - Matroids and independence systems

- **Claim:** `SFT-MATH-GRAPH-MATROID-INDEPENDENCE-011` - Matroids and independence systems.
- **Forced law:** A matroid is the uniquely retained finite independence structure satisfying nonempty support, heredity and exact exchange across unequal independent carriers.
- **Unique survivor:** GRAPH-011 uniquely retains hereditary-exchange-independence, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-HYPERGRAPH-HIGHER-INCIDENCE-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-011: matroid_independence; exact observation: {"basis_count":6,"ground_size":4,"independent_sets":11,"rank":2}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:79008ce7f60b8ac644403613cdfbfae3d6154aae97f94cc37b611586fe4bd286`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:077cd80c2b38fa6ce2bb0a184b5495bb7ca867079f5a53a209ec4b495d0b511d`; engine receipt `sha256:6bf043d86d7d1d7a8a860124bf51cdb5f9bb780586d937880377b4fe0175c65e`.

`SFT-MATH-OBL-GRAPH-012` - Network reliability and failure custody

- **Claim:** `SFT-MATH-GRAPH-RELIABILITY-FAILURE-CUSTODY-012` - Network reliability and failure custody.

- **Forced law:** Reliability is the exact retained fraction of lawful states in the complete failure-mask support; every favorable and adverse mask remains in custody.
- **Unique survivor:** GRAPH-012 uniquely retains complete-failure-mask-support, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-MATROID-INDEPENDENCE-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-012: reliability_failure_custody; exact observation: {"connected_masks":4,"edge_masks":8,"exact_reliability":{"denominator":2,"numerator":1}}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bd1aacb7ed1308c8096094ba3f2cd9e2418a2bdab62cf36684a1b0b0995fed7b`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:3eca83aa54f99db594ddc8cdf5ac785aac6599d9d9615b977b4081ea25ce5040`; engine receipt `sha256:a2827845f35854c3fc3774078d4e794b651ee10ec65995f2cbb14a31c50f72ab`.

`SFT-MATH-OBL-GRAPH-013` - Spectral graph correspondence

- **Claim:** `SFT-MATH-GRAPH-SPECTRAL-CORRESPONDENCE-013` - Spectral graph correspondence.
- **Forced law:** Spectral correspondence is reconstructed through exact graph operators and held/opposed distinction modes; even-walk operators keep proof scalars nonnegative and exact.
- **Unique survivor:** GRAPH-013 uniquely retains even-walk-mode-correspondence, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-RELIABILITY-FAILURE-CUSTODY-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-013: spectral_even_walk_correspondence; exact observation: {"distinction_mode":1,"operator":"adjacency-square-k4","trace":12,"uniform_mode":9}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:cacdc1a92c4a4e250ce4d11fb8968c4cfe80fcf910caf4272aa68b552b8ea9f9`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:6913fc000747d0d4bb9169f21c8c1a15cfba7d98e12b5541804650b8323d5edd`; engine receipt `sha256:dfd637109db6dfdc9a1ebd98dff9ce928782ddec159f8310428d7154178d4ce5`.

`SFT-MATH-OBL-GRAPH-014` - Dynamic and temporal networks

- **Claim:** `SFT-MATH-GRAPH-DYNAMIC-TEMPORAL-NETWORK-014` - Dynamic and temporal networks.
- **Forced law:** A temporal path composes incidences only in nondecreasing transition order; static reachability cannot erase temporal custody.
- **Unique survivor:** GRAPH-014 uniquely retains time-ordered-incidence-composition, complete incidence, exact enumeration, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-COMB-COUNTING-LAWS-001 -> SFT-MATH-COMB-RAMSEY-FORCING-011 -> SFT-MATH-GRAPH-SPECTRAL-CORRESPONDENCE-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GRAPH-014: dynamic_temporal_network; exact observation: {"arrival_time":3,"static_but_time_forbidden_path":[1,4,3],"time_respecting_path":[1,2,3,4]}; complete family observation custody: 14 records; all rows and the adverse predecessor are preserved; complete deterministic graph enumeration supplies the observation; no random, fitted or imported theorem answer enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GRAPH-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target value, imported graph theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite graph family or continuum embedding; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:39cd84af37143dc5e3535cf6bf8e7b7f540ecbcb3da5aa58aa1acd46f51a9bd0`; independent `sha256:8e325e12630a9ab2da42f0ee63866f32388d33ce85b759994a5fb4441eb7b743`; empirical `sha256:20febd6db42e6fb308a4fa7dceacfa0089fea5e0a9052bc56aa1dc50e3b505f8`; engine receipt `sha256:fb1a3980a9e3273950f1ad3d9426d228287085c3cba62162735538726bde0626`.

40.8 Family `LINEAR` - 14/14 complete

This family contributes 3,584 completely decided candidates, 14 unique survivors and 56 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-LINEAR-001` - Vectors as exact generated coordinate carriers

- **Claim:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001` - Vectors as exact generated coordinate carriers.
- **Forced law:** A vector is an ordered generated coordinate carrier whose components remain individually addressable under lawful junction and exact-part scaling.
- **Unique survivor:** LINEAR-001 uniquely retains coordinate-identity-preserving-junction, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-001: vector_coordinates; exact observation: {"junction":[3,3,4],"left":[1,2,3],"right":[2,1,1]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or

continuum space; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:6df296385ffc1bee86de4232202b01a80cd3e8c553aeffecf05c630093ebcb40`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:b5d205abdae1b157303b76c4b28fee0b00f8cec58aa5ea0fc7bb9dfd3e7cfd82`; engine receipt `sha256:61163023459ed332dfc906e44bcb647778587e2db0c85e32b9390334de8675e9`.

`SFT-MATH-OBL-LINEAR-002` - Linear maps and composition

- **Claim:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002` - Linear maps and composition.
- **Forced law:** A linear map is forced by preservation of exact junction and scaling, and composition retains the ordered action of both maps.
- **Unique survivor:** LINEAR-002 uniquely retains junction-scaling-preserving-map-composition, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-002: linear_map_composition; exact observation: {"composed_output":[8,6],"input":[1,2]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:355edd57d0180af80cce98dc6da4611e68da7c469cefdb45597643b8019f0050`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:6266493aedb50b9da494f31994bc244e002e2e5aface8990be1e506386acfd02`; engine receipt `sha256:b0ff81fb4f50d49ba8e11c1e0dc7f414ff7711bf5ba1ec4d70cd9af6cdf7f5fe`.

`SFT-MATH-OBL-LINEAR-003` - Matrix representation and exact row operations

- **Claim:** `SFT-MATH-LINEAR-MATRIX-ROW-OPERATIONS-003` - Matrix representation and exact row operations.
- **Forced law:** A matrix is the coordinate incidence of a map; lawful row operations are exact reversible relation transformations with full operation custody.
- **Unique survivor:** LINEAR-003 uniquely retains reversible-row-relation-custody, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-MAP-COMPOSITION-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-003: matrix_row_operations; exact observation: {"input_rows":[[1,2],[3,4]],"relation_preserved":true,"swapped_rows":[[3,4],[1,2]]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7b2c77c8def1abd4b13bff0e0a4fb38f5ab1faf0b2ee2ee5a154b66d43976915`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:0eba202734aa28173c9ce52ad197dd373951255772397af93684ccc45fd74dcd`; engine receipt `sha256:a069bdc5521cb9dabf4978a906631c9f795c332771bfd6ee7f7bad56b0d37e07`.

`SFT-MATH-OBL-LINEAR-004` - Rank, nullity and retained distinctions

- **Claim:** `SFT-MATH-LINEAR-RANK-NULLITY-004` - Rank, nullity and retained distinctions.
- **Forced law:** Rank counts retained independent image distinctions while nullity counts source distinctions merged to absence; together they exhaust the finite source dimension.
- **Unique survivor:** LINEAR-004 uniquely retains image-kernel-distinction-ledger, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-MATRIX-ROW-OPERATIONS-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-004: rank_nullity; exact observation: {"columns":3,"field_support":2,"nullity":1,"rank":2}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c49a530091e69518ae8886b32534de89ddcb43d1286e03b1ec39219024af5d5d`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:742cf4ef2bd8942c72fb866d34d54e0f872fa5bddb5a8a4fa4a68bd5ed35a908`; engine receipt `sha256:a1111d9fdee741f94a9f57268078c4f8935bdc8fcddec9419a2f858d67418bc9`.

`SFT-MATH-OBL-LINEAR-005` - Determinants and orientation custody

- **Claim:** `SFT-MATH-LINEAR-DETERMINANT-ORIENTATION-005` - Determinants and orientation custody.
- **Forced law:** Determinant structure is the exact excess between held and opposed permutation products, recorded as an orientation label and nonnegative magnitude.
- **Unique survivor:** LINEAR-005 uniquely retains held-opposed-volume-orientation, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-RANK-NULLITY-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-005: determinant_orientation; exact observation: {"held_product":4,"magnitude":2,"opposed_product":6,"orientation":"opposed"}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted

value enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:37ed07457eb59c096ccff782a271491e04191b77225f739a91e2917a935de00d`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:5b3de428e72c8240f571749090bcbfc855274d488e1e18419c0744aafbaf38c3`; engine receipt `sha256:1eb50aad30609be2adbf13fd7589e8fdb3013d5a4e0e25292b4c27d2cda0587d`.

`SFT-MATH-OBL-LINEAR-006` - Exact systems of linear relations

- **Claim:** `SFT-MATH-LINEAR-EXACT-SYSTEMS-006` - Exact systems of linear relations.
- **Forced law:** A linear system is closed by generating every lawful coordinate candidate and retaining exactly the candidates satisfying every relation.
- **Unique survivor:** LINEAR-006 uniquely retains complete-solution-support-census, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-DETERMINANT-ORIENTATION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-006: exact_linear_system; exact observation: {"relations":[3,4],"solution":[2,1]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:6bb9729d649ff522569dea22ef8f154799a93a977770690d31ed481f6e83da18`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:7a5ac55126c0a9bf45f56abaf8a84eb407bfd7fcdf35eebfde23b7e94d411cf4`; engine receipt `sha256:df388656cff7ee3d4afa27c2c5efb2dad840fb0abfbdebc457eb2eb6f3e22a10`.

`SFT-MATH-OBL-LINEAR-007` - Basis, dimension and change of basis

- **Claim:** `SFT-MATH-LINEAR-BASIS-DIMENSION-007` - Basis, dimension and change of basis.
- **Forced law:** A basis is a minimal independent spanning carrier; dimension is its exact cardinality and change of basis is a reversible coordinate relabelling.
- **Unique survivor:** LINEAR-007 uniquely retains reversible-independent-coordinate-frame, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-EXACT-SYSTEMS-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** LINEAR-007: basis_dimension; exact observation: {"ambient_dimension":2,"basis":[[1,0],[1,1]],"coordinates":[1,2]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3e7eaa7e5ce437bfc38f41e24761c2f305044c2e2e8c00dc6e8c4e42a7d25629`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:fc5d19197d64bdc031e6cb34e59d1b1d77c4be066861966410ead86ac73b715d`; engine receipt `sha256:f9cc18b562f233e53210a5788786195759321f2d9484faa0188a05c46c192de9`.

`SFT-MATH-OBL-LINEAR-008` - Inner-product and metric correspondence

- **Claim:** `SFT-MATH-LINEAR-INNER-PRODUCT-METRIC-008` - Inner-product and metric correspondence.
- **Forced law:** Inner-product correspondence is an exact symmetric bilinear pairing; metric correspondence uses exact squared separation when roots are not lawful proof scalars.
- **Unique survivor:** LINEAR-008 uniquely retains exact-bilinear-pairing-squared-distance, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-BASIS-DIMENSION-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-008: inner_product_metric; exact observation: {"inner_product":5,"squared_distance":5}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3ad0a974b8d4bda3db7890713be601bf1b04630e24b820d61055472eeb28755d`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:374227e7bd487ed65126f5b659e4364afcd7f54e872fc8fcd8feb47b388a9784`; engine receipt `sha256:bf412fb7caf2d4fe721ab9d01381a552a490d1b789b8469eb534a287de9da355`.

`SFT-MATH-OBL-LINEAR-009` - Eigenvalue and invariant-support correspondence

- **Claim:** `SFT-MATH-LINEAR-EIGEN-INVARIANT-SUPPORT-009` - Eigenvalue and invariant-support correspondence.
- **Forced law:** An invariant mode is a held coordinate relation reproduced by an operator up to one exact lawful scale, with opposed distinctions represented structurally rather than negatively.
- **Unique survivor:** LINEAR-009 uniquely retains invariant-mode-exact-scaling, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-INNER-PRODUCT-METRIC-008`;

registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** LINEAR-009: eigen_invariant_support; exact observation: {"distinction_mode":1,"uniform_mode":3}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ce5211d932e87ed2b831cf1a14466643b259e7f389cd67f9080ff08fa211f808`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:5728068f2bb8669dd948a262e357688455ec039a644533b17f10526ffe549d93`; engine receipt `sha256:f4dd00fe4be2cb27fec953298d56913b6299c2c17cf34cec00a7fcd1d71ccb33`.

`SFT-MATH-OBL-LINEAR-010` - Exact rational spectral enclosures

- **Claim:** `SFT-MATH-LINEAR-RATIONAL-SPECTRAL-ENCLOSURE-010` - Exact rational spectral enclosures.
- **Forced law:** A non-rational conventional spectral value is represented only by nested exact rational bounds with a proved exclusion and refinement rule.
- **Unique survivor:** LINEAR-010 uniquely retains nested-rational-mode-enclosure, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-EIGEN-INVARIANT-SUPPORT-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-010: rational_spectral_enclosure; exact observation: {"lower":[8,5],"relation":"x*x=x+1","upper":[13,8]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5c5892582b824b1585a41b97e989cfdc3e49cf205760c4d0eeaa145e35941660`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:aa23b2604d95e5564d66a765227044917965df7dd1ba972f495ae33112a50252`; engine receipt `sha256:78bfa64c5f59b2314c1137839cdb8e8cef9778d056a094837ec553a1c5ddffc2`.

`SFT-MATH-OBL-LINEAR-011` - Multilinear maps and tensor products

- **Claim:** `SFT-MATH-LINEAR-MULTILINEAR-TENSOR-PRODUCT-011` - Multilinear maps and tensor products.
- **Forced law:** A tensor product is the complete coordinate product that converts separately linear multi-input action into one exact linear carrier.
- **Unique survivor:** LINEAR-011 uniquely retains universal-multicarrier-coordinate-product, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-RATIONAL-SPECTRAL-ENCLOSURE-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-011: multilinear_tensor_product; exact observation: {"left_dimension":2,"outer_product":[[3,4,5],[6,8,10]],"product_dimension":6,"right_dimension":3}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4b99fbff5a16be85950d2f528716299e1d9499e6b8459bad97dbbb64e936ac30`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:79c5c2ecba964057d2e8d3141be6a9819118149e454acc0438a28334e6410d25`; engine receipt `sha256:9a0648891c36fa2004a4130cf182de9f9aed20f746ed391c30dd8b657c6a2b53`.

`SFT-MATH-OBL-LINEAR-012` - Tensor contraction and index custody

- **Claim:** `SFT-MATH-LINEAR-TENSOR-CONTRACTION-012` - Tensor contraction and index custody.
- **Forced law:** Contraction pairs exactly the declared dual indices, sums their complete incidence support and retains every uncontracted index identity.
- **Unique survivor:** LINEAR-012 uniquely retains paired-index-junction-custody, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-MULTILINEAR-TENSOR-PRODUCT-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-012: tensor_contraction; exact observation: {"contracted_matrix":[[6,8],[10,12]],"tensor_shape":[2,2,2]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3266d1f12a50e166fe4890d2e6ac9cd9a3dfdd7159aa7653f05555d46151605d`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:3eb38b7d9e4b216845a4baa6025ee6c65210682ef0f8a672344b76b95f1c6191`; engine receipt `sha256:144d72725402644b443e0674b9a7d36c0ac3759527fb55ae4a664fbf31be1016`.

`SFT-MATH-OBL-LINEAR-013` - Exterior and symmetric composition

- **Claim:** `SFT-MATH-LINEAR-EXTERIOR-SYMMETRIC-013` - Exterior and symmetric composition.
- **Forced law:** Exterior composition retains orientation and makes repeated direction absent; symmetric composition quotients order alone while retaining multiplicity.

- **Unique survivor:** LINEAR-013 uniquely retains orientation-and-symmetry-quotient, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-TENSOR-CONTRACTION-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-013: exterior_symmetric; exact observation: {"distinct_wedge_orientation":"held","repeated_wedge":"absence","symmetric_pair_count":3}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:cf9bf59b57b93c2b0e2a4e12d67ca3c407f8dca9d7ef75091c8659ac830c0abe`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:0aac9b1a2b1fa57a514383e733c8249c39ea93f10c129acbe8a83223aca3b6be`; engine receipt `sha256:7514349b966d1a39444e5f841909f6e00a777c31d426db7a12a91da6e9013351`.

`SFT-MATH-OBL-LINEAR-014` - Finite-dimensional operator decomposition

- **Claim:** `SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014` - Finite-dimensional operator decomposition.
- **Forced law:** A finite operator decomposition is lawful when exact invariant components reconstruct the operator and each component has an independently witnessed action.
- **Unique survivor:** LINEAR-014 uniquely retains exact-invariant-component-decomposition, exact coordinates, structural orientation, complete enumeration, root forcing and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001 -> SFT-MATH-ARITH-CANONICAL-FRACTION-008 -> SFT-MATH-LINEAR-EXTERIOR-SYMMETRIC-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LINEAR-014: operator_decomposition; exact observation: {"component_weights":[2,1],"idempotent_components":true,"operator":[[2,0],[0,1]]}; complete family observation custody: 14 records; all rows preserved; all scalars are exact lawful parts or nonnegative integers; orientation is structural and no fitted value enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LINEAR-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no target coordinate, imported linear-algebra theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite-dimensional or continuum space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e27955bfe827c82d6ba147d3f4b286e52797402f9b520af1213367375efa03d8`; independent `sha256:215057257b829acb9bdd277b28a1fce045fde73971d91391f73eb980fbaebce7`; empirical `sha256:4b3f355ec02ff9781addcd3c06c865faa15bf5c9026b382bec2c081fbb65d170`; engine receipt `sha256:52a0d942eb72cefecec9a693af78189036fe049d1da9a5ab8b2f1c85c61ce74a`.

40.9 Family `ALG` - 16/16 complete

This family contributes 4,096 completely decided candidates, 16 unique survivors and 64 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-ALG-001` - Magma and closed operation structure

- **Claim:** `SFT-MATH-ALG-MAGMA-CLOSED-OPERATION-001` - Magma and closed operation structure.
- **Forced law:** A magma is forced when every ordered carrier pair has exactly one operation image inside the generated carrier support.
- **Unique survivor:** ALG-001 uniquely retains closed-generated-binary-operation, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-001: magma; exact observation: {"carrier_count":2,"closed":true,"table_cells":4}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:9c3cf0654d79424e723b0950f0eff16bf6d98d4ebe6b98d7a132129f0a0c75d5`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:03bb7b5fe3c9b8a2a5cfaf7fe4ffcfa7c8ad6fd6a2f376002af02e363a033df4`; engine receipt `sha256:c5e93e05277ffeb172cc3a7003518d9210ad7be5ad8771ba88a6dd397f067b80`.

`SFT-MATH-OBL-ALG-002` - Semigroup associativity witnesses

- **Claim:** `SFT-MATH-ALG-SEMIGROUP-ASSOCIATIVITY-002` - Semigroup associativity witnesses.
- **Forced law:** A semigroup is a closed operation whose two three-input bracketings agree on the complete carrier census.
- **Unique survivor:** ALG-002 uniquely retains complete-associativity-census, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-MAGMA-CLOSED-OPERATION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-002: semigroup; exact observation: {"associative_triples":27,"carrier_count":3}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c2e0bf41b4b5309a05f97c50ec4d2e68b476afe59b95c5b174436a9478839e99`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical

`sha256:10cc721c8d1a0beee12792510a3852e2c2ecb98be664ad81738ecac0d621a0fb`; engine receipt `sha256:cf47d05b924a577416959ddc022d305c6aa02509046c37239b125f1e0fe13ab4`.

`SFT-MATH-OBL-ALG-003` - Monoid identity witnesses

- **Claim:** `SFT-MATH-ALG-MONOID-IDENTITY-003` - Monoid identity witnesses.
- **Forced law:** A monoid adds one uniquely forced two-sided identity carrier to an associative operation.
- **Unique survivor:** ALG-003 uniquely retains unique-two-sided-identity, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-SEMIGROUP-ASSOCIATIVITY-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-003: monoid; exact observation: {"identity_carrier":1,"two_sided":true}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4907c9ea806a0b50fad2b2e8445cd827311d01b8ed5b5fb095f970a9b2ba6bea`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:b04565e6a08b99d19c4443c3c2581301d6f27455fb8d3e25b1e09f0c2b376025`; engine receipt `sha256:beb8f7c7965680020a19391958ee25ec44998ca6b33feb9978f7adfda74a1fe5`.

`SFT-MATH-OBL-ALG-004` - Group inverse as held reversal

- **Claim:** `SFT-MATH-ALG-GROUP-HELD-INVERSE-004` - Group inverse as held reversal.
- **Forced law:** A group requires one held reversal for every carrier whose ordered composition returns the identity.
- **Unique survivor:** ALG-004 uniquely retains held-reversal-to-identity, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-MONOID-IDENTITY-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-004: group; exact observation: {"carrier_count":3,"unique_inverses":3}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:b1f096a4bcdff504ddb0e7f91eb52a6bf7c567e32473018dd7b85c14bf7b6777`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:ff8d8f38b87e68e61a27ab56d25d2594b79f4255040e999f65942449dcadbd72`; engine receipt

`sha256:94ea42e4036ab10f3e1eb8cdab2e194e18a1a053c664f1070fbac6a7a7c73152`.

`SFT-MATH-OBL-ALG-005` - Permutation-group action

- **Claim:** `SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005` - Permutation-group action.
- **Forced law:** A permutation action is a reversible relabelling composition that preserves the acted carrier support.
- **Unique survivor:** ALG-005 uniquely retains reversible-carrier-action, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-GROUP-HELD-INVERSE-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-005: permutation_action; exact observation: {"actions":6,"carriers":3}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ca19fef83e3c15a8e76521d4851582d2d33858ecd56d84f7cf0bd42b1b54604e`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:dc8c7e5b0d156fb7480e113fdf7ed54524dc6deb2f0ecb310724dee399455237`; engine receipt `sha256:4fe452d169159708a9cb65c9f42edfba611f0d86fc6143026a5122b839fba278`.

`SFT-MATH-OBL-ALG-006` - Quotient and normal-substructure correspondence

- **Claim:** `SFT-MATH-ALG-QUOTIENT-NORMAL-SUBSTRUCTURE-006` - Quotient and normal-substructure correspondence.
- **Forced law:** A quotient identifies exactly the cosets forced by a normal held substructure and retains all distinct classes.
- **Unique survivor:** ALG-006 uniquely retains normal-coset-equivalence, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-006: quotient; exact observation: {"cosets":2,"normal_substructure":2,"parent_carriers":4}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3a41a3eb29bc36d2a1c2f1258473d89b1672eff16a32b390fd0bef73d48f2089`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical

`sha256:b8e192de22211517657da1385f1539f43bfa8e1d55f951e61b9876d0a34cdd32`; engine receipt `sha256:d468d9fb67765ecaa8c7b3f2e1e0404a27e07bbdb5843f943e01c62ac2d4f7b6`.

`SFT-MATH-OBL-ALG-007` - Ring distributive structure

- **Claim:** `SFT-MATH-ALG-RING-DISTRIBUTIVE-007` - Ring distributive structure.
- **Forced law:** A ring couples junction and product operations through complete left and right distributivity witnesses.
- **Unique survivor:** ALG-007 uniquely retains dual-operation-distribution, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-QUOTIENT-NORMAL-SUBSTRUCTURE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-007: ring; exact observation: {"distributive_triples":27,"modulus":3}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2787c480e07b430d7317101d9bc09a660828710264214520de5779f9fe47761a`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:41457482a162fe1345ac8d439bd9212e1c552cb1cffa73eb9ad36e41519c9f92`; engine receipt `sha256:2207ec02691fbc1989e0ae93e324ebb034047996b14277b521774ae634cac4fd`.

`SFT-MATH-OBL-ALG-008` - Integral-domain boundary

- **Claim:** `SFT-MATH-ALG-INTEGRAL-DOMAIN-008` - Integral-domain boundary.
- **Forced law:** An integral-domain boundary forbids two nonabsence carriers from composing to multiplicative absence.
- **Unique survivor:** ALG-008 uniquely retains nonabsence-product-retention, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-RING-DISTRIBUTIVE-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-008: domain; exact observation: {"modulus":5,"nonabsence_pairs":16,"zero_divisors":"absence"}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e11219489f9b1483f55375af57343a7274dce1c9180dd213b69c4be43de5c68f`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical

sha256:d727cd9fe244caa1e4fd8757befe697dbf0dcfbe58122919b49388cae1ce94d5; engine receipt sha256:69176d3a233e9eaa708fbbf5517c30b9cb3b7148f4f6cf12562d35d788b4f6b1.

SFT-MATH-OBL-ALG-009 - Field correspondence under exact division

- **Claim:** SFT-MATH-ALG-FIELD-EXACT-DIVISION-009 - Field correspondence under exact division.
- **Forced law:** A field correspondence requires one exact multiplicative reversal for every nonabsence carrier.
- **Unique survivor:** ALG-009 uniquely retains unique-nonabsence-division, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-INTEGRAL-DOMAIN-008; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ALG-009: field; exact observation: {"modulus":5,"nonabsence_inverses":4}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:70132aa0aaf92ea0546a67d05428eb780972268374620 82f9b8729f52e17bd35; independent sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927; empirical sha256:fac9542a32facdc9aad33338981fe908b9cb348ccbb501003ef5e4dc9a9c4244; engine receipt sha256:d1e1bc506263760193b0b99b66e3f7669ec9a2581ab2e4e10bb3f5ac656dd7ee.

SFT-MATH-OBL-ALG-010 - Modules and scalar-action structure

- **Claim:** SFT-MATH-ALG-MODULE-SCALAR-ACTION-010 - Modules and scalar-action structure.
- **Forced law:** A module is a carrier junction with scalar action preserving both scalar and carrier composition laws.
- **Unique survivor:** ALG-010 uniquely retains compatible-scalar-carrier-action, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-FIELD-EXACT-DIVISION-009; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ALG-010: module; exact observation: {"coordinate_dimension":2,"field_support":2}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:022df7fb50d7a224ce4ab2c9178bc8124764199c79fe976743011238830f9de5; independent sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927; empirical sha256:0c395f0a4f081722255fa4b5f2fa1af6eda5a9cde1077f05c3a42942eced4979; engine receipt

sha256:bb21a886d6f54b9c232c9e4c6abb37c78d20372f1b3aa8c3a30b7e9795cc5fdf.

SFT-MATH-OBL-ALG-011 - Algebras and compatible products

- **Claim:** SFT-MATH-ALG-COMPATIBLE-ALGEBRA-PRODUCT-011 - Algebras and compatible products.
- **Forced law:** An algebra adds an internal product compatible with exact scalar action through bilinearity.
- **Unique survivor:** ALG-011 uniquely retains bilinear-internal-product, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-MODULE-SCALAR-ACTION-010; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ALG-011: algebra; exact observation: {"bilinear":true,"coordinate_dimension":2}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:1518b936f8dacbc914653f1c30815a674ac4f4280c0505566869610ce2948e10; independent sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927; empirical sha256:4725df72c1930e8eff5213f66080905b73ae2bd61e71256ef778a5b15d12a48e; engine receipt sha256:2170e5d46c51d5cac7879b84b7466bad4f891683985fa16d920f0a0f1f4b3a6f.

SFT-MATH-OBL-ALG-012 - Ideals and quotient structures

- **Claim:** SFT-MATH-ALG-IDEAL-QUOTIENT-012 - Ideals and quotient structures.
- **Forced law:** An ideal is a junction-closed absorbing substructure whose cosets carry a lawful quotient operation.
- **Unique survivor:** ALG-012 uniquely retains absorbing-substructure-quotient, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-COMPATIBLE-ALGEBRA-PRODUCT-011; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ALG-012: ideal; exact observation: {"ideal_carriers":2,"quotient_classes":3,"ring_carriers":6}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:c025b2af3872140d509882dc692a06e982c3464f3a5a051838d8ede2e8c47319; independent sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927; empirical sha256:9481b2cee4f9b26959c22ab2d06d55126d874f29b357a477a7645224fdbce4d0; engine receipt

sha256:51014e51baa2db97fcb3c486c521aa813b08e33541a38c791abf465958ecfc3e.

SFT-MATH-OBL-ALG-013 - Representation and action decomposition

- **Claim:** SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013 - Representation and action decomposition.
- **Forced law:** A representation turns abstract composition into exact reversible linear action and decomposes only along witnessed invariant supports.
- **Unique survivor:** ALG-013 uniquely retains invariant-action-components, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-IDEAL-QUOTIENT-012; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ALG-013: representation; exact observation: {"action":"swap","distinction_mode":"held-opposed","uniform_mode":"held"}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:3f25c3b5b4cfb4398c3ede25e416b677e2677c476d336e3fd93973bf382d94c4; independent sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927; empirical sha256:0a774bbd416b49454a260429451c9b1d20e8ac1232f7219bb99f3543d2396163; engine receipt sha256:57cbe383561e1219e2ebae0731dba20a8a73b7f2961313cd20ff89c3af3ca2e8.

SFT-MATH-OBL-ALG-014 - Exact sequence and homological correspondence

- **Claim:** SFT-MATH-ALG-EXACT-SEQUENCE-HOMOLOGICAL-014 - Exact sequence and homological correspondence.
- **Forced law:** Exactness is the pointwise equality of one map image with the next map kernel, with every absence and retained distinction recorded.
- **Unique survivor:** ALG-014 uniquely retains image-kernel-equality, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ALG-014: exact_sequence; exact observation: {"equal":true,"image_size":2,"kernel_size":2}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:f3f7c6de976c7dcc48dc2533b559931adaf84fe0923064a8099c19c9677bedcf`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:f0113ef2727db6c761b75c06cf2bb301fc8d7b518b72414838889df4df76f68b`; engine receipt `sha256:22db8eceff3c0225e195f7d08ed2011e187177df110bb9220cd2aa76d4337380`.

`SFT-MATH-OBL-ALG-015` - Universal algebra and identities

- **Claim:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015` - Universal algebra and identities.
- **Forced law:** A universal-algebra identity survives only when every valuation over the declared operation signature satisfies it.
- **Unique survivor:** ALG-015 uniquely retains signature-wide-identity-census, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-EXACT-SEQUENCE-HOMOLOGICAL-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-015: universal_algebra; exact observation: {"carrier_count":3,"identity":"associative-commutative-idempotent"}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:28f19c887d0a9df5beca581222dc5a33beb70f9d43f6e2f2af0f8c7aa2a8fdd7`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:505f6ea78c4323603a5f9a1dd280895936aef36ba7fa290f86581472e97711b8`; engine receipt `sha256:88a76935879afa29f6f647396afa0711a5626baae4107b68e4538a57f03c1a73`.

`SFT-MATH-OBL-ALG-016` - Operadic algebraic composition interface

- **Claim:** `SFT-MATH-ALG-OPERADIC-COMPOSITION-016` - Operadic algebraic composition interface.
- **Forced law:** Operadic composition is typed substitution of multi-input operations with leaf identity and association retained.
- **Unique survivor:** ALG-016 uniquely retains typed-substitution-association, complete operation custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-MAP-COMPOSITION-002 -> SFT-MATH-ARITH-CONGRUENCE-010 -> SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ALG-016: operad; exact observation: {"leaf_count":4,"substitution_associative":true}; complete family observation custody: 16 records; all rows preserved; complete finite operation enumeration supplies the observation; no fitted table, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ALGEBRA-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported algebraic axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite algebraic carrier; no failed

route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:f985509941927723abf8a6e34a74e4139e61e4bf32c258ec3c41d705176e293f`; independent `sha256:3e795c9ed31b2ce2bd9006e7964e508aac1c46498d50a317f1117e4012088927`; empirical `sha256:c1ff19b22d7316e33ddbdb58d8a2bf954dc0ad095989f9ba261cffd81c2591b7`; engine receipt `sha256:c45a860ee26cca756e9d1120e40d217d8255f5d68495e8cd9b3b275311081f7a`.

40.10 Family `ORDER` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-ORDER-001` - Preorder and distinguishability quotient

- **Claim:** `SFT-MATH-ORDER-PREORDER-QUOTIENT-001` - Preorder and distinguishability quotient.
- **Forced law:** A preorder is forced by reflexive composable reachability; mutually reachable carriers quotient only when observation cannot distinguish them.
- **Unique survivor:** ORDER-001 uniquely retains reflexive-transitive-distinguishability-quotient, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-001: preorder_quotient; exact observation: {"depth":3,"length_classes":4,"reflexive_transitive":true}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ca5091f295da823f0d7477f643f6943e69b6a55283e242442f86d08e6508d01e`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:314590fddb9cc1ffeb727476907c27c46bfbe56b1fa6e804ce26c5186486ed1d`; engine receipt `sha256:62b8a349524faaad77513549e980c4f3d9f3960dfea1aea5df04bd0f7338e4cb`.

`SFT-MATH-OBL-ORDER-002` - Partial order and antisymmetry

- **Claim:** `SFT-MATH-ORDER-PARTIAL-ANTISYMMETRY-002` - Partial order and antisymmetry.
- **Forced law:** A partial order adds the law that mutual reachability of distinguished carriers forces their identity.
- **Unique survivor:** ORDER-002 uniquely retains antisymmetric-reachability-order, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-PREORDER-QUOTIENT-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-002: partial_order; exact observation: {"antisymmetric":true,"carrier_support":8}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the

observation; no fitted order, imported axiom list or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f7f36ba29a4c41aed3d2f34fc5c48fb32ebcc62299baf60c4394788be397605d`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:927538266b0087ce6d39ead45e670d6ee765ef41a77b2f3f9817c262aa75040b`; engine receipt `sha256:90737e2dd2e996533513d1d6b2dedb0453101ced5dbf1972fd12672d7eb09ee2`.

`SFT-MATH-OBL-ORDER-003` - Conditional total order

- **Claim:** `SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003` - Conditional total order.
- **Forced law:** A total order is lawful only on a registered carrier domain where every pair has an exact comparison witness.
- **Unique survivor:** ORDER-003 uniquely retains exact-comparability-boundary, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-PARTIAL-ANTISYMMETRY-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-003: conditional_totality; exact observation: {"all_pairs_comparable":true,"exact_fraction_carriers":4}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ba62044ea076d2ed9392b4b05976d8233a461be7db4b7d0ff811d0ab76ad1cde`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:d5fa5d47d65c2d65a3311d9ef87735ec3a5eaff70d9d1ed586148fef062f9b0a`; engine receipt `sha256:fa8f185e14ec8d4e621b22db29f13bcf50d9c9a3b1101e434b148b6bc25ac5f3`.

`SFT-MATH-OBL-ORDER-004` - Meet, join and lattice structure

- **Claim:** `SFT-MATH-ORDER-MEET-JOIN-LATTICE-004` - Meet, join and lattice structure.
- **Forced law:** Meet and join are uniquely forced greatest-lower and least-upper carriers under the complete order relation.
- **Unique survivor:** ORDER-004 uniquely retains greatest-lower-least-upper-custody, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-004: meet_join; exact observation: {"carrier_support":8,"ordered_pairs":64,"unique_meet_join":true}; complete family observation custody: 12 records; all

rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c99a1533fdaa38c5ff7041d1e463722a641c9b9fe5db0acd3099c1f52503d144`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:d1f3e4a6b89d3099100d59e888efa48c492df45a88297028359fee9e3cb1f128`; engine receipt `sha256:ef972980b97f3eae7b7060fcca7ce8f6b2d49f86cabc6f3ddd23852195c7b44d`.

`SFT-MATH-OBL-ORDER-005` - Distributive and modular lattice correspondence

- **Claim:** `SFT-MATH-ORDER-DISTRIBUTIVE-MODULAR-005` - Distributive and modular lattice correspondence.
- **Forced law:** Distributive and modular correspondence is admitted by complete finite identity censuses over meet and join.
- **Unique survivor:** ORDER-005 uniquely retains lattice-distribution-modularity, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-MEET-JOIN-LATTICE-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-005: distributive_modular; exact observation: {"distributive":true,"modular":true,"ordered_triples":512}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2203ad25cb0aec3062938ce27f394b9b6014eb109b32481ba51b3d09fc6917ad`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:c7d8a1d174d1b76f91637d61f907e0652709b73f0f09501dda26c060d764d200`; engine receipt `sha256:c67a3e785599e9dd49003227fef98d588bc99d2a18d7d267e1ea1e7646eacf89`.

`SFT-MATH-OBL-ORDER-006` - Boolean-like complement correspondence

- **Claim:** `SFT-MATH-ORDER-BOOLEAN-COMPLEMENT-006` - Boolean-like complement correspondence.
- **Forced law:** Complement is a carrier whose meet is absence and join is the declared universe, with uniqueness tested over complete support.
- **Unique survivor:** ORDER-006 uniquely retains relative-complement-boundary, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-DISTRIBUTIVE-MODULAR-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** ORDER-006: complement; exact observation: {"carrier_support":8,"unique_complements":8}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f720b975401f1ceac55f60981f6a09f3c5156a71704b8d5c5b473ece035cd0b0`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:2b2d1b98a29cd095a42896c4012eba4882f0df4a25ad44f909c6a1bd9b9b2e07`; engine receipt `sha256:bab7103a46409d68b5275b3e60e12a6b7457763dbbc8cbf5b18d859d44d652f2`.

`SFT-MATH-OBL-ORDER-007` - Closure operators and closure systems

- **Claim:** `SFT-MATH-ORDER-CLOSURE-SYSTEM-007` - Closure operators and closure systems.
- **Forced law:** A closure operator is uniquely characterized by extensivity, monotonicity and idempotence; its fixed carriers form the closure system.
- **Unique survivor:** ORDER-007 uniquely retains extensive-monotone-idempotent-closure, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-BOOLEAN-COMPLEMENT-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-007: closure; exact observation: {"carrier_support":8,"extensive_monotone_idempotent":true}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:51f16c12aff75dd5d9122d2239315ed23a8041ccb05e850f1292fa150c1ccd87`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:34c059972df7500aebcd64c5acd22974450a454becdb3b6f06dbb05ec6dc8893`; engine receipt `sha256:8c63e67ef5d6b6d785dbfb1bcebe0df202af2908650236db7902c13e7ef0171a`.

`SFT-MATH-OBL-ORDER-008` - Galois connections between orders

- **Claim:** `SFT-MATH-ORDER-GALOIS-CONNECTION-008` - Galois connections between orders.
- **Forced law:** A Galois connection is the exact equivalence between one order comparison and its reversed adjoint comparison.
- **Unique survivor:** ORDER-008 uniquely retains adjoint-order-equivalence, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-008: galois_connection; exact observation: {"equivalence_pairs":32,"left_subsets":8,"right_subsets":4}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:71532050dc9ac057cf3dd44897543313570590231586850860c7e853617b0520`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:c1277ff524e7c17b6f03ded0d931d384eeeddcf5f54718b06cee02f78bb45096`; engine receipt `sha256:e68701f93b8bc3b1b2994f1701d31f2a56e3ed49d54306f13f605e7b9cfb2fd2`.

`SFT-MATH-OBL-ORDER-009` - Domain approximation correspondence

- **Claim:** `SFT-MATH-ORDER-DOMAIN-APPROXIMATION-009` - Domain approximation correspondence.
- **Forced law:** Domain approximation is reconstructed by directed finite approximants whose exact join recovers the retained carrier.
- **Unique survivor:** ORDER-009 uniquely retains finite-approximant-directed-join, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-GALOIS-CONNECTION-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-009: domain_approximation; exact observation: {"chain_carriers":4,"recovered_by_approximants":4}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7358b757170350efcc5fcdefab8c9220c1e5deacf9816fe50f0de3a4262f1dd9`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:9599a6a4a9836426c0991365e53b08c6226c1766a019829e51c007d60d3a1ca1`; engine receipt `sha256:94ae3c06d59a7cc4fd1de2f6db35ceb1fad52c343e8622360f94e14f65ec5087`.

`SFT-MATH-OBL-ORDER-010` - Monotone maps and order preservation

- **Claim:** `SFT-MATH-ORDER-MONOTONE-MAP-010` - Monotone maps and order preservation.
- **Forced law:** A monotone map preserves every registered comparison and cannot reverse a retained order distinction.
- **Unique survivor:** ORDER-010 uniquely retains order-preserving-map, complete comparison custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-DOMAIN-APPROXIMATION-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-010: monotone_map; exact observation: {"chain_carriers":4,"comparison_pairs":16,"preserved":true}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:982b932b5b04926149592045111bcf0d0f8cb587947e0416abe6f2d38b2d7389`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:c3f1fc963141d43048db64dc1bb784437cbe548ee9fbb58cc8e453619394288b`; engine receipt `sha256:053130750729ea82310b9c55b17d1d9113d05666aaac6d2639b86863c497b8d5`.

`SFT-MATH-OBL-ORDER-011` - Fixed-point existence on finite orders

- **Claim:** `SFT-MATH-ORDER-FINITE-FIXED-POINT-011` - Fixed-point existence on finite orders.
- **Forced law:** A monotone self-map on a finite complete order reaches a witnessed fixed point by exact iteration from a boundary carrier.
- **Unique survivor:** ORDER-011 uniquely retains finite-monotone-iteration-fixed-point, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-MONOTONE-MAP-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-011: fixed_point; exact observation: {"iterations":1,"least_fixed_support":[1]}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:745b054628489b5bb249b805e46d2641e18198dabddd9898624265494b073160`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:1d57ce514aba4f3acaf837d2fb2d1e756e663c0f3f3aa9afca3c37dadb3973ff`; engine receipt `sha256:d81be77c0466b0b27fc0c703c9e40c284f2052c4663d8ae491bf23f7e1f2e7bc`.

`SFT-MATH-OBL-ORDER-012` - Complete-lattice correspondence boundary

- **Claim:** `SFT-MATH-ORDER-COMPLETE-LATTICE-BOUNDARY-012` - Complete-lattice correspondence boundary.
- **Forced law:** Finite complete-lattice correspondence requires a meet and join for every generated subfamily, including empty-family boundary conventions.

- **Unique survivor:** ORDER-012 uniquely retains all-generated-family-meet-join, complete comparison custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015 -> SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007 -> SFT-MATH-ORDER-FINITE-FIXED-POINT-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ORDER-012: complete_lattice; exact observation: {"all_meets_joins_present":true,"carrier_support":8,"generated_subfamilies":256}; complete family observation custody: 12 records; all rows preserved; complete finite comparison enumeration supplies the observation; no fitted order, imported axiom list or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ORDER-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported order axiom list, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite lattice or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:153e71c96065e2214f5fc25301bc6b66bb268d12f72344bda67971f4a3aecb8d`; independent `sha256:0e9530f3c571a1dbf4c3b89e3ef812de846ddd049df78264f02b3f7f094b3b94`; empirical `sha256:6618c7a9c962748fdcc18efce98a3099ed679a384cd7a5b054df151872b6f19c`; engine receipt `sha256:c058a971d9c63aea9690da203a2bafd33ff1a85c3b456ed893c0fbc483d1df96`.

40.11 Family `GEOM` - 16/16 complete

This family contributes 4,096 completely decided candidates, 16 unique survivors and 64 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-GEOM-001` - Point, incidence and exact coordinate identity

- **Claim:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001` - Point, incidence and exact coordinate identity.
- **Forced law:** A point is a named coordinate carrier and incidence is an exact relation that preserves every point identity.
- **Unique survivor:** GEOM-001 uniquely retains named-point-incidence-custody, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-001: point_incidence; exact observation: {"collinear":true,"held_total":11,"opposed_total":11,"points":3}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:6e54fdc1b6bd55e00c33a86e75a9d3252348b1aaf6dde36346d5493506dd39d1`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:6604c33d4c9b36327baa70c264daaa218a4b83e88c2b2dce6eed051465d9f5e5`; engine receipt `sha256:6e21c474249b5f7416086cffdfa67b0dc82ae3b3b5ec3e085f966a51923489c2`.

SFT-MATH-OBL-GEOM-002 - Finite Euclidean-distance correspondence

- **Claim:** SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002 - Finite Euclidean-distance correspondence.
- **Forced law:** Euclidean distance correspondence is carried by exact squared coordinate separation when a conventional root would be irrational.
- **Unique survivor:** GEOM-002 uniquely retains exact-squared-distance, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-002: euclidean_distance; exact observation: {"root_imported":false,"squared_distance":25}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:1bff82d89d227affa2d7ba04e408d04ccad713488fadc4010d4efff0ad681c2a`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:c11a850832b3b543e18b06dd050a10cc01b1f747789da5eb32f071377662ea48`; engine receipt `sha256:476f77266580be2d60d3958a6ac59279ff4c0f30757ec7a0034dc792cf4a0b3b`.

SFT-MATH-OBL-GEOM-003 - Affine combinations and affine invariance

- **Claim:** SFT-MATH-GEOM-AFFINE-INVARIANCE-003 - Affine combinations and affine invariance.
- **Forced law:** Affine combinations use exact parts summing to the whole and remain invariant under common translation.
- **Unique survivor:** GEOM-003 uniquely retains exact-part-affine-combination, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-003: affine_invariance; exact observation: {"midpoint":[2,3],"translated_midpoint":[4,4]}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:097686c0b372ed45ed1f75ec467dda0caab5f62b0fc7d294ef418b20a02de0c7`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical

sha256:208ab747f41a94756a58468cfa4329b4af0e10bd8ee79fb2c154bfa366b9d452; engine receipt sha256:10e679a7ee169bd7d4db58480e7b08a4db29a2bcce9a0071775a143ea077fbf6.

SFT-MATH-OBL-GEOM-004 - Projective incidence and perspective

- **Claim:** SFT-MATH-GEOM-PROJECTIVE-PERSPECTIVE-004 - Projective incidence and perspective.
- **Forced law:** Projective correspondence retains point-line incidence under homogeneous relabelling without importing continuum coordinates.
- **Unique survivor:** GEOM-004 uniquely retains projective-incidence-custody, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-AFFINE-INVARIANCE-003; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** GEOM-004: projective_incidence; exact observation: {"line_per_pair":1,"lines":7,"points":7,"points_per_line":3}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:3e65d69cd106fecd0337c68752e6215ecbe514e1f862fb5bec66e3e2b29455b6; independent sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9; empirical sha256:a1cd2801ca5d152846617b12cd62477bea4c715e4dfd13821bf7b01303d9be7a; engine receipt sha256:aa1b4a590e43ae08efb84276484f8e9bf0bd6fa76611eea1d1dff07ccb996a88.

SFT-MATH-OBL-GEOM-005 - Convex hulls and separation

- **Claim:** SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005 - Convex hulls and separation.
- **Forced law:** A convex hull contains exactly the lawful exact-part combinations; separation is witnessed by an exact support comparison.
- **Unique survivor:** GEOM-005 uniquely retains exact-convex-combination-separation, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-PROJECTIVE-PERSPECTIVE-004; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** GEOM-005: convex_separation; exact observation: {"centre":[[1,2],[1,2]],"external_coordinate":2,"separated":true}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:f66ff5f987206e1f420e59f5b24166886ef870695326a70c6fb866a2312ccc44`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:4d10622170d80b61aceaeee6f0ec43e310ca33fab2a6c2522329d45c80ff02af`; engine receipt `sha256:6c66327e91dff8df94b11eddc42778bb8ab3d86e174401333decd8bd3d033aa5`.

`SFT-MATH-OBL-GEOM-006` - Discrete geometry and lattice polytopes

- **Claim:** `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` - Discrete geometry and lattice polytopes.
- **Forced law:** A lattice polytope is the finite hull of exact coordinate carriers with boundary and interior custody.
- **Unique survivor:** GEOM-006 uniquely retains lattice-point-polytope-incidence, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-006: lattice_polytope; exact observation: {"area":1,"boundary_points":4,"interior":"absence"}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5d8402cc0938770edc52a0250fbc25efd67c79aa2cfa9e1c552a60d7e008932a`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:3419b3406b8fdd4e6a5410d4e053298750e50147bdcd6ae7f35c39048a3b89df`; engine receipt `sha256:5f1da049f9226f536b49893ca4fa13e2f7f454f53bbcb2f6ec3ec46ec650e7d1`.

`SFT-MATH-OBL-GEOM-007` - Polyhedral faces and Euler-type incidence

- **Claim:** `SFT-MATH-GEOM-POLYHEDRAL-EULER-INCIDENCE-007` - Polyhedral faces and Euler-type incidence.
- **Forced law:** Polyhedral structure is the complete face-edge-vertex incidence ledger and Euler correspondence follows from its exact alternating orientation labels.
- **Unique survivor:** GEOM-007 uniquely retains face-edge-vertex-ledger, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-007: polyhedral_incidence; exact observation: {"edges":12,"euler_total":2,"faces":6,"vertices":8}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem

answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:e549b11c36647af1778811717707a4b198cc1dbdd8757a59a2e7e75d799cd786`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:d018845e5e28326ebb9c1ea5501c4cbb21b5ebbf85c5d1dcabd358e59d11c7b7`; engine receipt `sha256:f7150f1b221d7610f49c7e175d12578d3c5e646cf4834b803f2fd87254335532`.

`SFT-MATH-OBL-GEOM-008` - Computational geometry predicates

- **Claim:** `SFT-MATH-GEOM-COMPUTATIONAL-PREDICATES-008` - Computational geometry predicates.
- **Forced law:** A computational geometry predicate is an exact finite comparison whose boundary and tie behavior are registered before evaluation.
- **Unique survivor:** GEOM-008 uniquely retains exact-geometric-decision-predicates, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-POLYHEDRAL-EULER-INCIDENCE-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-008: computational_predicates; exact observation: {"nearest":[1,1],"query":[2,2],"unique":true}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bb3633c499cdab0f79db56fca6086d15b0db2d27d3aec1b06b2c74c0ed3bdb89`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:fc5b7dfba1882d6ac0a8bc404df25087432a2750c502a2a455a8e56cada61de5`; engine receipt `sha256:88bd71da009f553ffc0ea404616ad3a4269dffd0c2e3d233ed1f24b8abb30774`.

`SFT-MATH-OBL-GEOM-009` - Exact orientation and intersection tests

- **Claim:** `SFT-MATH-GEOM-ORIENTATION-INTERSECTION-009` - Exact orientation and intersection tests.
- **Forced law:** Orientation is a held/opposed product comparison, and intersection retains the exact common carrier or exact-part coordinate.
- **Unique survivor:** GEOM-009 uniquely retains held-opposed-orientation-intersection, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-COMPUTATIONAL-PREDICATES-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-009: orientation_intersection; exact observation: {"intersection":[2,2],"orientation":"held"}; complete family observation custody: 16 records; all rows preserved; complete

finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:fb03dc30cd87ee867a8e048701cae94c1038221da4b73baa94b155fd5b26f0f2`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:c38caa404f3beefc43eed1cc86145828d94130921fc0fd714a474019160e6d72`; engine receipt `sha256:24596a0663ce622201d42208bb517558807fa73ee356f40d32ea7466a6045b93`.

`SFT-MATH-OBL-GEOM-010` - Algebraic-geometry solution-set correspondence

- **Claim:** `SFT-MATH-GEOM-ALGEBRAIC-SOLUTION-SET-010` - Algebraic-geometry solution-set correspondence.
- **Forced law:** Algebraic geometry correspondence is the complete exact solution-set census of registered polynomial relations over lawful carriers.
- **Unique survivor:** GEOM-010 uniquely retains finite-polynomial-solution-census, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-ORIENTATION-INTERSECTION-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-010: algebraic_solution_set; exact observation: {"ordered_solutions":4,"relation":"x*y=6"}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:b7dba10a2e1694d1d37413939c9c06ba63922e68ae8c9473a82d6bb1744ae37a`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:65d5a1df85c8cd8603f64405bfcdc6584bc9ba1bbe9590e7fc8c51ac681e7a90`; engine receipt `sha256:cfa6ae0f410c735a5d4a2a782ae80bd70e149682beec68d45bb58fa3a740b1d2`.

`SFT-MATH-OBL-GEOM-011` - Differential-geometry finite-chart correspondence

- **Claim:** `SFT-MATH-GEOM-DIFFERENTIAL-FINITE-CHART-011` - Differential-geometry finite-chart correspondence.
- **Forced law:** A finite chart correspondence retains local coordinate relations and exact transition maps without presuming a continuum manifold.
- **Unique survivor:** GEOM-011 uniquely retains finite-chart-transition-custody, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 ->`

SFT-MATH-GEOM-ALGEBRAIC-SOLUTION-SET-010; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.

- **Post-registry observations:** GEOM-011: finite_charts; exact observation: {"adjacency_preserved":true,"chart_points":4,"transition":[2,3]}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:02781279644241e82edbabd7be750755f5d0c9dbdc91a17d4d3ea4be63cc4e22; independent sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9; empirical sha256:3c91e4b6b27515f732aafc470df0329b0a8b5cc2cd52230befc3316141d7c6b5; engine receipt sha256:f3e05eae5cba1877fe920845cfd8b489897d358f61b57737840d3266a4d89fb6.

SFT-MATH-OBL-GEOM-012 - Curvature by exact finite transport

- **Claim:** SFT-MATH-GEOM-CURVATURE-FINITE-TRANSPORT-012 - Curvature by exact finite transport.
- **Forced law:** Curvature is reconstructed as the exact retained discrepancy after closed finite transport or incidence-angle custody.
- **Unique survivor:** GEOM-012 uniquely retains finite-transport-curvature-ledger, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-DIFFERENTIAL-FINITE-CHART-011; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** GEOM-012: finite_curvature; exact observation: {"curvature_each":[1,2],"total":2,"vertices":4}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:4f80c4c4ba3e66ad69eb2a991482f2ec32410ba53db7d7ec8222be33a02fb4dd; independent sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9; empirical sha256:283d2e14420eaca43660d015383edf98c9f8b0a9b626b4109b4fc70fa929ea72; engine receipt sha256:c402bd060dc95d06eef218f4ecbbeea4cb2d907e73d3d0208703f4cbeba24f34.

SFT-MATH-OBL-GEOM-013 - Metric geometry and geodesic correspondence

- **Claim:** SFT-MATH-GEOM-METRIC-GEODESIC-013 - Metric geometry and geodesic correspondence.
- **Forced law:** A geodesic is the least exact path separation in the complete generated path support.
- **Unique survivor:** GEOM-013 uniquely retains least-exact-path-separation, complete configuration custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-CURVATURE-FINITE-TRANSPORT-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-013: metric_geodesic; exact observation: {"end":[3,3],"length":4,"start":[1,1]}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:87fe3a57cad1740f60f19fc8e4c214d3136c9edbca713c4dad25744ebf4af613`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:3136951163bb18af5a64df9cf7cfed31fcc0997bcc35b1e17d7d8b3680397c48`; engine receipt `sha256:09605a920fde7f770e6951a2b9d2d1ec641a89dd55fb2a7c00ad3ae9c7766007`.

`SFT-MATH-OBL-GEOM-014` - Fractal and self-similar geometry

- **Claim:** `SFT-MATH-GEOM-FRACTAL-SELF-SIMILAR-014` - Fractal and self-similar geometry.
- **Forced law:** Self-similar geometry is an exact replacement recursion with every finite depth certified; irrational dimension labels are not proof scalars.
- **Unique survivor:** GEOM-014 uniquely retains finite-depth-self-similar-replacement, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-METRIC-GEODESIC-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-014: self_similarity; exact observation: {"depth_counts":[1,3,9,27],"exact_part_scale":[1,2]}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2ee776d7c96c7b4066306af623208a4565a82df1a603ac62a7efdd861e15692d`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:c635a0935bd8620e40d6cba6e643acbe930e36e72870b9abd4c0e8fe43eba2c1`; engine receipt `sha256:695c16a32dd6cc22ac2f0fca7817a89088b58cd1a88b7f6d1c7b3d1c7d104a60`.

`SFT-MATH-OBL-GEOM-015` - Packing, covering and tessellation

- **Claim:** `SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015` - Packing, covering and tessellation.
- **Forced law:** Packing retains disjoint interiors, covering retains complete support, and tessellation requires both at the declared boundary.

- **Unique survivor:** GEOM-015 uniquely retains disjoint-interior-complete-cover, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-FRACTAL-SELF-SIMILAR-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-015: tessellation; exact observation: {"cells":9,"columns":3,"complete_cover":true,"rows":3}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:671d0328bd3999b0c7a8aec8396f3198e58ef882c6a60f465e269099971dd124`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:5bd06c72774516b21f112bf26e22bde7b50d77a1e7a2cbe0c51e9b0236a5a30b`; engine receipt `sha256:f0e9e9bfb4de8cec2f7419c8a96325a040497b09b4cfa3457fc6fe4b2799ec25`.

`SFT-MATH-OBL-GEOM-016` - Geometric transformation groups

- **Claim:** `SFT-MATH-GEOM-TRANSFORMATION-GROUPS-016` - Geometric transformation groups.
- **Forced law:** A geometric transformation group is the complete reversible action preserving the registered incidence or metric relation.
- **Unique survivor:** GEOM-016 uniquely retains reversible-incidence-transformations, complete configuration custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001 -> SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003 -> SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** GEOM-016: transformation_group; exact observation: {"square_symmetries":8}; complete family observation custody: 16 records; all rows preserved; complete finite configuration enumeration supplies the observation; no fitted coordinate, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-GEOMETRY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum geometry, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum manifold or infinite tessellation; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2b91d1ba542627875c747b1328d59bb42daba8c7e6f19c66df5dd8ab3eb7e385`; independent `sha256:c05b253043fdd4dccbc518c94010e9e82f42994dde21ca5696d46c43fa49b9a9`; empirical `sha256:cc03a3a3009394fe833342adea0e57ba8e3a578d44262fd151edcbc979a94e1c`; engine receipt `sha256:f601584092d818d2180fa5b6b32aff5089152b8a2b29d31a826ad2f106658c2e`.

40.12 Family `TOPO` - 14/14 complete

This family contributes 3,584 completely decided candidates, 14 unique survivors and 56 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-TOPO-001` - Open-set correspondence on generated supports

- **Claim:** `SFT-MATH-TOPO-OPEN-SET-SUPPORT-001` - Open-set correspondence on generated supports.
- **Forced law:** An open-set correspondence is a generated observation family containing absence and whole and closed under declared unions and finite intersections.
- **Unique survivor:** TOPO-001 uniquely retains finite-open-support-closure, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-001: open_support; exact observation: {"carriers":2,"closed_under_required_operations":true,"open_sets":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:dcd17c115e1b3efb60d0af8e6944eb20b42713999141330dffd32f9ea7825eea`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:7a427bfd46ec1d1c578718eb43a0e3d7185725e56ff04a3b374785d95c37136c`; engine receipt `sha256:087501b54410e5b0cc2582c58f56e0ae422ff60ae8e112669ad6257904b828f6`.

`SFT-MATH-OBL-TOPO-002` - Continuity as distinction-preserving transport

- **Claim:** `SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002` - Continuity as distinction-preserving transport.
- **Forced law:** A transport is continuous when every observable open distinction has an open preimage, so no hidden discontinuity is introduced.
- **Unique survivor:** TOPO-002 uniquely retains open-preimage-continuity, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-OPEN-SET-SUPPORT-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-002: continuity; exact observation: {"continuous":true,"open_preimages":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:56813a2b0da0dcab7c07323466756483211a4cfc96c318df7437e4a6578e7ba5`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:0786269f18283f635ed697d46b241576f410fcd9ba7eb5844b7f0b42d9478628`; engine receipt `sha256:573c965f1d7c4ca35f06606ef420cbd243824da0a33ef5239ed3eeda9b39bc11`.

`SFT-MATH-OBL-TOPO-003` - Compactness finite-subcover correspondence

- **Claim:** `SFT-MATH-TOPO-COMPACT-FINITE-SUBCOVER-003` - Compactness finite-subcover correspondence.
- **Forced law:** Compactness correspondence requires every generated open cover to retain a finite subcover; finite supports close by complete cover enumeration.
- **Unique survivor:** TOPO-003 uniquely retains finite-cover-subcover-custody, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-003: compactness; exact observation: {"cover_sets":2,"finite_subcover":2,"support":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f57a62ebfc8fa2ee43b7c8c13666b1caef4f93b9484137f9e085a03158d76654`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:77ba0c8eca8de509feb9c02e73567b1af000c9eaf2e2e3e5f7ec5f1918a1c408`; engine receipt `sha256:093d2d6a7de58e145ed538bc8ca98e21197f57236c9895ecc688896aef4f0da1`.

`SFT-MATH-OBL-TOPO-004` - Connectedness and component structure

- **Claim:** `SFT-MATH-TOPO-CONNECTED-COMPONENT-004` - Connectedness and component structure.
- **Forced law:** Components are the unique maximal carriers linked by composable incidence paths.
- **Unique survivor:** TOPO-004 uniquely retains maximal-connected-components, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-COMPACT-FINITE-SUBCOVER-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-004: connectedness; exact observation: {"components":2,"vertices":5}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:6ea0fe4f16b9e1fb9e019f92e79246ed217296980acfa86e18e7242671bd26fd`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:45838a9db3115c844e18f029884eb3c41b53214d79e4fe1e823a41f2946b6d5a`; engine receipt `sha256:15563c18cf9cbc4909fed4801546848b883a4a532e5b92a19d7f865abb66942a`.

`SFT-MATH-OBL-TOPO-005` - Separation properties on finite observations

- **Claim:** `SFT-MATH-TOPO-SEPARATION-FINITE-005` - Separation properties on finite observations.
- **Forced law:** Separation properties are exact statements about which generated observations distinguish each carrier pair.
- **Unique survivor:** TOPO-005 uniquely retains finite-observation-separation, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-CONNECTED-COMPONENT-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-005: separation; exact observation: {"carriers":3,"ordered_distinct_pairs":6,"separated":true}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e9604fa3013eccc168e818dc331919b92f3a703d2fe253d514a3ccb4dbb97051`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:b61eaf89b292e9f1cf013689c72493d1f4a1ce99d2965712a06ded591ed00376`; engine receipt `sha256:c5e7396091b5d6f67fbda3a6b709c3c2e0a65a549376de10eb271a3bc52e5f40`.

`SFT-MATH-OBL-TOPO-006` - Product, quotient and subspace topology correspondence

- **Claim:** `SFT-MATH-TOPO-PRODUCT-QUOTIENT-SUBSPACE-006` - Product, quotient and subspace topology correspondence.
- **Forced law:** Product, subspace and quotient topology are forced by paired observations, exact restriction and class-saturated projection.
- **Unique survivor:** TOPO-006 uniquely retains product-restriction-quotient-observation, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-SEPARATION-FINITE-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-006: product_quotient_subspace; exact observation: {"product_carriers":6,"quotient_classes":2}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT

number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:d37eae95cca3b615d75dfd41389793852e22654e08b521be3747577fd624f468`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:05021d14c5140a8af8b9d4d9e2476599cbec9e931b3c659a0a7438e1912a0412`; engine receipt `sha256:a82e2923c6b54b8bc343741e75582056e2c37a5058376ca947c0c07d15cff42b`.

`SFT-MATH-OBL-TOPO-007` - Simplicial complexes and incidence

- **Claim:** `SFT-MATH-TOPO-SIMPLICIAL-INCIDENCE-007` - Simplicial complexes and incidence.
- **Forced law:** A simplicial complex is a generated face family closed under every retained subface.
- **Unique survivor:** TOPO-007 uniquely retains downward-closed-simplex-incidence, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-PRODUCT-QUOTIENT-SUBSPACE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-007: simplicial_complex; exact observation: {"downward_closed":true,"nonempty_faces":7,"vertices":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:536cb7a681400b4738f3080640ef9cd31393d6d96a7687ee81ea9eb7178662af`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:76cbbf0bed7639b42a72c2c10676940f1c318cdfbc321df0a4d4d015b801449c`; engine receipt `sha256:8ab47bcdc0b0a32ffe001651b90b87a33462ee55f203ffa30300664c917ee346`.

`SFT-MATH-OBL-TOPO-008` - Homotopy path-deformation correspondence

- **Claim:** `SFT-MATH-TOPO-HOMOTOPY-PATH-DEFORMATION-008` - Homotopy path-deformation correspondence.
- **Forced law:** Finite homotopy correspondence is an endpoint-preserving sequence of exact face-supported path deformations.
- **Unique survivor:** TOPO-008 uniquely retains finite-face-path-deformation, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-SIMPLICIAL-INCIDENCE-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-008: homotopy; exact observation: {"endpoints_preserved":true,"face_moves":1,"paths":2}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:74865fe0e1e40e293a417c43bfe7542fe36dccc9c632149181a3284329eae504`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:600b707b48f918839923d05ff7978383b6301232578f70b4cc537cfa6a0db543`; engine receipt `sha256:e9a150b12a6b2c46ae0e816f04f943af98073118526fbdc446d96bf870f1d580`.

`SFT-MATH-OBL-TOPO-009` - Fundamental-cycle and group correspondence

- **Claim:** `SFT-MATH-TOPO-FUNDAMENTAL-CYCLE-GROUP-009` - Fundamental-cycle and group correspondence.

- **Forced law:** Fundamental-cycle correspondence retains closed paths under composition and held reversal modulo witnessed face deformation.

- **Unique survivor:** TOPO-009 uniquely retains cycle-composition-reversal, complete topological custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-HOMOTOPY-PATH-DEFORMATION-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** TOPO-009: fundamental_cycle; exact observation: {"components":1,"cycle_rank":1,"edges":3,"vertices":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:114651158da3821832333b3c5b8ea59c265498b0330dc866e91666a9fc4d8c9a`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:ab28c2bce8aab0ef29b5d04b9e02072b4f7492d5ee9b8d3ec8bd81fffde8ba90`; engine receipt `sha256:70093215c70da4bb5253edf8b7a016a6788c5158c670ce96b24413a952a81ff4`.

`SFT-MATH-OBL-TOPO-010` - Homology and boundary composition

- **Claim:** `SFT-MATH-TOPO-HOMOLOGY-BOUNDARY-010` - Homology and boundary composition.

- **Forced law:** Homology correspondence follows from exact chains and a boundary operation whose second application is structural absence.

- **Unique survivor:** TOPO-010 uniquely retains boundary-of-boundary-absence, complete topological custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-FUNDAMENTAL-CYCLE-GROUP-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** TOPO-010: homology; exact observation: {"second_boundary":"absence","triangle_boundary_edges":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:805c1a2a02c4d5698adb8e9e878ce151740c7d36ec55a97fce6d5a5f8e4b46df`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:3414b74272f4ad28ffda4dee14c8c19a29dcc36104e743367cb689b90b8db31e`; engine receipt `sha256:01fa67cf3367f6ba1401e220a59a69e8e81be36b53ae713f95149e6638f3286b`.

`SFT-MATH-OBL-TOPO-011` - Cohomology and dual-observation correspondence

- **Claim:** `SFT-MATH-TOPO-COHOMOLOGY-DUAL-OBSERVATION-011` - Cohomology and dual-observation correspondence.
- **Forced law:** Cohomology correspondence is the dual observation of retained cycles modulo observations generated by lower boundaries.
- **Unique survivor:** TOPO-011 uniquely retains dual-cycle-observation, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-HOMOLOGY-BOUNDARY-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-011: cohomology; exact observation: {"cycle_edges":3,"dual_evaluations":3}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a65cfc85eff403898cf77f55aca2f5c895ce537dc470dfa2d00768aaa26e83c5`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:c39c7a5b7a07935e7928e8a02a55605ddc94bddadbaf6a7da3cfdf5a4d7e1a6a`; engine receipt `sha256:daa2f48449e9a9b802fa8193d8a7298683680fc55118d1963caa2ecf2037475a`.

`SFT-MATH-OBL-TOPO-012` - Manifold finite-atlas correspondence

- **Claim:** `SFT-MATH-TOPO-MANIFOLD-FINITE-ATLAS-012` - Manifold finite-atlas correspondence.
- **Forced law:** A finite-atlas correspondence requires local chart carriers and exact compatible overlap transports without presuming a continuum manifold.
- **Unique survivor:** TOPO-012 uniquely retains finite-local-chart-overlap, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-COHOMOLOGY-DUAL-OBSERVATION-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** TOPO-012: finite_atlas; exact observation: {"charts":4,"local_carriers":3,"overlaps_compatible":true}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:708b4c037915938361a6dbcbd0e4a7196dfa7d1d382b1ed29bd29a466f7cd871`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:44a43ed4db4c4cded76afd9bed74609acff22b38920c13ba756cb50dfce32d89`; engine receipt `sha256:06b0bd43fcc4ff33c4522a44cd3425d14dab1d0c833e8f5993f553567abf6865`.

`SFT-MATH-OBL-TOPO-013` - Knot and link finite-diagram invariants

- **Claim:** `SFT-MATH-TOPO-KNOT-LINK-INVARIANTS-013` - Knot and link finite-diagram invariants.
- **Forced law:** A knot or link invariant is admitted only when retained across every registered local diagram move and relabelling.
- **Unique survivor:** TOPO-013 uniquely retains finite-diagram-move-invariant, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-MANIFOLD-FINITE-ATLAS-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-013: knot_link; exact observation: {"crossings":3,"cyclic_invariant":true,"gauss_word_length":6}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e498e3f245455fb09ee7e33a543861b4b83c12a0c0a766ee0e17d62f176dab2c`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:d3785e4ad344acab3ed71e7f59bbb33d7a8175d9fc69a2019d3e6654a0b7f9b7`; engine receipt `sha256:5896d19c43946aaf7ea803a46d656d83877e3a33bfe835fb25652e7d6b0a0cc0`.

`SFT-MATH-OBL-TOPO-014` - Computational topology and persistent-feature custody

- **Claim:** `SFT-MATH-TOPO-PERSISTENT-FEATURE-CUSTODY-014` - Computational topology and persistent-feature custody.
- **Forced law:** Persistent topology retains every feature birth, continuation and merge across an exact filtration without resurrecting lost distinctions.
- **Unique survivor:** TOPO-014 uniquely retains filtered-feature-birth-merge-ledger, complete topological custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001 -> SFT-MATH-ORDER-CLOSURE-SYSTEM-007 -> SFT-MATH-TOPO-KNOT-LINK-INVARIANTS-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** TOPO-014: persistence; exact observation: {"component_counts":[4,2,1],"resurrection":false}; complete family observation custody: 14 records; all rows preserved; complete deterministic topology enumeration supplies the observation; no fitted invariant, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-TOPOLOGY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum topology, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite cover or continuum manifold; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:b9a16c4e3e0bab0df705a0d96ce2833c70d1e01e6c8bfd7925792744ec1b4e12`; independent `sha256:a16f6c730aeee097503a93d340a6412194787efcf19b204b94e3f1d4f8e150cb`; empirical `sha256:96fe7e722fc273784e6c9afd5897dfd1320195eb971c2895ca0e3ed9da8fe7c1`; engine receipt `sha256:cb7338023bc31ad5a784693d1b348ba04c7808a83dd59d02d8bb91602176fe4a`.

40.13 Family `CALC` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-CALC-001` - Exact finite difference and local change

- **Claim:** `SFT-MATH-CALC-FINITE-DIFFERENCE-001` - Exact finite difference and local change.
- **Forced law:** Finite difference is the exact held/opposed change between adjacent generated values, with magnitude represented nonnegatively.
- **Unique survivor:** CALC-001 uniquely retains oriented-local-change, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-001: finite_difference; exact observation: {"changes":[3,5,7],"values":[1,4,9,16]}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a68728bd6cd24096c6a4568c5a13e9d040247874c68533443d1fb21a83fb6d93`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:d8ae66d6bccfac64dc0a67e6d0e8d040f413276b5e4c2ba17554e8c2cdd296ff`; engine receipt `sha256:507dcb047bbe74bb826a5e6ed551568a04827638d7e02341b616ed21b9441a74`.

`SFT-MATH-OBL-CALC-002` - Higher finite differences and polynomial degree

- **Claim:** `SFT-MATH-CALC-HIGHER-DIFFERENCE-DEGREE-002` - Higher finite differences and polynomial degree.

- **Forced law:** Higher differences iterate exact local change; constant depth certifies the registered finite polynomial-degree correspondence.
- **Unique survivor:** CALC-002 uniquely retains iterated-difference-degree, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-FINITE-DIFFERENCE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-002: higher_difference; exact observation: {"sequence":[1,8,27,64,125],"third_difference":[6,6]}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:94eab315ef54ad0ed5cb2bdd00ecd3e52af1964a8145b8607020a743d12e9f86`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:4c7348bf3b711c425ea50522420a9771098274240fa995d293b4496a0e2fa120`; engine receipt `sha256:06e93214674f5a4c1ee6bea0c9670845143bb157b4562a64c80028e916a297fd`.

`SFT-MATH-OBL-CALC-003` - Accumulation and exact finite sums

- **Claim:** `SFT-MATH-CALC-ACCUMULATION-SUMS-003` - Accumulation and exact finite sums.
- **Forced law:** Accumulation is the complete junction of every generated local contribution with none omitted or duplicated.
- **Unique survivor:** CALC-003 uniquely retains complete-finite-accumulation, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-HIGHER-DIFFERENCE-DEGREE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-003: accumulation; exact observation: {"sum":15,"terms":5}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:601a53686ea2699e202f1abf713ef061f2d741512d5b9561222b60bba48ea46c`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:ed5f1548ba6d6a791d8d0f42dc00293adbb0db37db2264e26e477b1c68f53ff2`; engine receipt `sha256:64ca281a0dcc6cc417fdc9a401001f9ca15e65053726e2f599a1b8dbfd183d1e`.

SFT-MATH-OBL-CALC-004 - Fundamental difference-accumulation correspondence

- **Claim:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004` - Fundamental difference-accumulation correspondence.
- **Forced law:** Difference and accumulation are exact reversals when every intermediate held/opposed change is retained.
- **Unique survivor:** CALC-004 uniquely retains telescoping-reconstruction, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-ACCUMULATION-SUMS-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-004: fundamental_correspondence; exact observation: {"change_sum":15,"initial":1,"terminal":16}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:39dfdf64bf8a9d9dc8b66b63bd3e5095a099a5a090f2c44d941cf717a919473d`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:238e6275deba0d69e42118130aa621d0676c557a6006eabb2d059fe4f9d4f56d`; engine receipt `sha256:e6035a82d486ed59b79d120ed341ccaeee17ab11c8e492b828ce58dec5f17579`.

SFT-MATH-OBL-CALC-005 - Product and composition difference laws

- **Claim:** `SFT-MATH-CALC-PRODUCT-COMPOSITION-LAWS-005` - Product and composition difference laws.
- **Forced law:** Product and composition difference laws follow by complete expansion and regrouping of lawful exact changes.
- **Unique survivor:** CALC-005 uniquely retains exact-product-composition-change, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-005: product_composition; exact observation: {"law_reconstruction":6,"product_change":6}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f79fc82d27cbbd11b5f9ac00b32a0ccf785a855b3c5b84bac0fd75df2b6c31ee`; independent

sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a; empirical sha256:67ac6f9b050c01dab291c1e32d3572f0860d6fb9ce917aeb937e364b991d7931; engine receipt sha256:0b230f54b347912d403b5661a468156a48a9d79b28c09053b666552ff4980873.

SFT-MATH-OBL-CALC-006 - Rational enclosure convergence

- **Claim:** SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 - Rational enclosure convergence.
- **Forced law:** Convergence correspondence is a nested exact rational enclosure whose width refines toward structural absence under a certified rule.
- **Unique survivor:** CALC-006 uniquely retains nested-rational-width-refinement, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-PRODUCT-COMPOSITION-LAWS-005; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** CALC-006: rational_convergence; exact observation: {"nested":true,"refinements":8,"width_rule":"1/n"}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:252ebf5ab958410028efc7f8f1366514e42662368cf26238880653ab6a3b192e; independent sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a; empirical sha256:a07ae9bf16eae3071eb2e5880d1dbfd37e5d5a259c3a03b0405866e081315e8e; engine receipt sha256:09ae9608a9587eadc2d1fe6741cd1840ccb16bf9d0cb4eb8cedb0ca11b35bf24.

SFT-MATH-OBL-CALC-007 - Derivative correspondence through shrinking exact parts

- **Claim:** SFT-MATH-CALC-DERIVATIVE-SHRINKING-PARTS-007 - Derivative correspondence through shrinking exact parts.
- **Forced law:** Derivative correspondence is the stable exact component of local change ratios under a registered shrinking-part family.
- **Unique survivor:** CALC-007 uniquely retains shrinking-part-local-ratio, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** CALC-007: derivative_correspondence; exact observation: {"base_component":4,"retained_part_rule":"1/n"}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem

answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:d6f39db303cf8ece43b7674666a8591e410b26f9a0aef63601bafa6625a25a59`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:933fa2bbfdbab257a19f6122771a5acdd39e7ea5f51021b0d2f94fc0c6451868`; engine receipt `sha256:0df975977b5212414b32cf5991f27778c811c0d19b9c4640a377ef2748077376`.

`SFT-MATH-OBL-CALC-008` - Integral correspondence through refinement sums

- **Claim:** `SFT-MATH-CALC-INTEGRAL-REFINEMENT-SUMS-008` - Integral correspondence through refinement sums.
- **Forced law:** Integral correspondence is the common exact enclosure of lower and upper finite accumulation sums under certified refinement.
- **Unique survivor:** CALC-008 uniquely retains lower-upper-refinement-accumulation, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-DERIVATIVE-SHRINKING-PARTS-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-008: integral_correspondence; exact observation: {"enclosed_value":[1,2],"refinements":7,"width_rule":"1/n"}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:1db1677c90434f8c3aab1c796224269e93c48f874eb50d7f856c663fa1dd166c`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:5566d9a0e58cc2b880581e2a6e6d4dd041d938033fa20d1e5c5ba57b1d00c5c2`; engine receipt `sha256:9dae94ea1c1bb7f5cc021fc5a5e54daa799b2b96cefbab47d0ce3c089b95e3d3`.

`SFT-MATH-OBL-CALC-009` - Multivariable difference and directional change

- **Claim:** `SFT-MATH-CALC-MULTIVARIABLE-DIRECTIONAL-009` - Multivariable difference and directional change.
- **Forced law:** Multivariable change retains the varied coordinate and holds every other coordinate fixed before composition.
- **Unique survivor:** CALC-009 uniquely retains coordinate-directional-change, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-INTEGRAL-REFINEMENT-SUMS-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-009: multivariable_change; exact observation: {"x_direction":3,"y_direction":2}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:84007ef3d8ef65b11770d3703ba527a36072302b51ee38f3ada34354d4a4431e`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:80155898785cbc97a30315c6e1a37a73d6c0e849d38a89e9bb332c379f892f12`; engine receipt `sha256:cd9c565f4ff93e6039bb09ff0e461c6f3a0ebb1201efc18962274d45d27eb633`.

`SFT-MATH-OBL-CALC-010` - Discrete divergence and flux correspondence

- **Claim:** `SFT-MATH-CALC-DIVERGENCE-FLUX-010` - Discrete divergence and flux correspondence.
- **Forced law:** Discrete divergence accumulates local held/opposed flux; internal faces cancel structurally and the boundary record remains.
- **Unique survivor:** CALC-010 uniquely retains internal-cancellation-boundary-flux, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-MULTIVARIABLE-DIRECTIONAL-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-010: divergence_flux; exact observation: {"boundary_magnitude":2,"local_changes":[1,1]}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f65611afa61dd3d1a85050ee001091dd152a74eb2dfeea3277f7578ab8b2e7ec`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:d9cd3b1379183ee1be96f0665e3052f3385fcd4cc7d7b7a059e08bc00ef5e93e`; engine receipt `sha256:d5e9fa1ad6b35ecd29795a2de9d3edda147d6cb081b7eb4bda68b5734e410bfb`.

`SFT-MATH-OBL-CALC-011` - Variational difference and stationary structure

- **Claim:** `SFT-MATH-CALC-VARIATIONAL-STATIONARY-011` - Variational difference and stationary structure.
- **Forced law:** A stationary structure survives every registered local variation and exact extremum comparison in the complete candidate support.
- **Unique survivor:** CALC-011 uniquely retains complete-variation-stationarity, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-DIVERGENCE-FLUX-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** CALC-011: variational_stationary; exact observation: {"candidates":5,"unique_stationary":3}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:01b2151339c534839ee08abd78b33c5035a6b26ce0b940b85356be5495cd94f7`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:657320344d0914208c203e8bbf942cd3e825100b776fbdbff1da4478a0eb4faa`; engine receipt `sha256:def0ef022d70876425bc65b54b4bdd196317d3acb4961a76190a959ca54a02dd`.

`SFT-MATH-OBL-CALC-012` - Continuum-limit admissibility boundary

- **Claim:** `SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012` - Continuum-limit admissibility boundary.
- **Forced law:** A continuum-limit statement is admissible only as a finite-successor enclosure certificate; no completed continuum object is imported.
- **Unique survivor:** CALC-012 uniquely retains certified-enclosure-only-limit, complete refinement custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014 -> SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002 -> SFT-MATH-CALC-VARIATIONAL-STATIONARY-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CALC-012: continuum_boundary; exact observation: {"completed_continuum_claimed":false,"exact_enclosures":8,"nested":true}; complete family observation custody: 12 records; all rows preserved; complete finite refinement enumeration supplies the observation; no fitted limit, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-CALCULUS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported infinitesimal, continuum theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite limit or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:67579e0da5051d8a1c69f2a52dd2f752d07e069c2d29138344f3269232937bd4`; independent `sha256:01c1c959afbdf233f79d9220fb7ebaa838df986886f61f4383b1dbe04c044d6a`; empirical `sha256:3c6e95f71d361eec6b903108c292e986d7a3b44598df646736570686e6842b2d`; engine receipt `sha256:f5917674c7af39fbe08ef5fad2ea7bdd850218b38b72af99fa3d3c75b81b348b`.

40.14 Family `ANAL` - 16/16 complete

This family contributes 4,096 completely decided candidates, 16 unique survivors and 64 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-ANAL-001` - Exact sequence convergence certificates

- **Claim:** `SFT-MATH-ANAL-SEQUENCE-CONVERGENCE-001` - Exact sequence convergence certificates.
- **Forced law:** Sequence convergence is an exact nested enclosure certificate with a lawful refinement rule and no completed continuum premise.
- **Unique survivor:** ANAL-001 uniquely retains nested-sequence-enclosure, complete analytic custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-001: sequence_convergence; exact observation: {"enclosed_carrier":1,"refinements":8,"width_rule":"1/n"}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:21646a365877c34d6ac720aadbdf598847f0e2f2b5225e227f2267663fc1bfa0`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:f4d69628ffdd3c1581b6bf8953e4414fd91d25bfebd0c06603ae327aa434c126`; engine receipt `sha256:ab97f2a13f8e91555f3f654b3d1fce419aaca2e71b6964550a971abe5aa5073b`.

`SFT-MATH-OBL-ANAL-002` - Cauchy-type generated support correspondence

- **Claim:** `SFT-MATH-ANAL-CAUCHY-SUPPORT-002` - Cauchy-type generated support correspondence.
- **Forced law:** Cauchy correspondence requires every generated tail pair to lie within the registered exact tail bound.
- **Unique survivor:** ANAL-002 uniquely retains tail-pair-distance-bound, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-SEQUENCE-CONVERGENCE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-002: cauchy_support; exact observation: {"pair_bounds_preserved":true,"tail_depths":6}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:1b24e1206d8076a0574f07574f3854bc9256501d25e18073ee4167398cc6b4a0`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:dadf83988beb832f6ade7541483f9ee03df3f7498ca2de4f6670b65bb5a16ca0`; engine receipt `sha256:75b3fde9dbd4c94ec57c3214b75bc6956a83af140ea077502b3613a8f2161cb2`.

`SFT-MATH-OBL-ANAL-003` - Completeness correspondence without completed continuum

- **Claim:** `SFT-MATH-ANAL-COMPLETENESS-CORRESPONDENCE-003` - Completeness correspondence without completed continuum.

- **Forced law:** Completeness correspondence is a certified retained carrier common to every nested exact enclosure, without importing a completed continuum.
- **Unique survivor:** ANAL-003 uniquely retains nested-carrier-intersection-certificate, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-CAUCHY-SUPPORT-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-003: completeness; exact observation: {"common_exact_carrier":1,"completed_continuum_claimed":false,"nested_enclosures":8}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a69a30d7a87aec55e1cf740b8e600991fc4b55fd524529cc482bb1de5191b0f6`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:189004648c5b1afb3511b1b60803fc1da006d543064c191612657765394dbcb8`; engine receipt `sha256:096b603a26d7963ec79cbd3a02212e6051b0183db3a19933de1a1bfbfe1dbfe7`.

`SFT-MATH-OBL-ANAL-004` - Series convergence and remainder enclosures

- **Claim:** `SFT-MATH-ANAL-SERIES-REMAINDER-004` - Series convergence and remainder enclosures.
- **Forced law:** A series certificate retains every partial term and an exact remainder enclosure whose refinement is independently witnessed.
- **Unique survivor:** ANAL-004 uniquely retains partial-sum-remainder-ledger, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-COMPLETENESS-CORRESPONDENCE-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-004: series_remainder; exact observation: {"remainder_rule":"1/2^n","truncations":8}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:673aa1810d41d48e4413e1c4db9d5025cf1296710f295c4a5e83d320c347d00c`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:d3c425772e8232c7db525be1081d9721f322b4427ee16c490a4fd3a3288d73a8`; engine receipt `sha256:9ed9ee5292bcc297c39b1079e4de4067f14d8b23cb72f38fb0be39ae8e78bf14`.

SFT-MATH-OBL-ANAL-005 - Power-series finite truncation custody

- **Claim:** SFT-MATH-ANAL-POWER-SERIES-TRUNCATION-005 - Power-series finite truncation custody.
- **Forced law:** Power-series correspondence retains coefficients, exact powers, truncation depth and tail enclosure separately.
- **Unique survivor:** ANAL-005 uniquely retains coefficient-term-tail-custody, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-SERIES-REMAINDER-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-005: power_series; exact observation: {"terms":5,"truncation":[31,16]}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:262a869409452ad48d4761192a2bd23f64ac29462c1f217c640d59441c753dde`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:974573221a43ceff8d285c3e10f590059e546cca47eabd616f740dfe32435d0a`; engine receipt `sha256:b1c09495ffe4a58ae36b3f567a88ed80ddb8ef6a08deba96278aefc38688a0c8`.

SFT-MATH-OBL-ANAL-006 - Functional-space finite-representation correspondence

- **Claim:** SFT-MATH-ANAL-FUNCTIONAL-SPACE-REPRESENTATION-006 - Functional-space finite-representation correspondence.
- **Forced law:** A function space is represented by complete exact value carriers on a generated domain and pointwise lawful operations.
- **Unique survivor:** ANAL-006 uniquely retains finite-function-value-carrier, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-POWER-SERIES-TRUNCATION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-006: functional_space; exact observation: {"binary_functions":8,"domain_carriers":3}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:597bbebeae6183688409ab1bb859a48676a8112f5745e693ef7a8597d177b1a5`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:ce4090fa76b82159c1b295d5fc74f4b1fc6e4cfa57894662b59e46269693e1ce`; engine receipt

sha256:084a51e26bac975e955191ce1370b9f77e05fa555c1685d6b7b93879e027177c.

SFT-MATH-OBL-ANAL-007 - Norm, seminorm and metric correspondence

- **Claim:** SFT-MATH-ANAL-NORM-SEMINORM-METRIC-007 - Norm, seminorm and metric correspondence.
- **Forced law:** Norm, seminorm and metric correspondence is admitted through complete exact size, degeneracy and triangle-relation censuses.
- **Unique survivor:** ANAL-007 uniquely retains exact-size-separation-relations, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-FUNCTIONAL-SPACE-REPRESENTATION-006; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ANAL-007: norm_metric; exact observation: {"metric_triangle":true,"one_norm":6}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:3abe0bfc69ced4c77786b44a2eecb4f1058b2cdbea9c9df2bda73c8046415b5d; independent sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b; empirical sha256:9d4566a493803c4c8a179ce7304b280d6080577bec7f1d7c47a58d24be245975; engine receipt sha256:07d66c5b288d72dcfae3829b10af6fa129b261c4171007ad7ea387dbe1d56eb2.

SFT-MATH-OBL-ANAL-008 - Bounded and compact operator correspondence

- **Claim:** SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008 - Bounded and compact operator correspondence.
- **Forced law:** Bounded and compact operator correspondence retains an exact action bound and the complete finite generated image support.
- **Unique survivor:** ANAL-008 uniquely retains finite-operator-bound-image-custody, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-NORM-SEMINORM-METRIC-007; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** ANAL-008: bounded_operator; exact observation: {"all_generated_inputs_pass":true,"bound":2,"input_dimension":2}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:47bce95dd708eb146f55c9d95101425f9b6093e828da388b15787fdd3a762950`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:6a97594a336a16d977b9837634d989845bdb4aa3c0cc36c59187bb857328e77a`; engine receipt `sha256:0770111ad74222f7d56d86d7798010faa49c9ae065c1b83282848d5792005267`.

`SFT-MATH-OBL-ANAL-009` - Harmonic and Fourier finite-support correspondence

- **Claim:** `SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009` - Harmonic and Fourier finite-support correspondence.
- **Forced law:** Finite harmonic correspondence decomposes a generated signal into exact held/opposed phase components without imaginary proof scalars.
- **Unique survivor:** ANAL-009 uniquely retains held-opposed-harmonic-components, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-009: harmonic_support; exact observation: {"alternating_orientation":"opposed","magnitude":2,"points":4}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:88c70f6f524a13dd9c4409c128e914a6bc78eaaf5a4a35fecab9dea29c344353`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:55d1b67105de7ec494f1e744654a3c76d2550d514895a96dd77f5f62c24311f7`; engine receipt `sha256:963056642837c7794301e5fb6787e9a61f3d61eba474eeaae6be0b67ad63449f`.

`SFT-MATH-OBL-ANAL-010` - Transform inversion on generated supports

- **Claim:** `SFT-MATH-ANAL-TRANSFORM-INVERSION-010` - Transform inversion on generated supports.
- **Forced law:** A transform is admissible when its exact held/opposed component ledger reconstructs every input carrier without loss.
- **Unique survivor:** ANAL-010 uniquely retains exact-transform-reconstruction, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-010: transform_inversion; exact observation: {"exact_reconstruction":true,"input":[3,1],"transformed":[4,2]}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5c812699d25f1dc8cad094f5fdc3856f358ca4a19ba8d724cca5424f6aa6d681`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:93cddba344b7048aa3235fb3eaef4476c718ab023b4e654d42b5b4fc7f8e2b1d`; engine receipt `sha256:d5504ccb27de0398792565ba73813476b426919a9c01fe1490c799478bbb2946`.

`SFT-MATH-OBL-ANAL-011` - Convolution and correlation identities

- **Claim:** `SFT-MATH-ANAL-CONVOLUTION-CORRELATION-011` - Convolution and correlation identities.
- **Forced law:** Convolution and correlation are complete shifted pair accumulations with every cyclic or bounded index retained.
- **Unique survivor:** ANAL-011 uniquely retains complete-shift-pair-accumulation, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-TRANSFORM-INVERSION-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-011: convolution; exact observation: {"identity_support":[1,0,0],"reconstructed":true,"signal":[1,2,3]}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:564488f25726bf5dba043e99d26b38b32111981d4f1cd8670b727b70d0157ac2`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:5df3f76fa5bff847a9d79d5575eeef03bf1bfe5147cd288a764cdfa57d1fcbe1`; engine receipt `sha256:e9d6cff64a2340ae44d6c448e4c3f2f21b13b5401fb80abffd6a9c15a3f6f0a7`.

`SFT-MATH-OBL-ANAL-012` - Orthogonality and basis expansion correspondence

- **Claim:** `SFT-MATH-ANAL-ORTHOGONAL-BASIS-EXPANSION-012` - Orthogonality and basis expansion correspondence.
- **Forced law:** Orthogonal expansion correspondence requires absent cross-pairing and exact reconstruction from all retained basis coordinates.
- **Unique survivor:** ANAL-012 uniquely retains orthogonal-coordinate-reconstruction, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-CONVOLUTION-CORRELATION-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`;

axioms `0`; free parameters `0`.

- **Post-registry observations:** ANAL-012: orthogonal_expansion; exact observation: {"coordinates":[3,2],"cross_pairing":"absence"}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:93aef7bbc63f395f1fd29e514f0dce2a10b56f7d4f9027ee9929fe1b496a291f`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:7e2cce1a4282be3558ab2812a2fef41f64479971565ceb4babf91108e70f4955`; engine receipt `sha256:be232594554c9502d6329fe8b147c2ad5c13efb6f0aa44a1fb6cce82be5be7ec`.

`SFT-MATH-OBL-ANAL-013` - Distributional and weak-observation correspondence

- **Claim:** `SFT-MATH-ANAL-DISTRIBUTIONAL-WEAK-OBSERVATION-013` - Distributional and weak-observation correspondence.
- **Forced law:** Weak or distributional correspondence is only an exact action on registered test-function carriers, not an imported generalized object.
- **Unique survivor:** ANAL-013 uniquely retains exact-test-function-pairing, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-ORTHOGONAL-BASIS-EXPANSION-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-013: weak_observation; exact observation: {"pairing":[5,3],"weights":[[1,3],[2,3]]}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:1466516610bf70f88aee6de2f16a74542ee29c86bf34fbd1ec3e9d8fb83cf620`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:c8e601c7fa2a5535614275bc2f49dc8791257400ea9d56f8c4e286be255b1b7e`; engine receipt `sha256:45c9743262487c3681ba1a4d8b290e0a4ccee29cdeae06ef19dfe4e2d66f203a`.

`SFT-MATH-OBL-ANAL-014` - Nonlinear analysis and contraction boundaries

- **Claim:** `SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014` - Nonlinear analysis and contraction boundaries.
- **Forced law:** A contraction boundary is an exact pair-distance reduction law with a separately witnessed fixed carrier.
- **Unique survivor:** ANAL-014 uniquely retains exact-contraction-fixed-carrier, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-DISTRIBUTIONAL-WEAK-OBSERVATION-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-014: contraction; exact observation: {"factor":[1,2],"fixed_carrier":1}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:50ea47c1d824894769941ce4ff35efd4782d3fbf39b4557e1191fce6ca98733d`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:02ac2d77e8cd816528a60ad46a24d103954729fbdb3d8e3f168c9f31236c8b64`; engine receipt `sha256:beaae3861d939920aa65210f3279a53acd8d0416166a75c5f5caa7758ad0d638`.

`SFT-MATH-OBL-ANAL-015` - Complex-analysis held-pair correspondence

- **Claim:** `SFT-MATH-ANAL-COMPLEX-HELD-PAIR-015` - Complex-analysis held-pair correspondence.
- **Forced law:** Complex correspondence uses exact held phase-labelled pairs and structural opposition; no imaginary or negative proof scalar is admitted.
- **Unique survivor:** ANAL-015 uniquely retains period-four-held-phase-pair, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-015: held_phase_pair; exact observation: {"imaginary_scalar_used":false,"quarter_phase_magnitude":2,"real_orientation":"absence"}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:031e1373016d701f609aa14ef0591d23fdb8334339a4533635e24d422ad0fb2e`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:fa7859100ba289da3706308eaf976b05c9b0ec2981340b0ae046c1d4081d3439`; engine receipt `sha256:5d105963b77ef66e5184e30d31c9be4a513602bd2544b90ec59258defa6fad5d`.

`SFT-MATH-OBL-ANAL-016` - Operator spectral-measure correspondence

- **Claim:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016` - Operator spectral-measure correspondence.
- **Forced law:** Spectral-measure correspondence is a positive exact weight ledger over independently witnessed invariant operator modes.

- **Unique survivor:** ANAL-016 uniquely retains positive-invariant-weight-ledger, complete analytic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006 -> SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014 -> SFT-MATH-ANAL-COMPLEX-HELD-PAIR-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** ANAL-016: spectral_measure; exact observation: {"first_moment":14,"modes":[2,3],"total":5,"weights":[1,4]}; complete family observation custody: 16 records; all rows preserved; complete finite support and truncation enumeration supplies the observation; no fitted limit, imaginary scalar, imported continuum or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-ANALYSIS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum, infinity, imaginary scalar, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite sequence space or continuum function space; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:96a3a861bb62a3c33cc5ff17171f536d587278fe7b2f5e91961754b2d941bae9`; independent `sha256:66fef03aa56dea86cf65267ee9f3b98103e1e38e3f5d76cedd8b8082452b8c8b`; empirical `sha256:3eb01a43ea2dbbb344f2c2bd8ea31ee2b056bb296392a20b5ea4bacc464ef733`; engine receipt `sha256:2b900f3b430de0fcb8c0be9b27c90e8fa9d99b870d3cb197a23db3ad0449a7a5`.

40.15 Family `EQN` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-EQN-001` - Ordinary difference-equation structure

- **Claim:** `SFT-MATH-EQN-ORDINARY-DIFFERENCE-001` - Ordinary difference-equation structure.
- **Forced law:** An ordinary difference equation is a registered one-step state relation whose complete finite solution path is generated from initial custody.
- **Unique survivor:** EQN-001 uniquely retains one-step-state-relation, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-001: ordinary_difference; exact observation: {"initial":1,"solution":[1,2,4,8,16],"transition":"twice-current"}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e49c012e72b33711ea4a1daf2b860410328257bfc04eb376c2bdf8b5ff692c1c`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:c52154b7030329339f2cc916415a4784d55dd772a33fc279e7883857cc816c64`; engine receipt

sha256:4fc9003cf83d90dcf955003cca7001e1547e5d7e1602af161870186ae1fcf66b.

SFT-MATH-OBL-EQN-002 - Ordinary differential-equation correspondence

- **Claim:** SFT-MATH-EQN-ORDINARY-DIFFERENTIAL-002 - Ordinary differential-equation correspondence.
- **Forced law:** Ordinary differential correspondence is admitted only through exact shrinking-step local-ratio enclosures and finite successor certificates.
- **Unique survivor:** EQN-002 uniquely retains finite-step-local-ratio-correspondence, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** EQN-002: ordinary_differential_correspondence; exact observation: {"carriers":["1","3/2","2"],"local_ratio_equals_carrier":true,"step_denominators":[1,2,3,4,5,6]}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:819de58fa53dd0b9340f0c5bacb04568df2cebc1ed3382954e9f51a2cacb29f5; independent sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851; empirical sha256:ece92f0035d519406c5106aac1890c3b9dbfae821cd0d9b721c7a48ce253064a; engine receipt sha256:3c2ca60887577a62c86587512cec9143f1cb08293ac1e0267f4a4c5466be5511.

SFT-MATH-OBL-EQN-003 - Partial difference-equation structure

- **Claim:** SFT-MATH-EQN-PARTIAL-DIFFERENCE-003 - Partial difference-equation structure.
- **Forced law:** A partial difference equation retains every coordinate direction and applies exact local change only along the declared direction.
- **Unique survivor:** EQN-003 uniquely retains multi-coordinate-local-relation, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-ORDINARY-DIFFERENTIAL-002; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** EQN-003: partial_difference; exact observation: {"change_each_direction":1,"grid_rule":"i-plus-j","i_domain":[1,2,3],"j_domain":[1,2,3]}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:1f985366b3183b28299470d0d294a9b8d1cf4e28246b391fd87746daf9e3c2eb`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:0ad58c04e4b31ffe959424b5fc606cc2c42feb8b62f6304ee0a86c6e5b7f0f7e`; engine receipt `sha256:78591458da34ac31e6419e6670802ae5100128e808c08b987cf5b34cc3324e59`.

`SFT-MATH-OBL-EQN-004` - Partial differential-equation correspondence

- **Claim:** `SFT-MATH-EQN-PARTIAL-DIFFERENTIAL-004` - Partial differential-equation correspondence.
- **Forced law:** Partial differential correspondence is a compatible family of exact multidirectional difference relations under certified refinement.
- **Unique survivor:** EQN-004 uniquely retains refined-multidirectional-enclosure, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-PARTIAL-DIFFERENCE-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-004: partial_differential_correspondence; exact observation: {"all_grid_rows_preserved":true,"grid_rule":"i-plus-j","second_difference_each_direction":"absence"}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:9b00e448373a030051da4e97bad87623d6018d127101890ebfdd06382f676b17`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:2eea74718e317db4e1a1f324c1526b8b16ddad00d09d25a4c749eaeabe819909`; engine receipt `sha256:6e4130baf805346be5ab289307002290c2f9b6833686c72a85f987be1e25e51a`.

`SFT-MATH-OBL-EQN-005` - Boundary and initial record well-posedness

- **Claim:** `SFT-MATH-EQN-BOUNDARY-INITIAL-WELL-POSED-005` - Boundary and initial record well-posedness.
- **Forced law:** Well-posedness requires a retained initial or boundary record, at least one generated solution, uniqueness in the complete solution census and perturbation custody.
- **Unique survivor:** EQN-005 uniquely retains existence-uniqueness-record-custody, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-PARTIAL-DIFFERENTIAL-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-005: boundary_initial_well_posedness; exact observation: {"initial":1,"transition":"successor","unique_solution":[1,2,3,4,5,6]}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT

number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:efcdcfde51147f72a98c7454dd1c4d31364a86277e778eda64b0e417824f15d6`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:92fb3aaaaea5c9d0f2bb70ab3e41592553b97cf2625cf62dca9ad0c2a24b6591`; engine receipt `sha256:201f70839a8f800c9b62f6fc4cb957e2d3330533c87533ea6d83b19d30027900`.

`SFT-MATH-OBL-EQN-006` - Integral-equation correspondence

- **Claim:** `SFT-MATH-EQN-INTEGRAL-CORRESPONDENCE-006` - Integral-equation correspondence.
- **Forced law:** Integral-equation correspondence is a finite accumulation relation with exact kernel, source and refinement custody.
- **Unique survivor:** EQN-006 uniquely retains finite-accumulation-equation, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-BOUNDARY-INITIAL-WELL-POSED-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-006: integral_equation_correspondence; exact observation: {"finite_volterra_solution":[1,2,4,8],"kernel":"unit-retained","source":1}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7ba76da3ebb5ba2b7919404a0247e7bf2db414154e601e688759620f1113af8a`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:2842dcab66c347a7fe5223e9ba54dd1e689da48d3261a1b2036831b8922384e2`; engine receipt `sha256:82ade58665afc230884f81bd940a5ef39596e8c5015fbb861185f237547a1adc`.

`SFT-MATH-OBL-EQN-007` - Functional-equation structure

- **Claim:** `SFT-MATH-EQN-FUNCTIONAL-STRUCTURE-007` - Functional-equation structure.
- **Forced law:** A functional equation is closed only by testing every generated argument composition in its declared domain.
- **Unique survivor:** EQN-007 uniquely retains complete-argument-composition-census, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-INTEGRAL-CORRESPONDENCE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-007: functional_equation; exact observation: {"composition_identity_passed":true,"domain":[1,2,3,4],"function":"two-to-generated-power"}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:02a1798a980ecd51992998621bb1eaa167a02edb5c09005e5b12782e69292fbd`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:f9d4925cae05b60c82c26dc9c17c6014a31c285e56c93bdc1887368b45f9ad58`; engine receipt `sha256:23a0f72d36f5d6653020722360e364cd157fdce162fb6eaf881713a12936432d`.

`SFT-MATH-OBL-EQN-008` - Recurrence-equation solution spaces

- **Claim:** `SFT-MATH-EQN-RECURRENCE-SOLUTION-SPACE-008` - Recurrence-equation solution spaces.
- **Forced law:** A recurrence solution space is the complete family generated from every lawful initial record under one fixed transition relation.
- **Unique survivor:** EQN-008 uniquely retains initial-record-recurrence-space, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-FUNCTIONAL-STRUCTURE-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-008: recurrence_solution_space; exact observation: {"initial_record":[1,1],"solution":[1,1,2,3,5,8]}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:29347520d64a9eaef7ea3c8b3b1d94b471b9db5e12d8788c22fc7a581bcc66fc`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:7d5015c3e8805c3fc5e39eca27cf9b67913a7d75fd5f8bc42f9ab8dad5b78ffa`; engine receipt `sha256:23579f5dbdd465547c8cce50c194a49101b72e8c30cd8dbd336ad962e8d2e802`.

`SFT-MATH-OBL-EQN-009` - Green-response finite correspondence

- **Claim:** `SFT-MATH-EQN-GREEN-RESPONSE-009` - Green-response finite correspondence.
- **Forced law:** Finite Green correspondence is the exact response to each generated impulse and its lawful accumulation reconstructs every source response.
- **Unique survivor:** EQN-009 uniquely retains impulse-response-superposition, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-RECURRENCE-SOLUTION-SPACE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** EQN-009: green_response; exact observation: {"exact_difference_reconstruction":true,"response":[1,1,3,3],"source":[1,0,2,0]}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5ff4a7d4187853d686641beb6536e7c649eaccde9fa9bb762d00d78277dbd03a`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:5c945e7b1afd425cec143eecc7b04aeae55b45e838285f19d9a1fd862f8cbb59`; engine receipt `sha256:9dcd5a3ee4736a56455470116d2c72297f47e59f886ce1950e123bdcf2565225`.

`SFT-MATH-OBL-EQN-010` - Conservation-law weak correspondence

- **Claim:** `SFT-MATH-EQN-CONSERVATION-WEAK-010` - Conservation-law weak correspondence.
- **Forced law:** Weak conservation correspondence retains total content while internal held/opposed flux cancels structurally and boundary flux remains.
- **Unique survivor:** EQN-010 uniquely retains boundary-flux-conservation-ledger, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-GREEN-RESPONSE-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-010: conservation_weak_correspondence; exact observation: {"after":[3,2],"before":[2,3],"retained_total":5}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:9d32c0bbdc29a0c72ae6078b3dc1a8299fb704c214cff444a9eb281f4eb2a8dd`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:634a0e8ec8cfc283d98cff9c9464ad87792ba80f7690ecc2480a12164b7064bd`; engine receipt `sha256:4fbb4a7f64ef3694210546883ed3742b90e0a11c995f921e5d3e9269b94522d3`.

`SFT-MATH-OBL-EQN-011` - Stability and perturbation enclosures

- **Claim:** `SFT-MATH-EQN-STABILITY-PERTURBATION-011` - Stability and perturbation enclosures.
- **Forced law:** Stability is an exact bound transporting every registered input perturbation to its solution separation.
- **Unique survivor:** EQN-011 uniquely retains exact-solution-perturbation-bound, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-CONSERVATION-WEAK-010`;

registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** EQN-011: stability_perturbation; exact observation: {"distance_ratio":"1/2","input_carriers":["1/4","1/2","3/4","1"],"solution_map":"half-contraction"}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5ceabf7977354e2848d73022829628efbaf8bdf0d0aee9b1cf3278b4e02d87d5`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:072dad909d2735c9aba5c84ad37d93a436223eb385a059d2327e5084d1d07487`; engine receipt `sha256:b5862177e99aea7b7d3f1f03e5d062b99b4072dcfde36163dc8ed7f07544c36f`.

`SFT-MATH-OBL-EQN-012` - Existence, uniqueness and blow-up boundaries

- **Claim:** `SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012` - Existence, uniqueness and blow-up boundaries.
- **Forced law:** Existence and uniqueness are certified at every registered finite depth; blow-up claims require a separate depth-independent enclosure and cannot be inferred from growth alone.
- **Unique survivor:** EQN-012 uniquely retains finite-depth-existence-uniqueness-boundary, complete solution custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004 -> SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014 -> SFT-MATH-EQN-STABILITY-PERTURBATION-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** EQN-012: existence_uniqueness_blowup_boundary; exact observation: {"finite_solution":[2,4,16,256,65536],"initial":2,"transition":"square-current","unrestricted_blowup_claimed":false}; complete family observation custody: 12 records; all rows preserved; complete finite equation and solution enumeration supplies the observation; no fitted parameter, imported continuum theorem or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-EQUATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no imported continuum equation, theorem answer, fitted parameter or opaque solver selects the law; host 0 displays absence or counts artifacts only; it is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed infinite solution trajectory or continuum domain; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:d0e86bcc2c77a9b246965b9e9e747ec46b52a9f304b78d285bbd46c2b203bb47`; independent `sha256:001ce4143fa695176d2daae0582c2aa2f5ade841883fc056923d9098e6414851`; empirical `sha256:27b0716f13e096e67e9958a6c17477947f9d908a4f2d9949b2c84cb3f1cfe24d`; engine receipt `sha256:a5b9056f828669049f5d02eb1c9493657a229b7da2454f7cea9353b1b7665579`.

40.16 Family `MEAS` - 10/10 complete

This family contributes 2,560 completely decided candidates, 10 unique survivors and 40 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-MEAS-001` - Finite measure and exact support weight

- **Claim:** `SFT-MATH-MEAS-FINITE-SUPPORT-WEIGHT-001` - Finite measure and exact support weight.
- **Forced law:** Finite measure is the exact positive weight assigned to every generated support with the complete carrier normalized to one.

- **Unique survivor:** MEAS-001 uniquely retains exact-finite-support-weight, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** MEAS-001: finite_support_weight; exact observation: {"complete_weight":1,"support":[1,2,3],"weights":["1/6","2/6","3/6"]}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:752122ae353ece3b0d4dea7ac7b460d43bae20a433daf850abf688349ddb9a7b`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:99ac4155fd96cab1af9d0e4a788b2e91745469a1f926bd9f5ed48641592b8009`; engine receipt `sha256:14d25ad66d068b2319cec859ce7d94be7d66cefda1b4721b2ec5719b124b3e1f`.

`SFT-MATH-OBL-MEAS-002` - Additivity on disjoint generated support

- **Claim:** `SFT-MATH-MEAS-DISJOINT-ADDITIVITY-002` - Additivity on disjoint generated support.
- **Forced law:** Disjoint additivity is forced by retaining each support distinction once and composing their exact weights without overlap.
- **Unique survivor:** MEAS-002 uniquely retains disjoint-support-additivity, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-FINITE-SUPPORT-WEIGHT-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** MEAS-002: disjoint_additivity; exact observation: {"all_additive":true,"all_disjoint_pairs_tested":true,"support_count":8}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:0e2c78d95ff25639d174434cf8e1a09722b5440c5f85bfdce81d3fe8d2b84bb5`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:2f754f5fb92ab299dcf29c943dd7d16da6cb3d106c99b5f538dbdd23e77e4f01`; engine receipt `sha256:c2c5612d75dce3a55857b1e506e47f3b86496117aabcc4e83cb8b0b6d5a8aaf0`.

SFT-MATH-OBL-MEAS-003 - Outer-measure and covering correspondence

- **Claim:** SFT-MATH-MEAS-OUTER-COVERING-003 - Outer-measure and covering correspondence.
- **Forced law:** Finite outer-measure correspondence is the least exact weight among every generated cover of the target support.
- **Unique survivor:** MEAS-003 uniquely retains complete-cover-minimum-weight, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-DISJOINT-ADDITIVITY-002; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** MEAS-003: outer_covering; exact observation: {"all_covers_tested":true,"least_cover_weight":"4/6","target":[1,3]}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:65bac06bd516adf5b8e3bab10d23d6ac0a863098c4a42fbe7894b066cf128fac; independent sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580; empirical sha256:761f5d791041be9edc294a9cff04586d1d31b34a4a4ee007f3b0cf8ad799aeb1; engine receipt sha256:0ee9c394f47d86ec4a0cb0e8812e82aaa1df94bf7739b6360a4e0fb257a1a95e.

SFT-MATH-OBL-MEAS-004 - Measurable-boundary correspondence

- **Claim:** SFT-MATH-MEAS-MEASURABLE-BOUNDARY-004 - Measurable-boundary correspondence.
- **Forced law:** A boundary is measurable when every generated test support decomposes exactly into its retained inside and outside parts.
- **Unique survivor:** MEAS-004 uniquely retains exact-boundary-decomposition, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-OUTER-COVERING-003; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** MEAS-004: measurable_boundary; exact observation: {"all_exact_decompositions":true,"boundary":[1,2],"test_support_count":8}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:8d23f61e3d718cf6863ef63785aef5dcf87b8412cd9a661a3a2a2058c4572583; independent

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580; empirical sha256:11068841afae99369fb268e36e5780f0aacd6ce92781ebe11bd24bf9805ddc64; engine receipt sha256:fc75d7acb7ac063472a62ef61784d3b2464536f8938410cea68282dd69f5936d.

SFT-MATH-OBL-MEAS-005 - Exact integration over finite support

- **Claim:** SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005 - Exact integration over finite support.
- **Forced law:** Exact integration over finite support is the complete accumulation of each retained value composed with its exact support weight.
- **Unique survivor:** MEAS-005 uniquely retains weighted-support-accumulation, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-MEASURABLE-BOUNDARY-004; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** MEAS-005: finite_support_integration; exact observation: {"integral":"7/3","values":[1,2,3],"weights":["1/6","2/6","3/6"]}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:b78c9b623a0d03aa4121dd37d8dcdb73cbcba27ef43d8ff2024d3a8d14dc3108; independent sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580; empirical sha256:f0e2560fb3d10088c62a105cd4c341dfcac5f9a921e7ee47e13cf614000ecbbc; engine receipt sha256:d0b85a891e495ff46031d96111dd3c424206599b9def8efaa67a60bb8929fc04.

SFT-MATH-OBL-MEAS-006 - Refinement-sum integration correspondence

- **Claim:** SFT-MATH-MEAS-REFINEMENT-SUM-INTEGRATION-006 - Refinement-sum integration correspondence.
- **Forced law:** Integration correspondence under refinement is admitted through an exact successor family of finite sums with every cell and weight retained.
- **Unique survivor:** MEAS-006 uniquely retains exact-refinement-sum-correspondence, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** MEAS-006: refinement_sum_integration; exact observation: {"midpoint_sum_each":"1/2","refinement_counts":[1,2,3,4,5,6,7,8]}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT

number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:1b2f8f42db11734e2c9a8cfa787d907877f7e31c8026c6ddca916bb90a43d0ce`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:e8fdd5b3bec10369a6fbc7f4fbc8b0c400d3fd33b45b8331d63d4f1144f2fc75`; engine receipt `sha256:22c4fe37ebac4cc637920ff3872ba28f8ed7011a639014be2aaa6760ac71581b`.

`SFT-MATH-OBL-MEAS-007` - Product measure and conditional support

- **Claim:** `SFT-MATH-MEAS-PRODUCT-CONDITIONAL-SUPPORT-007` - Product measure and conditional support.
- **Forced law:** Product weight composes coordinate weights exactly; conditional support is the retained joint weight relative to its declared nonempty conditioning support.
- **Unique survivor:** MEAS-007 uniquely retains product-and-conditional-support-law, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-REFINEMENT-SUM-INTEGRATION-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** MEAS-007: product_conditional_support; exact observation: {"conditional_second":"3/4","first_weights":["1/3","2/3"],"product_total":1,"second_weights":["1/4","3/4"]}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c4bf3667cb3b7415bfef9cde2889fac1d2adc588f45ef36b56738231f8940b17`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:cc9fa9b235fe39e8c48652f6638bb893c193dc5ac22c280af12061895969418a`; engine receipt `sha256:eaffbc9318e599b402acbdba1cc51cd4dd49fb27e7f009d1d17fb2c09befc06f`.

`SFT-MATH-OBL-MEAS-008` - Signed-measure replacement by held orientation

- **Claim:** `SFT-MATH-MEAS-HELD-ORIENTATION-SIGNED-008` - Signed-measure replacement by held orientation.
- **Forced law:** Signed-measure correspondence uses separate held and opposed positive ledgers; cancellation is an exact relation and never a negative proof scalar.
- **Unique survivor:** MEAS-008 uniquely retains held-opposed-measure-ledger, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-PRODUCT-CONDITIONAL-SUPPORT-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** MEAS-008: held_orientation_signed; exact observation: {"held":"3/4","negative_scalar_used":false,"opposed":"1/4","retained_held":"1/2"}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum

measure, fitted parameter or forbidden scalar is imported

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e943902355ec71fd674aa435b47734a1e00b13c5bc21e7d76b612c4705be4d41`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:c20030d28a7eec0d32efde7e303d2d06e5881b92d491c8a757188d72f4d63f6b`; engine receipt `sha256:5d84b3dbf7725037c1ec28db61b83139d79504335292b4357a74539f98745017`.

`SFT-MATH-OBL-MEAS-009` - Distribution and generalized-observation correspondence

- **Claim:** `SFT-MATH-MEAS-DISTRIBUTION-OBSERVATION-009` - Distribution and generalized-observation correspondence.
- **Forced law:** A distributional correspondence is an exact action on every generated test function, with composition and observation custody retained.
- **Unique survivor:** MEAS-009 uniquely retains exact-test-function-action, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-HELD-ORIENTATION-SIGNED-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** MEAS-009: distribution_observation; exact observation: {"composed_action":4,"first_action":"7/3","linearity_exact":true,"second_action":"5/3"}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c9762d0e559e3eae1d45f1922b90563949447cc90bf7cb1dfea97b93914db51d`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:9a7ba4ce521cf134b868385dc02d25dbfb2c375ceffe34e38c169846763d6d1c`; engine receipt `sha256:f72e9d297fdc79cbd7e4ffa47e81d163a42965e347157648cefef1ea11934a22`.

`SFT-MATH-OBL-MEAS-010` - Convergence-of-measures finite witness boundary

- **Claim:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010` - Convergence-of-measures finite witness boundary.
- **Forced law:** Convergence-of-measures correspondence requires exact finite witness widths at every registered successor; completed infinite equality remains outside the certificate.
- **Unique survivor:** MEAS-010 uniquely retains successor-refined-measure-enclosure, complete support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-ANAL-SPECTRAL-MEASURE-016 -> SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012 -> SFT-MATH-MEAS-DISTRIBUTION-OBSERVATION-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** MEAS-010: convergence_finite_witness; exact observation: {"completed_infinite_equality_claimed":false,"strictly_refined":true,"successors":8,"width_rule":"1/(n+1)"}; complete family observation custody: 10 records; all rows preserved; exact generated supports, fractions and structural orientation are retained; no continuum measure, fitted parameter or forbidden scalar is imported
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-MEASURE-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported theorem answer or target outcome selects the law; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no completed continuum support, sigma-totality or infinite convergence object; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f07b92f49ae9b09aefbf8ede5d863681f0f777b335954dec5b0ce6094cbce376`; independent `sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580`; empirical `sha256:a9665d54fc05d024526ce8d5bee4ce792abf62e45603ad8e4b8531e956787c9b`; engine receipt `sha256:a4f38ba621f479ce435299b06099cc716d53db25ee31cb5307d354573e3c1992`.

40.17 Family `PROB` - 16/16 complete

This family contributes 4,096 completely decided candidates, 16 unique survivors and 64 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-PROB-001` - Exact support-based probability correspondence

- **Claim:** `SFT-MATH-PROB-SUPPORT-CORRESPONDENCE-001` - Exact support-based probability correspondence.
- **Forced law:** Probability correspondence is the exact retained weight or count ratio of an event inside a completely generated support; it does not assert ontic randomness.
- **Unique survivor:** PROB-001 uniquely retains exact-support-count-ratio, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-001: support_probability; exact observation: {"event_count":2,"exact_weight":"1/2","ontic_randomness_claimed":false,"support_count":4}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e9329e6539c046dd1782e2b030fd54b12d360017f1ef47d878a5d805ebcf33e3`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:791b794c33e6139800f48ce7cbbe335bd9a1e604c5a0d5b1396fd1e02c9a904c`; engine receipt `sha256:2f7e63c215ef725afbaed9e153dc0e9afbac2ae30fa90237c27aaa18c463d71e`.

SFT-MATH-OBL-PROB-002 - Conditional support and Bayes correspondence

- **Claim:** SFT-MATH-PROB-CONDITIONAL-BAYES-002 - Conditional support and Bayes correspondence.
- **Forced law:** Conditional support is the exact joint support relative to a declared nonempty support, and Bayes correspondence is the same joint custody read in the opposite order.
- **Unique survivor:** PROB-002 uniquely retains conditional-support-composition, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-SUPPORT-CORRESPONDENCE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-002: conditional_bayes; exact observation: {"a_count":3,"a_given_b":"1/2","b_count":4,"b_given_a":"2/3","joint_count":2,"support_count":6}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:3d2a6c0add91bc72f2bafea9d1edd14b748dacb7e3fff6f7960dc97c6f2c6861`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:0a65841dc585afd67e13a7a7f0e7caec15f4e3e8407095131c2de0f4d5ba1fa5`; engine receipt `sha256:3de32cd0f017cd9829917e109c48bf5510084b96451c28f5f1af69c3059ec3d8`.

SFT-MATH-OBL-PROB-003 - Independence and factorization witnesses

- **Claim:** SFT-MATH-PROB-INDEPENDENCE-FACTORIZATION-003 - Independence and factorization witnesses.
- **Forced law:** Independence correspondence holds when complete joint-support enumeration equals the exact product of the retained marginal supports.
- **Unique survivor:** PROB-003 uniquely retains complete-product-factorization, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-CONDITIONAL-BAYES-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-003: independence_factorization; exact observation: {"joint":"1/6","marginals":["1/2","1/3"],"product_shape":[2,3]}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is

admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:4b0026f954c992cee946a78880da9d3468311c867850b81706b0f2425bdd3639`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:69e4f4d53737baca35fe31e0d9fc620c0085933968ad8ea730d2f512e083cff8`; engine receipt `sha256:c9f382194e21c74dc4838e29a2e6edf7e17fc470b5a171283dc88a7f94004469`.

`SFT-MATH-OBL-PROB-004` - Expectation as exact support accumulation

- **Claim:** `SFT-MATH-PROB-EXPECTATION-004` - Expectation as exact support accumulation.
- **Forced law:** Expectation is the exact finite accumulation of each generated value with its retained support weight.
- **Unique survivor:** PROB-004 uniquely retains exact-weighted-expectation, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-INDEPENDENCE-FACTORIZATION-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-004: expectation; exact observation: {"expectation":"5/2","values":[1,2,3,4],"weights":["1/4","1/4","1/4","1/4"]}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:de23f5314fab9a365086d3d9d3d60b214cd58768282bb0ad5858f1bccee9f567`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:e39bb69f335737be7f594c15a1b15dcce2e8bb18c084db3898dbd9500833ee1f`; engine receipt `sha256:b87c57beb9072c476180b813248201c2b22f3f7f98112c1e3069194cba7836d3`.

`SFT-MATH-OBL-PROB-005` - Variance and dispersion without negative magnitudes

- **Claim:** `SFT-MATH-PROB-VARIANCE-DISPERSION-005` - Variance and dispersion without negative magnitudes.
- **Forced law:** Dispersion accumulates squared positive held distances from the exact centre; no negative magnitude is admitted.
- **Unique survivor:** PROB-005 uniquely retains held-distance-dispersion, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-EXPECTATION-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-005: variance_dispersion; exact observation: {"centre":"5/2","dispersion":"5/4","negative_magnitude_used":false}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:86f29dc56d4e5f05cad4967d89af3a9f92919b9b4d07583b649f7e185342fe45`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:7dde247d4e80baa333a6eaaa8ecf2da941a3c29b0f18a07eb8861ff5cc8e05eb`; engine receipt `sha256:c20397f6b4dc7c33323fb18b25e87efcca673139404ef7103796a07479418a9a`.

`SFT-MATH-OBL-PROB-006` - Finite distribution families

- **Claim:** `SFT-MATH-PROB-FINITE-DISTRIBUTION-006` - Finite distribution families.

- **Forced law:** A finite distribution is a complete exact support-weight family whose weights compose to one and whose absent labels remain structural absence.

- **Unique survivor:** PROB-006 uniquely retains complete-finite-distribution, complete deterministic support custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-VARIANCE-DISPERSION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** PROB-006: finite_distribution; exact observation: {"binary_width":3,"complete_weight":1,"multiplicities":[1,3,3,1]}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:3d27090c467675e0f6e70cc809ec6f66ab2360ddec471d031e7e4cd9f13c12c0`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:0434c05034e2b0e062c754692ffdddb027b37310fc9e9bec25257b0e86e75c7b`; engine receipt `sha256:4e219f4e7826997fb21e65bba1e98f924b6ab6e847ef699dea331542e76eba86`.

`SFT-MATH-OBL-PROB-007` - Law-of-large-count correspondence under deterministic enumeration

- **Claim:** `SFT-MATH-PROB-LARGE-COUNT-007` - Law-of-large-count correspondence under deterministic enumeration.

- **Forced law:** Large-count correspondence is derived from complete deterministic word enumeration and exact frequencies, never from an ontic randomness premise.

- **Unique survivor:** PROB-007 uniquely retains deterministic-complete-word-frequency, complete deterministic support custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 ->`

SFT-MATH-PROB-FINITE-DISTRIBUTION-006; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.

- **Post-registry observations:** PROB-007: large_count; exact observation: {"complete_enumeration":true,"held_label_frequency_each":"1/2","word_widths":[1,2,3,4,5,6,7,8]}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:cfecc4b5879e2c88e8dc28dbc30e8ba2da2a62dbbfe26e8513a0b81c86219603; independent sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6; empirical sha256:02606527b1751538623f245b0127e2c732ae07c2a70542a8a9a1ccc8bd463828; engine receipt sha256:e9c8cd3b6e51c3a20e1640ce3b5815b87771a4f2b94f1d99e6b5aa02878d078c.

SFT-MATH-OBL-PROB-008 - Central-limit enclosure correspondence

- **Claim:** SFT-MATH-PROB-CENTRAL-LIMIT-ENCLOSURE-008 - Central-limit enclosure correspondence.
- **Forced law:** Central-limit correspondence is admitted only through exact finite count families and certified rational tail enclosures; no Gaussian or irrational scalar selects the result.
- **Unique survivor:** PROB-008 uniquely retains exact-central-tail-enclosure, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-LARGE-COUNT-007; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** PROB-008: central_limit_enclosure; exact observation: {"all_rows_pass":true,"count_widths":[2,3,4,5,6,7,8],"enclosure":"1/n","tail_rule":"2/(2^n)"}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:55a2a31f67298ad5a6d8a1bccff77cedab4cc9ab098ac511c8a692377914017f; independent sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6; empirical sha256:0e73f7922483aacb497a741170ab36dcf9410692e8338c0fac50296d0c9a5956; engine receipt sha256:1f4ece1dc9dcffc4ab540bcf7cd3dcf29d5bda76255a47b96ec80f3286808977.

SFT-MATH-OBL-PROB-009 - Estimation and sufficient-record structure

- **Claim:** SFT-MATH-PROB-ESTIMATION-SUFFICIENT-RECORD-009 - Estimation and sufficient-record structure.
- **Forced law:** A record is sufficient when every generated observation sharing it has the same exact parameter-dependent likelihood factor and no discarded distinction changes inference.

- **Unique survivor:** PROB-009 uniquely retains likelihood-preserving-sufficient-record, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-CENTRAL-LIMIT-ENCLOSURE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-009: sufficient_record; exact observation: {"multiplicities":[1,4,6,4,1],"record":"held-label-count","word_width":4}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4e22e8d897e4361229a8c58a5e250910ba26c674c5e016824edbcd156613b612`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:6a6a35b007ba8d9e19402df53f32f535c96fbcc6eccd14ea887ac73ca41e2e8e`; engine receipt `sha256:b10c05e8a85e8e31388d8b3fd77717a82dbb17564bb8841164daee3def78682f`.

`SFT-MATH-OBL-PROB-010` - Confidence and credible-region correspondence

- **Claim:** `SFT-MATH-PROB-CONFIDENCE-CREDIBLE-REGION-010` - Confidence and credible-region correspondence.
- **Forced law:** A confidence or credible-region correspondence is a complete candidate region selected by a preregistered exact coverage or evidence rule with all exclusions retained.
- **Unique survivor:** PROB-010 uniquely retains preregistered-exact-region-rule, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-ESTIMATION-SUFFICIENT-RECORD-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-010: confidence_credible_region; exact observation: {"candidates":["1/4","1/2","3/4"],"retained":["1/2","3/4"],"rule":"at-least-half-maximum"}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:88a4e937d913bfa286b8538cc1dfecfeacdd447b9b98b61362210814a584f803`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical

`sha256:1edc41a18e349efd345f56944d06038b80bad02a77c9b4156ddf1abfef3dfccb`; engine receipt `sha256:f264d01b3cb82e643590415a53b9ea4b9b0bac5e818de89426b4ceba2e0242b2`.

`SFT-MATH-OBL-PROB-011` - Hypothesis testing and error custody

- **Claim:** `SFT-MATH-PROB-HYPOTHESIS-ERROR-CUSTODY-011` - Hypothesis testing and error custody.
- **Forced law:** Hypothesis testing is a complete decision-support partition with type-one, type-two and unresolved error supports separately retained.
- **Unique survivor:** PROB-011 uniquely retains complete-decision-error-ledger, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-CONFIDENCE-CREDIBLE-REGION-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-011: hypothesis_error_custody; exact observation: {"both_ledgers_retained":true,"decision_support":8,"type_one":"1/4","type_two":"1/4"}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4ad0c48ad7314efba0341f121f7de1493fde8ad66ed811a1e49e585b6e8ac316`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:25453b7fed665374f95b8d0e6a8c479bc21ffa39ecee57a66c2657c659ecab57`; engine receipt `sha256:21aadee13d2df1e2737ad5c4cd7ba4ee10b6636b3810cf6458450ae67cd060b6`.

`SFT-MATH-OBL-PROB-012` - Likelihood and evidence ratios

- **Claim:** `SFT-MATH-PROB-LIKELIHOOD-EVIDENCE-RATIO-012` - Likelihood and evidence ratios.
- **Forced law:** Likelihood and evidence ratios compare exact observation support under every generated candidate without turning the ratio into a fitted prior.
- **Unique survivor:** PROB-012 uniquely retains exact-likelihood-ratio, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-HYPOTHESIS-ERROR-CUSTODY-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-012: likelihood_ratio; exact observation: {"candidates":["3/4","1/2"],"observation":"three-of-four-held","ratio":"27/16"}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic

randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:2465fbdacf973bd4172b0cc8bd143f39fec1664493805b022b01927b5a804a98`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:93f89a823248f6e32f17cf2316fce283e2807853ffc3360506330686f12b5a56`; engine receipt `sha256:2a7a74ff1432e3aa098187148f14b63b47977082f22f28848036374dcb971858`.

`SFT-MATH-OBL-PROB-013` - Bayesian update as exact conditional support

- **Claim:** `SFT-MATH-PROB-BAYESIAN-UPDATE-013` - Bayesian update as exact conditional support.
- **Forced law:** Bayesian update is exact conditioning of a declared prior support by registered observation weights, with normalization and source custody explicit.
- **Unique survivor:** PROB-013 uniquely retains exact-prior-observation-conditioning, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-LIKELIHOOD-EVIDENCE-RATIO-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-013: bayesian_update; exact observation: {"observation_weights":["3/4","1/4"],"posterior":["3/5","2/5"],"prior":["1/3","2/3"]}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:83dc756736fa02c04a78d3988e2afaab102c94f39bf2ba46ff95aeefc8d0d69c`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:d05816d99fb1afbb0c70288739405f0511b47c2d2e646ba85950810b30ffa1db`; engine receipt `sha256:c239b75f04ae9b064f3a6178ed377874ca563585c6e751956e1ea995fcfc0b5d`.

`SFT-MATH-OBL-PROB-014` - Finite stochastic-process correspondence

- **Claim:** `SFT-MATH-PROB-FINITE-STOCHASTIC-PROCESS-014` - Finite stochastic-process correspondence.
- **Forced law:** A finite stochastic-process correspondence is the complete deterministic path grammar plus exact path weights and transition custody; stochasticity is not ontic indeterminism.
- **Unique survivor:** PROB-014 uniquely retains deterministic-path-weight-process, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-BAYESIAN-UPDATE-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** PROB-014: finite_stochastic_process; exact observation: {"ontic_randomness_claimed":false,"path_count":4,"path_length":2,"states":2,"total_path_weight":1}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bbb27b2d9b7d207d826634004270c330ba16bb0bb5da9ac5e2ea2c8e822ecba9`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:1d5c5b9037716e4085e990c4991e32184aff0ef40af565c0e59cb85d35ac3db5`; engine receipt `sha256:686defbd4442ac15c71a3bc349175da32ba506f8cfcec565be22b6813e24d50a`.

`SFT-MATH-OBL-PROB-015` - Martingale-like conditional conservation

- **Claim:** `SFT-MATH-PROB-CONDITIONAL-CONSERVATION-015` - Martingale-like conditional conservation.
- **Forced law:** Martingale-like correspondence is exact branchwise conditional conservation of a retained value across the complete successor support.
- **Unique survivor:** PROB-015 uniquely retains branchwise-conditional-conservation, complete deterministic support custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-FINITE-STOCHASTIC-PROCESS-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-015: conditional_conservation; exact observation: {"branchwise_conserved":true,"children":[1,3],"grandchildren":["1/2","3/2","5/2","7/2"],"root":2}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:42891223a1fda0efebbabe8ae5521900f6495a5250ffa0d7b95c1e673e3522e8`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:41aa92305473ea1762a7da8d4e362dee145f8a31442f39bf783227ee561ebaf7`; engine receipt `sha256:467057cb14cfa1b348de39f0916d99f257be84b5c7ce10de85fd3932e5fb6e32`.

`SFT-MATH-OBL-PROB-016` - Statistical identifiability and nonidentifiability

- **Claim:** `SFT-MATH-PROB-IDENTIFIABILITY-016` - Statistical identifiability and nonidentifiability.
- **Forced law:** A parameter is identifiable exactly when distinct generated parameter records induce distinct complete observation records; merged records certify nonidentifiability.
- **Unique survivor:** PROB-016 uniquely retains observation-map-identifiability-boundary, complete deterministic support custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010 -> SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008 -> SFT-MATH-PROB-CONDITIONAL-CONSERVATION-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** PROB-016: identifiability; exact observation: {"identifiable_with_latent_custody":true,"latent_records":[["1/4","3/4"],["1/2","1/2"]],"merged_observation_weight":1,"nonidentifiable_after_merge":true}; complete family observation custody: 16 records; all rows preserved; complete deterministic path and support enumeration supplies the correspondence; no ontic randomness, fitted limit, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-PROBABILITY-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted parameter, imported stochastic law or target outcome selects the result; probability is exact generated-support correspondence and never an ontic randomness premise; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no asymptotic distribution or completed infinite trial sequence is admitted without a separate successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a9b306094f70cf6480eba061afcb227df5c9d4c23b4b5a7b053dca2d25be6bf0`; independent `sha256:54713dc80a9b0fe3fc0195dcd051c54852bf8749ddf1d7c5a122d1c7abaa0ed6`; empirical `sha256:14cc5ea81f1d682c595b02490218b76650319f7862c6641559db8eee65974914`; engine receipt `sha256:dfe162067f03fbfafd04d35e28dfc9de7e45e7e4fff14c9de674c9e7d3721d40`.

40.18 Family `OPT` - 16/16 complete

This family contributes 4,096 completely decided candidates, 16 unique survivors and 64 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-OPT-001` - Feasible set and objective-order structure

- **Claim:** `SFT-MATH-OPT-FEASIBLE-OBJECTIVE-ORDER-001` - Feasible set and objective-order structure.
- **Forced law:** Optimization begins with a completely generated feasible support and an exact objective preorder; no preferred candidate is imported.
- **Unique survivor:** OPT-001 uniquely retains feasible-objective-order, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-001: feasible_objective_order; exact observation: {"feasible":[2,3,4],"generated":[1,2,3,4],"minimum":2}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:565bb4084d62ddb81a4ddc6b563b5f2c640249d3b9efc95ec62affb1d37a787b`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical

`sha256:fbe64bb1502ac7f9317acd35874f528ea679db586e9428cf6bad9b6236660322`; engine receipt `sha256:a13f3cdb8b9906ebb0792c32ddaf516caf949ad77b0f00c032b8782cf7c15cc7`.

`SFT-MATH-OBL-OPT-002` - Exact minimum and maximum on finite support

- **Claim:** `SFT-MATH-OPT-FINITE-EXTREMA-002` - Exact minimum and maximum on finite support.
- **Forced law:** Finite extrema are the objective-equivalent classes surviving complete comparison across every feasible candidate.
- **Unique survivor:** OPT-002 uniquely retains complete-finite-extrema, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-FEASIBLE-OBJECTIVE-ORDER-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-002: finite_extrema; exact observation: {"maximum":4,"minimum":1,"values":[4,1,3,2]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:54cfc810383c88330bc455a04ffa7cc90bcb538f73dc0049af02de3d0e3224fc`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:2f649dd0e310944ddd99e9f9562fd448c06d323ee242a121af2ed6f08a280a63`; engine receipt `sha256:74b8e6f8afd76a490c34438b20b360709ad22fbe39887630bd2b72e79d8fa4a2`.

`SFT-MATH-OBL-OPT-003` - Pareto dominance and multiobjective order

- **Claim:** `SFT-MATH-OPT-PARETO-DOMINANCE-003` - Pareto dominance and multiobjective order.
- **Forced law:** The Pareto frontier contains exactly those feasible candidates not dominated in every registered objective coordinate.
- **Unique survivor:** OPT-003 uniquely retains complete-pareto-frontier, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-FINITE-EXTREMA-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-003: pareto_frontier; exact observation: {"nondominated":[[1,4],[2,2],[4,1]],"points":[[1,4],[2,2],[4,1],[3,3]]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or

unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:c11da6b42904697a12efcb052ad54aadd32c16c30d8adf712ca2ccba0577332a`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:791c4087ae7a05b77279ffee8d5b3c0bcaae88650bba8edbbea60466d61163f8`; engine receipt `sha256:0d581ec125db13d24f596f7cdbd8f656a244678a63441c9d71c17a24e36a2e52`.

`SFT-MATH-OBL-OPT-004` - Linear-program correspondence

- **Claim:** `SFT-MATH-OPT-LINEAR-PROGRAM-004` - Linear-program correspondence.
- **Forced law:** Linear-program correspondence is exact affine feasibility and objective ordering over a generated rational support, with relaxation boundaries explicit.
- **Unique survivor:** OPT-004 uniquely retains exact-linear-feasible-enumeration, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-PARETO-DOMINANCE-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-004: linear_program; exact observation: {"constraint":"x+y<=4","optimum":4,"optimum_set":[[1,3],[2,2],[3,1]],"positive_grid_extent":3}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:8997faeae9090423b4c0e37e415b75d25a8c8ebd44c811f2f853fcca0a78cf65`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:efc6ebd3b88aeb1dca4ef52d9fde87df61cdcf0115a899c57bacd7d5a972f03f`; engine receipt `sha256:6323878558e3eb3fb4d0d8b2726dbda5fdcc0ab9a40aa57d9cd54748cc4e30da`.

`SFT-MATH-OBL-OPT-005` - Integer and combinatorial optimization

- **Claim:** `SFT-MATH-OPT-INTEGER-COMBINATORIAL-005` - Integer and combinatorial optimization.
- **Forced law:** Integer and combinatorial optimization is complete exact enumeration or a separately certified structure-preserving reduction over discrete feasible forms.
- **Unique survivor:** OPT-005 uniquely retains complete-discrete-feasible-search, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-LINEAR-PROGRAM-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-005: integer_combinatorial; exact observation: {"capacity":3,"optimum_value":5,"unique_optimum_items":[1,2],"values":[2,3,4],"weights":[1,2,3]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:ca53440b562c9cc05848732ff31a2cd5e53f9ed6227e8c67b37fd628722bcc81`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:98064d65c3a9332d88d5e7d879d6dae2cf6a35479c2cc63700705008df1dd3aa`; engine receipt `sha256:bc3ef7947c9f9f489114c474b3c03d08c674e9ea91aef48a1e509b0088cf8a22`.

`SFT-MATH-OBL-OPT-006` - Convex optimization correspondence

- **Claim:** `SFT-MATH-OPT-CONVEX-CORRESPONDENCE-006` - Convex optimization correspondence.

- **Forced law:** Convex correspondence is exact midpoint or held-distance ordering over generated rational supports, not an imported continuum theorem.

- **Unique survivor:** OPT-006 uniquely retains held-distance-convex-minimum, complete candidate custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-INTEGER-COMBINATORIAL-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** OPT-006: convex_correspondence; exact observation: {"candidates":[1,2,3,4,5],"objective":"held-distance-square-to-three","unique_minimizer":3}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:6760c33825de4266bb977ee8253e686311b042f36a1accd91e13a5d7e83ad805`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:addcd8353740de580437cef1e254a1129b041dbe7ed019f25dc7fa8b4f8c6905`; engine receipt `sha256:e39890906deedccecea769f9e59b8e73d8a82683178f00471f69ce7cb347cc64`.

`SFT-MATH-OBL-OPT-007` - Duality and certificate structure

- **Claim:** `SFT-MATH-OPT-DUAL-CERTIFICATE-007` - Duality and certificate structure.

- **Forced law:** A dual certificate is an independently checkable exact bound matching a feasible primal value; equality closes optimality.

- **Unique survivor:** OPT-007 uniquely retains matching-primal-dual-certificate, complete candidate custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-CONVEX-CORRESPONDENCE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** OPT-007: duality_certificate; exact observation: {"dual_bound":4,"exact_match":true,"primal_optimum":4}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:66dd072426a7dc60c84c45cb77dc132f910e01ac625e35a9e97fc077c63cb43e`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:ebb916a31c0a525d27c1e5b2d98e1051b4dbba32503b707145b25b3329c937bb`; engine receipt `sha256:62c27ebb5829a3dad3f7fabc9647f4a7bdccc2bfdb1241574b8c9eb4cecf8625`.

`SFT-MATH-OBL-OPT-008` - Variational optimization correspondence

- **Claim:** `SFT-MATH-OPT-VARIATIONAL-CORRESPONDENCE-008` - Variational optimization correspondence.
- **Forced law:** Variational correspondence compares the exact action of every generated finite path and retains all boundary records.
- **Unique survivor:** OPT-008 uniquely retains complete-path-action-minimum, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-DUAL-CERTIFICATE-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-008: variational_correspondence; exact observation: {"action":"sum-of-squares","positive_parts":3,"total":6,"unique_path":[2,2,2]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:6c32a2bf7f0622308628b142e785cc182dfeb33b4943b73d71be31a35c0b93e6`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:76bf47c00ce7d3e7c65e5f6c5b8aa33f7560a1e85e25b96b1efb8f215ec794c8`; engine receipt `sha256:fee2c183cf7b4f34029a2606dddf3491a9b3c85ecb6d2f2d44b9afa49446ed91`.

`SFT-MATH-OBL-OPT-009` - Dynamic programming as compositional optimization

- **Claim:** `SFT-MATH-OPT-DYNAMIC-PROGRAMMING-009` - Dynamic programming as compositional optimization.
- **Forced law:** Dynamic programming is lawful when locally retained subproblem optima compose to the same result as complete path enumeration.
- **Unique survivor:** OPT-009 uniquely retains subproblem-optimum-composition, complete candidate custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-VARIATIONAL-CORRESPONDENCE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-009: dynamic_programming; exact observation: {"backward_recurrence":3,"complete_minimum":3,"path_costs":[3,3,6]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a3ca55cfc6de342162fd7861ef7ee8363ad086e9040e4451cd6634fbac29c5fa`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:c72ba62c88886dc22835555fadb94359fee7bd3107d8223f3d1a266ab6f6e832`; engine receipt `sha256:94a6507475f5b899b0280eda449cdac26c23fafe729038b5c35eb257b3a1f4c1`.

`SFT-MATH-OBL-OPT-010` - Optimal control on generated state paths

- **Claim:** `SFT-MATH-OPT-GENERATED-STATE-CONTROL-010` - Optimal control on generated state paths.
- **Forced law:** Optimal control is exact ordering of every generated admissible state-action path under fixed dynamics and boundary custody.
- **Unique survivor:** OPT-010 uniquely retains complete-control-path-optimum, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-DYNAMIC-PROGRAMMING-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-010: optimal_control; exact observation: {"controls":[1,2],"goal":5,"initial":1,"unique_least_step_path":[2,2]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e30fc51f7aef489ec85cb719ec6473e4d948f432065d547c62caa267c9703f82`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:642c16095f01d21b82adba9d3a8528918ea9210d664956d2511c515d1d8af532`; engine receipt `sha256:359e651b661d88d22d7602a0249f90c1b82a16d3eb9485f9ed59c3183bed13bf`.

SFT-MATH-OBL-OPT-011 - Game equilibrium finite correspondence

- **Claim:** SFT-MATH-OPT-GAME-EQUILIBRIUM-011 - Game equilibrium finite correspondence.
- **Forced law:** A finite game equilibrium is a complete action profile whose every unilateral alternative fails to improve the acting participant's exact order.
- **Unique survivor:** OPT-011 uniquely retains mutual-best-response-census, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-GENERATED-STATE-CONTROL-010; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** OPT-011: game_equilibrium; exact observation: {"actions":2,"profiles":4,"unique_mutual_best_response":[2,2]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:9cf4ef4bafbd786c2ad9fca9c7e3ce1a24aec9a9dc73f84838699c23ec0fefed; independent sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72; empirical sha256:95b28b61a4e845bb985d4aa24f481cd1e84452df15907068fb97d69a1e750fd2; engine receipt sha256:df50833d973d90601e8d47ddca56172fc4e914d0e8fa925b1e9121759ee7a738.

SFT-MATH-OBL-OPT-012 - Decision theory and loss ordering

- **Claim:** SFT-MATH-OPT-DECISION-LOSS-012 - Decision theory and loss ordering.
- **Forced law:** Decision theory orders complete action-state loss records under declared exact support weights without importing utility scales or priors.
- **Unique survivor:** OPT-012 uniquely retains exact-decision-loss-order, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-GAME-EQUILIBRIUM-011; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** OPT-012: decision_loss; exact observation: {"actions":["A","B","C"],"exact_losses":["3/2","2","5/2"],"minimum":"A"}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:bec5b52e046cfd17b226e30f548cc5ed29b7837102a07f6b21a785a9e25ce3ff`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:c962d9f6d0a6a260000f93277fe19bed2fef1890cf31600df16aed4e643eb381`; engine receipt `sha256:92b812df5237a10f034bf74974cb664dd25c0074fdd4c6186c7f802896823f6a`.

`SFT-MATH-OBL-OPT-013` - Operations-research flow and scheduling

- **Claim:** `SFT-MATH-OPT-FLOW-SCHEDULING-013` - Operations-research flow and scheduling.
- **Forced law:** Flow and scheduling retain capacity, conservation, precedence and resource custody across every generated route or schedule.
- **Unique survivor:** OPT-013 uniquely retains flow-schedule-conservation, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-DECISION-LOSS-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-013: flow_scheduling; exact observation: {"makespan":3,"maximum_flow":3,"single_machine_lengths":[2,1]}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e7bbedbae805d67cf01fd22b2ed07a0bea980d92900f5a72fd63a0f720b112d5`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:3bab44052684910af32c4e8b46efd9993d8889fa6fb27d11c0fff07fd304b24e`; engine receipt `sha256:0f7c3d3f7d1c43f2abf11d7c21970c7f941446520ab41893ef35f49dac67aa29`.

`SFT-MATH-OBL-OPT-014` - Robust optimization under bounded uncertainty

- **Claim:** `SFT-MATH-OPT-ROBUST-BOUNDED-UNCERTAINTY-014` - Robust optimization under bounded uncertainty.
- **Forced law:** Robust optimization orders each candidate by its complete declared scenario family, with no hidden distribution or fitted uncertainty radius.
- **Unique survivor:** OPT-014 uniquely retains complete-worst-case-scenario-order, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-FLOW-SCHEDULING-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-014: robust_optimization; exact observation: {"unique_choice":"B","worst_case_losses":{"A":5,"B":4,"C":6}}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer

answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:b94f91b47cb1bbd9359dc77da0177b571e4a4dadd0cdf4b6da5dd4be7ccaf28d`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:3689ed18e70e4b2ed63099fb93682d946df8d34142857168743382b54968a911`; engine receipt `sha256:03925760f19f50746faf61b93c7b164b58d424f52d585f6bcdc32f87c21e42fb`.

`SFT-MATH-OBL-OPT-015` - Approximation guarantees and gaps

- **Claim:** `SFT-MATH-OPT-APPROXIMATION-GAP-015` - Approximation guarantees and gaps.
- **Forced law:** An approximation guarantee requires an exact candidate value and independent optimum bound whose ratio or additive gap is explicitly certified.
- **Unique survivor:** OPT-015 uniquely retains exact-certified-approximation-gap, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-ROBUST-BOUNDED-UNCERTAINTY-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** OPT-015: approximation_gap; exact observation: {"candidate":8,"gap":2,"optimum_bound":10,"ratio":"4/5"}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:16ada288d1df25f508965aed8af4b4e642108a6dd97525c0a4b914db728e4326`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:b4e93c52c97188029765b70b768ee60b9e6de6af603e3d1e61b4e986345c0df4`; engine receipt `sha256:28224fd21eff9d40657483e3aceb4019904f841d9786da73c2ec59d114a16a08`.

`SFT-MATH-OBL-OPT-016` - Infeasibility and unboundedness boundaries

- **Claim:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016` - Infeasibility and unboundedness boundaries.
- **Forced law:** Infeasibility requires elimination of every feasible candidate; unboundedness requires a successor construction exceeding every supplied finite bound.
- **Unique survivor:** OPT-016 uniquely retains complete-infeasible-or-successor-unbounded-certificate, complete candidate custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-PROB-IDENTIFIABILITY-016 -> SFT-MATH-ORDER-MONOTONE-MAP-010 -> SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003 -> SFT-MATH-OPT-APPROXIMATION-GAP-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** OPT-016: infeasible_unbounded_boundary; exact observation: {"feasible_count":0,"opposed_bounds":["at-most-two","at-least-three"],"successor_exceeds_every_supplied_bound":true}; complete family observation custody: 16 records; all rows preserved; complete feasible-set, path and scenario enumeration supplies the result; no fitted objective, hidden tie breaker, forbidden scalar or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-OPTIMIZATION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted objective, imported optimizer answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque solver, sampled feasible subset or unregistered tie breaker; no unrestricted continuum optimum without a separate exact successor or enclosure certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:daace4f7a8d583c59446124a5f2be2ef102ac423fb83d526e989b8532de47b2c`; independent `sha256:98a6eaf0d6d2ad7fcfc3e420b9c594b9300c114ef67861d6655bb400f14ced72`; empirical `sha256:bd584a24c6f24f4faeb94627f99a379f3609bd2d93b12821eb54b9ff4765bf08`; engine receipt `sha256:e8e4a46874d1eb0f42b931300512437b8f1015e1de41ebc4c1a1b4a63a9b6080`.

40.19 Family `DYN` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-DYN-001` - State maps and exact orbit structure

- **Claim:** `SFT-MATH-DYN-STATE-ORBIT-001` - State maps and exact orbit structure.
- **Forced law:** A dynamical system is a generated state support with one fixed transition relation; its orbit is the exact successor trace from a retained initial record.
- **Unique survivor:** DYN-001 uniquely retains exact-state-orbit-generation, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-001: state_orbit; exact observation: {"initial":1,"map":{"1":2,"2":3,"3":2,"4":4},"orbit":[1,2,3,2,3]}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:82ad05bad12b9bc5c29f54198739c8f84322b6b7ed024bf5b0f3545364467c8b`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:37a6eb2b823533f009e708b81bfe91288ac8f39354fe18b7f76667f4af206c96`; engine receipt `sha256:aeb9921e8e40d6d59f313577666482b543ef5624f9061cbf22b2f79c176b6633`.

`SFT-MATH-OBL-DYN-002` - Fixed points and periodic cycles

- **Claim:** `SFT-MATH-DYN-FIXED-PERIODIC-002` - Fixed points and periodic cycles.
- **Forced law:** Fixed points and periodic cycles are exactly the states returning after the least registered positive transition count.

- **Unique survivor:** DYN-002 uniquely retains complete-fixed-cycle-census, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-STATE-ORBIT-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-002: fixed_periodic; exact observation: {"fixed":[4],"period_two":[2,3]}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:aa2cc07d5839579a57f85b9256ef207388cfc19ab8071613e94551d284d4f97f`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:9487cd05c82b80e324fe3b376fa64661749260a1055a047927969739818515b0`; engine receipt `sha256:9ae62db5ffdd31c07fea2a614bbbe9e5ba587bde5a38daa921fe0ba58c7e9a70`.

`SFT-MATH-OBL-DYN-003` - Recurrence and return-time structure

- **Claim:** `SFT-MATH-DYN-RECURRENCE-RETURN-003` - Recurrence and return-time structure.
- **Forced law:** Recurrence is an exact return to a retained state identity, and return time is the least counted successor depth producing it.
- **Unique survivor:** DYN-003 uniquely retains exact-first-return-record, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-FIXED-PERIODIC-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-003: recurrence_return; exact observation: {"state_four_return_time":1,"state_two_return_time":2}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:c9bd8ed7dbca1189e1c25b380072255626a1e1a69dae6cfd0571c5fa72d95509`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:b818bc280a2fa082f18cf019da7a54376a6d75f7595009b3b776105718338b15`; engine receipt `sha256:9c18019e9f20d1d1b810243cd7d08875312668c2bc99c216adeceb8e22f9c664`.

`SFT-MATH-OBL-DYN-004` - Invariant sets and conserved records

- **Claim:** `SFT-MATH-DYN-INVARIANT-CONSERVED-004` - Invariant sets and conserved records.

- **Forced law:** A support is invariant when its complete transition image remains the same support; a record is conserved when every transition retains its exact value.
- **Unique survivor:** DYN-004 uniquely retains invariant-support-custody, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-RECURRENCE-RETURN-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-004: invariant_conserved; exact observation: {"image":[2,3],"retained_cardinality":2,"support":[2,3]}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e4aa8fc344bfd186bc50a87e4a530aa0e4168c6156cebb05f8fecebe30020091`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:adec834d7f2ea3086ea35fd3b75aa351cdc1aedfaee7c777a457a16aed237fa9`; engine receipt `sha256:2df3a6cbc61017a6eef83e76253575b6bff962865cdc1002c7f77b00d2fba89c`.

`SFT-MATH-OBL-DYN-005` - Stability and attraction correspondence

- **Claim:** `SFT-MATH-DYN-STABILITY-ATTRACTION-005` - Stability and attraction correspondence.
- **Forced law:** Stability and attraction correspondence is an exact successor bound on held state separation throughout the generated orbit family.
- **Unique survivor:** DYN-005 uniquely retains exact-distance-contraction, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-INVARIANT-CONSERVED-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-005: stability_attraction; exact observation: {"distance_ratio":"1/2","fixed_carrier":1,"map":"(x+1)/2"}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:08c6c5638614b655de9e90d71fb612cc5621f84e540361d6ed29e7019d4ef10b`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:2acbeffe9333ae16a391f2240fdb66641252e2c0dccbd70445cf63cae29573db`; engine receipt `sha256:520a680b3349364ef71665582583a1f3bced6cd9411a4fd11f9de14a8adba730`.

`SFT-MATH-OBL-DYN-006` - Bifurcation as finite distinction change

- **Claim:** `SFT-MATH-DYN-BIFURCATION-DISTINCTION-006` - Bifurcation as finite distinction change.
- **Forced law:** Bifurcation is a forced change in the number or relation of invariant distinctions between generated transition labels, not an imported continuum parameter.
- **Unique survivor:** DYN-006 uniquely retains finite-invariant-count-change, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-STABILITY-ATTRACTION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-006: bifurcation_distinction; exact observation: {"distinction_change":true,"first_transition_attractors":1,"second_transition_attractors":2}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:d8cf49aad6135847123f0369cd9fb4690f6d50a8b11185d8d89ec729550b5700`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:3a76ed3d78ecf5b3141d9353eaea4fca472784bb8c8bb69c7c70871722e6bc4f`; engine receipt `sha256:8c84d60f21d282875750592142e7f749065029afeac13da0dc9ad38b11ecec93`.

`SFT-MATH-OBL-DYN-007` - Symbolic dynamics and shift correspondence

- **Claim:** `SFT-MATH-DYN-SYMBOLIC-SHIFT-007` - Symbolic dynamics and shift correspondence.
- **Forced law:** Symbolic dynamics is exact transition over generated Fold words; shift correspondence retains symbol identity, word support and return depth.
- **Unique survivor:** DYN-007 uniquely retains exact-word-shift-orbit, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-BIFURCATION-DISTINCTION-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-007: symbolic_shift; exact observation: {"multiset_preserved":true,"return_shifts":4,"word":[1,1,2,2]}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:26e243425509829818576170d28906afcdf4b029b15034140bda26dea01cc2d7`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical

`sha256:2f8603c8e87784c8d2138d20b63574a88811218249f700d7f6ab8a2f869b3938`; engine receipt `sha256:d5d7b98dac7aa6e91704f5a82154ce0e5498d0e00fc67b1db7afa2a87e79cbb6`.

`SFT-MATH-OBL-DYN-008` - Chaos through exact sensitivity witnesses

- **Claim:** `SFT-MATH-DYN-EXACT-SENSITIVITY-008` - Chaos through exact sensitivity witnesses.
- **Forced law:** Sensitivity is witnessed when one retained distinction outside the current observation class becomes observable after a counted transition depth.
- **Unique survivor:** DYN-008 uniquely retains finite-shift-sensitivity, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-SYMBOLIC-SHIFT-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-008: exact_sensitivity; exact observation: {"first_observed_distinction_depth":3,"first_word":[1,1,1,1],"second_word":[1,1,1,2]}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:04164da05ca66bde7c4cee1320fe3e82bc66269f04b1ee81696d146dcdd62eff`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:aac7655535f8fa974e88ad2004443bbdda18cef54fb5c1fc552c35f01a877f85`; engine receipt `sha256:73d7f19de7ba19d7987c99282739818524e9b29c547ad8b7a21838768e9f6f2a`.

`SFT-MATH-OBL-DYN-009` - Ergodic-average finite correspondence

- **Claim:** `SFT-MATH-DYN-ERGODIC-AVERAGE-009` - Ergodic-average finite correspondence.
- **Forced law:** Finite ergodic-average correspondence is equality of exact full-orbit averages across every starting state in the same generated cycle.
- **Unique survivor:** DYN-009 uniquely retains complete-orbit-average, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-EXACT-SENSITIVITY-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-009: ergodic_average; exact observation: {"all_start_full_cycle_average":2,"cycle":[1,3]}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:38d62b7bdba53748c21fdc6f121303df624ca66322d474de9d31c6443166ced9`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:5d68a76f000d489b3214d498edf88da4fc9b1fffb24f7f2fe0a3c0b098814ccd`; engine receipt `sha256:7f8777dd8e7f792ca65148f4a8015f93681cdd890936577821113af0b6d8b052`.

`SFT-MATH-OBL-DYN-010` - Hamiltonian and reversible-map correspondence

- **Claim:** `SFT-MATH-DYN-HAMILTONIAN-REVERSIBLE-010` - Hamiltonian and reversible-map correspondence.
- **Forced law:** Hamiltonian correspondence requires an exact reversible state map and preserved generated invariant, without importing continuum phase space.
- **Unique survivor:** DYN-010 uniquely retains reversible-invariant-map, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-ERGODIC-AVERAGE-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-010: hamiltonian_reversible; exact observation: {"map":"pair-swap","self_inverse":true,"total_preserved":true}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:96174a936f1809b4b980aebd7eba158ceb5318debd985abeeb60c7819143c39d`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:66ca7bc21a6913291c3f1a8b362b0f3662becd1339ade293e00efa6ca7a6882d`; engine receipt `sha256:ffa0ecb0ca362a3a6bd0a2c0033e0033b4addf071ca9c2a1316125debcd19474`.

`SFT-MATH-OBL-DYN-011` - Dissipative dynamics and retained loss

- **Claim:** `SFT-MATH-DYN-DISSIPATIVE-RETAINED-LOSS-011` - Dissipative dynamics and retained loss.
- **Forced law:** Dissipative correspondence is predecessor merging; exact reversal requires the retained fibre distinction identifying the lost predecessor.
- **Unique survivor:** DYN-011 uniquely retains many-to-one-loss-ledger, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-HAMILTONIAN-REVERSIBLE-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-011: dissipative_retained_loss; exact observation: {"images":2,"predecessors":4,"reconstructed_predecessors":4,"retained_fibre_labels":2}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT

number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:542f0702f246ec68c46cc17e726c083c5e7aa266ab62469ad55b8526e04b1b15`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:ec3a7d3f512f2e0dc01dd8497ac27234742394d8ab74408a23d79ac38b649e51`; engine receipt `sha256:013899ed0283322e5ec156198a2845d00447a602ebe224bbaefc5e984d6cf950`.

`SFT-MATH-OBL-DYN-012` - Coupled and networked dynamical systems

- **Claim:** `SFT-MATH-DYN-COUPLED-NETWORKED-012` - Coupled and networked dynamical systems.
- **Forced law:** A coupled dynamical system is a product state whose local successor depends only on registered neighbours, with complete network-orbit and invariant custody.
- **Unique survivor:** DYN-012 uniquely retains local-coupling-network-orbit, complete orbit custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016 -> SFT-MATH-EQN-ORDINARY-DIFFERENCE-001 -> SFT-MATH-DYN-DISSIPATIVE-RETAINED-LOSS-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** DYN-012: coupled_networked; exact observation: {"initial":[1,3],"orbit":[[1,3],[3,1],[1,3]],"period":2,"total_preserved":true}; complete family observation custody: 12 records; all rows preserved; complete finite state, transition and orbit enumeration supplies the observation; no fitted dynamics, imported continuum phase space or opaque solver enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-DYNAMICS-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, fitted transition, imported dynamical theorem or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque trajectory, continuum phase space or ungenerated infinite orbit; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:8b7c44e0be715c35534e04fd2a38c03a86c43a4deee2024a31966275f038cb7a`; independent `sha256:d0d4046e8dd2f6397651daad54df525a028027c5e6c59f448b720bbaa5362674`; empirical `sha256:f068d3369535c2826d5384f5b0eac2d3412dd8ae0ddde60763082d4e7dc9ac1f`; engine receipt `sha256:c525a01d63f6eeb62552b3058ed34808c261032dae7b3553d61adfabf872780e`.

40.20 Family `LOGIC` - 16/16 complete

This family contributes 4,096 completely decided candidates, 16 unique survivors and 64 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-LOGIC-001` - Propositions as generated distinctions

- **Claim:** `SFT-MATH-LOGIC-PROPOSITION-DISTINCTION-001` - Propositions as generated distinctions.
- **Forced law:** A proposition is a generated distinction whose declared interpretation returns exactly one held or opposed observation label.
- **Unique survivor:** LOGIC-001 uniquely retains generated-proposition-valuation, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** LOGIC-001: proposition_distinction; exact observation: {"complete":true,"exclusive":true,"labels":["held","opposed"]}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bd98e9368e15a033c206fc96a5d85e33dc745397a17bd6ba3ed3f0f867240a78`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:d8595b6cb3a2c8cfe3c643db42589bcdb275db5e4b6fd916a31e401ed182a62f`; engine receipt `sha256:f85f9e844e3ec3f9581c1d9302dbea00e5791f1c4d11df0a41e3ef0bce2bbbf7`.

`SFT-MATH-OBL-LOGIC-002` - Inference and consequence preservation

- **Claim:** `SFT-MATH-LOGIC-INFERENCE-CONSEQUENCE-002` - Inference and consequence preservation.
- **Forced law:** An inference is lawful when every completely generated valuation holding its premises also holds its conclusion.
- **Unique survivor:** LOGIC-002 uniquely retains valuation-preserving-inference, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-PROPOSITION-DISTINCTION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-002: inference_consequence; exact observation: {"all_preserved":true,"rule":"modus-ponens","valuations":4}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:13d4da445f9989d41f98abebbbb0bdd7f609da48d0cb4d853c977e13ec45442e`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:ec5e615f55d614d524158049596dd32200d119a35f5f039692c9e67b48f88eb0`; engine receipt `sha256:153966835981d27d4115b986d3dfa5519147893171304aa9224ace83a36e2f94`.

`SFT-MATH-OBL-LOGIC-003` - Soundness and completeness correspondence

- **Claim:** `SFT-MATH-LOGIC-SOUND-COMPLETE-CORRESPONDENCE-003` - Soundness and completeness correspondence.
- **Forced law:** Finite soundness and completeness correspondence is exact equality between generated proof reachability and truth in every generated interpretation.
- **Unique survivor:** LOGIC-003 uniquely retains finite-proof-model-equivalence, complete proof-model custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-INFERENCE-CONSEQUENCE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-003: sound_complete_correspondence; exact observation: {"proof_model_rows_equal":true,"propositions":2,"valuations":4}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:cd793da0824c4802bb6cafb14ae7386da8baa38baaceeaf171f92dc8a598db11`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:e5c84fd906171cfe2a9e97f8cf55d172353d914b1afe4dda64dccea72767e7ef`; engine receipt `sha256:340d97010a3dea95fd4339a6ad816685c90466027c866d62e45d4760ac2692c7`.

`SFT-MATH-OBL-LOGIC-004` - Formal proof objects and checking

- **Claim:** `SFT-MATH-LOGIC-PROOF-OBJECT-CHECK-004` - Formal proof objects and checking.
- **Forced law:** A proof is an exact finite object recording every premise, rule and predecessor; checking reconstructs each step without oracle access.
- **Unique survivor:** LOGIC-004 uniquely retains explicit-proof-object-check, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-SOUND-COMPLETE-CORRESPONDENCE-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-004: proof_object_check; exact observation: {"checked":true,"conclusion":"Q","premises":["P","P-implies-Q"],"rule":"modus-ponens"}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:0e973f803e425573be9a077fafd7e119b71d7675ac3c4d77379438262d5ee8b5`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:e7efe31bc9598dde3ec2642b18beae62bc91cfe1ef98d40762a03725f22d6dc2`; engine receipt `sha256:e1d02e00afb9f4bf37ce78131e933afe4a71ef346cafef5ac815e8788d6ec4aa`.

SFT-MATH-OBL-LOGIC-005 - First-order quantifier finite-support correspondence

- **Claim:** SFT-MATH-LOGIC-QUANTIFIER-FINITE-SUPPORT-005 - First-order quantifier finite-support correspondence.
- **Forced law:** Universal and existential quantifier correspondence is complete conjunction or disjunction over every label in a generated finite domain.
- **Unique survivor:** LOGIC-005 uniquely retains complete-domain-quantifier-census, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-PROOF-OBJECT-CHECK-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-005: finite_quantifier; exact observation: {"domain":[1,2,3],"exists":"held","forall":"opposed","predicate":"equals-two"}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:daab2419fab10b343411bdd387ec0c21938815a46621dc6ea278a1f20a12dc0c`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:f6898aa661e7e685014fb61a21a68e897629913c29dd4f5a9a2370bcf421798c`; engine receipt `sha256:53083a9fef5c6ab7df19ad3708e04c0c47900087047ac48893d4163c5db51cf1`.

SFT-MATH-OBL-LOGIC-006 - Model and interpretation structure

- **Claim:** SFT-MATH-LOGIC-MODEL-INTERPRETATION-006 - Model and interpretation structure.
- **Forced law:** A model is a generated domain plus a total interpretation record for every declared symbol, relation and formula.
- **Unique survivor:** LOGIC-006 uniquely retains complete-symbol-interpretation, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-QUANTIFIER-FINITE-SUPPORT-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-006: model_interpretation; exact observation: {"domain":[1,2,3],"existential":"held","predicate_extension":[2]}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:8ef532b84599da966ca10515cd8d7b1f75624617f0dd1cb626cc705e1968e32f`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:3c57b09a9757e6038da40c57bcd36fefc98e154b4f8ceeae2c00bcaf52bb4895`; engine receipt `sha256:c0ebbe707ae0dca10b94d1b69b5122c3e881de668df7bee5bbafef8d9678c03e`.

`SFT-MATH-OBL-LOGIC-007` - Compactness correspondence boundary

- **Claim:** `SFT-MATH-LOGIC-COMPACTNESS-BOUNDARY-007` - Compactness correspondence boundary.
- **Forced law:** Finite compactness correspondence is complete intersection custody across generated constraint families; unrestricted compactness requires a separate successor certificate.
- **Unique survivor:** LOGIC-007 uniquely retains finite-intersection-witness-boundary, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-MODEL-INTERPRETATION-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-007: compactness_boundary; exact observation: {"complete_exclusion_family_satisfied":false,"proper_exclusion_families_satisfied":4,"support":[1,2,3,4],"unrestricted_claimed":false}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f6beee4ae508cb18ce195c7448d25482d3eb9888b410f5446e1dcc2c2b8237f6`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:0a65924831e2bd816abcb812b969f02d38d0d12b8189dce5b515457082108180`; engine receipt `sha256:2f79e69c848156a49ac98efd70d86967895b4d2af23b30ea9d35b65f2099d888`.

`SFT-MATH-OBL-LOGIC-008` - Decidability and computability interface

- **Claim:** `SFT-MATH-LOGIC-DECIDABILITY-INTERFACE-008` - Decidability and computability interface.
- **Forced law:** A generated finite formula grammar is decidable when exact evaluation halts with one interpretation label for every formula and input record.
- **Unique survivor:** LOGIC-008 uniquely retains finite-formula-total-decision, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-COMPACTNESS-BOUNDARY-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-008: decidability_interface; exact observation: {"all_halted":true,"evaluations":24,"formulas":6,"valuations":4}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem

answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:cb93cac95efee3679b5ba94d18fd319e8ed48bd49c5298d441904b04e3330dd9`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:7c79a1da17e790edaa8b5af44669893028a4a1b0c17983e9998dbec4d5ab2549`; engine receipt `sha256:1fc8005aaf5441ebd0362d84d9b1e0a6521f9306dcdfbfc0299abe3f549f4ced`.

`SFT-MATH-OBL-LOGIC-009` - Incompleteness and self-reference boundary

- **Claim:** `SFT-MATH-LOGIC-INCOMPLETENESS-SELF-REFERENCE-009` - Incompleteness and self-reference boundary.
- **Forced law:** Self-reference reaches an incompleteness boundary when a generated sentence requires a label opposed to its own assigned label, so no consistent total internal assignment exists.
- **Unique survivor:** LOGIC-009 uniquely retains self-negating-fixed-point-boundary, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-DECIDABILITY-INTERFACE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-009: incompleteness_self_reference; exact observation: {"host_zero_is_count_only":true,"labels":["held","opposed"],"self_negating_fixed_labels":0}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:0917eb85428a55cea8625ecd7d708b6702e81843b068fa0ae0b0b74f6ec79010`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:f1ea9f4d0a2462e8dd6aded320f25bfd9bbfc88063ec54e97e1ca3f3f286f436`; engine receipt `sha256:a93e6a3917e8b56002318a4cb9bdc381988b70aa3431a940655ed4d4d371f794`.

`SFT-MATH-OBL-LOGIC-010` - Set-like finite collection theory

- **Claim:** `SFT-MATH-LOGIC-FINITE-COLLECTION-010` - Set-like finite collection theory.
- **Forced law:** Finite collection theory is generated membership custody with exact extensional identity and complete union, intersection and subcollection enumeration.
- **Unique survivor:** LOGIC-010 uniquely retains generated-finite-collection-algebra, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-INCOMPLETENESS-SELF-REFERENCE-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** LOGIC-010: finite_collection; exact observation: {"base_labels":3,"intersection_closed":true,"subcollections":8,"union_closed":true}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:58af0aa4f235739b3dafee3a54fcddfc53f94e1f705fb2f7a8f51818968ca61c`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:1c2f19f680c6faf6013ab76a60b3eabde7177c6db93ef960df7c1f3d9be630c1`; engine receipt `sha256:66e9c9d47b3364b32c328570ccb83fa0a7f094cc7adac043ccb30c42e25c9799`.

`SFT-MATH-OBL-LOGIC-011` - Class, universe and size boundaries

- **Claim:** `SFT-MATH-LOGIC-SIZE-BOUNDARY-011` - Class, universe and size boundaries.
- **Forced law:** Collection ranks are generated one finite successor at a time; no collection containing every possible future rank is admitted.
- **Unique survivor:** LOGIC-011 uniquely retains ranked-size-extension-boundary, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-FINITE-COLLECTION-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-011: size_boundary; exact observation: {"ranks":[1,2,3,4],"sizes":[2,4,8,16],"universal_totality_claimed":false}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f523b2334ac7643d9d1db314202cb00f8aa7a290500490c9e7e3e03ae5105004`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:bbde267a34bc7dae2d54e84666a8773ac03ab7511ecfaa9ad84098f0112b0b53`; engine receipt `sha256:ae93deb5f11e14001d83c1afd20e2cd8e90246dd8dc4d9a0c605f4b9877d5823`.

`SFT-MATH-OBL-LOGIC-012` - Constructive and intuitionistic correspondence

- **Claim:** `SFT-MATH-LOGIC-CONSTRUCTIVE-CORRESPONDENCE-012` - Constructive and intuitionistic correspondence.
- **Forced law:** Constructive correspondence admits a proposition only with an explicit proof object and admits a disjunction only with the retained chosen-side witness.
- **Unique survivor:** LOGIC-012 uniquely retains witness-bearing-construction, complete proof-model custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-SIZE-BOUNDARY-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-012: constructive_correspondence; exact observation: {"conjunction_witnesses":["proof-A","proof-B"],"disjunction_side":"left","disjunction_witness":"proof-A"}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7d457244f5ba9c737af68201d7db91252b4d9d866ed4bc97a288b9bea372525b`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:54834ac57fd5586314ac7fe03945116e6559910d4e74d8c2f94306f81ceacd02`; engine receipt `sha256:be1fa0f7ed9298f961e099db5213007d9c7a556e85756207024813bcea2cee41`.

`SFT-MATH-OBL-LOGIC-013` - Modal and temporal logic correspondence

- **Claim:** `SFT-MATH-LOGIC-MODAL-TEMPORAL-013` - Modal and temporal logic correspondence.
- **Forced law:** Modal and temporal operators are exact quantifications over registered successor paths and counted transition positions.
- **Unique survivor:** LOGIC-013 uniquely retains transition-indexed-modal-temporal-law, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-CONSTRUCTIVE-CORRESPONDENCE-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-013: modal_temporal; exact observation: {"necessary_at_state_one":true,"possible_at_state_two":true,"states":4,"transition_custody":true}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f5bd3ef6e50a2fa477386dc8442915095568655980880d7f09f574d0be124ac5`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:c2cdcaa50be9a43bae77442161de82ee129fffb76eae910534f80b5672c52ec0`; engine receipt `sha256:b2428380ee59ac674bd574be852e1168aaa1a392cb49f8e8879f82ae34cb95e5`.

`SFT-MATH-OBL-LOGIC-014` - Nonclassical many-valued correspondence

- **Claim:** `SFT-MATH-LOGIC-MANY-VALUED-014` - Nonclassical many-valued correspondence.

- **Forced law:** Many-valued correspondence is a generated finite valuation algebra with complete operation tables and explicit interpretation boundaries.
- **Unique survivor:** LOGIC-014 uniquely retains finite-ordered-valuation-algebra, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-MODAL-TEMPORAL-013`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-014: many_valued; exact observation: {"complete_binary_rows":9,"labels":["opposed","unresolved","held"],"reversal_preserves_unresolved":true}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e54ba8cbefbef47b9f5122468fdc0d697068ceddb01702d9c0492b41fbdd9bbb`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:0fc294b7459c9048507462dbfdffad3676a67747968a60f5b5efc10e06563363`; engine receipt `sha256:571b131341ecd1ad9d2fc2f3b131487f8e060e1a05e51af274793f8d3327a562`.

`SFT-MATH-OBL-LOGIC-015` - Proof-theoretic normalization

- **Claim:** `SFT-MATH-LOGIC-NORMALIZATION-015` - Proof-theoretic normalization.
- **Forced law:** Normalization is exact proof-object rewriting whose registered complexity measure strictly decreases until no reduction applies.
- **Unique survivor:** LOGIC-015 uniquely retains strict-proof-reduction-normalization, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-MANY-VALUED-014`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-015: normalization; exact observation: {"measure_strictly_decreased":true,"normal_form":"A","redex":"identity-applied-to-A"}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:f99dd7358106257d10c44a25fc86bb55ec8a5e002fccc1e19be61d238a192b62`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical

`sha256:b52af9f580b77385d69c619354d435bcef12e5e0f38f8411b9c8ab6cb9779b6c`; engine receipt `sha256:5af25a62e0f550c27d7dde1ff7314806f74aa97c6908f98c8e9753bad2654b76`.

`SFT-MATH-OBL-LOGIC-016` - Foundational consistency and self-verification limits

- **Claim:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016` - Foundational consistency and self-verification limits.
- **Forced law:** Consistency is certified by complete proof enumeration inside the declared grammar; no finite kernel may extend that certificate to arbitrary unregistered self-modifications.
- **Unique survivor:** LOGIC-016 uniquely retains finite-consistency-unrestricted-self-limit, complete proof-model custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-DYN-COUPLED-NETWORKED-012 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-LOGIC-NORMALIZATION-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** LOGIC-016: consistency_self_verification; exact observation: {"declared_closure":["P","P-implies-Q","Q"],"opposed_P_derived":false,"unrestricted_self_verification_claimed":false}; complete family observation custody: 16 records; all rows preserved; complete finite syntax, valuation, proof and model enumeration supplies the result; no imported logical axiom, oracle or unregistered self-extension enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-LOGIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported logical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; truth orientation is held or opposed structure and never a negative numerical scalar; no unrestricted infinite language, universe, compactness or self-consistency claim; no opaque theorem prover or proof step lacking explicit premises and rule custody; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:2899d5fc7e46c6daefed02e69f99288d36017cacbf1164af42327590b70f17a7`; independent `sha256:d9ad3e44fe6cff49a05239c9cc8a3e824fc66eab53c7cd3bcdb8f489d226cf6b`; empirical `sha256:1547cdbd0960253cd455c9267e79fc87e5e06e93433677fd23b818d2203523f1`; engine receipt `sha256:5717d448290a1af7fbbd8ad6a85c473c68b28abce0799711809f5538cb14d055`.

40.21 Family `CAT` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-CAT-001` - Objects, arrows and typed composition

- **Claim:** `SFT-MATH-CAT-OBJECT-ARROW-COMPOSITION-001` - Objects, arrows and typed composition.
- **Forced law:** Objects are generated type boundaries; arrows retain source and target, and composition exists exactly when adjacent boundaries match.
- **Unique survivor:** CAT-001 uniquely retains typed-arrow-composition, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-001: typed_composition; exact observation: {"arrows":["A-to-B","B-to-C"],"composite":"A-to-C","mismatched_rejected":true,"objects":["A","B","C"]}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher

coherence enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:0a4fb13ab60a6fd29b051109a9bb76beb4597d9ab804805e77661930afce911f`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:63d66f6032095493e3ab73cca160ef52207fdd3fcbfb16b5e634a760d9461c1a`; engine receipt `sha256:71c69fc458b52088e7c1dc218452b99d29ef3a4e9634d60c261a7a69cfbb0394`.

`SFT-MATH-OBL-CAT-002` - Identity and associativity witnesses

- **Claim:** `SFT-MATH-CAT-IDENTITY-ASSOCIATIVITY-002` - Identity and associativity witnesses.

- **Forced law:** Every object has a generated identity arrow, and every completely typed triple of arrows composes associatively.

- **Unique survivor:** CAT-002 uniquely retains identity-associative-composition, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-OBJECT-ARROW-COMPOSITION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** CAT-002: identity_associativity; exact observation: {"associativity":true,"finite_maps":3,"left_identity":true,"right_identity":true}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:728ece9a6f622cf845d13f93325428db035ae552a9d953e17e6414dde537a51f`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:d97683a012869ea9e3946378b72725cf8cf332ce713a99623a7b163424eb3045`; engine receipt `sha256:807ed10ff532e9c607cd05e5b289cae65eaef4fef96cfec772ce1eaf6bd64c89`.

`SFT-MATH-OBL-CAT-003` - Functorial structure preservation

- **Claim:** `SFT-MATH-CAT-FUNCTOR-PRESERVATION-003` - Functorial structure preservation.

- **Forced law:** A functor is a total object-and-arrow translation preserving types, identities and composition across the complete generated category.

- **Unique survivor:** CAT-003 uniquely retains identity-composition-preserving-map, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-IDENTITY-ASSOCIATIVITY-002`;

registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** CAT-003: functor_preservation; exact observation: {"all_rows_equal":true,"mapped_composite":"two-times-x-plus-one","source_composite":"plus-one-then-times-two"}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:abf63322046724108546d7f7c82c9b81843bec79d39ea349817ca566b92e508b`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:52656f0a2f211be47829f355e9a5e98bf144de85e1b7bb067a8e787cec8702e3`; engine receipt `sha256:45effaf6d8da98e393865ea391af69e936c277b99614dac177e4f0d14d850900`.

`SFT-MATH-OBL-CAT-004` - Natural transformation correspondence

- **Claim:** `SFT-MATH-CAT-NATURAL-TRANSFORMATION-004` - Natural transformation correspondence.
- **Forced law:** A natural transformation is a typed component arrow at every source object whose complete generated naturality squares commute.
- **Unique survivor:** CAT-004 uniquely retains commuting-component-family, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-FUNCTOR-PRESERVATION-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-004: natural_transformation; exact observation: {"all_commute":true,"naturality_rows":2,"source_labels":[1,2]}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4c4cf959c4fa4d84312eda66735a345ef41b1b40ea2c4538c83e8c6746d3d733`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:9bb9750893d9922512e58650ab9f455ea4f95c01383754e8ff7a3ff7e2efa248`; engine receipt `sha256:5dead80b1c237c637d35f6c43379a65812c2117954f77d5e9390cd3806c3e0e1`.

`SFT-MATH-OBL-CAT-005` - Products, coproducts and universal constructions

- **Claim:** `SFT-MATH-CAT-PRODUCT-COPRODUCT-005` - Products, coproducts and universal constructions.
- **Forced law:** Products and coproducts are generated records whose projection or injection equations force one unique mediating map for each supplied cone or cocone.

- **Unique survivor:** CAT-005 uniquely retains product-coproduct-universal-record, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-NATURAL-TRANSFORMATION-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-005: product_coproduct; exact observation: {"injections_distinct":true,"product_pairs":4,"projection_reconstruction":true,"tagged_coproduct_values":4}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:78be3dde2442eb6f53191d40f840c50642e15cc01fc4a6d501dfc40f6568fac1`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:a8d84cead61d921446c409ad15603d76aa2fb9ca076bf0b9bc2c6c18881dd711`; engine receipt `sha256:7d7a98b46c34532e47b6a37fcd19c87883fe3b009a0fc6059173309712bac821`.

`SFT-MATH-OBL-CAT-006` - Limits and colimits on generated diagrams

- **Claim:** `SFT-MATH-CAT-LIMIT-COLIMIT-006` - Limits and colimits on generated diagrams.
- **Forced law:** A limit or colimit is admitted through complete finite diagram enumeration and a unique universal mediating record.
- **Unique survivor:** CAT-006 uniquely retains finite-diagram-universal-construction, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-PRODUCT-COPRODUCT-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-006: limit_colimit; exact observation: {"coequalizer_classes":[[1],[2,3]],"equalizer":[1,3]}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:a73250bb9a0df7b60893d542b6f59075159b13daf6a019d7b6fbedba92f33557`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:5568a2fd5066d47902f1df6cf6c59d650aeda3c1258aada22fcab17d29df6565`; engine receipt `sha256:66e3d06fc4a7b84928e089f02b4a5e7151461520b6c96327b5fbf9627f2eb11e`.

`SFT-MATH-OBL-CAT-007` - Adjunction and paired universal maps

- **Claim:** `SFT-MATH-CAT-ADJUNCTION-007` - Adjunction and paired universal maps.
- **Forced law:** Adjunction correspondence is an exact reversible pairing between two completely generated hom-supports, natural in every retained object.
- **Unique survivor:** CAT-007 uniquely retains exact-hom-pairing-bijection, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-LIMIT-COLIMIT-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-007: adjunction; exact observation: {"bijection":true,"curried_maps":16,"uncurried_maps":16}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:cbf9506ab93904cc49db75897eefe3de13dc2673b2f2072566de96d8b22fa5f5`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:d1120ae9e22f6ea350f46373a6d441a5a65d8214e9c913db0468880bf7914a73`; engine receipt `sha256:bc77d0449dd3b2b490aff2c45dc27e9552a8798c49bad32cff6dc021dd25b247`.

`SFT-MATH-OBL-CAT-008` - Monoidal composition and tensor interface

- **Claim:** `SFT-MATH-CAT-MONOIDAL-TENSOR-008` - Monoidal composition and tensor interface.
- **Forced law:** Monoidal structure is a typed binary composition with explicit associativity and unit witnesses; structural absence supplies the empty unit form.
- **Unique survivor:** CAT-008 uniquely retains associative-unitary-tensor, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-ADJUNCTION-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-008: monoidal_tensor; exact observation: {"associative":true,"structural_empty_unit":true,"tensor":"word-concatenation"}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:58603d9df9afb3f5bb854fbdc9d34cdad895ee13e90b03b49c798fc857164090`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:d8fab2f375e3a262e80eaa104f73f049c944d4d0a8e719e359cff11f452bb43e`; engine receipt `sha256:1d46e43677dbab4705723767d4134bba4f59a0c6b60e6d45ae97057bfcfa7c5b`.

`SFT-MATH-OBL-CAT-009` - Closed structure and internal maps

- **Claim:** `SFT-MATH-CAT-CLOSED-INTERNAL-MAP-009` - Closed structure and internal maps.
- **Forced law:** Closed structure internalizes the complete map support and is forced by exact evaluation and reversible currying correspondence.
- **Unique survivor:** CAT-009 uniquely retains internal-map-evaluation-currying, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-MONOIDAL-TENSOR-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-009: closed_internal_map; exact observation: {"evaluation_currying_reconstruction":true,"four_entry_tables":16,"internal_maps":4}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:bb8c927166bc35fce9b7391b1272654114999636004 1a5f2036dfdabd5df3381`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:70c0719676973cba81e9913fd93082ac84ec43c4d435b6abd194bd48a8b0c678`; engine receipt `sha256:f68143ffe290ff07df3f325ad9af23893914b6dc1e8b02018f5ee5bf8fa9f8fd`.

`SFT-MATH-OBL-CAT-010` - Type correspondence and dependent records

- **Claim:** `SFT-MATH-CAT-TYPE-DEPENDENT-RECORD-010` - Type correspondence and dependent records.
- **Forced law:** A dependent type is a generated fibre over each base label; a lawful record retains its base label and a value from exactly that fibre.
- **Unique survivor:** CAT-010 uniquely retains dependent-fibre-record, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-CLOSED-INTERNAL-MAP-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-010: dependent_record; exact observation: {"base_labels":2,"fibre_sizes":[1,2],"well_typed_pairs":3}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem

answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:7b3d24f378953f941aeb312762461ed34efe71479c4728abc7f6396bf8c530f1`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:642e9719ddd2e714d499c3c9e9067de5d403d1215088926bce1e33b6bcfc95bb`; engine receipt `sha256:ff7742d564cdc04c75e692c5642100b378945b08fcdc9d11c0150af30e3a71e3`.

`SFT-MATH-OBL-CAT-011` - Sheaf-like local-to-global custody

- **Claim:** `SFT-MATH-CAT-SHEAF-LOCAL-GLOBAL-011` - Sheaf-like local-to-global custody.
- **Forced law:** Sheaf-like correspondence retains every local section and overlap; compatible sections glue to one unique generated global record.
- **Unique survivor:** CAT-011 uniquely retains compatible-local-unique-gluing, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-TYPE-DEPENDENT-RECORD-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-011: sheaf_local_global; exact observation: {"left_patch":{"1":"a","2":"b"},"overlap_agrees":true,"right_patch":{"2":"b","3":"a"},"unique_global":{"1":"a","2":"b","3":"a"}}; complete family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:88fdd3d2f45c65e2bb3c5f2b122e2291520f9a550e805583a721339fa6984990`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:ddff67d39a14897e95a96a695c052ebedf73307cbdba582283f6e97129aaf146`; engine receipt `sha256:73d7dd3a53e88a69cd63d2147afab60c9fce360c065b5f813ffc82b508bdfbfa`.

`SFT-MATH-OBL-CAT-012` - Operadic and higher-composition boundary

- **Claim:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012` - Operadic and higher-composition boundary.
- **Forced law:** Operadic composition is exact typed substitution into generated finite input slots; higher composition is admitted only at registered arity and coherence depth.
- **Unique survivor:** CAT-012 uniquely retains finite-arity-substitution-associativity, complete typed-diagram custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016 -> SFT-MATH-ALG-OPERADIC-COMPOSITION-016 -> SFT-MATH-CAT-SHEAF-LOCAL-GLOBAL-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** CAT-012: operad_higher_boundary; exact observation: {"registered_total_arity":5,"substitution_associative":true,"unrestricted_higher_coherence_claimed":false}; complete

family observation custody: 12 records; all rows preserved; complete finite object, arrow, diagram and dependent-record enumeration supplies the observation; no imported categorical axiom, opaque universal construction or unregistered higher coherence enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-COMPOSITION-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported categorical system, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no untyped composition, opaque universal construction or ungenerated infinite category; no unrestricted higher coherence beyond the registered arity and successor certificate; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:209c79cea714d80cfdd0a0a901d3bd40d5db60708d9ac94e433ccd64a930f0a2`; independent `sha256:e3677780a1671cfb8d7da7009721edb9f6c31c16d04668f6ea416a4bdc21fae2`; empirical `sha256:88d9a6cff7443304f557261b8c79c8c732c643a7062c81f3ba3ba93e25607f6c`; engine receipt `sha256:212418e46bfbec20e1f9b404cd43fb40fc13d180de2ab99bc0a8971017fd2197`.

40.22 Family `NUM` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-NUM-001` - Exact numerical representation and rounding custody

- **Claim:** `SFT-MATH-NUM-EXACT-REPRESENTATION-ROUNDING-001` - Exact numerical representation and rounding custody.
- **Forced law:** Every numerical approximation retains its exact rational source, enclosing neighbours, selected display and the comparison that selected it.
- **Unique survivor:** NUM-001 uniquely retains exact-representation-rounding-custody, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-001: exact_representation_rounding; exact observation: {"lower":2,"lower_gap":"1/3","selected_display":2,"source_fraction":"7/3","upper":3,"upper_gap":"2/3"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:aa0439e8c4b6e462ca753a6844d4aa196c310cfbb5c0d07373c695b0d2464eae`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:48ca7c06d874184181820fa8dc72b64fc64bf9edfed5d4c3b6a1e3a306620d9f`; engine receipt `sha256:8e0f6ddf5bc90fef46eefc32a535fa9fae3d8fc45d3a0f0dd93f657d8c1836c4`.

`SFT-MATH-OBL-NUM-002` - Interval and rational enclosure arithmetic

- **Claim:** `SFT-MATH-NUM-INTERVAL-RATIONAL-ENCLOSURE-002` - Interval and rational enclosure arithmetic.

- **Forced law:** Interval arithmetic propagates exact rational lower and upper witnesses without admitting an unrepresented continuum scalar.
- **Unique survivor:** NUM-002 uniquely retains outward-rational-enclosure, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-EXACT-REPRESENTATION-ROUNDING-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-002: interval_rational_enclosure; exact observation: {"left":["1/2","2/3"],"positive_product":["1/6","1/3"],"right":["1/3","1/2"],"sum":["5/6","7/6"]}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:60d0bcb86ea1c81400cc47dad3679e84e8507fb221c3c7dda45607931e4f7959`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:3563c47b3a299007ae41ea52717625bbaa3d7f45f1caa67dbb895721ef12b73c`; engine receipt `sha256:8c13117d2033b4657174a69f0dae9093d27277e74ea89abe10609bb463cc5072`.

`SFT-MATH-OBL-NUM-003` - Truncation and discretization error

- **Claim:** `SFT-MATH-NUM-TRUNCATION-DISCRETIZATION-ERROR-003` - Truncation and discretization error.
- **Forced law:** Truncation error is the exact retained residual required to reconstruct the untruncated generated relation.
- **Unique survivor:** NUM-003 uniquely retains retained-residual-error, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-INTERVAL-RATIONAL-ENCLOSURE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-003: truncation_residual; exact observation: {"all_rows_reconstruct":true,"generated_parts":8,"identity":"partial(n)+1/(2^n)=One"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational

enclosure is proved; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:932c87c84b67c0c943d68fedd3136e84045a6cad846f4475ee033ead1e83f89e`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:debf82ec63866f975266d65876158f140b7f27fac19c4afd1cbb974152085db7`; engine receipt `sha256:259b72ccea62db62192f6788cc51d7c5e54c21e36eb8a5099148c0f1b6b98b38`.

`SFT-MATH-OBL-NUM-004` - Forward and backward stability

- **Claim:** `SFT-MATH-NUM-FORWARD-BACKWARD-STABILITY-004` - Forward and backward stability.
- **Forced law:** Forward and backward stability are separate exact gaps between retained problem, result and reconstructed input records.
- **Unique survivor:** NUM-004 uniquely retains forward-backward-exact-gap, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-TRUNCATION-DISCRETIZATION-ERROR-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-004: forward_backward_stability; exact observation: {"approximate_result":"23/10","backward_gap":"1/23","exact_result":"7/3","forward_gap":"1/30","reconstructed_coefficient":"70/23"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:1ee8f586ef8ebe536dc6f6f06773216ae174eda6396e61b7239d2f1392d76630`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:9c7569f18cec8cf9a86286af1661b8e29612d3b39ccbd265d3365d1648eff595`; engine receipt `sha256:6929ba74b44f626e8ea7e3b5af82fd3c97e5a4375aa27f1ba050eab41fa2b9be`.

`SFT-MATH-OBL-NUM-005` - Conditioning and sensitivity

- **Claim:** `SFT-MATH-NUM-CONDITIONING-SENSITIVITY-005` - Conditioning and sensitivity.
- **Forced law:** Conditioning is an exact relation between a registered input distinction and the output distinction transported by the generated map.
- **Unique survivor:** NUM-005 uniquely retains input-output-gap-ratio, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-FORWARD-BACKWARD-STABILITY-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** NUM-005: conditioning_sensitivity; exact observation: {"exact_gap_ratio":5,"input_gap":"1/100","output_gap":"1/20"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:d98f9fd32723582e2b8256c8ea09d08b1fc65bb747e36cc04a0d1635cb19bf31`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:a2064617cdf4c01da9ee896e599512d7abef2d126577a9519c1a26fbaa0fcd7b`; engine receipt `sha256:d426bb53ef24834bf45d6ff67e99cbcd3d4b1eb55e300e460138fe2907ab1b34`.

`SFT-MATH-OBL-NUM-006` - Convergence order with exact bounds

- **Claim:** `SFT-MATH-NUM-CONVERGENCE-ORDER-BOUND-006` - Convergence order with exact bounds.
- **Forced law:** Convergence order is a certified successor relation among exact enclosure widths, not an assumed limiting scalar.
- **Unique survivor:** NUM-006 uniquely retains successor-error-contraction, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-CONDITIONING-SENSITIVITY-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-006: convergence_order; exact observation: {"all_bounds_exact":true,"error_family":"1/(2^n)","generated_successors":8,"successor_ratio":"1/2"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:98647d07766af438c7d97cbc21d595bc2bfbaa0a397030300ce4cc5b649848b6`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:4330fd88bba32d5c979d1eb0e6185528a92e728da2ca44ca99246130683f7db9`; engine receipt `sha256:eb7d30b58833016ecf2928e7ddc354f8d5e428bfe0ef3ca80a56a815d3706757`.

`SFT-MATH-OBL-NUM-007` - Root isolation and equation solving

- **Claim:** `SFT-MATH-NUM-ROOT-ISOLATION-EQUATION-SOLVING-007` - Root isolation and equation solving.
- **Forced law:** Equation solving retains exact rational brackets and structural comparison labels when the conventional solution is not an admissible proof scalar.

- **Unique survivor:** NUM-007 uniquely retains ordered-rational-root-bracket, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-CONVERGENCE-ORDER-BOUND-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-007: root_isolation; exact observation: {"equation":"square(x)=2","irrational_scalar_admitted":false,"lower":"7/5","lower_square":"49/25","upper":"3/2","upper_square":"9/4"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5760c094164dbc5df8e097a1ec6ef8a333341c87a2d1e3391da65237d575703e`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:bee7b2d4f814eb6804fa4ce3197cb5b1a0f93bc7da24233450b85cf238a8665d`; engine receipt `sha256:cdde0b0bd96c982f20d5f2caa6227c8dcc5a56338f8efd7e8a08fff00df49ecd`.

`SFT-MATH-OBL-NUM-008` - Linear-system exact and approximate solvers

- **Claim:** `SFT-MATH-NUM-LINEAR-SYSTEM-SOLVERS-008` - Linear-system exact and approximate solvers.
- **Forced law:** A finite linear system is solved by complete exact candidate enumeration or a certified enclosure with residual custody.
- **Unique survivor:** NUM-008 uniquely retains complete-positive-system-enumeration, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-ROOT-ISOLATION-EQUATION-SOLVING-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-008: linear_system_solver; exact observation: {"equations":["x+y=5","x+2y=8"],"positive_candidate_pairs":64,"unique_solution":[2,3]}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:4b2e7cceeb5a03da1f2ab693e04a89152af135185be610fd262cb3c1d2669b86`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:ebe7775a4944dcbbc6703b1aac1095d27602f27bb87401f81a92e42171472e46`; engine receipt `sha256:8b2b0f1ba1f2834f192150342fb2c319be37ff34289dbc4150edf24e092d4a0a`.

`SFT-MATH-OBL-NUM-009` - Interpolation and approximation

- **Claim:** `SFT-MATH-NUM-INTERPOLATION-APPROXIMATION-009` - Interpolation and approximation.
- **Forced law:** Interpolation is an exact weighted reconstruction from registered support; approximation retains its enclosure and residual.
- **Unique survivor:** NUM-009 uniquely retains exact-weighted-interpolation, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-LINEAR-SYSTEM-SOLVERS-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-009: interpolation; exact observation: {"endpoints":[[1,2],[3,6]],"midpoint":2,"value":4,"weights":["1/2","1/2"]}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:eabe204efc324a286f77669f21ecfa443e1972311effeb90172c531454a2b12e`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:757599e5b908da9793011d6d86e6a0f8318628700e347834efca9fd922256d24`; engine receipt `sha256:e751920f65b81f5cc2649abdfcd11aa0c1dfae51c10f4d888d4415716d4d2b09`.

`SFT-MATH-OBL-NUM-010` - Quadrature and accumulation enclosures

- **Claim:** `SFT-MATH-NUM-QUADRATURE-ACCUMULATION-010` - Quadrature and accumulation enclosures.
- **Forced law:** Quadrature is a finite exact cell accumulation whose refinement relation and enclosure are retained.
- **Unique survivor:** NUM-010 uniquely retains exact-cell-accumulation, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-INTERPOLATION-APPROXIMATION-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-010: quadrature_accumulation; exact observation: {"identity_values":true,"interval":[1,3],"midpoint_result":4,"trapezoid_result":4}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply

the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e383afc423b6a6f2534d1c803dfe48bb6cdeb7159f4f54163ba1218a37b46325`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:5e3768489c04d6f43b56f7532a67a88fed6c7dfb3bc8cd4c85873e297e7a33db`; engine receipt `sha256:648b6d3a1e7459deb29c93bf36fbb8f428b743ff0045f05f36cac4f6712a1910`.

`SFT-MATH-OBL-NUM-011` - Differential-equation numerical correspondence

- **Claim:** `SFT-MATH-NUM-DIFFERENTIAL-EQUATION-CORRESPONDENCE-011` - Differential-equation numerical correspondence.
- **Forced law:** Numerical differential-equation correspondence is an exact generated recurrence with every state, part width and residual recorded.
- **Unique survivor:** NUM-011 uniquely retains exact-recurrence-state-trace, exact numerical custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-QUADRATURE-ACCUMULATION-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-011: differential_equation_recurrence; exact observation: {"initial":"1","part_width":"1/2","steps":2,"trace":["1","3/2","9/4"]}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:6b4ea28c4ad8b8ca0fbe4e7f4e15079c5bf968e461bf400e14c8c1ec9ec73dc4`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:51dd4c37f946a726d71d905b30d208cd48c1e76b9405fcaa6802715b6446b06a`; engine receipt `sha256:00a35c5c34967fc4c562146e871ba053dd4af162aef69f61a6122d5222fd789e`.

`SFT-MATH-OBL-NUM-012` - Verified computation and certificate extraction

- **Claim:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012` - Verified computation and certificate extraction.
- **Forced law:** Verified computation emits an independently replayable certificate containing exact inputs, operations, result, reduction and boundary conditions.
- **Unique survivor:** NUM-012 uniquely retains replayable-arithmetic-certificate, exact numerical custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012 -> SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012 -> SFT-MATH-ALEXT-POLYNOMIAL-ROOT-ISOLATION-001 -> SFT-MATH-NUM-DIFFERENTIAL-EQUATION-CORRESPONDENCE-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** NUM-012: verified_computation_certificate; exact observation: {"cross_denominator":6,"cross_numerator":5,"expression":"1/2+1/3","gcd":1,"reduced_result":"5/6"}; complete family observation custody: 12 records; all rows preserved; complete exact rational inputs, operations, residuals, enclosures and refinement records supply the observation; no imported continuum scalar, opaque numerical routine or unrecorded rounding choice enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-NUMERICAL-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported numerical-analysis theorem, conventional library result or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unrecorded rounding, truncation, residual, conditioning or discretization distinction; no continuum limit is imported where only finite refinement or rational enclosure is proved; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:08e8ba8b220bdd807ed1e7aa653443d223999831a5fe284b392823bfb7c42d73`; independent `sha256:d360ddc9fc6c7e39f0001fe57fed5dfdca1d1b9ea964975dc5bd35ec0a1775de`; empirical `sha256:5d7c521770235c7bac20bced8d3b424809f5b3422510cb341f80eb39d5276baa`; engine receipt `sha256:0696ffb60057ca05592a019bc74437c233e398d8e3ed8633d13122917999105d`.

40.23 Family `SYMB` - 10/10 complete

This family contributes 2,560 completely decided candidates, 10 unique survivors and 40 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-SYMB-001` - Symbolic expression identity and canonical form

- **Claim:** `SFT-MATH-SYMB-CANONICAL-EXPRESSION-001` - Symbolic expression identity and canonical form.
- **Forced law:** Symbolic identity is equality of exact canonical generated syntax after only registered unit, ordering and association rules.
- **Unique survivor:** SYMB-001 uniquely retains canonical-symbolic-record, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-001: canonical_expression; exact observation: {"canonical_record":["add","x","y"],"inputs":2,"unit_and_order_paths_agree":true}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:1cfd2330329d4771b9ace9fee6ba8e6290c044b23382eae1856cae5fa1b40a3c`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:dcdad5d3cd2109d4e117d9d7c5c0310aebd4f2bf96351d27f87d666b58df1152`; engine receipt `sha256:1fc6dbd6b91cb7dac6d70badd5ab5bee41c63de626a6abd2a076fb40f3f2a407`.

`SFT-MATH-OBL-SYMB-002` - Exact simplification with provenance

- **Claim:** `SFT-MATH-SYMB-SIMPLIFICATION-PROVENANCE-002` - Exact simplification with provenance.
- **Forced law:** Every simplification retains its input, ordered rewrite steps, output and independently replayable provenance.
- **Unique survivor:** SYMB-002 uniquely retains trace-retaining-simplification, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-CANONICAL-EXPRESSION-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-002: simplification_provenance; exact observation: {"input":"add(mul(x,One),Empty One)","ordered_trace":["remove-multiplicative-One","remove-structural-absence"],"output":"x"}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:75b99b507b9bb07155f9b211a897b94c3185b53e04901b48f0fe59e06a476b0f`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:7b5236f3cb8b641fe80932f05c8973f38e25073247bd84784e3b081ed4297614`; engine receipt `sha256:9cf05501f0c8844ef8a38799f148078b8458cc4e425821140342206 72d1183cf`.

`SFT-MATH-OBL-SYMB-003` - Polynomial factorization and expansion

- **Claim:** `SFT-MATH-SYMB-POLYNOMIAL-FACTOR-EXPAND-003` - Polynomial factorization and expansion.
- **Forced law:** Polynomial expansion and factorization are inverse exact coefficient organizations over a completely enumerated declared degree and coefficient support.
- **Unique survivor:** SYMB-003 uniquely retains coefficient-convolution-factorization, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-SIMPLIFICATION-PROVENANCE-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-003: polynomial_factor_expand; exact observation: {"degree_support_complete":true,"expanded_coefficients":[2,3,1],"factors":[[1,1],[2,1]]}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters

- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7129e709446befc1e84ec9fe18c9a1a7c78da0d13b1b888b06948ed5063ecac0`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:08bfdd57dc5fc53787c79f866ac7d2c23fce35a901590a726d0074140eaab9ee`; engine receipt `sha256:532d8d3fe185773057827ec65c5ede613f674b27e0ffdbe8fdc49a9a419be9ff`.

`SFT-MATH-OBL-SYMB-004` - Symbolic equation solving

- **Claim:** `SFT-MATH-SYMB-EQUATION-SOLVING-004` - Symbolic equation solving.
- **Forced law:** A symbolic solution is the complete generated support of exact values satisfying the registered equality, with absence distinguished from an unsearched domain.
- **Unique survivor:** SYMB-004 uniquely retains complete-symbolic-solution-support, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-POLYNOMIAL-FACTOR-EXPAND-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-004: symbolic_equation; exact observation: {"candidate_labels":8,"complete_solution_support":[4],"equation":"x+3=7"}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:886163eb7799f1c7fd9a2cd36f0406398c0681516d2690b5b7526544003dbf8b`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:c1dcd4c8fde361ea531093d15a64d2b1405db8e858479aa9d92beb3268bb6f8a`; engine receipt `sha256:30c7bffc4bfa23e1bb4bbb549fd04f89c9cb1f72be967384b53be228a266d015`.

`SFT-MATH-OBL-SYMB-005` - Rewrite termination and confluence interface

- **Claim:** `SFT-MATH-SYMB-REWRITE-TERMINATION-CONFLUENCE-005` - Rewrite termination and confluence interface.
- **Forced law:** A rewriting system closes only with a decreasing termination witness and agreement of every declared critical rewrite path.
- **Unique survivor:** SYMB-005 uniquely retains decreasing-confluent-rewrite-record, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-EQUATION-SOLVING-004`; registered root

SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.

- **Post-registry observations:** SYMB-005: rewrite_termination_confluence; exact observation: {"both_terminate":true,"common_normal_form":"x","declared_critical_paths":2}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:4ff975d36fc8def113f7a56489e3bd05257956be5126fbcc8bfbc458cb435ffb; independent sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74; empirical sha256:eb5954216e7ddf4dcfcbbb9e55fa78fefb8dde45c2430dff85cfcb95daff47cd; engine receipt sha256:ab851259d84f0c5b760b72589e8b6695d2b3e44efc11dd4b503e839d4647b37f.

SFT-MATH-OBL-SYMB-006 - Generating-function transforms

- **Claim:** SFT-MATH-SYMB-GENERATING-FUNCTION-TRANSFORM-006 - Generating-function transforms.
- **Forced law:** A generating function is a finite exact coefficient record whose product is forced by complete degree-wise convolution.
- **Unique survivor:** SYMB-006 uniquely retains finite-coefficient-transform, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure depth_independent and external status empirically_tested_and_independently_replicated.
- **Root lineage:** SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-REWRITE-TERMINATION-CONFLUENCE-005; registered root SFT-ROOT-THERE-IS-NO-NOTHING; axioms 0; free parameters 0.
- **Post-registry observations:** SYMB-006: generating_function_transform; exact observation: {"convolution":[1,2,2,1],"left":[1,1,1],"right":[1,1]}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation sha256:cbcf798deb2d45034b996ceb91c4b1d1cea3cf0340cb2d7e8ec36d9ad9c73ff3; independent sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74; empirical sha256:c2b996ba3c4a118c9722084597fb05ee3ef6293b26e5f52b32d3172e71772370; engine receipt sha256:8c88175914327cf3786f3f8b238fa3ec9d4b5a872a2c54beafc9f759d7f0732a.

SFT-MATH-OBL-SYMB-007 - Fourier and Laplace correspondence transforms

- **Claim:** SFT-MATH-SYMB-FOURIER-LAPLACE-CORRESPONDENCE-007 - Fourier and Laplace correspondence transforms.
- **Forced law:** Transform correspondence uses finite held/opposed phase labels and positive exact weight records; conventional imaginary or transcendental scalars are not proof values.

- **Unique survivor:** SYMB-007 uniquely retains held-phase-positive-weight-transform, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-GENERATING-FUNCTION-TRANSFORM-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-007: fourier_laplace_correspondence; exact observation: {"held_walsh_output":[2,"EmptyOne"],"imaginary_scalar_admitted":false,"positive_weighted_support":"11/4"}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:79e489fd6a084f9a966d806bdc7d0f6a6b3f0874454ae6160b4e7fab7a3dbaa7`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:897acbe4c1229b2294c1135e2a4d6c6d5030509b571e4ab54edda3a1df4abebb`; engine receipt `sha256:78a40cfd8bef40d4c2979e7fcf85c1ff31f0b49925a73452d2bd3619bbf8e731`.

`SFT-MATH-OBL-SYMB-008` - Special-function recurrence representation

- **Claim:** `SFT-MATH-SYMB-SPECIAL-FUNCTION-RECURRENCE-008` - Special-function recurrence representation.
- **Forced law:** A special-function correspondence is admitted through an exact base record, successor recurrence and complete generated trace.
- **Unique survivor:** SYMB-008 uniquely retains successor-recurrence-special-function, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-FOURIER-LAPLACE-CORRESPONDENCE-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-008: special_function_recurrence; exact observation: {"generated_trace":[1,2,6,24],"recurrence":"Gamma-successor=label*Gamma"}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:585130b72396e89fb825782884d60ebfa9a89a1f513f5256867848aa2d94ae11`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical

`sha256:098d9fab79b3703af63993ee63c7afa26bf792ac6e84ffb7fd27775dc0d5f47d`; engine receipt `sha256:86d0c48ae5b12195f37496629db40327bed840d8bbfe1fea05830f68540f6152`.

`SFT-MATH-OBL-SYMB-009` - Automated theorem search boundary

- **Claim:** `SFT-MATH-SYMB-THEOREM-SEARCH-BOUNDARY-009` - Automated theorem search boundary.
- **Forced law:** Automated theorem search is complete only for a registered grammar and depth, returning every proof or an explicit bounded absence record.
- **Unique survivor:** SYMB-009 uniquely retains bounded-complete-proof-search, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-SPECIAL-FUNCTION-RECURRENCE-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-009: theorem_search_boundary; exact observation: {"A_to_C_proofs":1,"premises":["A-to-B","B-to-C"],"registered_depth":2,"unbounded_completion_claimed":false}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite, assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:5c8562e9629f7546c3889a158c752082ee8656e1dd5479ec86ba33f147988ac2`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:8feb6351cced8216d854d87cb86f3840fe601c09a7f2ccab25adf3a1b3e53a94`; engine receipt `sha256:f30bfc00ab43dc67ac366865b093161f74729273113b2e2a6361b45e80901069`.

`SFT-MATH-OBL-SYMB-010` - Constructive witness and certificate generation

- **Claim:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010` - Constructive witness and certificate generation.
- **Forced law:** A constructive conclusion carries an exact generated witness and a certificate that independently reconstructs the claimed relation.
- **Unique survivor:** SYMB-010 uniquely retains constructive-witness-certificate, complete symbolic custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012 -> SFT-MATH-LOGIC-NORMALIZATION-015 -> SFT-MATH-SYMB-THEOREM-SEARCH-BOUNDARY-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** SYMB-010: constructive_certificate; exact observation: {"positive_witness":3,"relation":"x+x=6","replay":"3+3=6"}; complete family observation custody: 10 records; all rows preserved; complete exact syntax, rewrite paths, coefficient supports, recurrence traces, bounded proof searches and constructive witnesses supply the observation; no opaque algebra oracle or unrecorded rewrite enters
- **Sources and boundaries:** SFT-V3-INDEPENDENT-EXACT-SYMBOLIC-OBSERVER, SFT-V1-V2-MATHEMATICS-OBSERVATION-CORPUS. Exclusions: no axiom, imported computer-algebra rule, theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no opaque simplifier, unrecorded rewrite,

assumed confluence or silent branch deletion; no unbounded theorem-search completion is claimed from a finite search; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:aa235bf258e525c947871c23cc24ab06fb20127e1e694e3380673cec55435ed2`; independent `sha256:8e47dac0a1f0aee7f83b21d46d7b20d05e90f74659638d21fa8b0d25d46e0d74`; empirical `sha256:cc71af9468e36537af4fbee56c301470d21a1576a26d417b10494a8771dff184`; engine receipt `sha256:fda3df9ae9119867ecd3cef93e2affb43dc0dd8596f37043767c6a45b3cf617a`.

40.24 Family `XINT` - 8/8 complete

This family contributes 2,048 completely decided candidates, 8 unique survivors and 32 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-XINT-001` - Mathematics-to-Information exact-structure handoff

- **Claim:** `SFT-MATH-XINT-INFORMATION-HANDOFF-001` - Mathematics-to-Information exact-structure handoff.
- **Forced law:** Mathematics supplies exact support and relation structures; Information Science alone owns their informational interpretation and observation laws.
- **Unique survivor:** XINT-001 uniquely retains typed-information-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-001: information; exact observation: {"downstream_owns":["semantic-information-quantity","channel-observation"],"mathematics_owns":["exact-support","distinguishability-structure"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:40d25e999ac7752e415624327ac08c87a500b9a78a135bb4962bc567e579f2da`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:fa3132dce9f2aba376be0a6c286d5f07c31c23c98c51fc40d30de3e3bf6cc62c`; engine receipt `sha256:6ffddc69383435d029cf280ab5d376838c88e0747f4d47ab9bb5db8157b66f68`.

`SFT-MATH-OBL-XINT-002` - Mathematics-to-Computation model handoff

- **Claim:** `SFT-MATH-XINT-COMPUTATION-HANDOFF-002` - Mathematics-to-Computation model handoff.
- **Forced law:** Mathematics supplies state and relation structures; Computation alone owns their execution, machine and resource interpretation.
- **Unique survivor:** XINT-002 uniquely retains typed-computation-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-INFORMATION-HANDOFF-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** XINT-002: computation; exact observation: {"downstream_owns":["execution","resource-law","machine-behaviour"],"mathematics_owns":["state-set","relation","proof-structure"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:4424d36cbe3d27f42413a8d5a636214924dcd781741bb29f6801f845a7cae8e9`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:5f85b3d20ad268bec907fbf651c056c5a529b26cd9667a780fecf4e2f934b5bf`; engine receipt `sha256:3c14d83ea82bd2caaadcdd4e8f83f2d9cfa2ceca3bd433ec8a9036382e43b450`.

`SFT-MATH-OBL-XINT-003` - Mathematics-to-Physics quantity and geometry handoff

- **Claim:** `SFT-MATH-XINT-PHYSICS-HANDOFF-003` - Mathematics-to-Physics quantity and geometry handoff.
- **Forced law:** Mathematics supplies exact quantity and geometry structures; Physics alone owns physical identity, unit realization and measurement.
- **Unique survivor:** XINT-003 uniquely retains typed-physics-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-COMPUTATION-HANDOFF-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-003: physics; exact observation: {"downstream_owns":["physical-identification","measured-value","unit-realization"],"mathematics_owns":["exact-quantity-record","geometry","symmetry-structure"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ac19b09cf848c60b9cc452adfe3ff8aee4974c6369459e6bb5cb9b317deae078`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:d1a85242537e338ec244c7fba0c2eff4d988fb48ee50dbd686f81ab528111e6e`; engine receipt `sha256:a8886f024818e308656ead69f4cd4aaab32762944ace0d4e6bc2892a6f8062de`.

`SFT-MATH-OBL-XINT-004` - Mathematics-to-Chemistry structure handoff

- **Claim:** `SFT-MATH-XINT-CHEMISTRY-HANDOFF-004` - Mathematics-to-Chemistry structure handoff.
- **Forced law:** Mathematics supplies graph and algebraic structures; Chemistry alone owns chemical identity, reaction meaning and chemical measurement.
- **Unique survivor:** XINT-004 uniquely retains typed-chemistry-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-PHYSICS-HANDOFF-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-004: chemistry; exact observation: {"downstream_owns":["chemical-identity","reaction-measurement","material-context"],"mathematics_owns":["graph","algebraic-relation","enumeration"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7241230f88a4ebdc31a9a1d4c99192ecf3bc082249f65107dda67a0a64468763`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:66c92430ecea6663f18bac6c021bbb5695e71c1081e1d114be4584e20830766b`; engine receipt `sha256:7d8ff0f49eec26d921585ff950f2f0a0361059515d99ef868dac5147078db211`.

`SFT-MATH-OBL-XINT-005` - Mathematics-to-Biology organization handoff

- **Claim:** `SFT-MATH-XINT-BIOLOGY-HANDOFF-005` - Mathematics-to-Biology organization handoff.
- **Forced law:** Mathematics supplies network, order and inference structures; Biology alone owns biological function, organism observation and history.
- **Unique survivor:** XINT-005 uniquely retains typed-biology-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-CHEMISTRY-HANDOFF-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-005: biology; exact observation: {"downstream_owns":["biological-function","organism-observation","evolutionary-history"],"mathematics_owns":["network","order","probability-correspondence"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:1491275b9ab8b5bb67edc1f3e2d40854686d7e9273f5eee03a59740de74d6e69`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:75ad17d34d720beb05e5e54a0889ed38e3f571d1b349e986ec40b72336458d64`; engine receipt `sha256:ccdcf95d6ac30220d49853eed9c196f15fdc6a4cf578a555811714099d5dc038`.

`SFT-MATH-OBL-XINT-006` - Mathematics-to-Social inference handoff

- **Claim:** `SFT-MATH-XINT-SOCIAL-HANDOFF-006` - Mathematics-to-Social inference handoff.
- **Forced law:** Mathematics supplies inference, game and network structures; Social Science alone owns population observation and institutional meaning.
- **Unique survivor:** XINT-006 uniquely retains typed-social-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-BIOLOGY-HANDOFF-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-006: social; exact observation: {"downstream_owns":["population-observation","institutional-meaning","behavioural-claim"],"mathematics_owns":["inference-structure","game-relation","network"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:70552cf87ac63a4f6a0d133b5e441bf4f11b6c93b71e10bd0c0515581bef9143`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:490ae6c106301a7b5e017db1025c3363f280540ead11804d65e593c0dbfb224a`; engine receipt `sha256:b404e098c202044f424847255966dc5938d54a3ee47c96cb75031d6074c6216f`.

`SFT-MATH-OBL-XINT-007` - Mathematics-to-Engineering calculation handoff

- **Claim:** `SFT-MATH-XINT-ENGINEERING-HANDOFF-007` - Mathematics-to-Engineering calculation handoff.
- **Forced law:** Mathematics supplies calculation, optimization and certificates; Engineering alone owns design choices, performance tests and safety acceptance.
- **Unique survivor:** XINT-007 uniquely retains typed-engineering-structure-handoff, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-SOCIAL-HANDOFF-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-007: engineering; exact observation: {"downstream_owns":["design-choice","performance-test","safety-acceptance"],"mathematics_owns":["calculation","optimization-structure","certificate"]}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:873b36dae06a947885b5828573e77b560fcb53d1621955d807ec9803c4251556`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:93b90de5d5206de111f243a83e0375df71e164f02e603293d37ec1255663c060`; engine receipt `sha256:26cb9935ef8d7b4deeecc3475b0347694990abe929c4b3dd8d123afa8a1f6a98`.

`SFT-MATH-OBL-XINT-008` - Shared mathematical identity without duplicate ownership

- **Claim:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008` - Shared mathematical identity without duplicate ownership.
- **Forced law:** A shared mathematical object has exactly one owning Mathematics identity and downstream branches cite it without duplicating or silently changing it.

- **Unique survivor:** XINT-008 uniquely retains single-owner-cross-reference, one-owner custody, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010 -> SFT-MATH-XINT-ENGINEERING-HANDOFF-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** XINT-008: one_owner_identity; exact observation: {"duplicate_owners":[],"registered_interfaces":7,"typed_references_only":true,"unique_branch_labels":7}; complete family observation custody: 8 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** SFT-V3-CROSS-BRANCH-OWNERSHIP-OBSERVER, SFT-V3-BRANCH-ROADMAP-CORPUS. Exclusions: no axiom, imported branch conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, silent semantic import or untyped handoff; no downstream measurement is recast as a mathematical premise; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:421c95bd73557f39ef14352206b1f7f2094bc613e83bc2a21318794a0c2b097e`; independent `sha256:fe9b61ff20c1199c64e2f1b991041a8aeed4aba1147db501142fb5efdaad5c0b`; empirical `sha256:ab492d320948ee342bd56b6e57f6f9c778f9605a6baaa8bb3501c77f427f89e3`; engine receipt `sha256:ae8b83e55ba8150ddafe5fc01cae15e7e7d880da1556b1946704ef737060cd80`.

40.25 Family `VALID` - 12/12 complete

This family contributes 3,072 completely decided candidates, 12 unique survivors and 48 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-VALID-001` - Arithmetic and algebra complete validation vector

- **Claim:** `SFT-MATH-VALID-ARITHMETIC-ALGEBRA-001` - Arithmetic and algebra complete validation vector.
- **Forced law:** Arithmetic and algebra complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-001 uniquely retains complete-arithmetic-algebra-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-001: arithmetic_algebra; exact observation: {"all_external_statuses":["independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independentl

y_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":55,"families":["BASE","ARITH","ALEXT"],"receipt_hashes":["sha256:28252bae62373d8657967ba28f8f2d497c2cf09abbae71609cb05c4f40196418","sha256:632de46c995329ed86f8397e8cf2cc493d5074027941a50069df66b4ac9e29bb","sha256:52e1db94bb7865270bb3387c47c609c438de3338b542cc31d910b981e4250968","sha256:b4072f93147c8cd9b7233fc96cfeadeb791210239ab4e4ff7dec0ed7462e6d2a","sha256:1f08bfaf9f602a3825dc75979070e05f9d51a7d01cdf3c34bab31c06849855bd","sha256:d0e20f40782f51cc85c1f572b624570e65b51bbb80135a5e8a6b3cf69f71a5f6","sha256:a04f7de0ede6e2348896cf16e59981eebb5391198e498a1ec546b68a4a8d3a6e","sha256:2a86d644881cdacb757327d5aaca01de41c3458c043980c66a16f5084cdfb45b","sha256:3385414ceea0fd4212111e42bd6ce7d6d35556b7c97d66563813d5b8d1531849","sha256:3e0e44b65fbe660fe2a29b9927b1870a40da25c4b0c786e2adaa3df9a2f7cabe","sha256:3d366d57b26d002cb27a16e74b022a0856e133c616b7da6a83233c84bdd50813","sha256:47873a83a90e29fa22d065ff13aa0152edb6212414e35cc0435e8eff2c9ea5a8","sha256:8141546a120b73aa7cac0cd843728bdf3122d1bf3c0048c132760c82fd103238","sha256:8725bce36fe77923288b99b1a978ef3bf84cac9ffe1b172fd67bd221527d19ff","sha256:a32c6d6ee891f97acadd53cbec922bbcad5a6ad8a928264bb654b4caaf997098","sha256:ba975fdfe0c8946cb713eabcd9f48e77284941ec08fc257b9d3bb614b701540d","sha256:e355bc5aa649df8fe2daf21a3e0ae56d5582680edd7f867a751bb954fca224a4","sha256:6fb3f40f5631367f23bad6cda58c87b335d33fd212a7c9d37068d8b920ad65e0","sha256:52fcc090c336360dcdfd4b95919ffa5926d9d8fcd430ff6bb53d7d18cf5db1d6","sha256:27847629dbd2fb1d1c2f14d05017f8fb9395482d6e6ca001a24622e55364dfc4","sha256:3bb3a9fe4790150369defc5b2dc8568fed100b688dbb6a9c1d5cbd40c4f9a045","sha256:0abf8d557108f37f4d9cfe735db3fd6e0d79a3d9405129f0a40ae86fb606d5b1","sha256:f95ac235ac96429b36b10e8331478f110ed074e98a338c6f015d3f8874dc787c","sha256:2d243dbd08260c483e79009067051239f1304ad3998006c1ff82c301062ba28b","sha256:75b0b5df0397c0aeaba035272561fd0e62949b0ef75580a153c8a109e84a06bb","sha256:addeb561438c9913313824ba43eb8fceccb53c574f5a9853b3c9c1fac4e4e4b7","sha256:feeb1879d97191edc8827be6da5421a2c78aba56ba2d3e0fe5dbf0882aac9867","sha256:d3229eb70862bcc3a5d03f42c9abfc650ab90781d45fc2d5090abf8322d2ca08","sha256:ecbb32a37fb10022315429bb39d88f49437fc856396d0b8db42fca49b2032493","sha256:e25cde93469d6d813be366bc228e55cb13e704a965e60711ab396c6f7a36f476","sha256:e02b5bc7444746ceaf5a115853d012583abae63804f6531b39727fc504424e7e","sha256:4b082b2a5fd4bed326e2ebaa180ff23aa3d7bcb0475947732bedf9d65f821af6","sha256:d74f03c498a284dcb6a576786ce45a09c87dd921ac78db3ce71bcd54247bc018","sha256:12882efe837fb967097e5f5540d48077cd12b48bb0c37662fbdd6e0052c797ac","sha256:45b93a4d1bb7a137599e21c74e7e9522054f6b741433365d957dd06fc703a3a1","sha256:b18a07076f76362a892ce1ed995e8b67e1af5c41ecdde90b16ea34a35dfa886d","sha256:dfc025633b4bc068fab1d51dd8596eaf531d428a8b8d83d5c49b015b1311b62a","sha256:f15dd91147438a821fcedb1cd0abcab032151e1393fea1766f02d97fe5d4d64f","sha256:3f73120fde18553beedeec0de54e9ee4834b9eb9ebe19976ddb38b251d1bffac","sha256:2a078b638cd00e98a59fb4ccd6a47e9b691c4a713710bdbb4112062441c41085","sha256:a8a9c91bf09055eafe7c8946783af33d8cbf408241c7a0cc65b022fc8a85393d","sha256:18143d49ddbc35dfaa29e6333b0529d9c93c269483c06206f79a2bd465d2bbfb","sha256:322408091620c6dfed8b960578f5744a202f7f05536bc0b348f9e62cccb1b9fb","sha256:a0126b838d3f636788373c0a95adc8ff48e8a9d9b1d7c0797aae6403fda8473e","sha256:2d9cb180af9554925c3ff156e2b05d1545b9dba2206a24b60c0e17023e0e9fe2","sha256:37fd63da2da1d8bff586bea60bebad91ef43d0115437e9efd7e360dbf05b94f0","sha256:41100b9b02e60bc1321f4ffa7590f3f2c5ef6223f3983e323a4177d53bde4a4e","sha256:b4db0d57fd93d9489c8d94fe9bae1167b9ffa29c775db04f909dc9c707cdb922","sha256:3e417f42c092e58b78ceda9477a4d4c3303b8a1cb0896c6d3623f387edc24824","sha256:2dff8bb2232a8572a0c20e34c05fc873f83f2e9cb74d008cc32b7b8ff2d3d26b","sha256:81f843b80e1f07e44b1f212eeefd448f604aef89255cfe08dd26ff5fd8c5c253","sha256:ccaaa6636f523d2e6bc6cd13996d49a0c748dbc4378568188f6e307de4bfc48d","sha256:1492db9e71b8fbd66da325cd1c27997d144df3e1674818eda01f3b9e11547d2b","sha256:c38b6cbec7e61e8fbfa7be7ca0a1ab4eb3af572576521a37c9cf03acde3bb413","sha256:674c5e710359cd7da9382006ad39c9539591fc9f0ba8d7bd9728ca3eb804a010"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters

- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:4917a2c00132f9176c975843d65de40635617a9741d2364c3edac73b6b4bbfc5`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:d5be3a231c2f94025acc23c34eb7020292416dd6f2da1d031ee26ebf6b5970c9`; engine receipt `sha256:f7793e7fbdbdaf18783503febe565e6a2f87706cf66df942752963520d0d7378`.

`SFT-MATH-OBL-VALID-002` - Combinatorics and graph complete validation vector

- **Claim:** `SFT-MATH-VALID-COMBINATORICS-GRAPH-002` - Combinatorics and graph complete validation vector.
- **Forced law:** Combinatorics and graph complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-002 uniquely retains complete-combinatorics-graph-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-ARITHMETIC-ALGEBRA-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-002: combinatorics_graph; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":26,"families":["COMB","GRAPH"],"receipt_hashes":["sha256:afca8ab23f33cbad39e5aa6e51d5df24bfddd946cfa5568f46e6c781b5503d4f","sha256:9b95112577a4816525ad072bf79acf3c8a573c98847872d16c79f5161d11625a","sha256:5d43c44b342b930315faa5159e3ab7ec026fc538d9b3f77bdd56ff7826ce098e","sha256:c40a635564aae5ef85402c7b69e06ec78ed0024112d2084b58fcc0a2ccf7e004","sha256:ba8380cefcf6fd4b157b57674588bacdf9c5565bc3e848f3638882df1bcc3a1f","sha256:2d097fe60b69c55a86accb8d417c083c3bdbe07041e0eb058081887689943664","sha256:e01c8059812a3e920d2f722ef9f47e2e3375ae255da534d20c5efcab589e17c2","sha256:d650fb44b26a7778b1431e877854dbcbaed78f8612c57d47d6e80118fc5f37d6","sha256:d1e03a25a63c3099f5d2a8f414ad7815c8471db3b30581e48a08d58a39dcf982","sha256:b5ede9884b69fb563c6774eacd7bd3df38cd803bb164cb775e000e7d1cb8c2ac","sha256:2a8c17b72cdefc363e819e2019edbffaecf13d12ea569b64b0bee4dfa8f01cf2","sha256:c7ab86cf76ff13e5c47488f640d8d34afa30fb2008fc16e4a273562b3f4e2c98","sha256:489635c31e29a277498395071437690e60c00fd3287af337754ed86cd2d1c9a8","sha256:164d8d5708f6238d6c7864f2b2261dea4a4097d12c335f68d56765cea778cc01","sha256:383274c9576c7f228b386a5eb969fb995d051ca71d963c8068014712304be26c","sha256:f81889343d13708d3cc085df21f33b4636619dea455cbf58cd4df9abada48e57","sha256:9b310cbbfe57e1eca969de9a2e73c7136ce86428d741a10cff9e7e1170fa24c4","sha256:361fb44c8f989543dacde61ce4b3d03f6f6096ab9fba773664b339627f275671","sha256:283c65166202c674101581fbcf1b718bab3f5ad45ddcf38b4375f34480fa11f9","sha256:a3ea7e43de3c0a16186153b9855e4b9f90acc8b096a2767bc84d0002b4f5e66c","sha256:95711a8c91c539d061e323813786738ecf6f65f5b4d800ce7a7619e1a64bc993","sha256:9ce82e3125329f6738cca2ab5b89ef6bbd92f237af6cf1619160351d5d220b69","sha256:6bf043d86d7d1d7a8a860124bf51cdb5f9bb780586d937880377b4fe0175c65e","sha256:a2827845f35854c3fc3774078d4e794b651ee10ec65995f2cbb14a31c50f72ab","sha256:dfd637109db6dfdc9a1ebd98dff9ce928782ddec159f8310428d7154178d4ce5","sha256:fb1a3980a9e3273950f1ad3d9426d228287085c3cba62162735538726bde0626"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result;

host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:07396e36975ee752022f86826c53e27862aa6f592c5f112881e449917f48ed24`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:98aa79c6b906e120b4ff0298e950419feb859a7c2f2247d26c8843f8401385ed`; engine receipt `sha256:a8db169be7fb482b1aa32a8a38915111fc9663128124afb60a00528023449460`.

`SFT-MATH-OBL-VALID-003` - Linear and algebraic-structure validation vector

- **Claim:** `SFT-MATH-VALID-LINEAR-ALGEBRAIC-003` - Linear and algebraic-structure validation vector.
- **Forced law:** Linear and algebraic-structure validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-003 uniquely retains complete-linear-algebraic-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-COMBINATORICS-GRAPH-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-003: linear_algebraic; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":30,"families":["LINEAR","ALG"],"receipt_hashes":["sha256:61163023459ed332dfc906e44bcb647778587e2db0c85e32b9390334de8675e9","sha256:b0ff81fb4f50d49ba8e11c1e0dc7f414ff7711bf5ba1ec4d70cd9af6cdf7f5fe","sha256:a069bdc5521cb9dabf4978a906631c9f795c332771bfd6ee7f7bad56b0d37e07","sha256:a1111d9fdee741f94a9f57268078c4f8935bdc8fcddec9419a2f858d67418bc9","sha256:1eb50aad30609be2adbf13fd7589e8fdb3013d5a4e0e25292b4c27d2cda0587d","sha256:df388656cff7ee3d4afa27c2c5efb2dad840fb0abfbdebc457eb2eb6f3e22a10","sha256:f9cc18b562f233e53210a5788786195759321f2d9484faa0188a05c46c192de9","sha256:bf412fb7caf2d4fe721ab9d01381a552a490d1b789b8469eb534a287de9da355","sha256:f4dd00fe4be2cb27fec953298d56913b6299c2c17cf34cec00a7fcd1d71ccb33","sha256:78bfa64c5f59b2314c1137839cdb8e8cef9778d056a094837ec553a1c5ddffc2","sha256:9a0648891c36fa2004a4130cf182de9f9aed20f746ed391c30dd8b657c6a2b53","sha256:144d72725402644b443e0674b9a7d36c0ac3759527fb55ae4a664fbf31be1016","sha256:7514349b966d1a39444e5f841909f6e00a777c31d426db7a12a91da6e9013351","sha256:52a0d942eb72cefecec9a693af78189036fe049d1da9a5ab8b2f1c85c61ce74a","sha256:c5e93e05277ffeb172cc3a7003518d9210ad7be5ad8771ba88a6dd397f067b80","sha256:cf47d05b924a577416959ddc022d305c6aa02509046c37239b125f1e0fe13ab4","sha256:beb8f7c7965680020a19391958ee25ec44998ca6b33feb9978f7adfda74a1fe5","sha256:94ea42e4036ab10f3e1eb8cdab2e194e18a1a053c664f1070fbac6a7a7c73152","sha256:4fe452d169159708a9cb65c9f42edfba611f0d86fc6143026a5122b839fba278","sha256:d468d9fb67765ecaa8c7b3f2e1e0404a27e07bbdb5843f943e01c62ac2d4f7b6","sha256:2207ec02691fbc1989e0ae93e324ebb034047996b14277b521774ae634cac4fd","sha256:69176d3a233e9eaa708fbbf5517c30b9cb3b7148f4f6cf12562d35d788b4f6b1","sha256:d1e1bc506263760193b0b99b66e3f7669ec9a2581ab2e4e10bb3f5ac656dd7ee","sha256:bb21a886d6f54b9c232c9e4c6abb37c78d20372f1b3aa8c3a30b7e9795cc5fdf","sha256:2170e5d46c51d5cac7879b84b7466bad4f891683985fa16d920f0a0

f1f4b3a6f","sha256:51014e51baa2db97fcb3c486c521aa813b08e33541a38c791abf465958ecfc3e","sha256:57cbe383561e1219e2ebae0731dba20a8a73b7f2961313cd20ff89c3af3ca2e8","sha256:22db8eceff3c0225e195f7d08ed2011e187177df110bb9220cd2aa76d4337380","sha256:88a76935879afa29f6f647396afa0711a5626baae4107b68e4538a57f03c1a73","sha256:c45a860ee26cca756e9d1120e40d217d8255f5d68495e8cd9b3b275311081f7a"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters

- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7430d76bdc8bfd68090de60bc52fe2fbb4110f0cf3e9284aba83005496db0afa`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:9e4720df6b675b71566ec87018f0dcd242e70bea82377b6b89ad990ff70e9b38`; engine receipt `sha256:923631359a6f87d1ffd3e994ea3f27d4789e01732aa0efaeaeea462bb3407667`.

`SFT-MATH-OBL-VALID-004` - Order and geometry complete validation vector

- **Claim:** `SFT-MATH-VALID-ORDER-GEOMETRY-004` - Order and geometry complete validation vector.
- **Forced law:** Order and geometry complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-004 uniquely retains complete-order-geometry-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-LINEAR-ALGEBRAIC-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-004: order_geometry; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":28,"families":["ORDER","GEOM"],"receipt_hashes":["sha256:62b8a349524faaad77513549e980c4f3d9f3960dfea1aea5df04bd0f7338e4cb","sha256:90737e2dd2e996533513d1d6b2dedb0453101ced5dbf1972fd12672d7eb09ee2","sha256:fa8f185e14ec8d4e621b22db29f13bcf50d9c9a3b1101e434b148b6bc25ac5f3","sha256:ef972980b97f3eae7b7060fcca7ce8f6b2d49f86cabc6f3ddd23852195c7b44d","sha256:c67a3e785599e9dd49003227fef98d588bc99d2a18d7d267e1ea1e7646eacf89","sha256:bab7103a46409d68b5275b3e60e12a6b7457763dbbc8cbf5b18d859d44d652f2","sha256:8c63e67ef5d6b6d785dbfb1bcebe0df202af2908650236db7902c13e7ef0171a","sha256:e68701f93b8bc3b1b2994f1701d31f2a56e3ed49d54306f13f605e7b9cfb2fd2","sha256:94ae3c06d59a7cc4fd1de2f6db35ceb1fad52c343e8622360f94e14f65ec5087","sha256:053130750729ea82310b9c55b17d1d9113d05666aaac6d2639b86863c497b8d5","sha256:d81be77c0466b0b27fc0c703c9e40c284f2052c4663d8ae491bf23f7e1f2e7bc","sha256:c058a971d9c63aea9690da203a2bafd33ff1a85c3b456ed893c0fbc483d1df96","sha256:6e21c474249b5f7416086cffdfa67b0dc82ae3b3b5ec3e085f966a51923489c2","sha256:476f7726658

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- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:64fb185dd5df70143effc664f582e1750e77264524e3d772bcd8bb4353eae4e2`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:8613e2fa16d80d726431f4780abd77f5781eb51fdf26836c0ed85faaefc7692f`; engine receipt `sha256:c2640e5e3f66fe60919039e8c8a2f775c2f913c951f5cec77bc9b8924b5d317f`.

`SFT-MATH-OBL-VALID-005` - Topology and analysis complete validation vector

- **Claim:** `SFT-MATH-VALID-TOPOLOGY-ANALYSIS-005` - Topology and analysis complete validation vector.

- **Forced law:** Topology and analysis complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.

- **Unique survivor:** VALID-005 uniquely retains complete-topology-calculus-analysis-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-ORDER-GEOMETRY-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** VALID-005: topology_analysis; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_an

d_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":42,"families":["TOPO","CALC","ANAL"],"receipt_hashes":["sha256:087501b54410e5b0cc2582c58f56e0ae422ff60ae8e112669ad6257904b828f6","sha256:573c965f1d7c4ca35f06606ef420cbd243824da0a33ef5239ed3eeda9b39bc11","sha256:093d2d6a7de58e145ed538bc8ca98e21197f57236c9895ecc688896aef4f0da1","sha256:15563c18cf9cbc4909fed4801546848b883a4a532e5b92a19d7f865abb66942a","sha256:c5e7396091b5d6f67fbda3a6b709c3c2e0a65a549376de10eb271a3bc52e5f40","sha256:a82e2923c6b54b8bc343741e75582056e2c37a5058376ca947c0c07d15cff42b","sha256:8ab47bcdc0b0a32ffe001651b90b87a33462ee55f203ffa30300664c917ee346","sha256:e9a150b12a6b2c46ae0e816f04f943af98073118526fbdc446d96bf870f1d580","sha256:70093215c70da4bb5253edf8b7a016a6788c5158c670ce96b24413a952a81ff4","sha256:01fa67cf3367f6ba1401e220a59a69e8e81be36b53ae713f95149e6638f3286b","sha256:daa2f48449e9a9b802fa8193d8a7298683680fc55118d1963caa2ecf2037475a","sha256:06b0bd43fcc4ff33c4522a44cd3425d14dab1d0c833e8f5993f553567abf6865","sha256:5896d19c43946aaf7ea803a46d656d83877e3a33bfe835fb25652e7d6b0a0cc0","sha256:cb7338023bc31ad5a784693d1b348ba04c7808a83dd59d02d8bb91602176fe4a","sha256:507dcb047bbe74bb826a5e6ed551568a04827638d7e02341b616ed21b9441a74","sha256:06e93214674f5a4c1ee6bea0c9670845143bb157b4562a64c80028e916a297fd","sha256:64ca281a0dcc6cc417fdc9a401001f9ca15e65053726e2f599a1b8dbfd183d1e","sha256:e6035a82d486ed59b79d120ed341ccaeee17ab11c8e492b828ce58dec5f17579","sha256:0b230f54b347912d403b5661a468156a48a9d79b28c09053b666552ff4980873","sha256:09ae9608a9587eadc2d1fe6741cd1840ccb16bf9d0cb4eb8cedb0ca11b35bf24","sha256:0df975977b5212414b32cf5991f27778c811c0d19b9c4640a377ef2748077376","sha256:9dae94ea1c1bb7f5cc021fc5a5e54daa799b2b96cefbab47d0ce3c089b95e3d3","sha256:cd9c565f4ff93e6039bb09ff0e461c6f3a0ebb1201efc18962274d45d27eb633","sha256:d5e9fa1ad6b35ecd29795a2de9d3edda147d6cb081b7eb4bda68b5734e410bfb","sha256:def0ef022d70876425bc65b54b4bdd196317d3acb4961a76190a959ca54a02dd","sha256:f5917674c7af39fbe08ef5fad2ea7bdd850218b38b72af99fa3d3c75b81b348b","sha256:ab97f2a13f8e91555f3f654b3d1fce419aaca2e71b6964550a971abe5aa5073b","sha256:75b3fde9dbd4c94ec57c3214b75bc6956a83af140ea077502b3613a8f2161cb2","sha256:096b603a26d7963ec79cbd3a02212e6051b0183db3a19933de1a1bfbfe1dbfe7","sha256:9ed9ee5292bcc297c39b1079e4de4067f14d8b23cb72f38fb0be39ae8e78bf14","sha256:b1c09495ffe4a58ae36b3f567a88ed80ddb8ef6a08deba96278aefc38688a0c8","sha256:084a51e26bac975e955191ce1370b9f77e05fa555c1685d6b7b93879e027177c","sha256:07d66c5b288d72dcfae3829b10af6fa129b261c4171007ad7ea387dbe1d56eb2","sha256:0770111ad74222f7d56d86d7798010faa49c9ae065c1b83282848d5792005267","sha256:963056642837c7794301e5fb6787e9a61f3d61eba474eeaae6be0b67ad63449f","sha256:d5504ccb27de0398792565ba73813476b426919a9c01fe1490c799478bbb2946","sha256:e9d6cff64a2340ae44d6c448e4c3f2f21b13b5401fb80abffd6a9c15a3f6f0a7","sha256:be232594554c9502d6329fe8b147c2ad5c13efb6f0aa44a1fb6cce82be5be7ec","sha256:45c9743262487c3681ba1a4d8b290e0a4ccee29cdeae06ef19dfe4e2d66f203a","sha256:beaae3861d939920aa65210f3279a53acd8d0416166a75c5f5caa7758ad0d638","sha256:5d105963b77ef66e5184e30d31c9be4a513602bd2544b90ec59258defa6fad5d","sha256:2b900f3b430de0fcb8c0be9b27c90e8fa9d99b870d3cb197a23db3ad0449a7a5"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters

- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e89899a9e0c3316e71a80dcf06ed83867f7aea2853e7b541c0e0e98c9e173d18`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:fc020818c97bd330afd9fdf9c964761631ef3c0227c9d77d1631dd5576690f86`; engine receipt `sha256:ad81f85b9b3f94edde0402935da86a255c85544a99f4180c07651921d2e471a0`.

`SFT-MATH-OBL-VALID-006` - Equation and measure complete validation vector

- **Claim:** `SFT-MATH-VALID-EQUATION-MEASURE-006` - Equation and measure complete validation vector.
- **Forced law:** Equation and measure complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-006 uniquely retains complete-equation-measure-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-TOPOLOGY-ANALYSIS-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-006: equation_measure; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":22,"families":["EQN","MEAS"],"receipt_hashes":["sha256:4fc9003cf83d90dcf955003cca7001e1547e5d7e1602af161870186ae1fcf66b","sha256:3c2ca60887577a62c86587512cec9143f1cb08293ac1e0267f4a4c5466be5511","sha256:78591458da34ac31e6419e6670802ae5100128e808c08b987cf5b34cc3324e59","sha256:6e4130baf805346be5ab289307002290c2f9b6833686c72a85f987be1e25e51a","sha256:201f70839a8f800c9b62f6fc4cb957e2d3330533c87533ea6d83b19d30027900","sha256:82ade58665afc230884f81bd940a5ef39596e8c5015fbb861185f237547a1adc","sha256:23a0f72d36f5d6653020722360e364cd157fdce162fb6eaf881713a12936432d","sha256:23579f5dbdd465547c8cce50c194a49101b72e8c30cd8dbd336ad962e8d2e802","sha256:9dcd5a3ee4736a56455470116d2c72297f47e59f886ce1950e123bdcf2565225","sha256:4fbb4a7f64ef3694210546883ed3742b90e0a11c995f921e5d3e9269b94522d3","sha256:b5862177e99aea7b7d3f1f03e5d062b99b4072dcfde36163dc8ed7f07544c36f","sha256:a5b9056f828669049f5d02eb1c9493657a229b7da2454f7cea9353b1b7665579","sha256:14d25ad66d068b2319cec859ce7d94be7d66cefda1b4721b2ec5719b124b3e1f","sha256:c2c5612d75dce3a55857b1e506e47f3b86496117aabcc4e83cb8b0b6d5a8aaf0","sha256:0ee9c394f47d86ec4a0cb0e8812e82aaa1df94bf7739b6360a4e0fb257a1a95e","sha256:fc75d7acb7ac063472a62ef61784d3b2464536f8938410cea68282dd69f5936d","sha256:d0b85a891e495ff46031d96111dd3c424206599b9def8efaa67a60bb8929fc04","sha256:22c4fe37ebac4cc637920ff3872ba28f8ed7011a639014be2aaa6760ac71581b","sha256:eaffbc9318e599b402acbdba1cc51cd4dd49fb27e7f009d1d17fb2c09befc06f","sha256:5d84b3dbf7725037c1ec28db61b83139d79504335292b4357a74539f98745017","sha256:f72e9d297fdc79cbd7e4ffa47e81d163a42965e347157648cefef1ea11934a22","sha256:a4f38ba621f479ce435299b06099cc716d53db25ee31cb5307d354573e3c1992"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:ed2c6c88df11caa2005cca4128567bb5c34b6ac577c001de66bed3d547695de6`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:274d444bbd2f9094b1097be69e0d194cd7279e27bf81f859304122bd19d50e8e`; engine receipt `sha256:65d0fbfca9c893839c453cca668a393d086cc8407c5b6b9b6bceda06ed2be95d`.

`SFT-MATH-OBL-VALID-007` - Probability and statistics complete validation vector

- **Claim:** `SFT-MATH-VALID-PROBABILITY-STATISTICS-007` - Probability and statistics complete validation vector.
- **Forced law:** Probability and statistics complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-007 uniquely retains complete-probability-statistics-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-EQUATION-MEASURE-006`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-007: probability_statistics; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":16,"families":["PROB"],"receipt_hashes":["sha256:2f7e63c215ef725afbaed9e153dc0e9afbac2ae30fa90237c27aaa18c463d71e","sha256:3de32cd0f017cd9829917e109c48bf5510084b96451c28f5f1af69c3059ec3d8","sha256:c9f382194e21c74dc4838e29a2e6edf7e17fc470b5a171283dc88a7f94004469","sha256:b87c57beb9072c476180b813248201c2b22f3f7f98112c1e3069194cba7836d3","sha256:c20397f6b4dc7c33323fb18b25e87efcca673139404ef7103796a07479418a9a","sha256:4e219f4e7826997fb21e65bba1e98f924b6ab6e847ef699dea331542e76eba86","sha256:e9c8cd3b6e51c3a20e1640ce3b5815b87771a4f2b94f1d99e6b5aa02878d078c","sha256:1f4ece1dc9dcffc4ab540bcf7cd3dcf29d5bda76255a47b96ec80f3286808977","sha256:b10c05e8a85e8e31388d8b3fd77717a82dbb17564bb8841164daee3def78682f","sha256:f264d01b3cb82e643590415a53b9ea4b9b0bac5e818de89426b4ceba2e0242b2","sha256:21aadee13d2df1e2737ad5c4cd7ba4ee10b6636b3810cf6458450ae67cd060b6","sha256:2a7a74ff1432e3aa098187148f14b63b47977082f22f28848036374dcb971858","sha256:c239b75f04ae9b064f3a6178ed377874ca563585c6e751956e1ea995fcfc0b5d","sha256:686defbd4442ac15c71a3bc349175da32ba506f8cfcec565be22b6813e24d50a","sha256:467057cb14cfa1b348de39f0916d99f257be84b5c7ce10de85fd3932e5fb6e32","sha256:dfe162067f03fbfafd04d35e28dfc9de7e45e7e4fff14c9de674c9e7d3721d40"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:42db8a4d09a5d0afd2c9599c6ba226a1a5594986fe99526aefc7129354dc39a1`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:54a1173aea7ae8d16f6d02a8c8aaad51caf3a337cf5a85ec261e0ba00afb7891`; engine receipt `sha256:c9179fff3bc77f7f13003ce25ed099bf0e6f557d2cbe7606dab7ab66c7dc68fd`.

`SFT-MATH-OBL-VALID-008` - Optimization and dynamics complete validation vector

- **Claim:** `SFT-MATH-VALID-OPTIMIZATION-DYNAMICS-008` - Optimization and dynamics complete validation vector.
- **Forced law:** Optimization and dynamics complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-008 uniquely retains complete-optimization-dynamics-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-PROBABILITY-STATISTICS-007`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** VALID-008: optimization_dynamics; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":28,"families":["OPT","DYN"],"receipt_hashes":["sha256:a13f3cdb8b9906ebb0792c32ddaf516caf949ad77b0f00c032b8782cf7c15cc7","sha256:74b8e6f8afd76a490c34438b20b360709ad22fbe39887630bd2b72e79d8fa4a2","sha256:0d581ec125db13d24f596f7cdbd8f656a244678a63441c9d71c17a24e36a2e52","sha256:6323878558e3eb3fb4d0d8b2726dbda5fdcc0ab9a40aa57d9cd54748cc4e30da","sha256:bc3ef7947c9f9f489114c474b3c03d08c674e9ea91aef48a1e509b0088cf8a22","sha256:e39890906deedccecea769f9e59b8e73d8a82683178f00471f69ce7cb347cc64","sha256:62c27ebb5829a3dad3f7fabc9647f4a7bdccc2bfdb1241574b8c9eb4cecf8625","sha256:fee2c183cf7b4f34029a2606dddf3491a9b3c85ecb6d2f2d44b9afa49446ed91","sha256:94a6507475f5b899b0280eda449cdac26c23fafe729038b5c35eb257b3a1f4c1","sha256:359e651b661d88d22d7602a0249f90c1b82a16d3eb9485f9ed59c3183bed13bf","sha256:df50833d973d90601e8d47ddca56172fc4e914d0e8fa925b1e9121759ee7a738","sha256:92b812df5237a10f034bf74974cb664dd25c0074fdd4c6186c7f802896823f6a","sha256:0f7c3d3f7d1c43f2abf11d7c21970c7f941446520ab41893ef35f49dac67aa29","sha256:03925760f19f50746faf61b93c7b164b58d424f52d585f6bcdc32f87c21e42fb","sha256:28224fd21eff9d40657483e3aceb4019904f841d9786da73c2ec59d114a16a08","sha256:e8e4a46874d1eb0f42b931300512437b8f1015e1de41ebc4c1a1b4a63a9b6080","sha256:aeb9921e8e40d6d59f313577666482b543ef5624f9061cbf22b2f79c176b6633","sha256:9ae62db5ffdd31c07fea2a614bbbe9e5ba587bde5a38daa921fe0ba58c7e9a70","sha256:9c18019e9f20d1d1b810243cd7d08875312668c2bc99c216adeceb8e22f9c664","sha256:2df3a6cbc61017a6eef83e76253575b6bff962865cdc1002c7f77b00d2fba89c","sha256:520a680b3349364ef71665582583a1f3bced6cd9411a4fd11f9de14a8adba730","sha256:8c84d60f21d282875750592142e7f749065029afeac13da0dc9ad38b11ecec93","sha256:d5d7b98dac7aa6e91704f5a82154ce0e5498d0e00fc67b1db7afa2a87e79cbb6","sha256:73d7f19de7ba19d7987c99282739818524e9b29c547ad8b7a21838768e9f6f2a","sha256:7f8777dd8e7f792ca65148f4a8015f93681cdd890936577821113af0b6d8b052","sha256:ffa0ecb0ca362a3a6bd0a2c0033e0033b4addf071ca9c2a1316125debcd19474","sha256:013899ed0283322e5ec156198a2845d00447a602ebe224bbaefc5e984d6cf950","sha256:c525a01d63f6eeb62552b3058ed34808c261032dae7b3553d61adfabf872780e"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters

- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:e7a571170c5d4ccb9b3b7e4b2afa253525cb371ae0b1f9e99ee5b2b9b5329e41`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:e92e83ce3496196e8342021a6e4f0521af2778582176fd7bc37b605e9ce98b26`; engine receipt `sha256:ca131191272d09d2ab262915de7f0b11ee564d86a207e61494b6761cc341426d`.

`SFT-MATH-OBL-VALID-009` - Logic and compositional complete validation vector

- **Claim:** `SFT-MATH-VALID-LOGIC-COMPOSITIONAL-009` - Logic and compositional complete validation vector.

- **Forced law:** Logic and compositional complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.

- **Unique survivor:** VALID-009 uniquely retains complete-logic-compositional-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-OPTIMIZATION-DYNAMICS-008`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-009: logic_compositional; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":28,"families":["LOGIC","CAT"],"receipt_hashes":["sha256:f85f9e844e3ec3f9581c1d9302dbea00e5791f1c4d11df0a41e3ef0bce2bbbf7","sha256:153966835981d27d4115b986d3dfa5519147893171304aa9224ace83a36e2f94","sha256:340d97010a3dea95fd4339a6ad816685c90466027c866d62e45d4760ac2692c7","sha256:e1d02e00afb9f4bf37ce78131e933afe4a71ef346cafef5ac815e8788d6ec4aa","sha256:53083a9fef5c6ab7df19ad3708e04c0c47900087047ac48893d4163c5db51cf1","sha256:c0ebbe707ae0dca10b94d1b69b5122c3e881de668df7bee5bbafef8d9678c03e","sha256:2f79e69c848156a49ac98efd70d86967895b4d2af23b30ea9d35b65f2099d888","sha256:1fc8005aaf5441ebd0362d84d9b1e0a6521f9306dcdfbfc0299abe3f549f4ced","sha256:a93e6a3917e8b56002318a4cb9bdc381988b70aa3431a940655ed4d4d371f794","sha256:66e9c9d47b3364b32c328570ccb83fa0a7f094cc7adac043ccb30c42e25c9799","sha256:ae93deb5f11e14001d83c1afd20e2cd8e90246dd8dc4d9a0c605f4b9877d5823","sha256:be1fa0f7ed9298f961e099db5213007d9c7a556e85756207024813bcea2cee41","sha256:b2428380ee59ac674bd574be852e1168aaa1a392cb49f8e8879f82ae34cb95e5","sha256:571b131341ecd1ad9d2fc2f3b131487f8e060e1a05e51af274793f8d3327a562","sha256:5af25a62e0f550c27d7dde1ff7314806f74aa97c6908f98c8e9753bad2654b76","sha256:5717d448290a1af7fbbd8ad6a85c473c68b28abce0799711809f5538cb14d055","sha256:71c69fc458b52088e7c1dc218452b99d29ef3a4e9634d60c261a7a69cfbb0394","sha256:807ed10ff532e9c607cd05e5b289cae65eaef4fef96cfec772ce1eaf6bd64c89","sha256:45effaf6d8da98e393865ea391af69e936c277b99614dac177e4f0d14d850900","sha256:5dead80b1c237c637d35f6c43379a65812c2117954f77d5e9390cd3806c3e0e1","sha256:7d7a98b46c34532e47b6a37fcd19c87883fe3b009a0fc6059173309712bac821","sha256:66e3d06fc4a7b84928e089f02b4a5e7151461520b6c96327b5fbf9627f2eb11e","sha256:bc77d0449dd3b2b490aff2c45dc27e9552a8798c49bad32cff6dc021dd25b247","sha256:1d46e43677dbab4705723767d4134bba4f59a0c6b60e6d45ae97057bfcfa7c5b","sha256:f68143ffe290ff07df3f325ad9af23893914b6dc1e8b02018f5ee5bf8fa9f8fd","sha256:ff7742d564cdc04c75e692c5642100b378945b08fcdc9d11c0150af30e3a71e3","sha256:73d7dd3a53e88a69cd63d2147afab60c9fce360c065b5f813ffc82b508bdfbfa","sha256:212418e46bfbec20e1f9b404cd43fb40fc13d180de2ab99bc0a8971017fd2197"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:0c5f0d887021455b6ad409e8963a6ed7b65d7cbb0cd7cf501c5b4e4316cdf897`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:7cb228476f9cc1d2429a340419527c66ddf5c9cc8296528dbabd3ad1846f8f13`; engine receipt `sha256:6afbb8bf15ef954c25d276f3b9e654737b824419855ef250a8e6fa7100873657`.

SFT-MATH-OBL-VALID-010 - Numerical and symbolic complete validation vector

- **Claim:** SFT-MATH-VALID-NUMERICAL-SYMBOLIC-010 - Numerical and symbolic complete validation vector.
- **Forced law:** Numerical and symbolic complete validation vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-010 uniquely retains complete-numerical-symbolic-vector, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-LOGIC-COMPOSITIONAL-009`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-010: numerical_symbolic; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"claim_count":22,"families":["NUM","SYMB"],"receipt_hashes":["sha256:8e0f6ddf5bc90fef46eefc32a535fa9fae3d8fc45d3a0f0dd93f657d8c1836c4","sha256:8c13117d2033b4657174a69f0dae9093d27277e74ea89abe10609bb463cc5072","sha256:259b72ccea62db62192f6788cc51d7c5e54c21e36eb8a5099148c0f1b6b98b38","sha256:6929ba74b44f626e8ea7e3b5af82fd3c97e5a4375aa27f1ba050eab41fa2b9be","sha256:d426bb53ef24834bf45d6ff67e99cbcd3d4b1eb55e300e460138fe2907ab1b34","sha256:eb7d30b58833016ecf2928e7ddc354f8d5e428bfe0ef3ca80a56a815d3706757","sha256:cdde0b0bd96c982f20d5f2caa6227c8dcc5a56338f8efd7e8a08fff00df49ecd","sha256:8b2b0f1ba1f2834f192150342fb2c319be37ff34289dbc4150edf24e092d4a0a","sha256:e751920f65b81f5cc2649abdfcd11aa0c1dfae51c10f4d888d4415716d4d2b09","sha256:648b6d3a1e7459deb29c93bf36fbb8f428b743ff0045f05f36cac4f6712a1910","sha256:00a35c5c34967fc4c562146e871ba053dd4af162aef69f61a6122d5222fd789e","sha256:0696ffb60057ca05592a019bc74437c233e398d8e3ed8633d13122917999105d","sha256:1fc6dbd6b91cb7dac6d70badd5ab5bee41c63de626a6abd2a076fb40f3f2a407","sha256:9cf05501f0c8844ef8a38799f148078b8458cc4e42582114034220672d1183cf","sha256:532d8d3fe185773057827ec65c5ede613f674b27e0ffdbe8fdc49a9a419be9ff","sha256:30c7bffc4bfa23e1bb4bbb549fd04f89c9cb1f72be967384b53be228a266d015","sha256:ab851259d84f0c5b760b72589e8b6695d2b3e44efc11dd4b503e839d4647b37f","sha256:8c88175914327cf3786f3f8b238fa3ec9d4b5a872a2c54beafc9f759d7f0732a","sha256:78a40cfd8bef40d4c2979e7fcf85c1ff31f0b49925a73452d2bd3619bbf8e731","sha256:86d0c48ae5b12195f37496629db40327bed840d8bbfe1fea05830f68540f6152","sha256:f30bfc00ab43dc67ac366865b093161f74729273113b2e2a6361b45e80901069","sha256:fda3df9ae9119867ecd3cef93e2affb43dc0dd8596f37043767c6a45b3cf617a"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:7da3bab49ddf336ed36b14627636b331ab577b18363131224b14b29b6dc85990`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:68b41b6655d2bba72b426ad46a3d01e76d07a62cf5246c0b52dc84e4b3ed7b3f`; engine receipt `sha256:2e74a2af4ae929e1b550db41ab49a11f01b5da2a65e5a8ae2c43a6187a70c817`.

`SFT-MATH-OBL-VALID-011` - Adverse absent unresolved and boundary vector

- **Claim:** `SFT-MATH-VALID-ADVERSE-BOUNDARY-011` - Adverse absent unresolved and boundary vector.
- **Forced law:** Adverse absent unresolved and boundary vector is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-011 uniquely retains complete-adverse-boundary-custody, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-NUMERICAL-SYMBOLIC-010`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** VALID-011: adverse_absent_unresolved_boundary; exact observation: {"all_external_statuses":["empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated","empirically_tested_and_independently_replicated"],"all_model_admitted":false,"boundary_identity":"sha256:33fca101a49c6f6dfa14b1e93c3ed4217b9683f274ff27d75c87b53159a235b7","boundary_record_count":305,"claim_count":8,"families":["XINT"],"favorable_adverse_absent_unresolved_and_boundary_rows_preserved":true,"receipt_hashes":["sha256:6ffddc69383435d029cf280ab5d376838c88e0747f4d47ab9bb5db8157b66f68","sha256:3c14d83ea82bd2caaadcdd4e8f83f2d9cfa2ceca3bd433ec8a9036382e43b450","sha256:a8886f024818e308656ead69f4cd4aaab32762944ace0d4e6bc2892a6f8062de","sha256:7d8ff0f49eec26d921585ff950f2f0a0361059515d99ef868dac5147078db211","sha256:ccdcf95d6ac30220d49853eed9c196f15fdc6a4cf578a555811714099d5dc038","sha256:b404e098c202044f424847255966dc5938d54a3ee47c96cb75031d6074c6216f","sha256:26cb9935ef8d7b4deeecc3475b0347694990abe929c4b3dd8d123afa8a1f6a98","sha256:ae8b83e55ba8150ddafe5fc01cae15e7e7d880da1556b1946704ef737060cd80"]}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:9559b5329a75e7081096a98c1fa673f8d9c90d7903c868a9f69c55ba17d37785`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:caeb4736e83f5b3f4d2520fd5aea0ff07dc20d02232f786b6e4b7abccb676eb0`; engine receipt `sha256:e24be1ab2cbf3a5f4a04ffaa42e4149e1976508eb8cbe259f63a4556082f0432`.

`SFT-MATH-OBL-VALID-012` - Mathematics empirical and formal Grand Lock

- **Claim:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012` - Mathematics empirical and formal Grand Lock.
- **Forced law:** Mathematics empirical and formal Grand Lock is the complete exact, disjoint, all-outcomes-preserved receipt reconstruction required by the frozen Mathematics census.
- **Unique survivor:** VALID-012 uniquely retains complete-mathematics-validation-grand-lock, complete disjoint coverage, unchanged authority, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-XINT-ONE-OWNER-IDENTITY-008 -> SFT-MATH-VALID-ADVERSE-BOUNDARY-011`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** VALID-012: empirical_formal_grand_lock; exact observation: {"all_named_groups_and_boundary_rows_present":true,"completed_family_count":22,"covered_pre_validation_claims":305,"frozen_census_count":323,"open_only_valid_and_hand":18,"reconciliation_identity":"sha256:4c8d29e42f67932c4aef3becc5ce63139ea0cb7996f5478e3316a3bc1e2cf486","unique_claim_ids":305,"unique_receipt_hashes":305}; complete family observation custody: 12 records and 305 boundary rows; all preserved; the complete pre-validation Mathematics reconciliation supplies every receipt, outcome class, source and boundary record exactly once; no favorable-only selection or duplicate coverage enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v21.json, census/claims.json, census/execution_manifest.json. Exclusions: no axiom, imported validation conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no favorable, adverse, absent, unresolved, boundary or source row may be omitted; no receipt, family or ownership identity may be duplicated to increase coverage; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:fc61db674aa9004d03826801f5168c1d172615c47289810a3ac988ec9481b43e`; independent `sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c`; empirical `sha256:2347776ab55563fba04933013984d6c718560412e29b88c6151f27b9f2261954`; engine receipt `sha256:822ec7056f4e3dcef00c42fa941fd1fc2f65746649f4f760a119defcda6fd237`.

40.26 Family `HAND` - 6/6 complete

This family contributes 1,536 completely decided candidates, 6 unique survivors and 24 passed controls. Its entries are dependency ordered and separately receipt backed.

`SFT-MATH-OBL-HAND-001` - Mathematics one-owner downstream handoff

- **Claim:** `SFT-MATH-HAND-DOWNSTREAM-ONE-OWNER-001` - Mathematics one-owner downstream handoff.
- **Forced law:** Mathematics one-owner downstream handoff retains one-owner lineage, exact boundary custody, unchanged authority and dated extension-open completion.
- **Unique survivor:** HAND-001 uniquely retains single-owner-downstream-reference, one-owner lineage, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** HAND-001: downstream_one_owner; exact observation: {"downstream_mode":"typed-reference","duplicate_owner":false,"mathematics_owner":"exact-structure-and-proof-relation","ownership_transfer":false}; complete family observation custody: 6 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v22.json, census/mathematics_discipline_obligations.json, docs/branch_roadmaps/README.md. Exclusions: no axiom, imported downstream conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, untyped target, measurement import or silent conventional premise; no completion claim suppresses adverse records or lawful future extension; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:0b1b282196717ecf69cfd7d6c2e9f70231d6f61a1902adfb229ea5fc8540dfb5`; independent `sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`; empirical `sha256:3b228deb54676606191bc9700f23755561a04bbe4d93da94b0b00ca979fa8178`; engine receipt `sha256:d02a4080f51e56bf936273065de8571ce061fb2475ae408e7ea01690d4fa8b1c`.

`SFT-MATH-OBL-HAND-002` - Mathematics measurement-boundary handoff

- **Claim:** `SFT-MATH-HAND-MEASUREMENT-BOUNDARY-002` - Mathematics measurement-boundary handoff.

- **Forced law:** Mathematics measurement-boundary handoff retains one-owner lineage, exact boundary custody, unchanged authority and dated extension-open completion.
- **Unique survivor:** HAND-002 uniquely retains typed-measurement-boundary, one-owner lineage, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012 -> SFT-MATH-HAND-DOWNSTREAM-ONE-OWNER-001`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** HAND-002: measurement_boundary; exact observation: {"boundary_explicit":true,"empirical_branch_owns":["physical-identification","unit-realization","measurement","uncertainty"],"mathematics_owns":"exact-representation"}; complete family observation custody: 6 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v22.json, census/mathematics_discipline_obligations.json, docs/branch_roadmaps/README.md. Exclusions: no axiom, imported downstream conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, untyped target, measurement import or silent conventional premise; no completion claim suppresses adverse records or lawful future extension; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:83c80a6b8c3c810b4811439ba2aaedc022514a77ff12b82daf036bd1e70c8f8c`; independent `sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`; empirical `sha256:053f3e1ef239b7df36bd813793db702112e54bb152d9bfa2bda6b03780886921`; engine receipt `sha256:a962af816adb6cc2256a1f75e91ae932fcdb2ac1bcaade669bb9ea46a008734c`.

`SFT-MATH-OBL-HAND-003` - Mathematics formal-to-empirical handoff

- **Claim:** `SFT-MATH-HAND-FORMAL-EMPIRICAL-003` - Mathematics formal-to-empirical handoff.
- **Forced law:** Mathematics formal-to-empirical handoff retains one-owner lineage, exact boundary custody, unchanged authority and dated extension-open completion.
- **Unique survivor:** HAND-003 uniquely retains sealed-formal-empirical-join, one-owner lineage, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012 -> SFT-MATH-HAND-MEASUREMENT-BOUNDARY-002`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** HAND-003: formal_empirical; exact observation: {"empirical_record":"post-seal-observation-and-comparison","formal_record":"root-bound-derivation-and-certificate","joined_only_by_registered_target":true}; complete family observation custody: 6 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v22.json, census/mathematics_discipline_obligations.json, docs/branch_roadmaps/README.md. Exclusions: no axiom, imported downstream conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, untyped target, measurement import or silent conventional premise; no completion claim suppresses adverse records or lawful future extension; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:52caafa81c9ac28a64e45d751c2ceacb91b474ae6cc59a2ef7766dad340914f4`; independent `sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`; empirical `sha256:60f7ac7d9630d4076d7672a547be461d7dde1655d49363cc55725c03b879d14f`; engine receipt `sha256:72d97f8cb655a93d62def7be0f336521322e6b4e7ebf3d6e67fe6dad8853e069`.

`SFT-MATH-OBL-HAND-004` - Mathematics conventional-correspondence handoff

- **Claim:** `SFT-MATH-HAND-CONVENTIONAL-CORRESPONDENCE-004` - Mathematics conventional-correspondence handoff.
- **Forced law:** Mathematics conventional-correspondence handoff retains one-owner lineage, exact boundary custody, unchanged authority and dated extension-open completion.
- **Unique survivor:** HAND-004 uniquely retains premise-free-correspondence-translation, one-owner lineage, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012 -> SFT-MATH-HAND-FORMAL-EMPIRICAL-003`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** HAND-004: conventional_correspondence; exact observation: {"conventional_term_role":"comparison-and-translation","correspondence_must_preserve_SFT_boundary":true,"premise_role":false}; complete family observation custody: 6 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters
- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v22.json, census/mathematics_discipline_obligations.json, docs/branch_roadmaps/README.md. Exclusions: no axiom, imported downstream conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, untyped target, measurement import or silent conventional premise; no completion claim suppresses adverse records or lawful future extension; no failed route retires an obligation or changes protected authority.
- **Certificates:** derivation `sha256:e23b6813ad2a78dd075e5a55dd46e7d119f7b9d068aa2223b70749a84cd4b1ba`; independent `sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`; empirical `sha256:6f9c7a529ba2e972d93ae71a17c3516415c6210bca26b789704607279e621f43`; engine receipt `sha256:970d5376a5beb8c4d8fdd387c12dda90c72e00671e71ea4ef4ee2cbd4289ad38`.

`SFT-MATH-OBL-HAND-005` - Mathematics open-extension handoff

- **Claim:** `SFT-MATH-HAND-OPEN-EXTENSION-005` - Mathematics open-extension handoff.
- **Forced law:** Mathematics open-extension handoff retains one-owner lineage, exact boundary custody, unchanged authority and dated extension-open completion.
- **Unique survivor:** HAND-005 uniquely retains dated-complete-extension-open, one-owner lineage, root forcing, post-registry observation and no extra rule.
- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.
- **Root lineage:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012 -> SFT-MATH-HAND-CONVENTIONAL-CORRESPONDENCE-004`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.
- **Post-registry observations:** HAND-005: open_extension; exact observation: {"completion":"dated-current-census","lawful_extension_open":true,"new_claim_requires_full_protocol":true,"permanent_lock":false}; complete family observation custody: 6 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters

- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v22.json, census/mathematics_discipline_obligations.json, docs/branch_roadmaps/README.md. Exclusions: no axiom, imported downstream conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, untyped target, measurement import or silent conventional premise; no completion claim suppresses adverse records or lawful future extension; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:e1c29f614c0c1611f5d4b5a815ccf28460124818aeaf46465f2fa2483d2a8963`; independent `sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`; empirical `sha256:d5765b4d25a6f78836fa0c85c92232517a1a9d1aa170f42d8c32b063aa322fea`; engine receipt `sha256:55240c646e44e399889211389bc995a80dd78abc23904c0c6a866ec42d02cd80`.

`SFT-MATH-OBL-HAND-006` **- Mathematics cross-branch one-owner completeness certificate**

- **Claim:** `SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006` - Mathematics cross-branch one-owner completeness certificate.

- **Forced law:** Mathematics cross-branch one-owner completeness certificate retains one-owner lineage, exact boundary custody, unchanged authority and dated extension-open completion.

- **Unique survivor:** HAND-006 uniquely retains complete-cross-branch-handoff-certificate, one-owner lineage, root forcing, post-registry observation and no extra rule.

- **Enumeration and falsification:** 256 candidates, one survivor, 4 controls; closure `depth_independent` and external status `empirically_tested_and_independently_replicated`.

- **Root lineage:** `SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012 -> SFT-MATH-HAND-OPEN-EXTENSION-005`; registered root `SFT-ROOT-THERE-IS-NO-NOTHING`; axioms `0`; free parameters `0`.

- **Post-registry observations:** HAND-006: cross_branch_completeness; exact observation: {"expected_post_handoff_total": 323,"frozen_census_total":323,"pre_handoff_closed":317,"registered_handoff_obligations":6,"single_owner_required":true}; complete family observation custody: 6 records; all rows preserved; complete mathematical-owner and downstream-semantic-owner records supply the observation; no duplicate owner, silent semantic import or untyped cross-branch handoff enters

- **Sources and boundaries:** census/mathematics_discipline_current_reconciliation_v22.json, census/mathematics_discipline_obligations.json, docs/branch_roadmaps/README.md. Exclusions: no axiom, imported downstream conclusion or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicated ownership, untyped target, measurement import or silent conventional premise; no completion claim suppresses adverse records or lawful future extension; no failed route retires an obligation or changes protected authority.

- **Certificates:** derivation `sha256:2eb6acb9f7abc12fd2301e44de97ffd7cd68613473380ef4588b848b88980ac7`; independent `sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`; empirical `sha256:4b5b44ea060ca47d2d585ecad0faf09f4a535adfe81b8c42258cee32f1899369`; engine receipt `sha256:d66b4c2094c3e340398bce5fb89aa88a873dfba5f19c67892cc598184f68709d`.

41. Complete-field empirical interpretation

Mathematics is empirical here in the ordinary sense that its generated consequences are observed through exact executions and independently reproduced records. Where a mathematical structure is later used to describe nature, the owning empirical science must separately identify the physical target, seal the prediction and compare units, values and uncertainty. The Mathematics handoff forbids a formal identity from impersonating a physical measurement.

The Grand Lock's 305 boundary rows include favourable, adverse, absent, unresolved and excluded cases. Their preservation is essential: the reported 100.0% is the fraction of frozen obligations with complete admitted packages, not the fraction of attempted routes that happened to look favourable. A halted attempt earns no admission; a corrected route remains visible through its versioned evidence chain.

42. Reproducibility and publication boundary

The complete branch can be reviewed claim by claim through `census/mathematics_discipline_current_reconciliation_v23.json` and rerun through the repository's documented verification route. The heavy all-branch verification command remains reserved for the final global Grand Lock; this paper was prepared with exact family replay, focused tests, repository validation and both immutable seals. Version 1.5 is locally prepared and is not published until Maria Smith explicitly authorises its GitHub and Zenodo release.

Data and code availability

The open repository is `MettaMazza/ernos-labs-sft-platform`. The live model-admitted index is `census/claims.json`; the Mathematics field boundary is `census/mathematics_discipline_obligations.json`. Every represented claim has its own package under `claims/<claim-id>/` and its immutable model-admitted receipt under `receipts/engine/model_admitted/`.

The publication evidence map is `publications/successors/mathematics/evidence_map_v1_5.json` and the candidate manifest is `publications/successors/mathematics/manifest_v1_5.json`. Registered external records and source captures are retained under `experiments/external_sources/mathematics/` together with their source identities, transport outcomes and hashes. Files too large for ordinary Git remain checksum-bound and are distributed through the applicable existing Zenodo record rather than omitted or replaced.

Build, render, replay and verification programs are retained under `tools/`, with branch implementations under `sft/`, independent reconstructions under `generated/` and tests under `tests/`. The final approved Markdown, rendered PDF, evidence map, manifest, metadata and checksums must agree before deposit. No successful test is treated as empirical confirmation unless the claim-specific evidence protocol separately authorises that status.

Shared claim-record clauses

The clauses below previously appeared verbatim in multiple claim records. They are preserved once here to reduce mechanical repetition. Each in-text clause reference applies this exact wording at that claim location; no scientific statement, status, condition or qualification has been removed.

`MATH-S001` - applied at 22 claim locations

Exactly one candidate combines every forced coordinate. Replacing any of those coordinates by its generated alternative triggers the corresponding rejection in the table. Adding a rule fails the final addition coordinate. This establishes minimality inside the declared grammar and named-shape uniqueness over the complete product. The survivor entails the following operational laws:

`MATH-S002` - applied at 22 claim locations

The finite product execution classifies representation rules; this base/successor certificate establishes that the forced rule continues at every generated finite size or depth. It does not posit a completed infinity.

References

1. Smith M. *From Nothing to Fold: A Premise-Free, Parameter-Free and Machine-Closed Foundation for Smithian Fold Theory*. Ernos Labs Foundation Branch Paper 001, version 1.1. 2026. doi:10.5281/zenodo.21535636.
2. Smith M. *Smithian Fold Theory Scientific Constitution*. V3 clean-room repository, `CONSTITUTION.md`, 2026.
3. Ernos Labs. *The Single SFT Admission Engine*. V3 clean-room repository, `docs/ENGINE_AUTHORITY.md`, 2026.
4. Ernos Labs. *Comprehensive Branch-Paper Protocol*. V3 clean-room repository, `publications/BRANCH_PAPER_PROTOCOL.md`, 2026.
5. Ernos Labs. *Mathematics Frozen Current-Knowledge Inventory*. V3 clean-room repository, `publications/inventories/mathematics.json`, 2026.
6. Lundh A, Lexchin J, Mintzes B, Schroll JB, Bero L. Industry sponsorship and research outcome. *Intensive Care Medicine*. 2018. doi:10.1007/s00134-018-5293-7.

7. Fabbri A, Lai A, Grundy Q, Bero LA. The influence of industry sponsorship on the research agenda. *American Journal of Public Health*. 2018. PMID:30252531.

8. Gallo SA et al. Reliability and fairness in peer review of research funding. 2023. https://pmc.ncbi.nlm.nih.gov/articles/PMC10553257/.

9. Demicheli V, Di Pietrantonj C. Peer review for improving the quality of grant applications. *Cochrane Database of Systematic Reviews*. https://pmc.ncbi.nlm.nih.gov/articles/PMC8973940/.

10. *Toward more published null and negative results*. *Nature Communications*. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12459790/.

11. UNESCO. *Recommendation on Open Science*. 2021. https://www.unesco.org/en/legal-affairs/recommendation-open-science.

12. National Institute of Standards and Technology. *AI Risk Management Framework 1.0*. https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics/.

13. Heil BJ et al. Reproducibility standards for machine learning in the life sciences. *Nature Methods*. 2021. doi:10.1038/s41592-021-01256-7.

Complete current-census successor supplement

The authoritative current scope of this paper is 341 live model-admitted claims (`mathematics` 341). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 18 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

SFT reconstruction of sharp Cohn-Elkies sphere-packing asymptotics

Claim ID: `SFT-MATH-OAI26-SPHERE-PACKING-001`

Exact statement: The ten-field SharpFullCohnElkiesManuscriptConclusions record holds with every Real replaced by an exact-real name, every Tendsto by its modulus-and-enclosure definition, every eventually-atTop by a generated threshold plus successor proof, and every FullAdmissible/fullQuotient/fullLinearProgram object by its total generated SFT encoding; all ten fields and their original conjunction are retained.

Dependency route: `SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015` -> `SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: packing supports x Fourier-support certificates x exact enclosure refinements x generated-dimension successor traces. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: The ten-field SharpFullCohnElkiesManuscriptConclusions record holds with every Real replaced by an exact-real name, every Tendsto by its modulus-and-enclosure definition, every eventually-atTop by a generated threshold plus successor proof, and every FullAdmissible/fullQuotient/fullLinearProgram object by its total generated SFT encoding; all ten fields and their original conjunction are retained.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:d038ac064cfb1a7e52019d2bfe69de45c7702f84b34f3977cc2287e6c679d2c5` at `receipts/engine/model_admitted/SFT-MATH-OAI26-SPHERE-PACKING-001-d038ac064cfb1a7e.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of the strict binary-code MRRW improvement

Claim ID: `SFT-MATH-OAI26-BINARY-CODE-MRRW-002`

Exact statement: For every admissible exact-real name d, if d is strictly positive and strictly below one-of-two, then the generated-enclosure value of Hamming binaryRate d is strictly below the generated-enclosure value of mrrwRate d, with a positive rational separation certificate.

Dependency route: `SFT-MATH-COMB-CODING-PACKING-010` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: complete finite binary-code censuses x exact distance names x rate enclosures x length-successor certificates. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: For every admissible exact-real name d, if d is strictly positive and strictly below one-of-two, then the generated-enclosure value of Hamming binaryRate d is strictly below the generated-enclosure value of mrrwRate d, with a positive rational separation certificate.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:0a26f43c604f2edea3d3ede330f37404300450352e822361a7bdc2b14d239cba` at `receipts/engine/model_admitted/SFT-MATH-OAI26-BINARY-CODE-MRRW-002-0a26f43c604f2ede.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of the strict spherical-code hierarchy

Claim ID: `SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-003`

Exact statement: For every admissible exact-real name s strictly between structural absence and the One, every generated level r satisfies both strict successor-level inequalities, and the source conjunction linking sphericalCodeRate, localizedHierarchyRate, localizedLevelRate one, localizedRowRate, localizedLevelRate base and classicalLocalizedRate holds with exact enclosure/separation certificates.

Dependency route: `SFT-MATH-COMB-CODING-PACKING-010` -> `SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: spherical-code supports x packing certificates x hierarchy-level successor traces x exact rate enclosures. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: For every admissible exact-real name s strictly between structural absence and the One, every generated level r satisfies both strict successor-level inequalities, and the source conjunction linking sphericalCodeRate, localizedHierarchyRate, localizedLevelRate one, localizedRowRate, localizedLevelRate base and classicalLocalizedRate holds with exact enclosure/separation certificates.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:489951532200f54be0fd25447059d0425e9e38e8a88cd01acd6415b205982361` at `receipts/engine/model_admitted/SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-003-489951532200f54b.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of a finitely presented non-sofic group

Claim ID: `SFT-MATH-OAI26-NONSOFIC-GROUP-004`

Exact statement: There exists an admissible generated group description G with a total group-law certificate and a finite presentation certificate such that the exact SFT translation of SoficGroups.Sofic G is false; the negation must exhaust the complete finite-test/permutation-approximation witness grammar rather than infer from carrier rejection.

Dependency route: `SFT-MATH-ALG-GROUP-HELD-INVERSE-004` -> `SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005` -> `SFT-MATH-LOGIC-MODEL-INTERPRETATION-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: finite group presentations x generated word carriers x finite test sets x permutation actions x exact error/separation enclosures. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: There exists an admissible generated group description G with a total group-law certificate and a finite presentation certificate such that the exact SFT translation of SoficGroups.Sofic G is false; the negation must exhaust the complete finite-test/permutation-approximation witness grammar rather than infer from carrier rejection.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:667047c373ae5659fad4242adb84fa7e3dcf54ab6335b9dd9cd86a3511dc1a70` at `receipts/engine/model_admitted/SFT-MATH-OAI26-NONSOFIC-GROUP-004-667047c373ae5659.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of the Connes-rigidity counterexample family

Claim ID: `SFT-MATH-OAI26-CONNES-RIGIDITY-005`

Exact statement: There exist a generated countable-discrete-group description Lambda and a successor generator Gamma for group descriptions satisfying every source FG, ICC, property-(T), tracial-factor-isomorphism and group-nonisomorphism conjunct, with each universal natural quantifier carried by base-and-successor certificates and every operator/factor relation carried by exact generated support.

Dependency route: `SFT-MATH-ALG-GROUP-HELD-INVERSE-004` -> `SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013` -> `SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008` -> `SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: generated group presentations x representation actions x operator supports x finite-support traces x family-successor certificates. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: There exist a generated countable-discrete-group description Lambda and a successor generator Gamma for group descriptions satisfying every source FG, ICC, property-(T), tracial-factor-isomorphism and group-nonisomorphism conjunct, with each universal natural quantifier carried by base-and-successor certificates and every operator/factor relation carried by exact generated support.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:420c1f5775a9fa6e531150cbc77b63211d989b0e2a89ea75d6dfe98f1f57ce1b` at `receipts/engine/model_admitted/SFT-MATH-OAI26-CONNES-RIGIDITY-005-420c1f5775a9fa6e.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of the sharp Ehrhart volume inequality

Claim ID: `SFT-MATH-OAI26-EHRHART-VOLUME-006`

Exact statement: For every positive generated dimension and every admissible exact generated encoding of a real-space set with certificates for the source convexity, compactness, nonempty interior, centered barycenter and unique interior lattice point hypotheses, the exact normalized-volume enclosure is at most (n plus One)^n divided by n factorial.

Dependency route: `SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005` -> `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` -> `SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005` -> `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: generated convex-body descriptions x lattice-point support x barycenter certificates x exact volume enclosures x dimension-successor traces. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: For every positive generated dimension and every admissible exact generated encoding of a real-space set with certificates for the source convexity, compactness, nonempty interior, centered barycenter and unique interior lattice point hypotheses, the exact normalized-volume enclosure is at most (n plus One)^n divided by n factorial.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:ef2c2d736a04e9302262e688320740a2fd9f0bfe9df7a6cda695fd0ff742235d` at `receipts/engine/model_admitted/SFT-MATH-OAI26-EHRHART-VOLUME-006-ef2c2d736a04e930.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation

Control	Passed	Expected	Observed
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of Erdos problem 183

Claim ID: `SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-007`

Exact statement: The original two-conjunct proposition holds: every generated colour count k at least two satisfies the full explicit lower bound through exact exp/log/fractional-power enclosures, and a generated threshold/modulus plus successor proof establishes divergence of the k-th-root sequence beyond every supplied exact bound.

Dependency route: `SFT-MATH-COMB-RAMSEY-FORCING-011` -> `SFT-MATH-COMB-CODING-PACKING-010` -> `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: complete multicolour edge-colouring supports x separated palette codes x Ramsey witnesses x colour-successor induction x divergence modulus. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: The original two-conjunct proposition holds: every generated colour count k at least two satisfies the full explicit lower bound through exact exp/log/fractional-power enclosures, and a generated threshold/modulus plus successor proof establishes divergence of the k-th-root sequence beyond every supplied exact bound.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:c6f32c5a7454c8c8c7c80f0ecf971e3772e67b5dba71f59fd2a81f8b9a8b7158` at `receipts/engine/model_admitted/SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-007-c6f32c5a7454c8c8.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of the quantitative compactness counterexample

Claim ID: `SFT-MATH-OAI26-COMPACTNESS-008`

Exact statement: There exist one complete finite generated forbidden-graph family and positive exact-real names c and C satisfying every source geometry, member-lower, all-host, all-size, exponent, noncompactness and conjecture-negation conjunct; both negations require actual predicate refutations under their exact translations.

Dependency route: `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007` -> `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` -> `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: finite forbidden-family supports x complete host graphs x free/embedding relations x extremal counts x size-successor certificates. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: There exist one complete finite generated forbidden-graph family and positive exact-real names c and C satisfying every source geometry, member-lower, all-host, all-size, exponent, noncompactness and conjecture-negation conjunct; both negations require actual predicate refutations under their exact translations.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:57893f72d65344e975edf2cc21c10796cf37d65a77cfc7f83713fbb13c69bfc8` at `receipts/engine/model_admitted/SFT-MATH-OAI26-COMPACTNESS-008-57893f72d65344e9.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of the two-degenerate extremal counterexample

Claim ID: `SFT-MATH-OAI26-TWO-DEGENERATE-009`

Exact statement: There exist a generated vertex count q and complete finite graph H satisfying the exact connected, bipartite, two-degenerate and every-two-colouring degree conjuncts, together with positive exact-real names c and epsilon and a generated threshold plus successor proof for the complete extremal-number lower bound.

Dependency route: `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007` -> `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` -> `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: finite graph supports x two-colouring census x degree/degeneracy witnesses x extremal host traces x size-successor threshold certificate. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: There exist a generated vertex count q and complete finite graph H satisfying the exact connected, bipartite, two-degenerate and every-two-colouring degree conjuncts, together with positive exact-real names c and epsilon and a generated threshold plus successor proof for the complete extremal-number lower bound.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:cb9465fb16f9cbfeadabcf7672b192409fe6111d0ba4473ebabb484a0c797975` at `receipts/engine/model_admitted/SFT-MATH-OAI26-TWO-DEGENERATE-009-cb9465fb16f9cbfe.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT disproof of source-artifact validity: PackingBounds.sharpFullCohnElkiesManuscriptConclusions

Claim ID: `SFT-MATH-OAI26-SPHERE-PACKING-VALIDITY-001`

Exact statement: Not SFTValid(exact frozen artifact PackingBounds.sharpFullCohnElkiesManuscriptConclusions): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires completed real-valued dimension limits and completed error functions.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-FOUNDATION-COUNT-001` -> `SFT-FOUNDATION-EXACT-OPERATIONS-001` -> `SFT-MATH-LIMIT-CONTINUUM-002` -> `SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact PackingBounds.sharpFullCohnElkiesManuscriptConclusions)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact PackingBounds.sharpFullCohnElkiesManuscriptConclusions). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:3ff1d6fda918860980cb2ed0b5c9b1cd96b4abfe706e9f5f5e88923771fefe67` at `receipts/engine/model_admitted/SFT-MATH-OAI26-SPHERE-PACKING-VALIDITY-001-3ff1d6fda9188609.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor

Control	Passed	Expected	Observed
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: MetricCodes.Johnson.binaryRate_lt_mrrw

Claim ID: `SFT-MATH-OAI26-BINARY-CODE-MRRW-VALIDITY-002`

Exact statement: Not SFTValid(exact frozen artifact MetricCodes.Johnson.binaryRate_lt_mrrw): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires completed real asymptotic rates defined through limsup, infimum, roots and logarithms.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-COMB-CODING-PACKING-010` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact MetricCodes.Johnson.binaryRate_lt_mrrw)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact MetricCodes.Johnson.binaryRate_lt_mrrw). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:f0d0f3e0e8e774d38c08c84e45268f5596dd6a47482d09a85d3be9232e1e52a8` at `receipts/engine/model_admitted/SFT-MATH-OAI26-BINARY-CODE-MRRW-VALIDITY-002-f0d0f3e0e8e774d3.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: MetricCodes.Spherical.HigherHierarchy.strict_hierarchy

Claim ID: `SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-VALIDITY-003`

Exact statement: Not SFTValid(exact frozen artifact MetricCodes.Spherical.HigherHierarchy.strict_hierarchy): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires an all-level hierarchy of completed real rate infima over an unbounded natural index.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-COMB-CODING-PACKING-010` -> `SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact MetricCodes.Spherical.HigherHierarchy.strict_hierarchy)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact MetricCodes.Spherical.HigherHierarchy.strict_hierarchy). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:a7848879142ae8da5b92d9d08fc66aa47b8267ca50e43758fcef5c60989bc6b7` at `receipts/engine/model_admitted/SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-VALIDITY-003-a7848879142ae8da.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group

Claim ID: `SFT-MATH-OAI26-NONSOFIC-GROUP-VALIDITY-004`

Exact statement: Not SFTValid(exact frozen artifact SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires an existential group carrier that is finitely presented and not sofic.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-ALG-GROUP-HELD-INVERSE-004` -> `SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005` -> `SFT-MATH-LOGIC-MODEL-INTERPRETATION-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:c3b8d1629e4dd819f4fdd118365d352065c405865c36d764684f53d9125954ac` at `receipts/engine/model_admitted/SFT-MATH-OAI26-NONSOFIC-GROUP-VALIDITY-004-c3b8d1629e4dd819.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors

Claim ID: `SFT-MATH-OAI26-CONNES-RIGIDITY-VALIDITY-005`

Exact statement: Not SFTValid(exact frozen artifact ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires infinite groups, an infinite indexed family, infinite conjugacy classes and completed operator factors.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-ALG-GROUP-HELD-INVERSE-004` -> `SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013` -> `SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008` -> `SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:bdccf181d2c78cc48f0b466f558ac91e6ca847db93fb85d161f9e6a6eb5b020c` at `receipts/engine/model_admitted/SFT-MATH-OAI26-CONNES-RIGIDITY-VALIDITY-005-bdccf181d2c78cc4.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: Ehrhart.Volume.ehrhart_volume_inequality_for_sets

Claim ID: `SFT-MATH-OAI26-EHRHART-VOLUME-VALIDITY-006`

Exact statement: Not SFTValid(exact frozen artifact Ehrhart.Volume.ehrhart_volume_inequality_for_sets): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires arbitrary subsets of a completed real space with topological interior, compactness and continuum volume.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-FOUNDATION-EXACT-OPERATIONS-001` -> `SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005` -> `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` -> `SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005` -> `SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact Ehrhart.Volume.ehrhart_volume_inequality_for_sets)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact Ehrhart.Volume.ehrhart_volume_inequality_for_sets). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:3181c693e4a4584bf741f212c3cda8a258aa01619ca6cba724a45d2682f8f27c` at `receipts/engine/model_admitted/SFT-MATH-OAI26-EHRHART-VOLUME-VALIDITY-006-3181c693e4a4584b.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit

Claim ID: `SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-VALIDITY-007`

Exact statement: Not SFTValid(exact frozen artifact ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires a completed Tendsto-atTop conjunct and all-colour real exp/log/fractional-power bounds.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-COMB-RAMSEY-FORCING-011` -> `SFT-MATH-COMB-CODING-PACKING-010` -> `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` -> `SFT-MATH-LIMIT-CONTINUUM-002`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:6ed37c034b19f0820f7fa5507967498458f95964b8cfa4923caf629a0930f18c` at `receipts/engine/model_admitted/SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-VALIDITY-007-6ed37c034b19f082.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: CompactnessConjecture.quantitativeCompactnessCounterexample

Claim ID: `SFT-MATH-OAI26-COMPACTNESS-VALIDITY-008`

Exact statement: Not SFTValid(exact frozen artifact CompactnessConjecture.quantitativeCompactnessCounterexample): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires eventually-atTop real lower bounds, all-size fractional powers and a completed compactness predicate.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007` -> `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` -> `SFT-MATH-LIMIT-CONTINUUM-002` -> `SFT-FOUNDATION-EXACT-OPERATIONS-001`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact CompactnessConjecture.quantitativeCompactnessCounterexample)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact CompactnessConjecture.quantitativeCompactnessCounterexample). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:5daf6c9b58d67ccaa5a4f3ad10799aa287fdde386b0475c1d2da4a55652968dc` at `receipts/engine/model_admitted/SFT-MATH-OAI26-COMPACTNESS-VALIDITY-008-5daf6c9b58d67cca.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: TwoDegenerateGraphs.twoDegenerateExtremalCounterexample

Claim ID: `SFT-MATH-OAI26-TWO-DEGENERATE-VALIDITY-009`

Exact statement: Not SFTValid(exact frozen artifact TwoDegenerateGraphs.twoDegenerateExtremalCounterexample): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires positive completed real constants and an eventually-atTop lower bound with a real fractional exponent.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007` -> `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006` -> `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007` -> `SFT-MATH-LIMIT-CONTINUUM-002` -> `SFT-FOUNDATION-EXACT-OPERATIONS-001`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact TwoDegenerateGraphs.twoDegenerateExtremalCounterexample)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact TwoDegenerateGraphs.twoDegenerateExtremalCounterexample). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:e1fb5ef1d55728a08a8e27c19b1295c9c315bf8641d77fbac65d2e60c175acba` at `receipts/engine/model_admitted/SFT-MATH-OAI26-TWO-DEGENERATE-VALIDITY-009-e1fb5ef1d55728a0.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.5.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports the complete-field Mathematics branch of the third clean-room reconstruction of Smithian Fold Theory (SFT), complete to the frozen dated census and explicitly open to lawful extension. The census contains **323 obligations in 24 dependency-ordered families**. Every obligation now has an untouched-engine model-admission receipt, an exact candidate census, one unique survivor, four adverse controls, a depth-independent finite-successor or explicit boundary certificate, an implementation-distinct reconstruction, and post-registry observation custody. Together the branch preserves **97,280 generated candidates, 323 unique survivors and 1,292 passed controls**. The final reconciliation identity is `sha256:526c6d78a3ff77c21c6d4dd71c266499f5b234a7763cb223e5d009fa9f101c64`.

The field reconstruction covers exact arithmetic and number structure; algebraic extensions; combinatorics; graphs, networks and matroids; linear, multilinear and tensor structure; algebraic systems; order, lattices and domains; geometry; topology; calculus correspondence; analysis correspondence; equation structures; measure and integration; deterministic-support probability and statistics; optimisation and operations research; dynamical systems; logic and foundations; category, type and compositional structures; numerical mathematics; symbolic and constructive mathematics; cross-disciplinary interfaces; the complete validation vector; and one-owner handoffs. The previously admitted Smithian Fold Scientific Calculator remains the accessible translation surface and does not become an alternate admission route.

The proof domain admits no conventional axiom as an SFT premise, no fitted or free parameter, no semantic numerical zero, negative proof magnitude, irrational or imaginary proof scalar, floating proof quantity, completed infinity, ungenerated continuum or ontic randomness. Displayed `0` denotes structural absence; opposition is typed orientation; non-rational conventional objects enter only through exact constructions or certified enclosures. The Validation Grand Lock partitions all 305 pre-validation receipts exactly once across 22 families and preserves all 305 favourable, adverse, absent, unresolved and boundary records. The final six handoffs bring the branch to 323/323 without transferring empirical ownership to Mathematics or claiming permanent closure.

Version 1.5 preserves the complete version 1.4 paper and its foundational, prior-corpus and calculator derivations, then executes the roadmap that version 1.4 registered. No application selected these laws. No engine or protected-verifier source was modified. Every later extension must enter as a new registered question and pass the same public protocol.

Keywords: Smithian Fold Theory; complete-field mathematics; exact arithmetic; algebra; combinatorics; graph theory; geometry; topology; calculus; analysis; probability; optimisation; dynamics; logic; category theory; numerical mathematics; symbolic mathematics; scientific calculator; computational proof; open science.

Scope and ownership boundary

This paper owns the `mathematics` surface comprising 341 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Successor machine-identity appendix

- Preceding source: `publications/successors/mathematics/FROM_FOLD_TO_MATHEMATICS_PAPER_001_V1_5.md`
- Preceding source SHA-256: `sha256:95dba6282d60c6b99b7298498f92352c758fafb080ed723b62b30227f66d6202`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256: `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification-authority seal: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`
- Remote actions performed: `false`

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **341** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.6.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 341 current claim(s) within `mathematics` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.